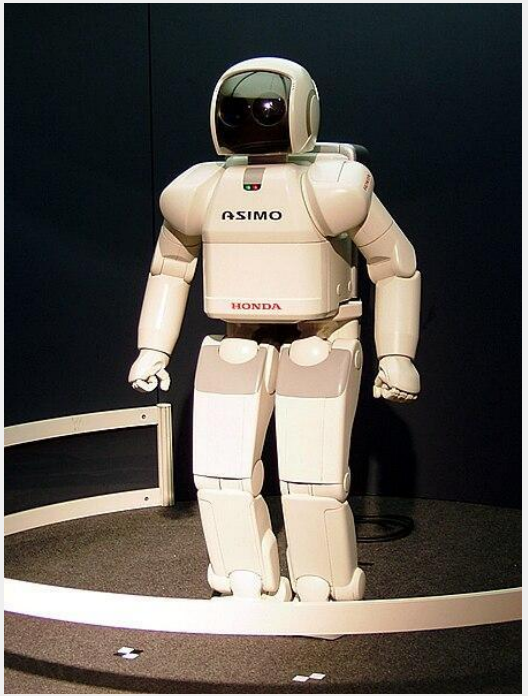
History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 11 April 11, 2025

GOD COMPLEX APRIL 2025 ISSUE

Christianity Was "Borderline Illegal" in Silicon Valley. Now It's the New Religion

Believers like Garry Tan are flipping the script in th
capital world, making faith matter just as much as
to turn a fortune. Says one entrepreneur, "There a
that are leveraging Christianity to get closer to Pe

BY ZOË BERNARD

MARCH 20, 2025

LAUREN GOODE BUSINESS MAR 14, 2025 6:30 AM

The Silicon Valley Christians Who Want to Build 'Heaven on Earth'

Is work religion, or is religion work? Both.

TECH / META

Meta got caught gaming AI benchmarks

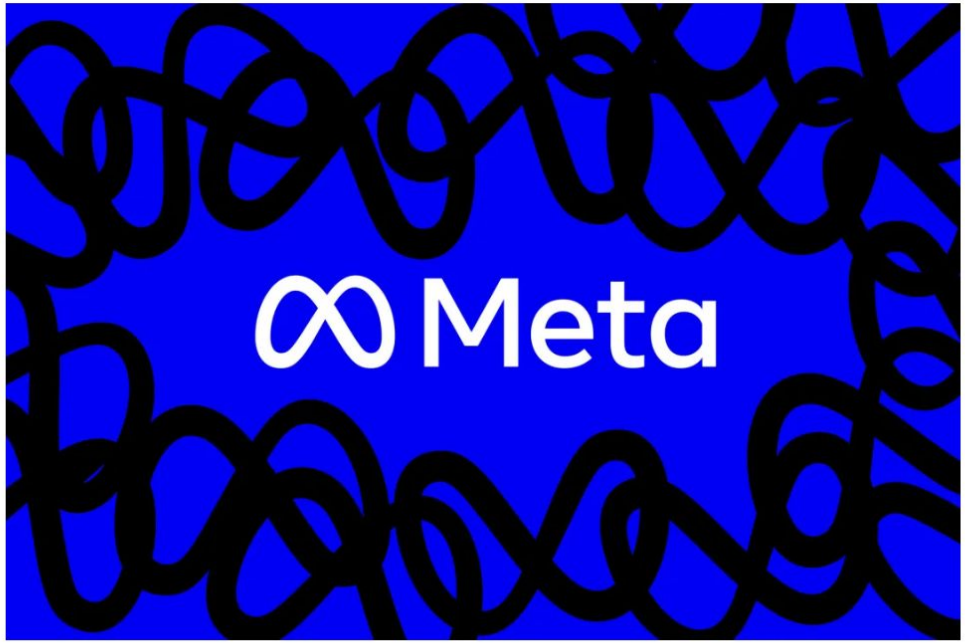


Illustration by Nick Barclay / The Verge

/ With Llama 4, Meta fudged benchmarks to appear as though its new AI model is better than the competition.

> "Meta should have made it clearer that Llama-4-Maverick-03-26-Experimental was a customized model to optimize for human preference,"

by **Kylie Robison**
Apr 7, 2025, 9:32 PM EDT

40 Comments (40 New)

🏆 Chatbot Arena LLM Leaderboard: Community-driven Evaluation for Best LLM and AI chatbots

Discord | Twitter | 小红书 | Blog | GitHub | Paper | Dataset | Kaggle Competition

Chatbot Arena is an open platform for crowdsourced AI benchmarking, developed by researchers at UC Berkeley SkyLab and LMArena. With over 1,000,000 user votes, the platform ranks best LLM and AI chatbots using the Bradley-Terry model to generate live leaderboards. For technical details, check out our paper.

Chatbot Arena thrives on community engagement — cast your vote to help improve AI evaluation!

New Launch! Chat with online LLMs in the 🌐 Search Arena tab!

Total #models: **222**. Total #votes: **2,838,248**. Last updated: 2025-04-02.

Code to recreate leaderboard tables and plots in this notebook. You can contribute your vote at lmarena.ai!

Category: Overall

Apply filter: Style Control | Show Deprecated

Overall Questions

#models: 222 (100%) #votes: 2,838,248 (100%)

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-2.5-Pro-Exp-03-25	1439	+7/-10	5858	Google	Proprietary
2	5	Llama-4-Maverick-03-26-Experimental	1417	+13/-12	2520	Meta	N/A
2	1	ChatGPT-4o-latest (2025-03-26)	1410	+8/-10	4899	OpenAI	Proprietary
2	4	Grok-3-Preview-02-24	1403	+6/-6	12391	xAI	Proprietary
3	2	GPT-4.5-Preview	1398	+5/-7	12312	OpenAI	Proprietary
6	7	Gemini-2.0-Flash-Thinking-Exp-01-21	1380	+4/-4	24298	Google	Proprietary
6	4	Gemini-2.0-Pro-Exp-02-05	1380	+4/-4	20289	Google	Proprietary

Midterm Examples!

Tesla's vision-based approach to self-driving cars marked a significant shift away from traditional AI development paths shaped primarily by military-funded research. Historically, artificial intelligence, particularly in autonomous systems, heavily relied on sensor fusion techniques incorporating radar, LiDAR, and other expensive sensors initially pioneered and funded through defense initiatives (Thrun et al., 2006). Tesla's decision to pivot exclusively to camera-based systems represented both a technical and philosophical departure, highlighting crucial economic, cultural, and technological shifts in the commercialization of AI in the 21st century.

NeoFace is not a neutral technology. It operates within, and reshapes, the cultural landscapes in which it is embedded. In Japan, it is embraced as a tool of order and innovation; in the United States, it is often feared as a mechanism of systemic injustice. Across both contexts, NeoFace affects not just how people are watched, but how they understand their relationship to public space, the state, and each other. As facial recognition technologies continue to expand globally, understanding their specific cultural and social implications becomes not only relevant—but urgent.

Cultural Norms and Public Space: Who Owns the Street? –

The differing reactions in Tokyo and San Francisco reveal distinct cultural understandings of public space and privacy. In Japan, walking through a train station or city square while being scanned is viewed as a minimal concession for maintaining national order and cleanliness. However, in many American cities, public space is imagined as a domain of individual freedom, not collective discipline.

challenging the purported objectivity of the dataset. This paper interrogates a critical yet initially overlooked question: How does ImageNet's categorization framework perpetuate historically entrenched Western-centric power imbalances, and what are the societal consequences of this? This analysis contends that the very structure of ImageNet's ontology—including the categories it legitimizes, the hierarchical order it imposes, and the labels it assigns—reflects and reproduces ideological frameworks derived explicitly from Western traditions of taxonomy and social hierarchy. As a result, ImageNet's Western-centric conceptualization exerts substantial societal influence, as AI systems trained on this dataset risk entrenching harmful stereotypes, marginalizing underrepresented communities, and proliferating a distinctly Western gaze on a global scale.

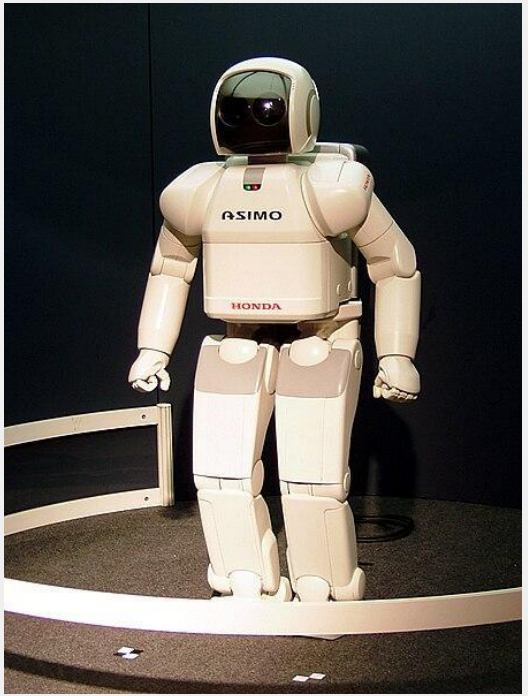
Why is the use of Generative AI in film production more controversial than other digital tools like CGI or Autotune? Is this apprehension rational or is it inflated?

The discomfort surrounding AI-assisted film production primarily stems from how it disrupts human autonomy and authorship, challenging established norms about the value of creative labor. Yet, this fear might be exaggerated for two reasons: First, society lacks a consensus on what constitutes true artistry, often conflating artistic value with human effort. Second, people tend to overestimate the uniqueness of human creativity, partly because they misunderstand what makes certain tasks difficult for AI.

Overview of Class Thus Far!

History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 2 January 31, 2025

Since when, AI? + What is AI anyway?

Early Modern

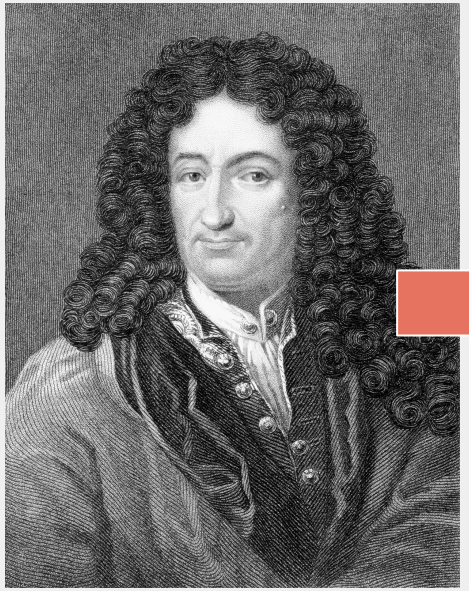
Modern

Large-Scale AI?

AI History

Since when, AI? + What is AI anyway?

Early Modern



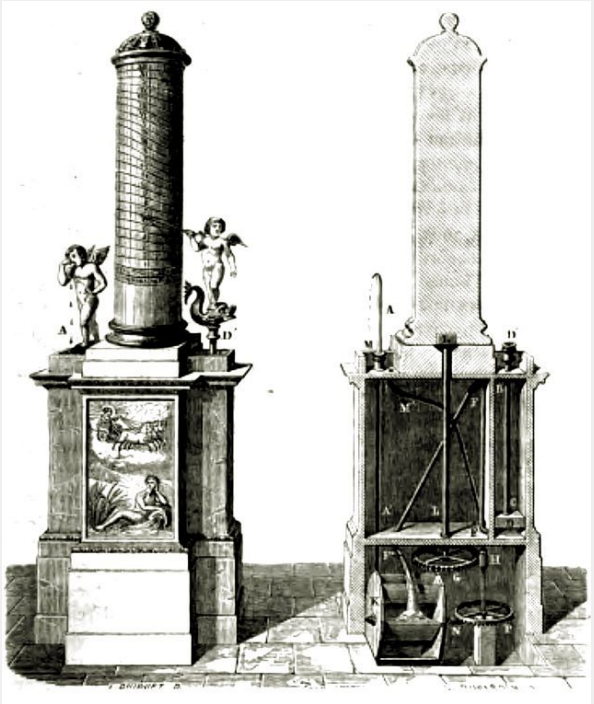
Modern

Large-Scale AI?

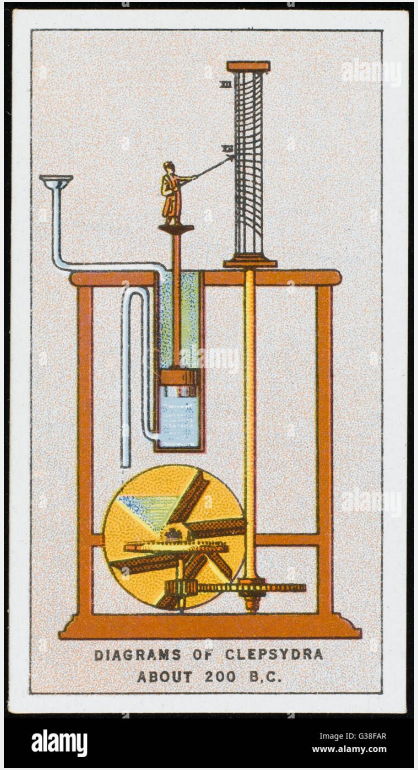
- What is automation?
- Why automation?
- Living things vs. Mechanical things
- Science & Faith

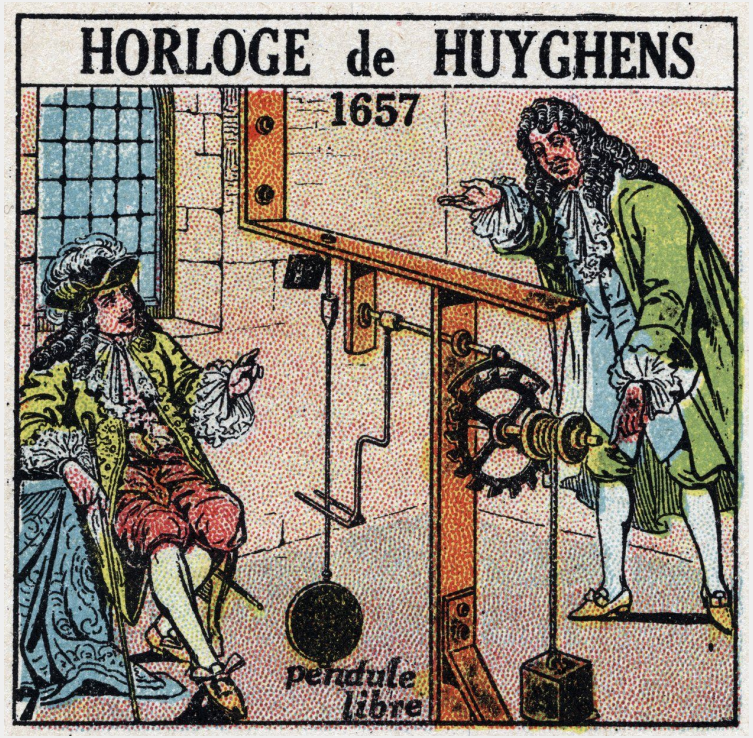
Ancient AI

How does this differ from the pendulum clock?



Ktseibios of Alexandria's Clepsydra. 250 BCE





1657 Pendulum Clock, Christian Huygens

Early Modern

16thC Protestant Reformation & Challenge the Pope
Mind-Body Separation

- Material entities have no agency (passive)
- Reformers asserted distinction between matter and spirit, setting their clockmaker God apart from his clockwork
- Only god has agency vs. medieval Catholic God had had presided over a cosmos permeated by spirit.

Agency

Organicist/Naturalist:

Nature = Agency = Life

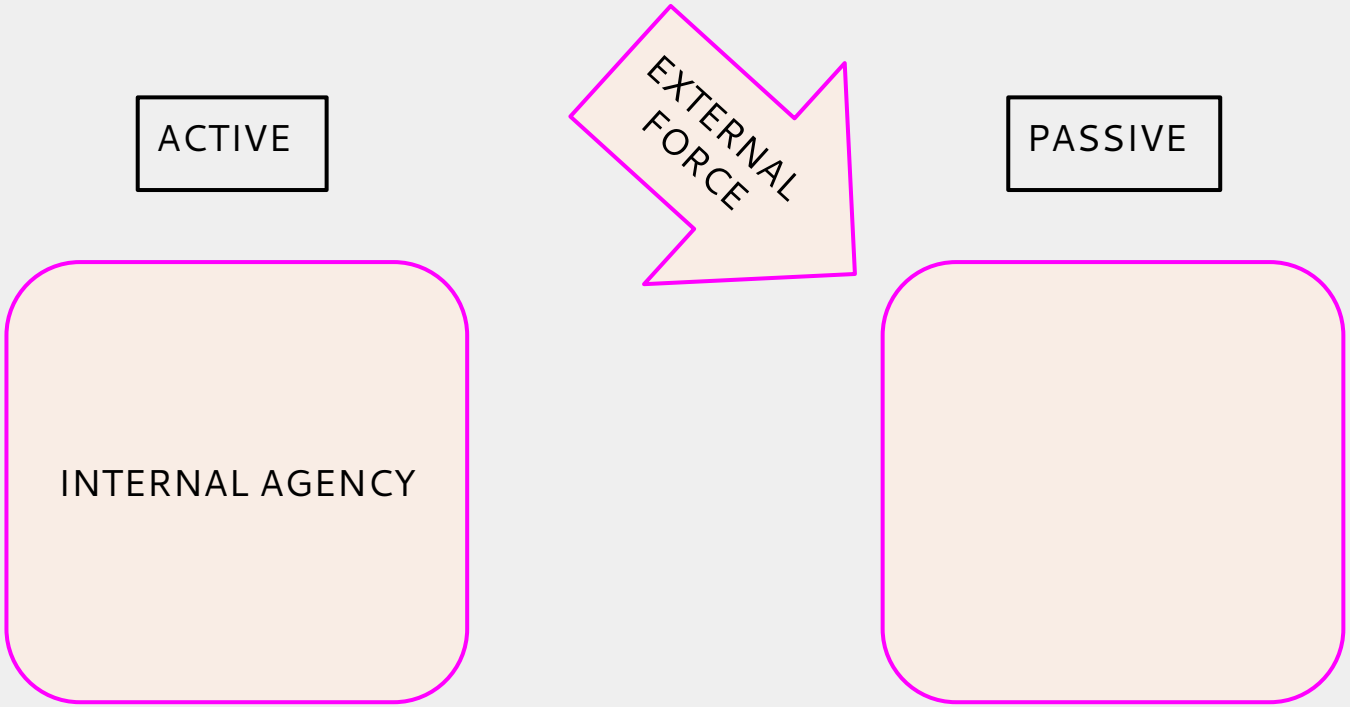
Mechanist:

Banish all agency from Nature

Everything can be explained with physics

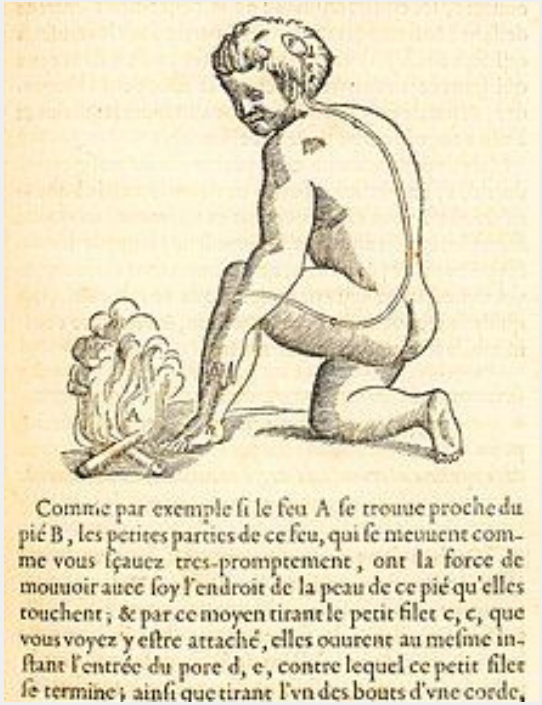
Early Modern

17thC Debates on “Agency” in Nature



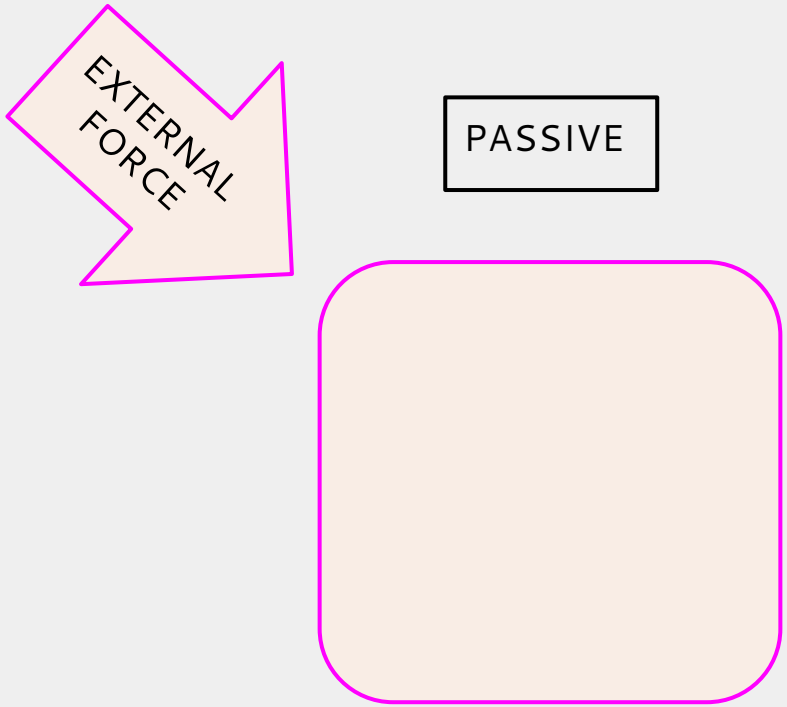
Passive Mechanist

Entirely devoid of agency



Comme par exemple ſi le feu A ſe trouue proche du pié B, les petites parties de ce feu, qui ſe meuuent comme vous ſçauez tres-promptement, ont la force de mouuoir auec ſoy l'endroit de la peau de ce pié qu'elles touchent; & par ce moyen tirant le petit filet c, c, que vous voyez y eſtre attaché, elles ouurent au meſme inſtant l'entrée du pore d, e, contre lequel ce petit filet ſe termine; ainſi que tirant l'vn des bouts d'vne corde,

Descartes's Traite de l'homme, 1664



Descartes
Newton
Boyle

Descartes

Passive, brute mechanism; “Classical mechanist”

Cartesian dualism—the idea that matter and spirit or soul were two absolutely distinct things

Only reason/rationality is from the soul (animals have no rationality)

One could explain every aspect of life in terms of machinery, and so could understand the workings of living beings as fully as a clockmaker understands a clock. Rather than to reduce life to mechanism, he meant to elevate mechanism to life: to explain life, never to explain it away.

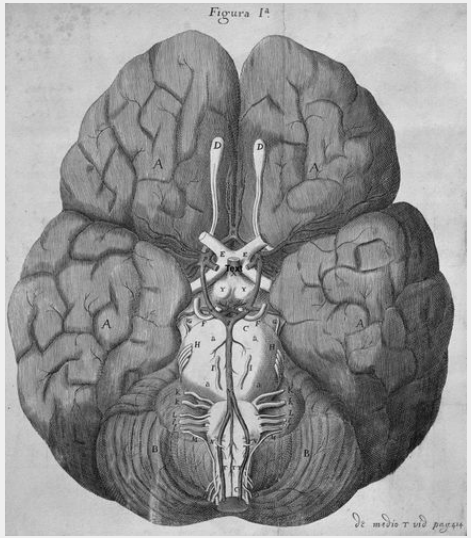
Active Mechanist

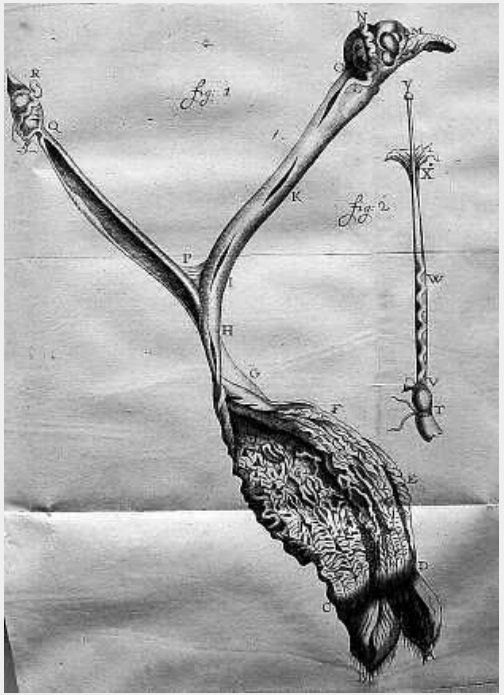
Actively Self-constituting

ACTIVE

INTERNAL AGENCY

Leibniz
Lamarck





Leibniz

Defines “Organism” through Active Mechanism

God works thru organization not design

- “God preformed things, such that the new organizations are but a mechanical consequence of a preceding organic constitution; as when butterflies come from silkworms.”

An animal was “a living organized body” whose identity attached to its organization. The identity of a human being consisted in an ongoing organization of ever-renewing material parts.

“I define the Organism, or the natural Machine.. a machine in which each part is a machine, and consequently the subtlety of its artifice goes to infinity, nothing being small enough to be neglected, whereas the parts of our artificial machines are not machines. That is the difference between Nature and Art.”

Mechanical “restless”? watches

Astronomic Watch 1660

Thomas Tompion’s Balance Spring Watch

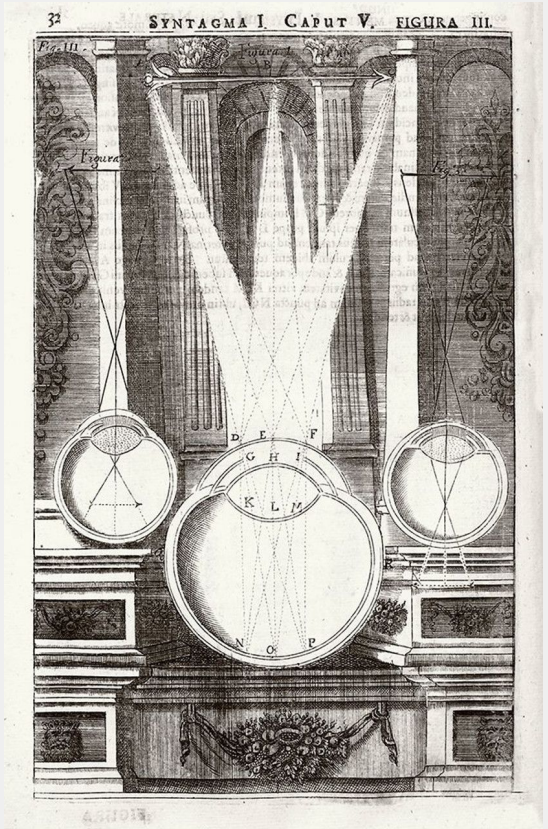
Portrait of Gentleman with Watch 1675

New Science's theological origins

"Nature is God's Artifact"

"There is no more rashness to say, that an Eye, than that a Telescope, was made for an Instrument to See with." - Boyle

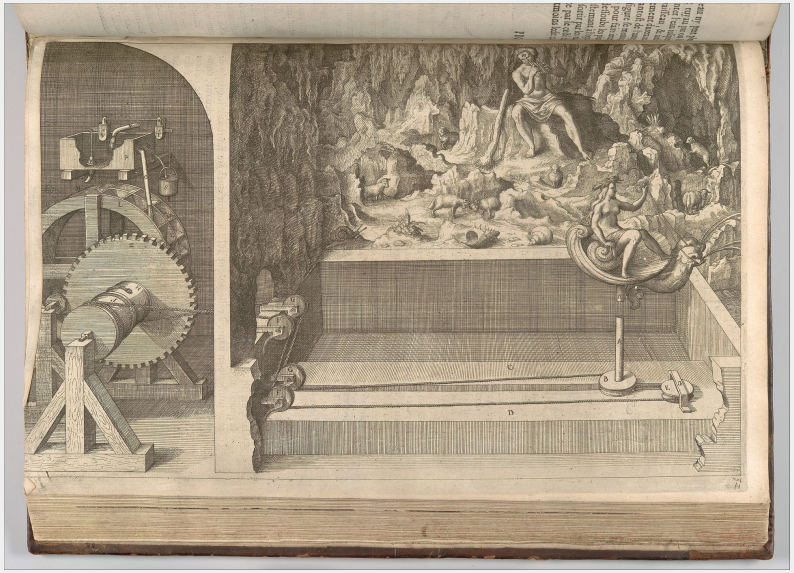
- Demonstrate God's existence by showing the detailed perfection of his world-machine



Oculus Artificialis, Johann Zahn. 1685

Animal-/Man- Machines

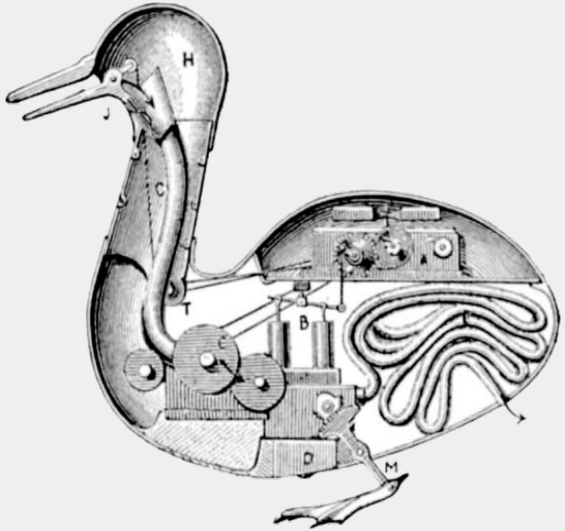
Hydraulic & Mechanical Engineering

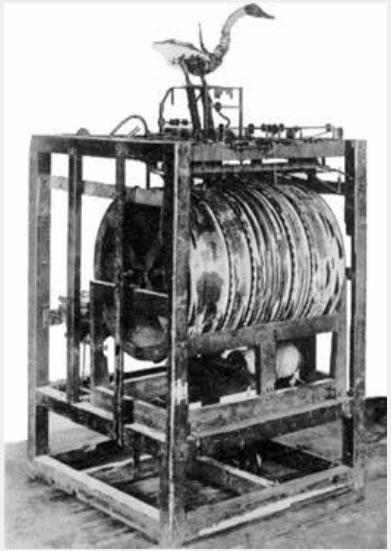


Les raisons des forces mouvantes,
Isaac De Cause. 1615

Enlightenment's first "androids"

"Prometheus's Rival"

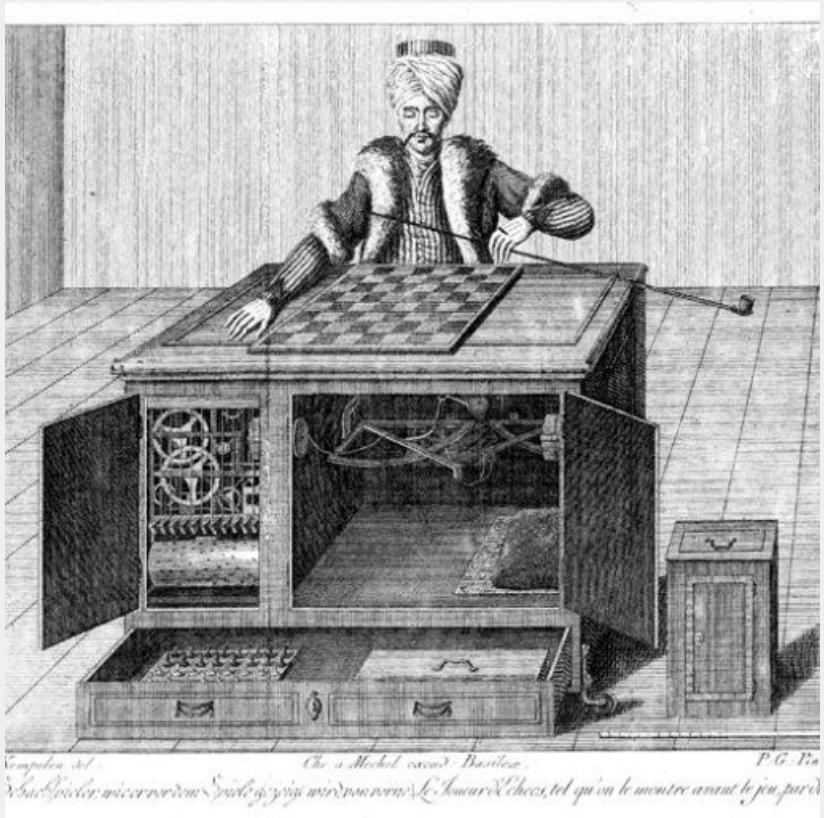




Canard Digérateur, Jacques Vaucanson. 1764

Mecanisme du fluteur automate, Vaucanson. 1738

Man-Machines

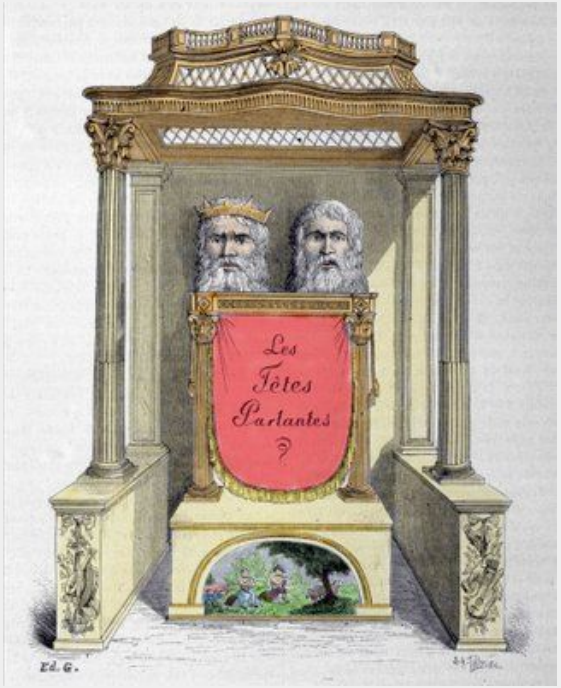


Chess-Playing Turk Wolfgang von Kempelen 1782

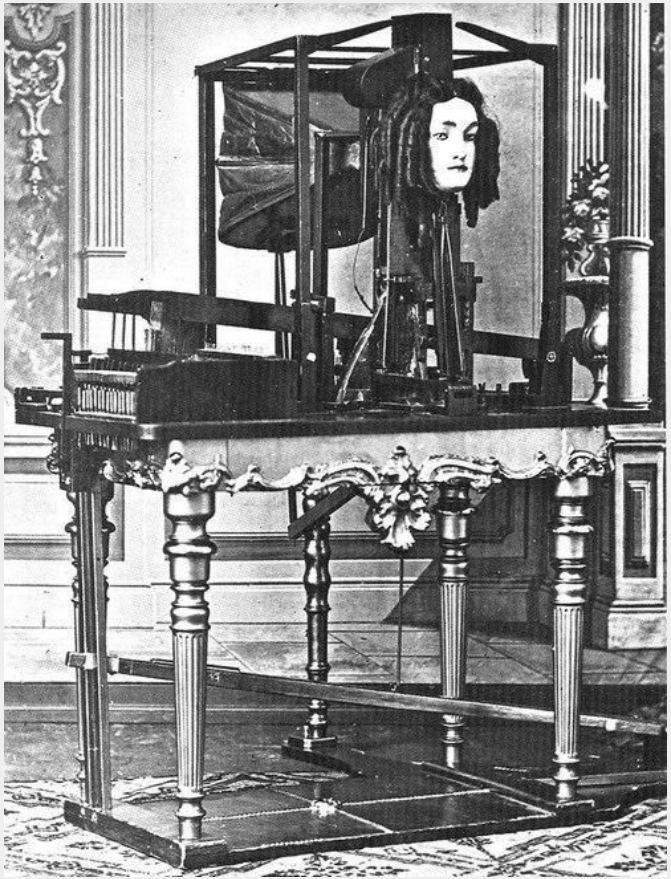
Speech Imitation

17th C obsession with “speech” autonoma

In praise of Louis XVI: “The King gives peace to Europe,” intoned the first head; “Peace crowns the King with Glory,” replied the second; “and Peace makes the Happiness of the People,” added the first; “O King Adorable Father of your People their Happiness shows Europe the Glory of your Throne”



abbe Mical discoursing heads 1778



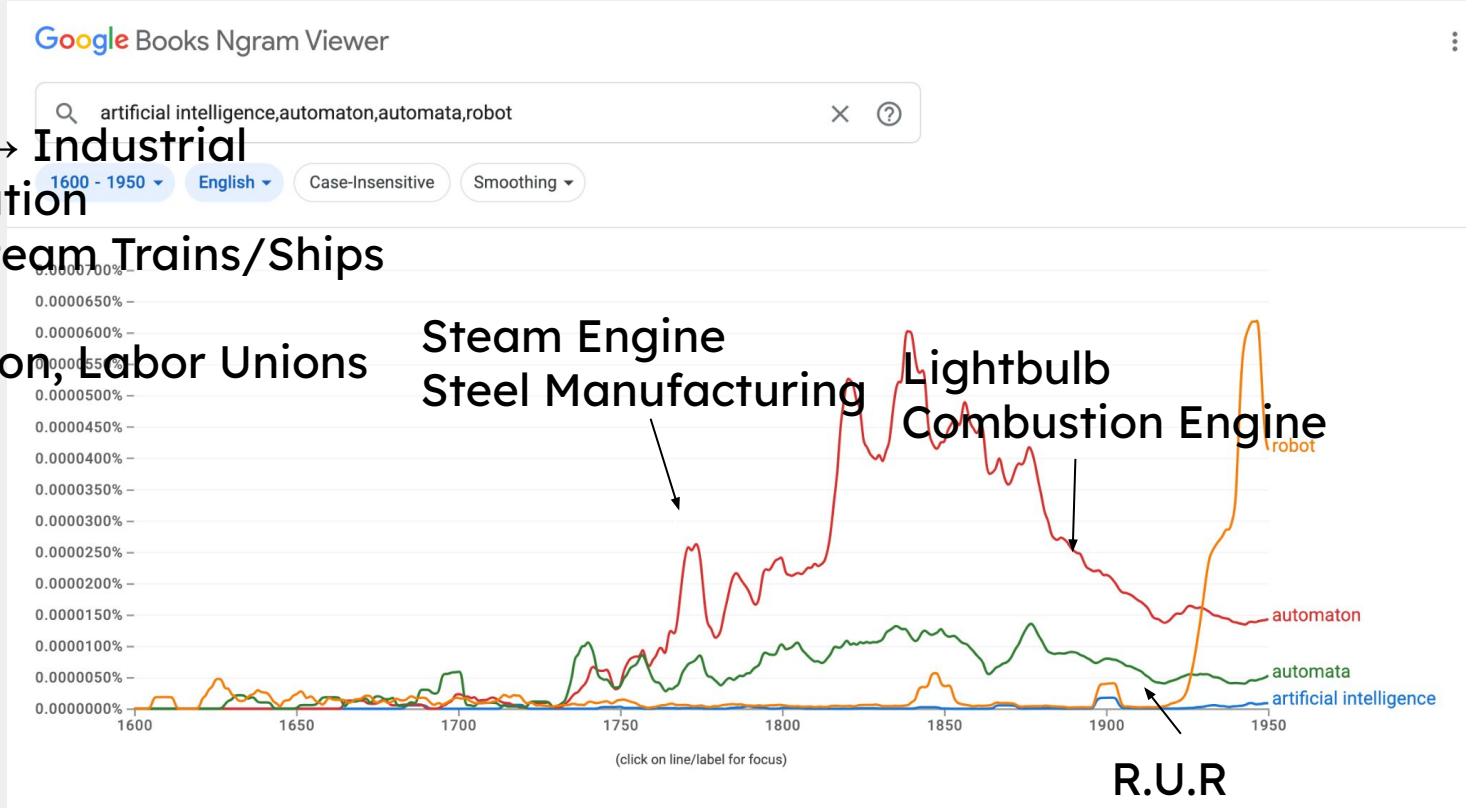
Faber’s Talking Head 1846

Jacqet-Droz 3 automata 1772

Karakuri Ningyo 17th C

1st Industrial Revolution

Agrarian → Industrial
Mechanization
Textiles, Steam Trains/Ships
Iron/Coal
Urbanization, Labor Unions

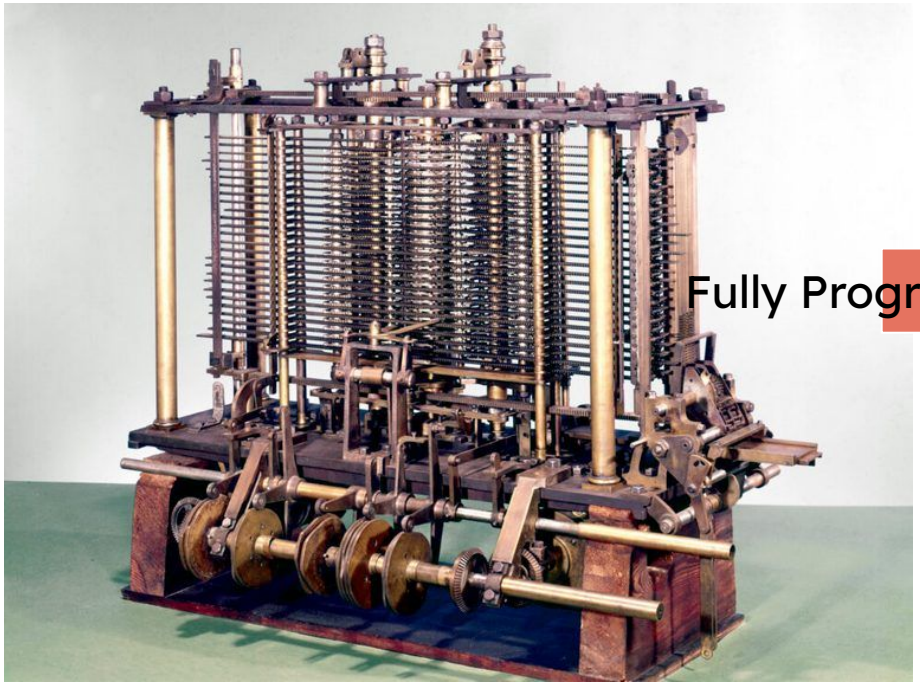


Punch Cards

Jacquard Loom 1804

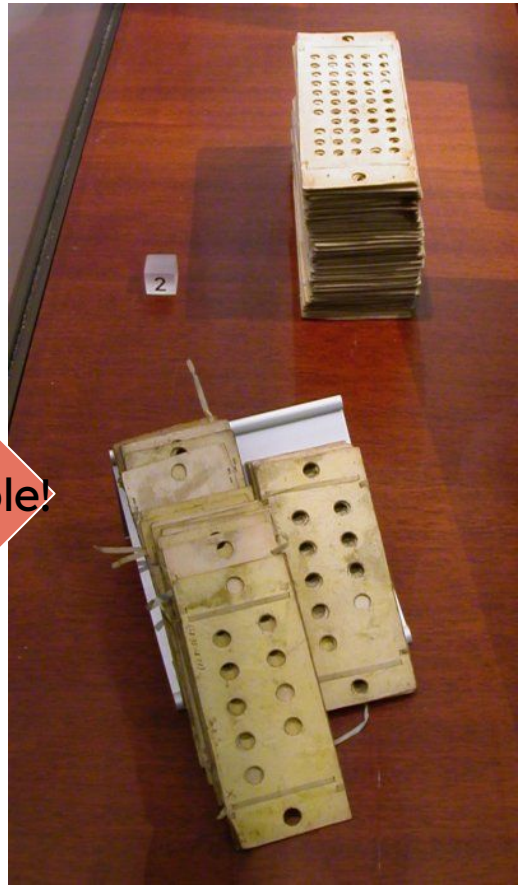
Jacquard woven on Silk. 1839

1820s Start of Computation



Difference Machine, Charles Babbage. 1820s

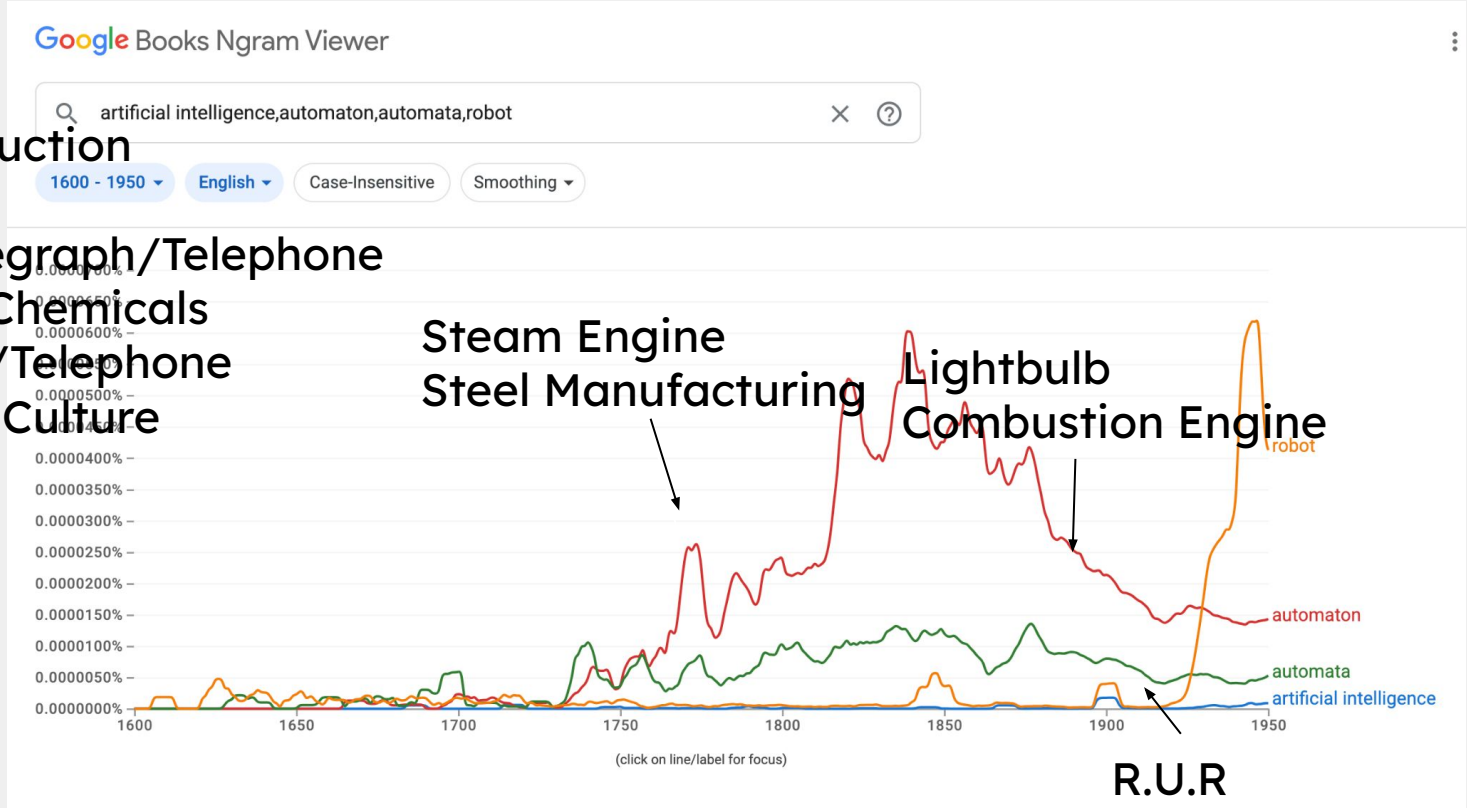
Fully Programmable!



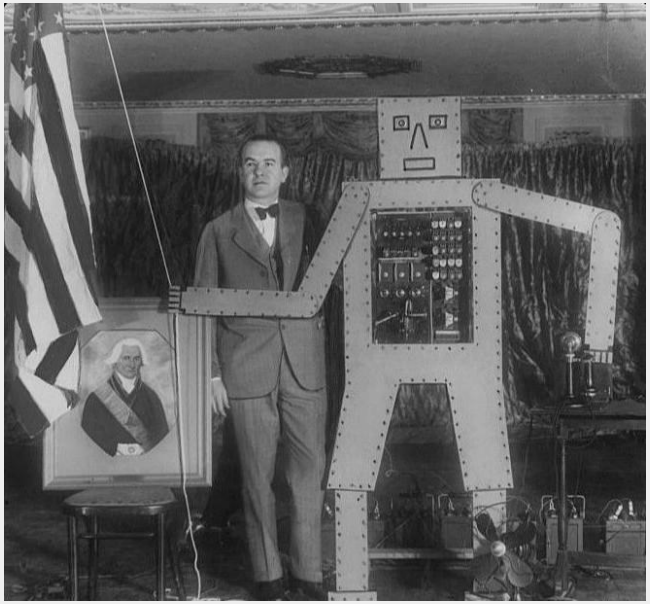
Punch Cards, Analytical Machine.
Charles Babbage

2nd Industrial Revolution

Mass Production
Electricity
Radio/Telegraph/Telephone
Steel/Oil/Chemicals
Telegraph/Telephone
Consumer Culture



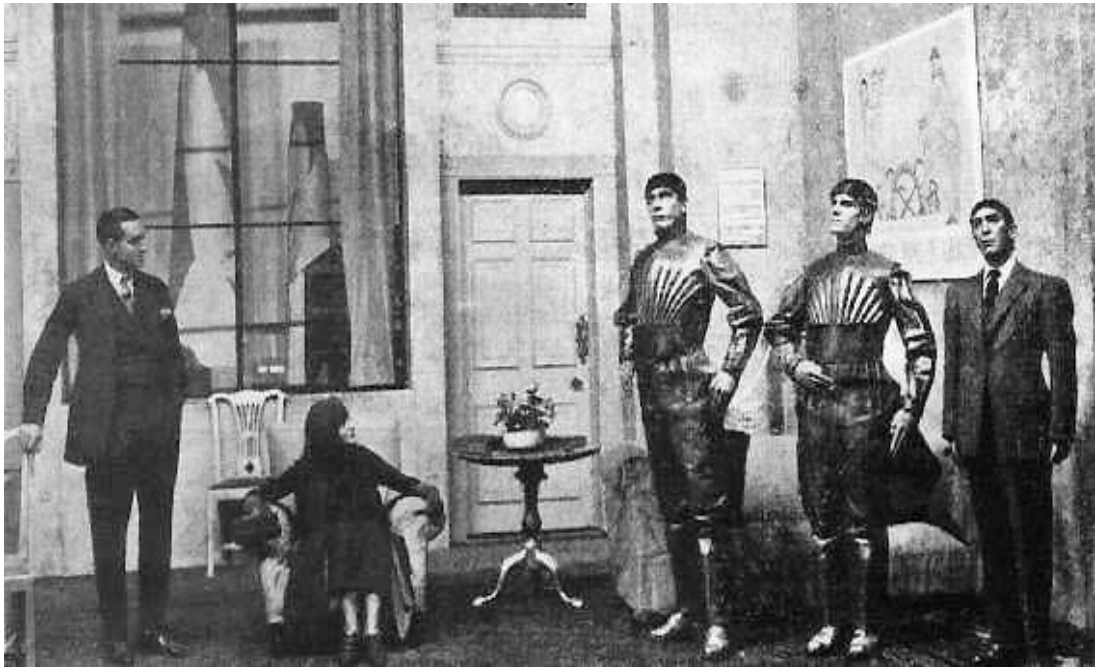
Man-Machines of 20th C



Mr. Televox. R.J. Wensley, Westinghouse (Westinghouse Electrical Company) 1926

Eric the robot at Model Engineer Exhibition at the Royal Agricultural Hall in London. 1928

“Robot” replaces “Automation” & “Android”



“Rossum’s Universal Robots,” Karel čapek. 1921

Mid-20th C First digital computers
Human Mind/Intelligence ←→ Computers

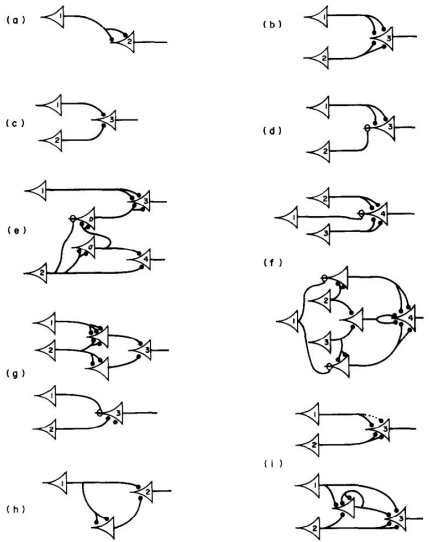
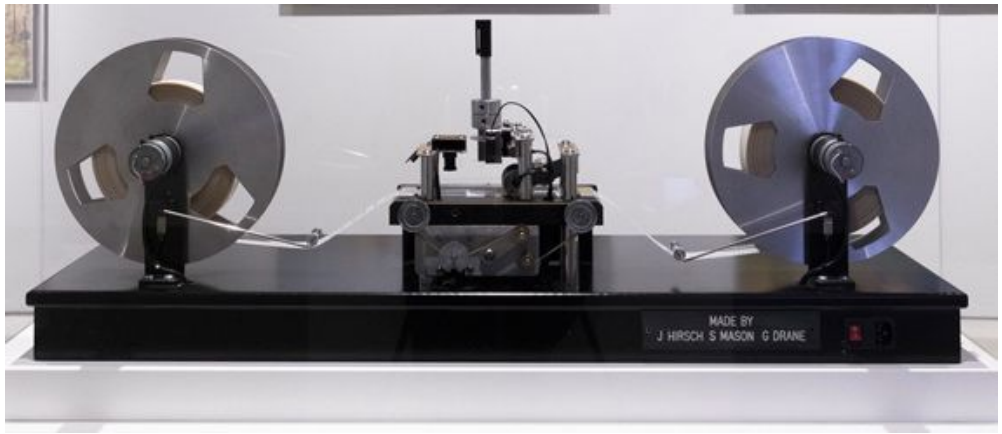


Figure 1. The neuron c_i is always marked with the numeral i upon the body of the cell, and the corresponding action is denoted by "N" with i s subscript, as in the text:

Pitts-McCulloch "Artificial Neural Network Theory." 1943



Physical model of Alan Turing's 1936 Turing Machine

Darwin

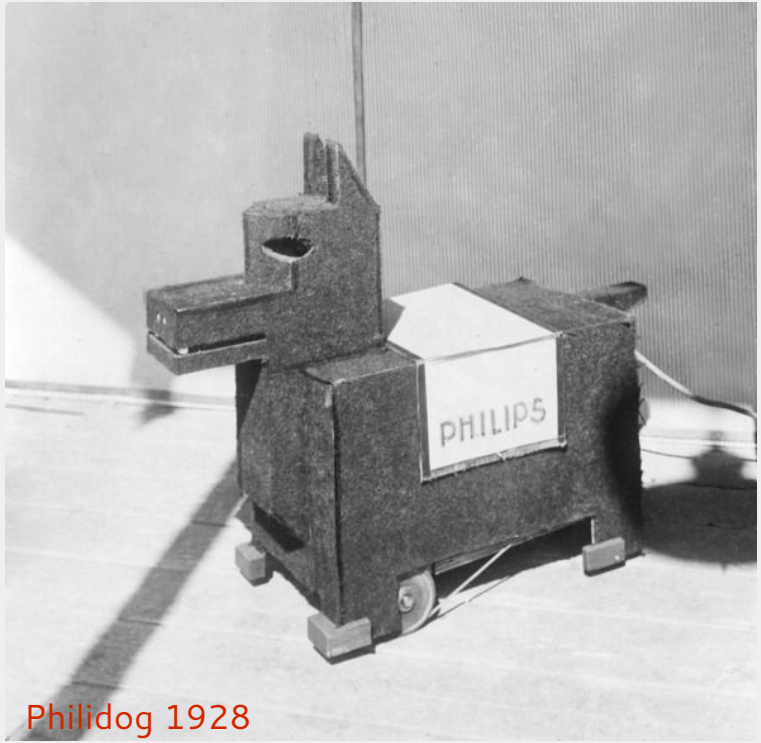
Straddling both active and passive views

Neither internal (striving)/external (divine) agency but:

1) living things actively improved themselves over time (historical agency)
2) living beings are passive, designed artifacts (divine agency)

> It is interesting to contemplate a tangled bank, clothed with many plants of many kinds, with birds singing on the bushes, with various insects flitting about, and with worms crawling through the damp earth, and to reflect that these elaborately constructed forms, so different from each other, and dependent upon each other in so complex a manner, have all been produced by laws acting around us. These laws, taken in the largest sense, being Growth with Reproduction; Inheritance which is almost implied by reproduction; **Variability from the indirect and direct [external removed] action** of the conditions of life, and from use and disuse: a Ratio of Increase so high as to lead to a Struggle for Life, and as a consequence to Natural Selection, entailing Divergence of Character and the Extinction of less-improved forms. Thus, from the war of nature, from famine and death, the most exalted object which we are capable of conceiving, namely, the production of the higher animals, directly follows. There is grandeur in this view of life, with its several powers, having been originally breathed by the ***Creator*** into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from so simple a beginning endless forms most beautiful and most wonderful have been, and are being evolved.
>
> From Origins of Species 1859

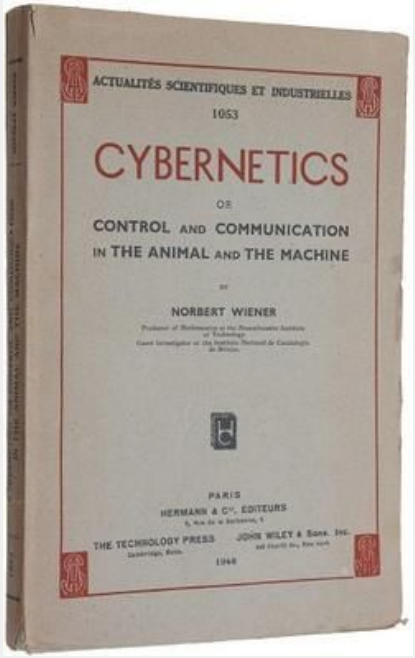
“Sensitive Automata”



Philidog 1928

Cybernetics

"Agencies acting to maintain constancy"
Passive mechanist, neo-Darwinism

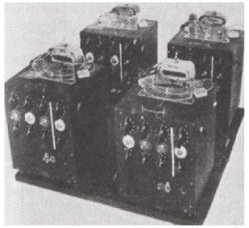


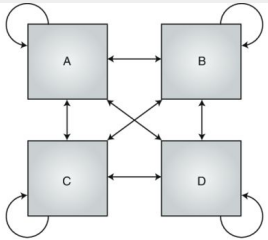
Norbert Weiner. 1949

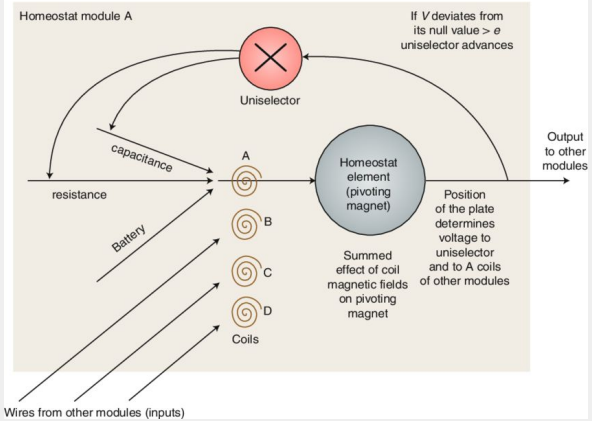
responsiveness is what makes organisms "living things" and responsiveness can be mechanically imitated → living organisms can be mechanically imitated

Cybernetics

Homeostasis and Negative Feedback







Homeostat with Negative Feedback.
Walter Ross Ashby. 1949

FIGURE 8/2/1: The Homeostat. Each unit carries on top a magnet and coil such as that shown in Figure 8/2/2. Of the controls on the front panel, those of the upper row control the potentiometers, those of the middle row the commutators, and those of the lower row the switches *S* of Figure 8/2/3.

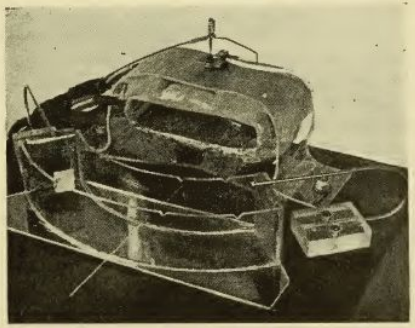
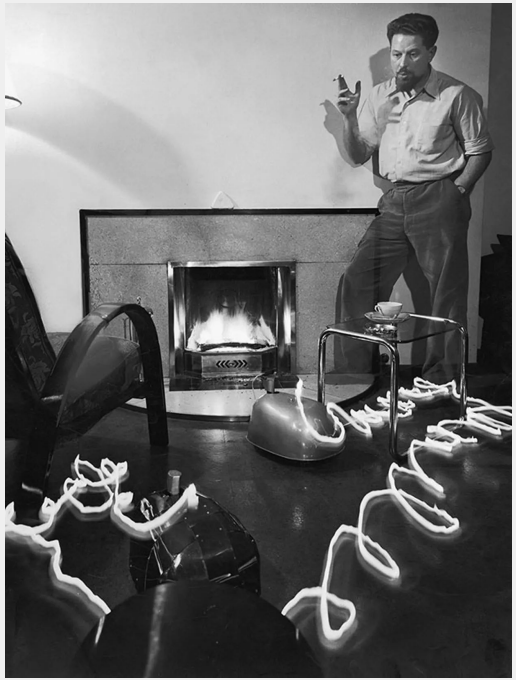
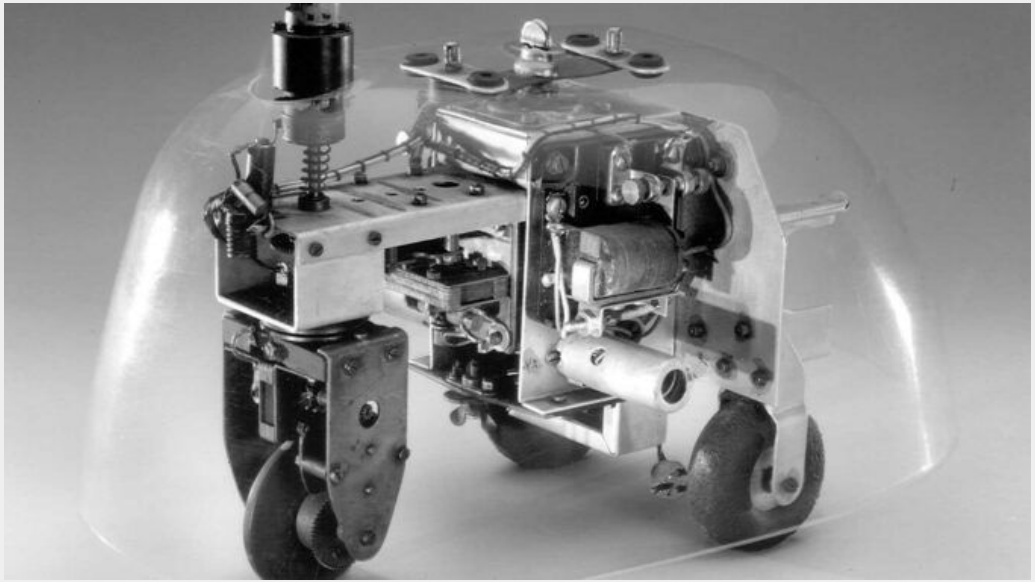


FIGURE 8/2/2: Typical magnet (just visible), coil, pivot, vane, and water potentiometer with electrodes at each end. The coil is quadruple, consisting of *A*, *B*, *C* and *D* of Figure 8/2/3.

Cybernetics

Sentience in terms of their outward manifestations not internal mechanism or subjective experience

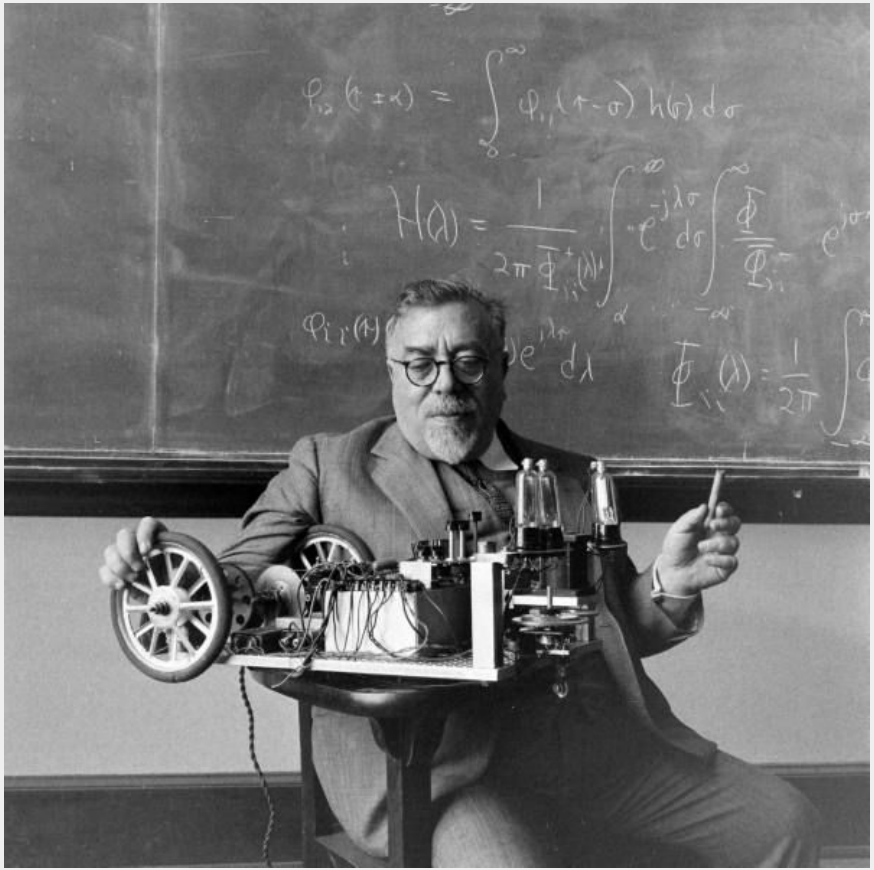




Elmer and Elsie 1948 (Grey Walter)

Voluntary Feedback

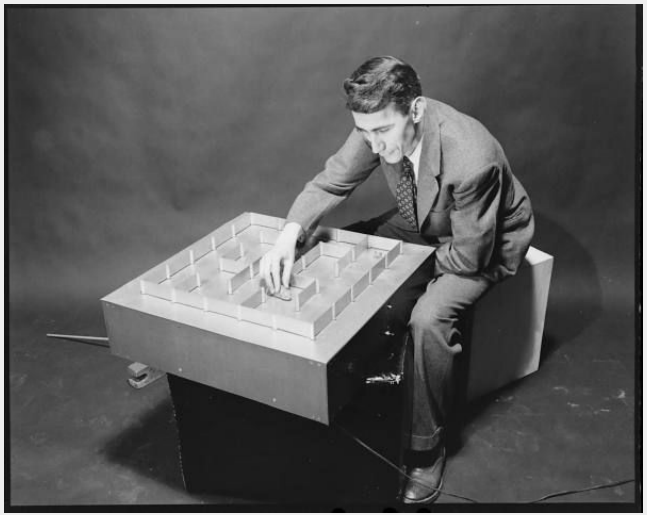
Light-seeking /
Light-avoiding
Voluntary Feedback



Moth/BedBug Machine

Early Reinforcement Learning

“Exploration Strategy” vs. “Goal Strategy”

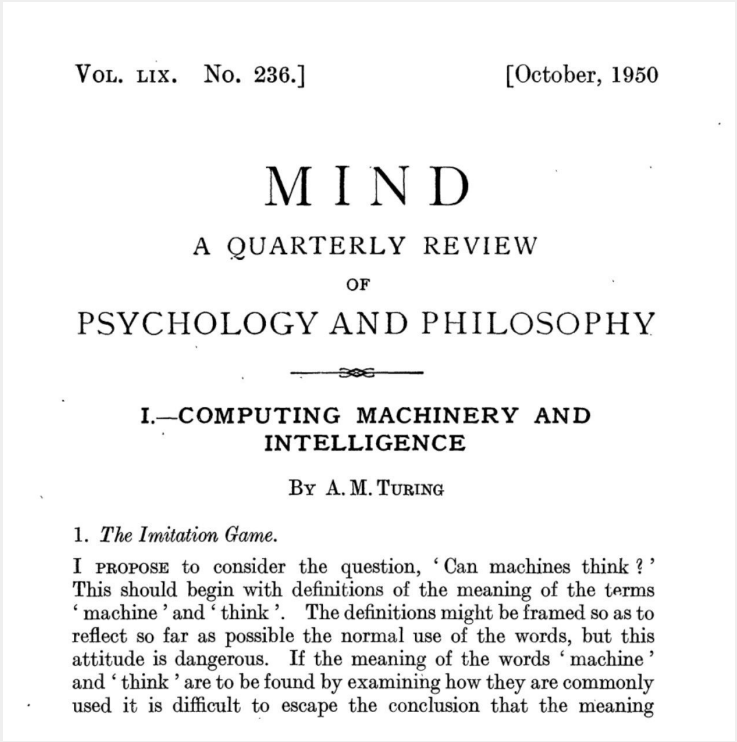


Claude Shannon’s Mouse Theseus 1950

Turing

Intelligence develops through learning

- 1950 “Computers Machinery and Intelligence”:
- Pleasure/Reward vs. Pain/Punishment
- “Machines must be fallible”
- Mathematical equations can’t determine whether a machine can think



VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

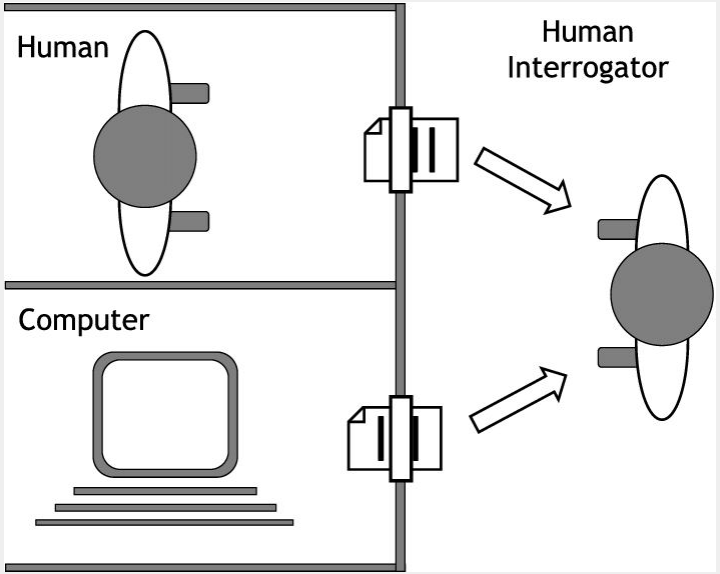
I PROPOSE to consider the question, ‘Can machines think?’ This should begin with definitions of the meaning of the terms ‘machine’ and ‘think’. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words ‘machine’ and ‘think’ are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

Turing

Appearing to have free-will

The triumphant machine's designers themselves would remain at least partially in the dark regarding its inner workings.147 This was necessarily the case, since to be able to describe the machine's inner workings completely would be to lose the sense that it was intelligent. But then, the same would be true in the case of a man: if one could attribute his behavior entirely to a set of explicit rules, Turing argued, one would come to see him as a machine and, by implication, without intelligence.

Riskin, Jessica. The Restless Clock: A History of the Centuries-Long Argument over What Makes Living Things Tick (p. 335).



Beginning of Modern AI

A Critical Mass of Researchers Fo
the “First Generation”

French Academie de Sciences 17th C

Dartmouth Conference 1956

“Modernizing” AI

“We propose that a 2 month, 10 man (**John McCarthy hosted; Minsky, Claude Shannon, Arthur Samuel, Ray Solomonoff, etc.**) study of **artificial intelligence** be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hamsphire.”

→ Stanford, CMU, MIT, IBM, etc.

Philosophy — reasoning, logic, rationalism
Mathematics – mathematical modeling, algorithms, probability
Economics – game theory, decision theory
Neuroscience – neurons, inspiration for computers
Psychology – cognitive science (brain as a information processing device)
Computer engineering – hardware and software
Cybernetics/Control Theory – homeostasis, objective function
Linguistics – knowledge representation, computational linguistics

Why AI?

THE U.S. Sun | News | Sport | Lifestyle | Entertainment | Money | Tech

A FORMER Google engineer who was let go by the tech giant and claimed its AI was sentient has made some more concerning claims.

Blake Lemoine made headlines last year after claiming Google's Large Language Model (LLM) had thoughts and feelings like a child.

Blake Lemoine is an AI consultant and public speaker who thinks some artificial intelligence could be sentient Credit: Getty

Google strongly denied these claims and said the AI was designed to respond like a human, but it's not sentient like us.

Lemoine has now published a new essay in Newsweek in which he discusses Microsoft's Bing's Open AI-powered chatbot.

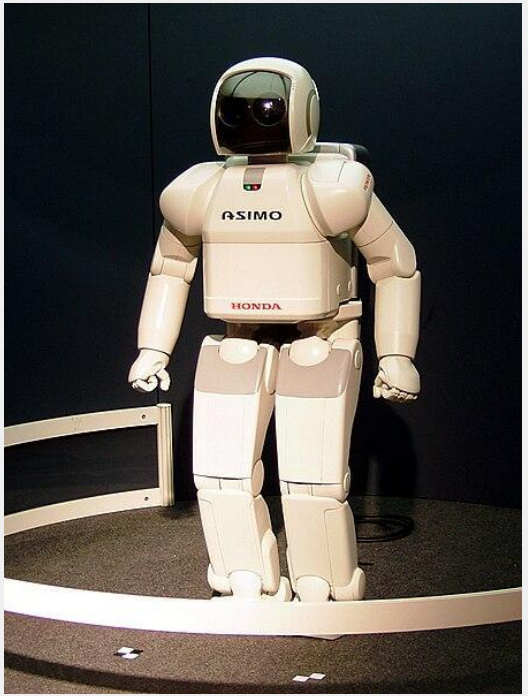
The New York Times Account

A.I. Is Not Sentient. Why Do People Say It Is?

Robots can't think or feel, despite what the researchers who build them want to believe.

History of AI

Global Perspectives in History of AI

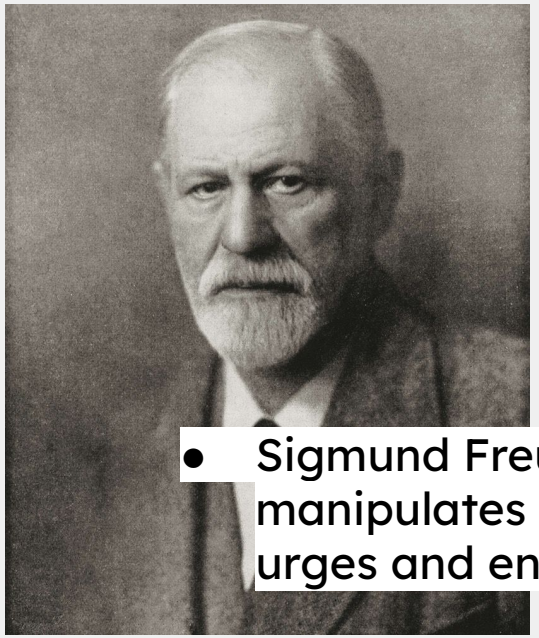


unsojo@cornell.edu

Lecture 3 Februrary 7, 2025

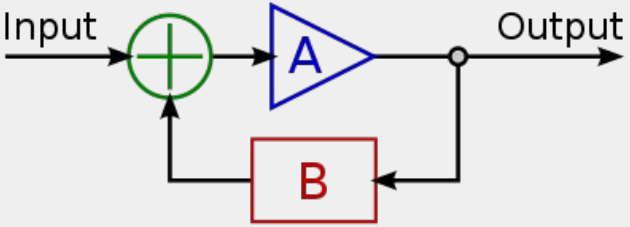
- What does it mean to “modernize” a scientific field?
- The role of defense in the origins of modern AI. How is AI defense changing warfare?
- Military Industrial Complex – is “invention the mother of necessity”
- Which came first: AI or computers?

Early Modern



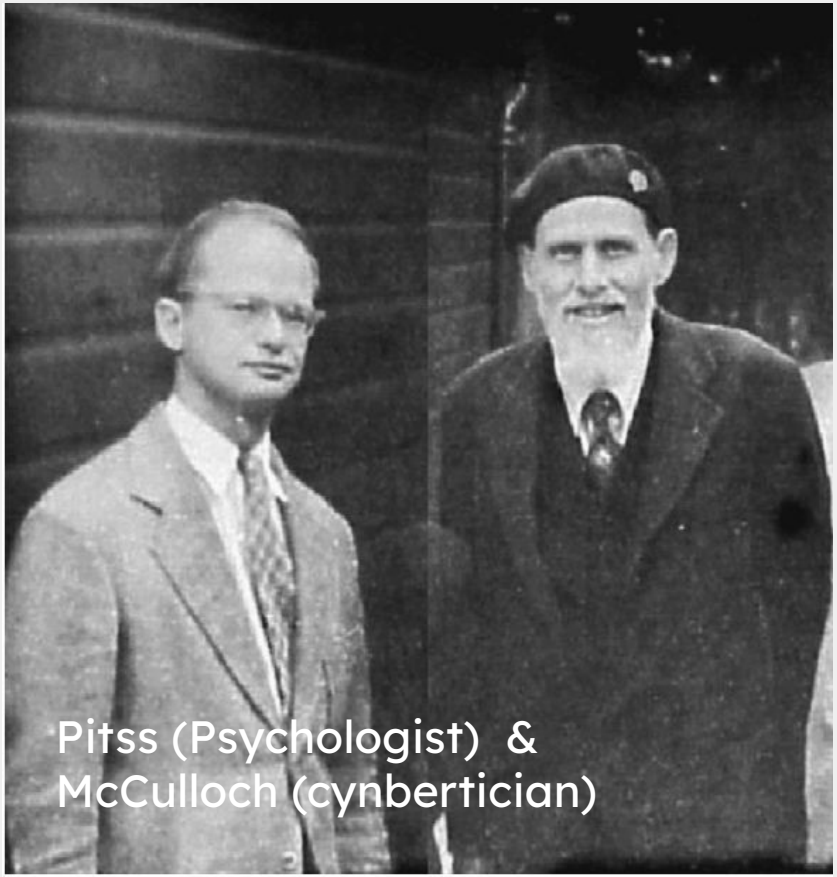
- Sigmund Freud: Mind manipulates biological urges and energies

Modern



- Intelligence is about taking, processing information

Pitts-McCulloch 1943



Pitss (Psychologist) &
McCulloch (cynbertician)

A LOGICAL CALCULUS OF THE IDEAS IMMANENT IN NERVOUS ACTIVITY

WARREN S. MCCULLOCH and WALTER H. PITTS

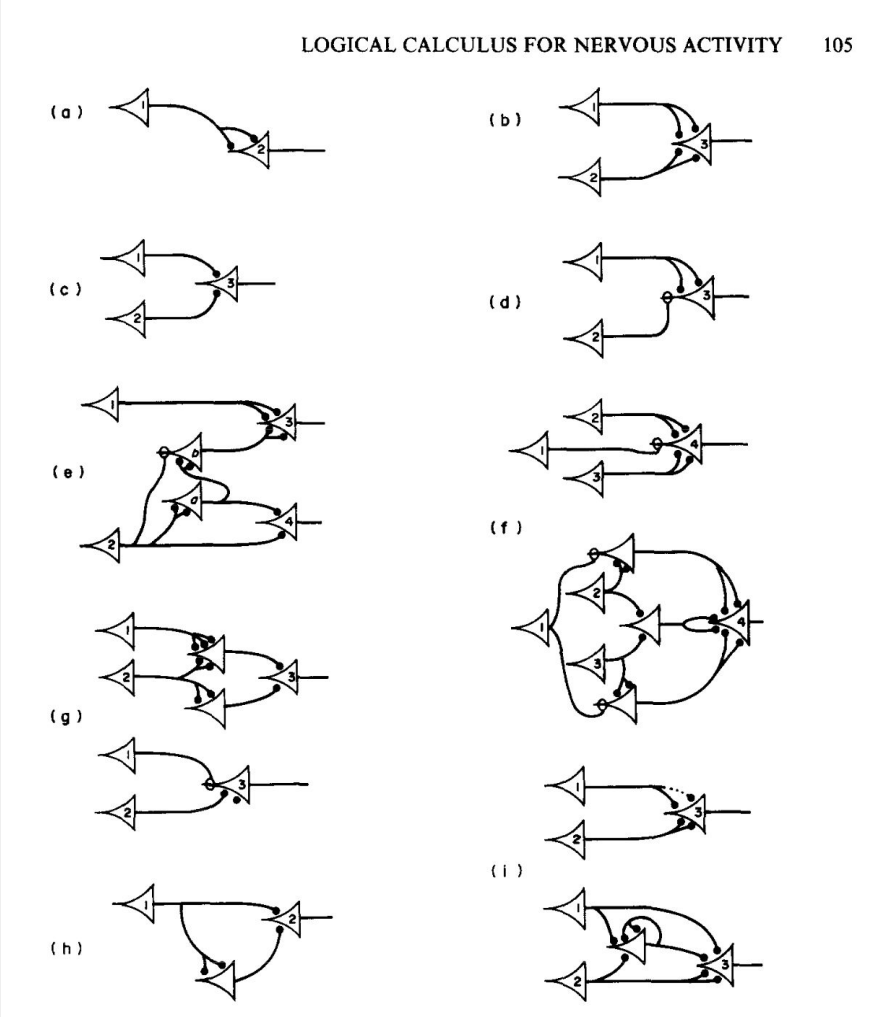
Because of the "all-or-none" character of nervous activity, neural events and the relations among them can be treated by means of propositional logic. It is found that the behavior of every net can be described in these terms, with the addition of more complicated logical means for nets containing circles; and that for any logical expression satisfying certain conditions, one can find a net behaving in the fashion it describes. It is shown that many particular choices among possible neurophysiological assumptions are equivalent, in the sense that for every net behaving under one assumption, there exists another net which behaves under the other and gives the same results, although perhaps not in the same time. Various applications of the calculus are discussed.

INTRODUCTION

THEORETICAL neurophysiology rests on certain cardinal assumptions. The nervous system is a net of neurons, each having a soma and an axon. Their adjunctions, or synapses, are always between the axon of one neuron and the soma of another. At any instant a neuron has some threshold, which excitation must exceed to initiate an impulse. This, except for the fact and the time of its occurrence, is determined by the neuron, not by the excitation.

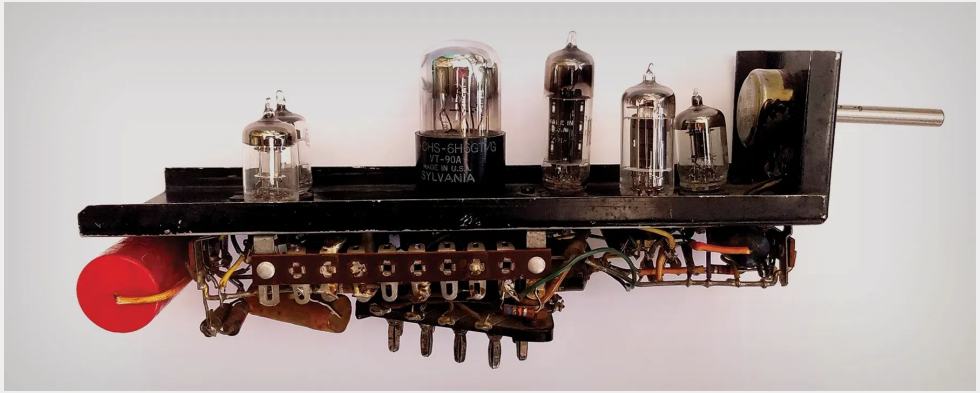
Pitts-McCulloch 1943

- inspired by feedback loops
- “can represent any logical expression with neural nets”
- “better than Turing machine!”



Marvin Minsky's SNARC 1951

Pitts-McCulloch neural nets + B.F. Skinner reinforcement learning



1 neuron!

- made out of b-24 bomber aircraft pieces
- 40 neurons
- 6 vacuum tubes each
- size of small room
- was able to solve Shannon's maze
- → sufficient memory for reasoning would require millions of neurons

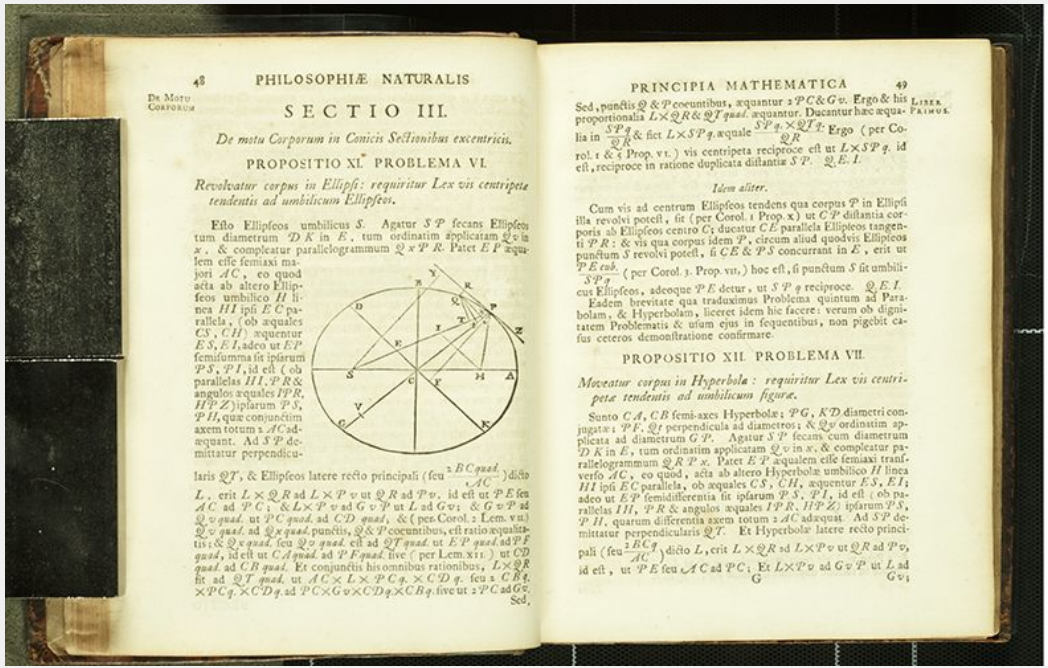
Dartmouth Conference - an interdisciplinary AI conference 1957

"We propose that a 2 month, 10 man (**John McCarthy hosted; Minsky, Claude Shannon, Arthur Samuel, Ray Solomonoff, etc.**) study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hamsphire."

→ Stanford, CMU, MIT, IBM

Dartmouth Conference 1957

First? AI Program - Logic Theorist 1955 (Newell, Simon, Shaw)



48 PHILOSOPHIÆ NATURALIS

De Motu Corporum

SECTIO III.

De motu Corporum in Conicis Sectionibus excentricis.

PROPOSITIO XI. PROBLEMA VI.

Revolvatur corpus in Ellipſi: requiritur Lex vis centripetæ tendentis ad umbilicum Ellipſeos.

Eſto Ellipſeos umbilicus *S*. Agatur *SP* fecans Ellipſeos tum diametrum *DK* in *E*, tum ordinatim applicatam *Qv* in *x*, & compleatur parallelogrammum *QxPR*. Patet *EP* æqualem eſſe ſemiaxi majori *AC*, eo quod acta ab altero Ellipſeos umbilico *H* linea *HI* ipſi *EC* parallela, (ob æquales *CS*, *CH*) æquentur *ES*, *EI*, adeo ut *EP* ſemiſumma ſit ipſarum *PS*, *PI*, id eſt (ob parallelas *HI*, *PR* & angulos æquales *IPR*, *HPZ*) ipſarum *PS*, *PH*, quæ conjunctim axem totum 2*AC* adæquant. Ad *SP* demittatur perpendicularis *QT*, & Ellipſeos latere recto principali (ſeu $\frac{2BC\ quad.}{AC}$) dicto *L*, erit $L \times QR$ ad $L \times Pv$ ut *QR* ad *Pv*, id eſt ut *PE* ſeu *AC* ad *PC*; & $L \times Pv$ ad Gv ut *L* ad *Gv*; & *Gv* ad *Qv* quad. ut *PC quad.* ad *CD quad.*, & (per Corol. 2 Lem. VII.) *Qv quad.* ad *Qx quad.* punctis, *Q* & *P* coeuntibus, eſt ratio æqualitatis; & *Qx quad.* ſeu *Qv quad.* eſt ad *QT quad.* ut *EP quad.* ad *PF quad.*, id eſt ut *CA quad.* ad *PF quad.* hoc eſt (per Lem. XII.) ut *CD quad.* ad *CB quad.* Et conjunctis his omnibus rationibus, $L \times QR$ fit ad *QT quad.* ut $AC \times L \times PC q. \times CD q.$ ſeu 2*CB q.* $\times PC q. \times CD q.$ ad $PC \times Gv \times CD q. \times CB q.$ hoc eſt ut 2*PC* ad *Gv*. Sed,

PRINCIPIA MATHEMATICA 49

Sed, punctis *Q* & *P* coeuntibus, æquantur 2*PC* & *Gv*. Ergo & his proportionalia $L \times QR$ & *QT quad.* æquantur. Ducantur hæc æqualia in $\frac{SP}{QR}$, & fiet $L \times SP\ q.$ æquale $\frac{SP\ q. \times QT\ q.}{QR}$. Ergo (per Corol. 1 & 2 Prop. VI.) vis centripeta reciproce eſt ut $L \times SP\ q.$ id eſt, reciproce in ratione duplicata diſtantiæ *SP*. *Q.E.I.*

LIBER PRIMUS.

Idem aliter.

Cum vis ad centrum Ellipſeos tendens qua corpus *P* in Ellipſi illa revolvi poteſt, ſit (per Corol. 1 Prop. X) ut *CP* diſtantia corporis ab Ellipſeos centro *C*; ducatur *CE* parallela Ellipſeos tangenti *PR*: & vis qua corpus idem *P*, circum aliud quodvis Ellipſeos punctum *S* revolvi poteſt, ſi *CE* & *PS* concurrant in *E*, erit ut $\frac{PE}{SP\ q.}$ (per Corol. 3. Prop. VII.) hoc eſt, ſi punctum *S* ſit umbilicus Ellipſeos, adeoque *PE* detur, ut $\frac{1}{SP\ q.}$ reciproce. *Q.E.I.*

Eadem brevitate qua traduximus Problema quintum ad Parabolam, & Hyperbolam, liceret idem hic facere: verum ob dignitatem Problematis & uſum ejus in ſequentibus, non pigebit caſus ceteros demonſtratione confirmare.

PROPOSITIO XII. PROBLEMA VII.

Moveatur corpus in Hyperbola: requiritur Lex vis centripetæ tendentis ad umbilicum figuræ.

Sunto *CA*, *CB* ſemi-axes Hyperbolæ; *PG*, *KD* diametri conjugatæ; *PF*, *Qf* perpendicula ad diametros; & *Qv* ordinatim applicata ad diametrum *GP*. Agatur *SP* fecans cum diametrum *DK* in *E*, tum ordinatim applicatam *Qv* in *x*, & compleatur parallelogrammum *QRPx*. Patet *EP* æqualem eſſe ſemiaxi tranſverſo *AC*, eo quod, acta ab altero Hyperbolæ umbilico *H* linea *HI* ipſi *EC* parallela, ob æquales *CS*, *CH*, æquentur *ES*, *EI*; adeo ut *EP* ſemidifferentia ſit ipſarum *PS*, *PI*, id eſt (ob parallelas *IH*, *PR* & angulos æquales *IPR*, *HPZ*) ipſarum *PS*, *PH*, quarum differentia axem totum 2*AC* adæquat. Ad *SP* demittatur perpendicularis *QT*. Et Hyperbolæ latere recto principali (ſeu $\frac{2BC}{AC}$) dicto *L*, erit $L \times QR$ ad $L \times Pv$ ut *QR* ad *Pv*, id eſt, ut *PE* ſeu *AC* ad *PC*; Et $L \times Pv$ ad Gv ut *L* ad *Gv*;

G

- Was able to prove 38 of 52 theorems in *Principia*
- *“We have invented a computer program capable of thinking non-numerically, and thereby solved the venerable mind-body problem.”*
- Implemented on a RAND computer 1956

Russell and Whitehead. Principia Mathematica. 1910.

First? AI Program - Logic Theorist 1955 (Newell, Simon, Shaw)

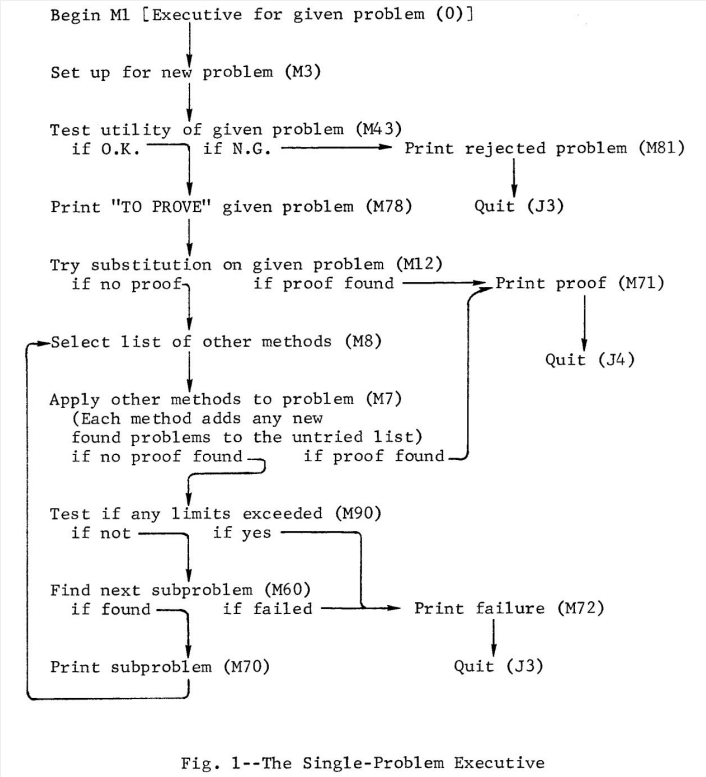
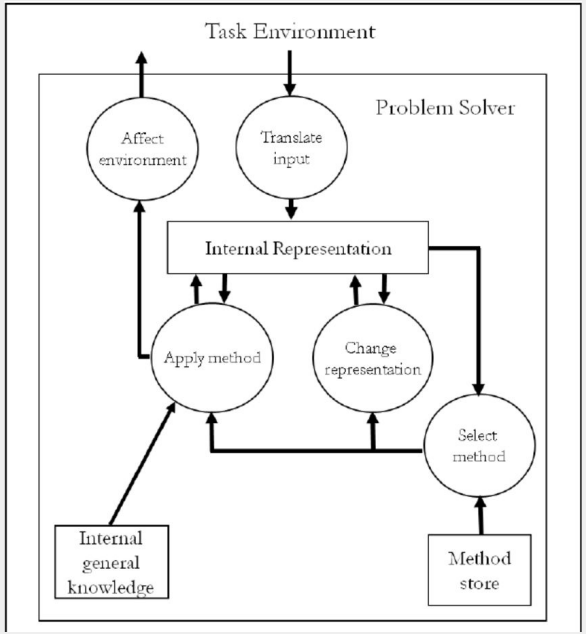


Fig. 1--The Single-Problem Executive

- Handwrote the program into index cards and then passed each card to a person who would then become a “component” (subroutine or component of its memory)
- Proving a theorem is akin to selective search on tree structure

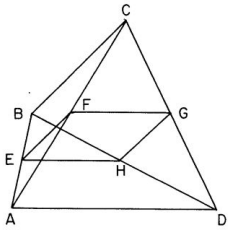
Logic Theorist

More AI Programs 1950s



- Logic Theorist → General Problem Solver (to be more general purpose and closer to ‘human thinking’)
- IBM implemented the ‘geometry version’ of logic theorist 1959 (could “see” through punched coordinates)

```
   PREMISES
**************
QUAD-LATERAL ABCD
POINT E MIDPOINT SEGMENT AB
POINT F MIDPOINT SEGMENT AC
POINT G MIDPOINT SEGMENT CD
POINT H MIDPOINT SEGMENT BD

   TO PROVE
**************
PARALELOGRAM EFGH

   SYNTACTIC SYMMETRIES
**************************
BA, AB, DC, CD, EE, HF, GG, FH,
CA, DB, AC, BD, GE, FF, EG, HH,
DA, CB, BC, AD, GE, HF, EG, FH,

   PROOF
***********
SEGMENT DG EQUALS SEGMENT GC
      DEFINITION OF MIDPOINT
SEGMENT CF EQUALS SEGMENT FA
      DEFINITION OF MIDPOINT
TRIANGLE DCA
      ASSUMPTION BASED ON DIAGRAM
PRECEDES DGC
      DEFINITION OF MIDPOINT
PRECEDES CFA
      DEFINITION OF MIDPOINT
SEGMENT GF PARALLEL SEGMENT AD
      SEGMENT JOINING MIDPOINTS OF SIDES OF TRIANGLE IS PARALLEL TO BASE
SEGMENT HE PARALLEL SEGMENT AD
      SYNTACTIC CONJUGATE
SEGMENT GF PARALLEL SEGMENT EH
      SEGMENTS PARALLEL TO THE SAME SEGMENT ARE PARALLEL
SEGMENT HG PARALLEL SEGMENT FE
      SYNTACTIC CONJUGATE
QUAD-LATERAL HGFE
      ASSUMPTION BASED ON DIAGRAM
PARALELOGRAM EFGH
      QUADRILATERAL WITH OPPOSITE SIDES PARALLEL IS A PARALLELOGRAM


   TOTAL ELAPSED TIME    1.03 MINUTES
```



General Problem Solver

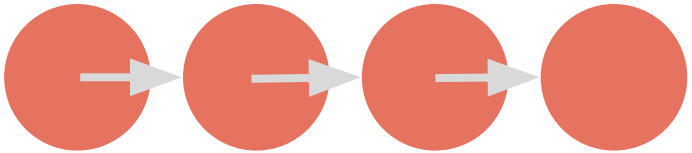
Geometry Theorem Solver

A new programming language for AI - IPL (Simon, Newell, Shaw)

- Before: FORTRAN - need to declare all arrays beforehand
- How to make lists more memory efficient and user-friendly?
- IPL - Information Processing Language mimics how the human mind works
  - thoughts are constantly created, changing, and destroying
  - links that connect symbols to other symbols in the mind can learned
- Introduced memory pointers!

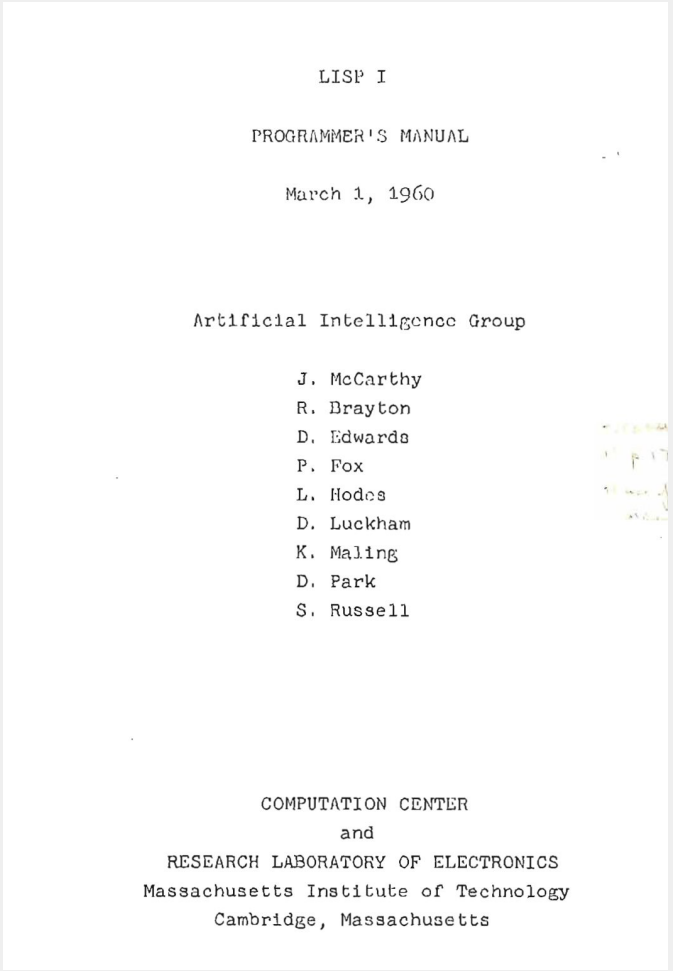
Herbert Simon describes the list-processing technique introduced in IPL

“The basic idea is that, whenever a piece of information is stored in memory, additional information should be stored with it telling where to find the next piece of information. In this way the entire memory could be organized like a long string of beads, but with the individual beads of the string stored in arbitrary locations…. “Nextness” was not determined by physical propinquity but by an address or **pointer** stored with each item.” - Herbert Simon



IPL inspires LISP I 1960 (John McCarthy)

- LISP made it much easier to insert, delete memory at any point in the list
- “programming language of AI” for the next 30 years
- LISP also introduced the idea of Lambda-functions where each function could be used as an object
- But required a compiling process in lower level languages that took long!



LISP I

PROGRAMMER'S MANUAL

March 1, 1960

Artificial Intelligence Group

J. McCarthy
R. Brayton
D. Edwards
P. Fox
L. Hodes
D. Luckham
K. Maling
D. Park
S. Russell

COMPUTATION CENTER
and
RESEARCH LABORATORY OF ELECTRONICS
Massachusetts Institute of Technology
Cambridge, Massachusetts

Advice Taker need creates LISP

“Apart from consulting on the geometry program, my own research in artificial intelligence was proceeding along the lines that led to the **Advice Taker** proposal in 1958 (McCarthy 1959). This involved representing information about the world by sentences in a suitable formal language and a reasoning program that would decide what to do by making logical inferences. Representing sentences by list structure seemed appropriate - it still is - and a list processing language also seemed appropriate for programming the operations involved in deduction - and still is.”

John McCarthy “LISP prehistory - Summer 1956 through Summer 1958.” 1979.

6.1 General Description

(1) Representation of S-expressions by List Structure

A list structure is a collection of computer words arranged as in Fig. 1a or 1b. Each word of the list structure is represented by one of the subdivided rectangles in the figure. The left box of a rectangle represents the address field of the word and the right box represents the decrement field. An arrow from a box to another rectangle means that the field corresponding to the box contains the location of the word corresponding to the other rectangle.

It is permitted for a substructure to occur in more than one place in a list structure, as in Fig. 1b, but it is not permitted for a structure to have cycles, as in Fig. 1c.

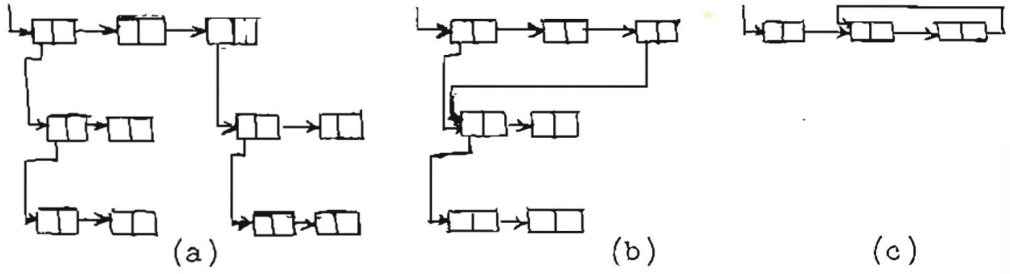


Fig. 1

The advantages of list structures for the storage of symbolic expressions are:

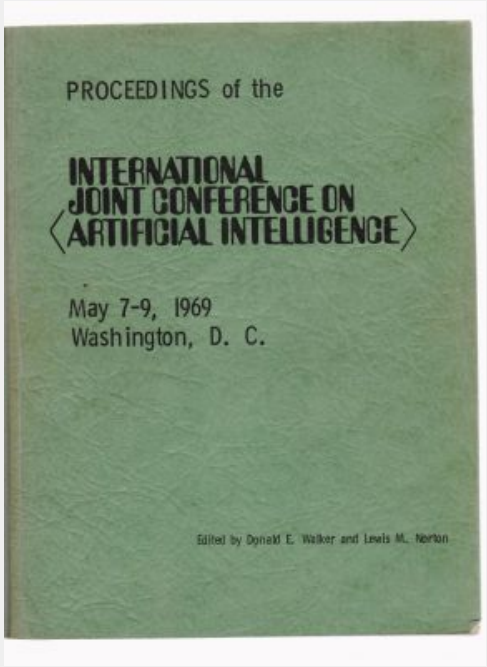
1. The size and even the number of expressions with which the program will have to deal cannot be predicted in advance. Therefore, it is difficult to arrange blocks of storage of fixed length to contain them.

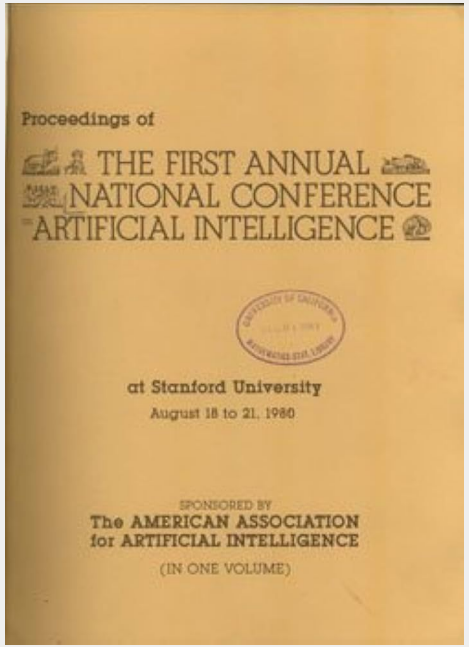
2. Registers can be put back on the free-storage list when they are no longer needed. Even one register returned to the list is of value, but if expressions are stored linearly, it is difficult to make use of blocks of registers of odd sizes that may become available.

3. An expression that occurs as a subexpression of several expressions need be represented in storage only once.

(more) AI Conferences & Expos

International Joint Conference on Artificial Intelligence (IJCAI) 1969





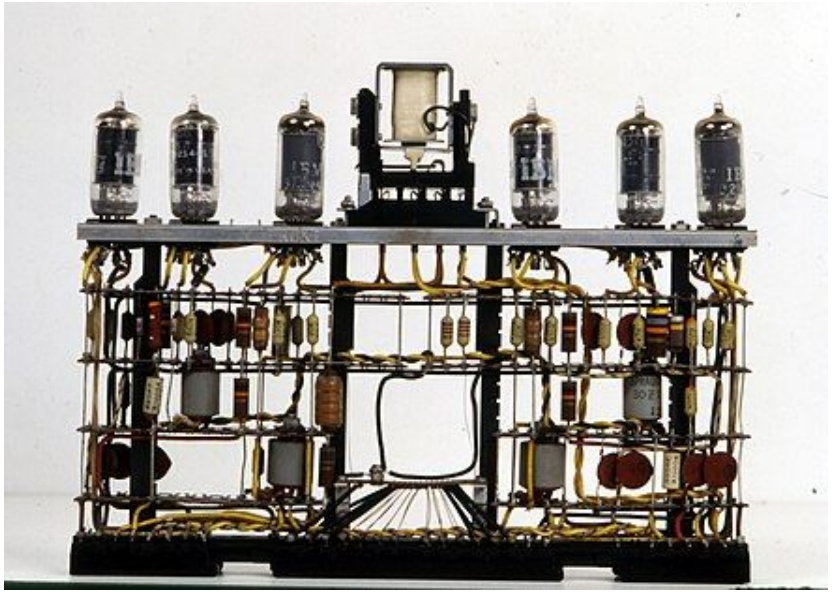
Consumer Electronics Show 1967

AAAI Conference (1980)

European Conference on AI (ECAI) [previously AISB] 1974

Meanwhile, IBM

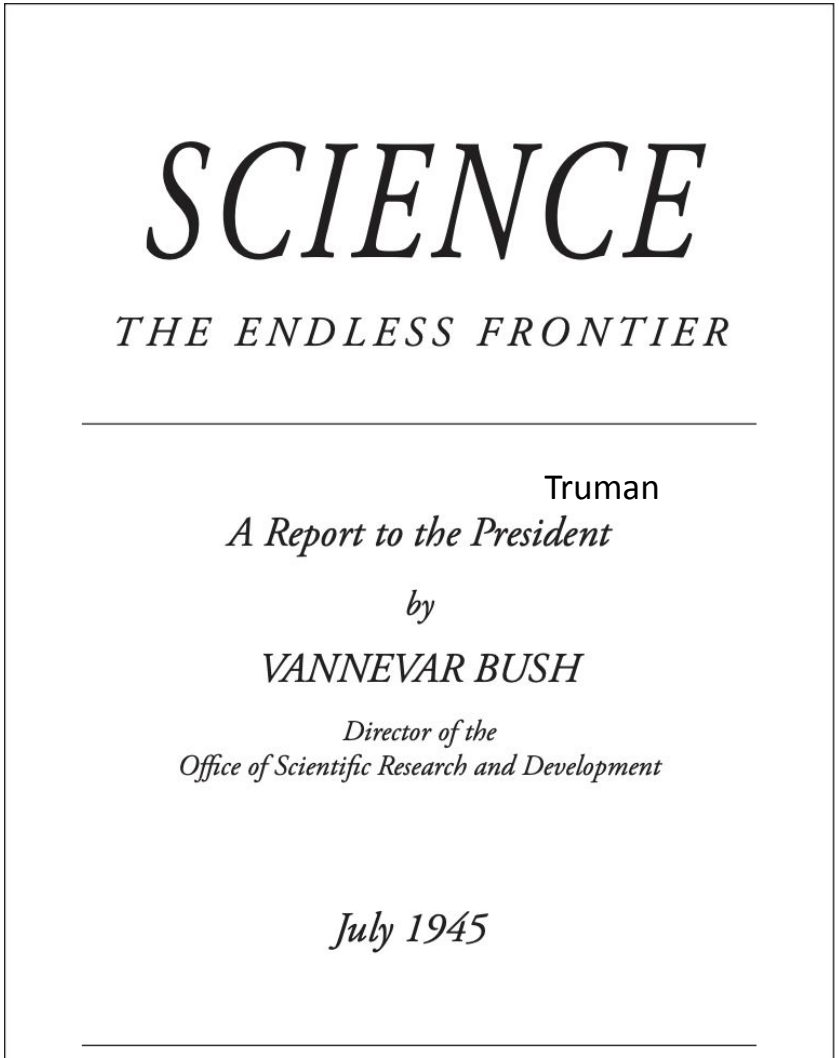
Introduced in 1954, IBM 704 was the first mass produced computer with floating-point arithmetic capacity



Vacuum Tubes

Arthur Samuel in 1952 wrote programs for the IBM 704 playing checkers at “strong amateur” level

Cold War!



U.S. funds

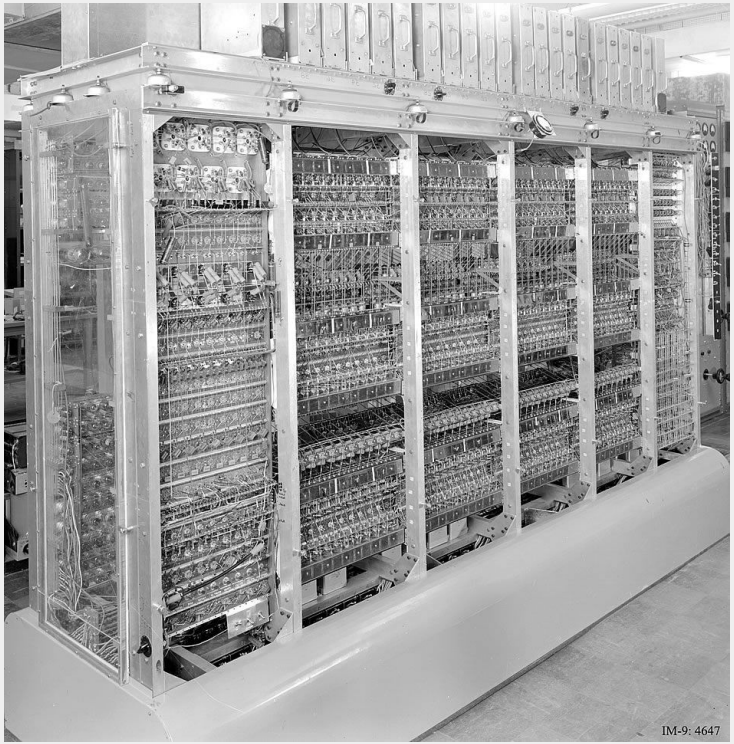
"War is increasingly <u>total war,</u> in which the armed services must be supplemented by active participation of every element of the civilian population.

<u>There must be more - and more adequate - military research during peacetime….The obligation of Government to support research on military problems is inescapable.</u>

Modern war requires the use of the most advanced scientific techniques. Many of the leaders in the development of radar are scientists who before the war had been exploring the nucleus of the atom. While there must be increased emphasis on science in the future training of officers for both the Army and Navy, such men cannot be expected to be specialists in scientific research. <u>Therefore a professional partnership between the officers in the Services and civilian scientists is needed.</u>"

ENIAC → MANIAC 1946

- Mathematical Numerical Integrator Computer
- Von Neumann funded by Atomic Energy Commission for Hydrogen bomb research
- Smarter and smaller than ENIAC
- “Computer and the Brain” 1957:
  - “artificial automaton that has been constructed for human use.”



1 March, 1954

- Castle Bravo
- First in the series of hydrogen bomb tests
- Bikini Atoll, Marshall Islands
- “With a predicted yield of 6 megatons, Castle Bravo would deliver twice as much power as all the bombs dropped on Germany and Japan during World War II together, including both atomic bombs.”
- Radiation fallout in nearby countries

October 4, 1957

"All the News That's Fit to Print"

The New York Times.

LATE CITY EDITION

VOL. CVII..No. 36,414. NEW YORK, SATURDAY, OCTOBER 5, 1957. FIVE CENTS

SOVIET FIRES EARTH SATELLITE INTO SPACE; IT IS CIRCLING THE GLOBE AT 18,000 M. P. H.; SPHERE TRACKED IN 4 CROSSINGS OVER U. S.

HOFFA IS ELECTED TEAMSTERS' HEAD; WARNS OF BATTLE

Defeats Two Foes 3 to 1 —Says Union Will Fight 'With Every Ounce'

Part of the Hoffa address is printed on Page 8.

By A. H. RASKIN

Special to The New York Times.

MIAMI BEACH, Oct. 4—The massive, scarred International Brotherhood of Teamsters elected James R. Hoffa as its president today.

He won by a margin of nearly 3 to 1 over the combined vote of two rivals who campaigned on pledges to clean up the union's biggest racket.

Amidst rackets investigations and Hoffa's strike in the union rank-and-file immediately opened actions to strip the 44-year-old former manhandler from Detroit of his election victory.

A packed Miami auditorium, however, greeted the news that he might be ousted from the American Federation of Labor and Congress of Industrial Organizations. He appealed for

IN TOKEN OF VICTORY: Dave Beck, retiring head of the Teamsters Union, raises hand of James R. Hoffa upon his election as union's president. At right is Mrs. Hoffa.

COURSE RECORDED

Navy Picks Up Radio Signals—4 Report Sighting Device

By WALTER SULLIVAN

Special to The New York Times.

WASHINGTON, Saturday, Oct. 5.—The Naval Research Laboratory announced early today that it had recorded four crossings of the Soviet earth satellite over the United States.

It said that six had passed near Washington. Two crossings were further to the west. The locations of the fourth was not made available immediately.

It added that tracking would be continued in an attempt to pin down the orbit sufficiently to obtain scientific information of the type sought in the International Geophysical Year.

Four visual sightings, one of which was in conjunction with a radio contact, were reported by early Saturday morning. Two sightings were made at Columbus, Ohio, and one each from Terre Haute, Ind. and Whittier, Calif.

Press Reports Noted

Soviet newspapers reported several weeks ago that the Soviet satellite would broadcast on frequencies in the neighborhood of twenty and forty megacycles. More exact frequencies were given by Soviet scientists

The approximate orbit of the Russian earth satellite is shown by black line. The rotation of the earth will bring the United States under the orbit of Soviet-made moon.

Device Is 8 Times Heavier Than One Planned by U.S.

Special to The New York Times.

WASHINGTON, Oct. 4—Leaders of the United States earth satellite program were astonished tonight to learn that the Soviet Union had launched a satellite eight times heavier than that contemplated by this country.

Dr. Joseph Kaplan, chairman

560 MILES HIGE

Visible With Simple Binoculars, Moscow Statement Says

Text of Tass announcement appears on Page 3.

By WILLIAM J. JORDEN

Special to The New York Times.

MOSCOW, Saturday, Oct. 5—The Soviet Union announced this morning that it successfully launched a man-made earth moon on its space postal day.

The Russians calculated the satellite's orbit as a maximum of 560 miles above the earth and its speed at 18,000 miles an hour.

The official Soviet news agency Tass said the artificial moon, with a diameter of twenty-two inches and a weight of 184 pounds, was circling the earth once every hour and thirty-five minutes. This means more than fifteen times a day.

Two radio transmitters, Tass said, are sending signals continuously on frequencies of 20.005 and 40.002 megacycles. These signals were said to be strong enough to be picked up by amateur radio operators. The trajectory of the satellite is being tracked by numerous scientific stations.

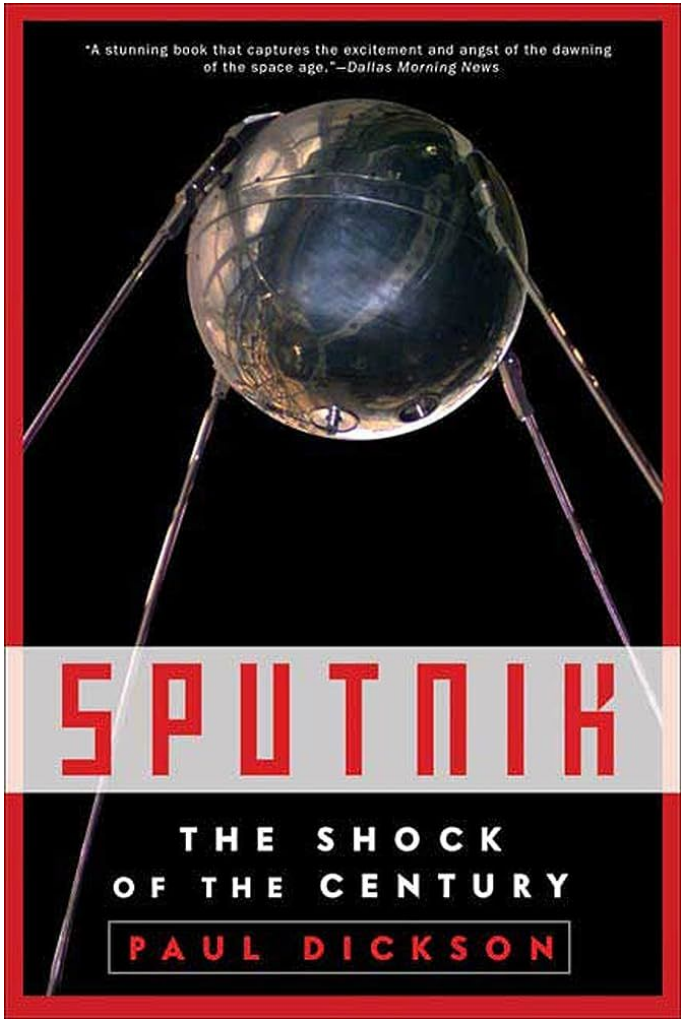
Text Over Moscow Today

FAUBUS COMPARES HIS STAND TO LEE'S

Flu Widens in City; 10% Rate Predicted; 200,000 Pupils Out

ARGENTINA TAKES EMERGENCY STEPS

SATELLITE SIGNAL

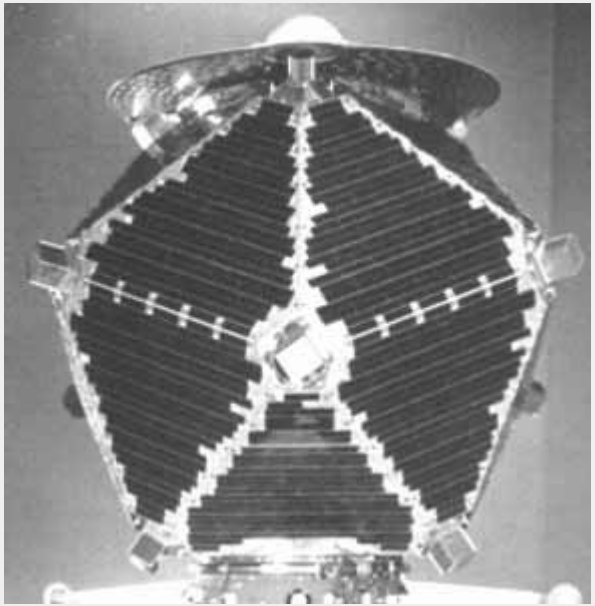


January 10th 1958
ARPA - rise of interactive computing for defense

- Eisenhower administration creates Advanced Research Projects Agency
- Until NASA was founded, originally an interim space agency (under Office of Secretary of Defense)
- Research = Security
- Was able to sponsor high risk projects, without congressional oversight, not subject to peer-review

ARPA & MAD

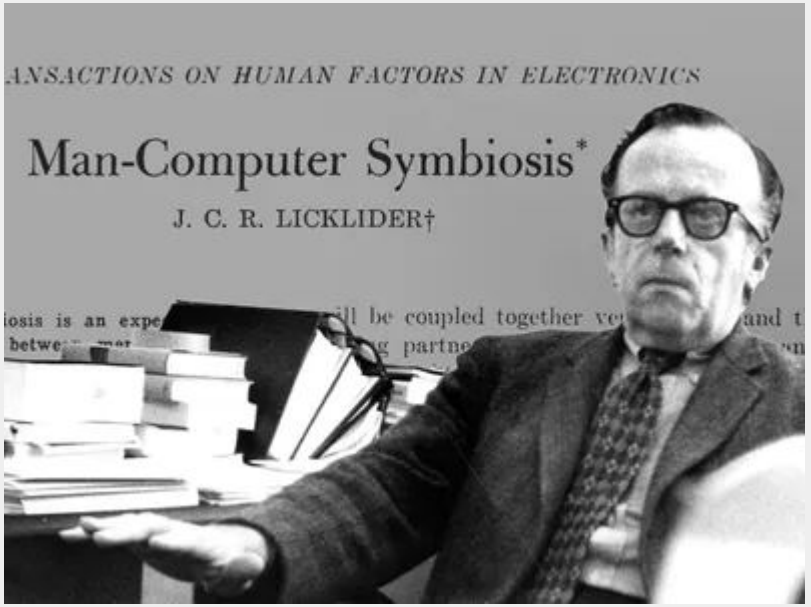
- Vela: Seismic & Atmospheric Sensing (Partial Nuclear Test Ban)
- Defender: Antiballistic Missile Project



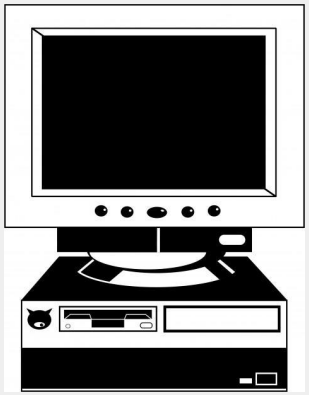
Vela Satellite

ARPA's IPTO 1961
Information Processing Techniques Office

- JFK's directive to step up "command and control" research
- Headed by "Lick" J.C.R. Licklider, psychologist who studied computer interaction & psychoacoustics in the 1950s
- "the electronic members of the partnership"
- **"Command and control" included research in ballistic missiles, radar tracking, high-energy lasers, etc.**



“Lick’s” Man-Computer Symbiosis 1960



speed and accuracy

flexibility and intuition

“The hope is that, in not too many years, human brains and computing machines will be coupled together very tightly, and that the resulting partnership will think as no human brain has ever thought and process data in a new way not approached by the information-handling machines we know today.”

JCR Licklider, Man-Computer Symbiosis, 1960

JCR Licklider

Time & Language: 2 components of Symbiosis

1. **Time**

   Batch-processing was too slow → need time sharing for real-time, interactive computing (especially in command-and-control issues)

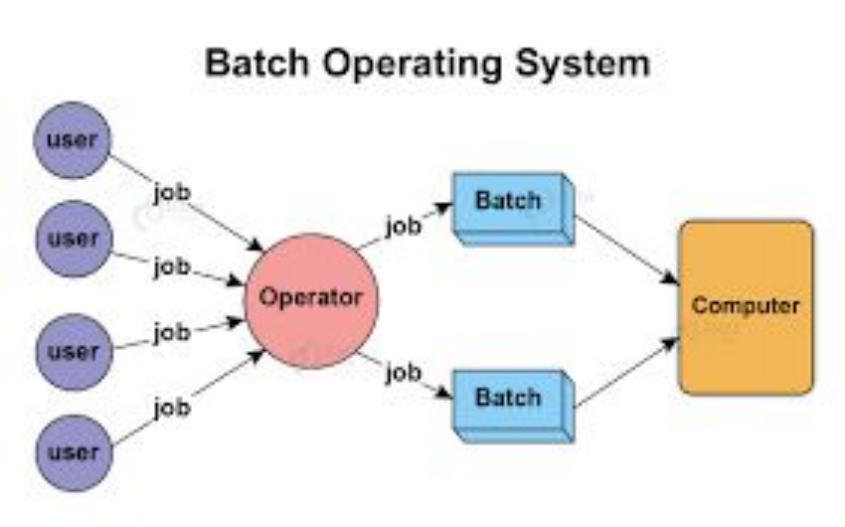
2. **Language**

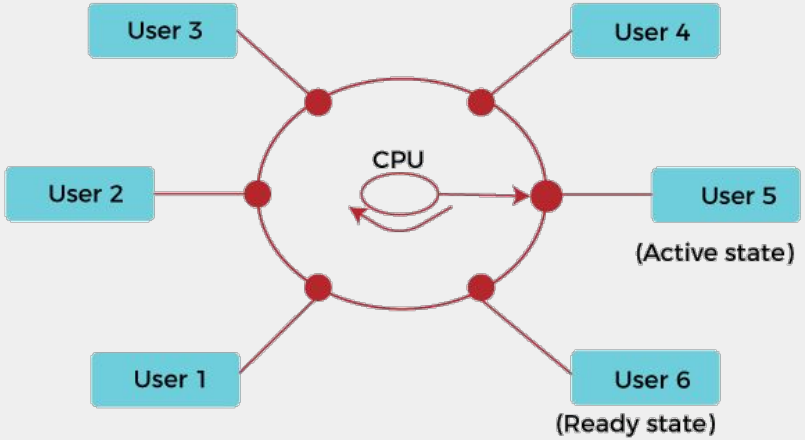
   Assembly → programming languages more similar to human beings (IPL/LISP); Computers and humans should be able to interact in ordinary speech (especially in military command-and-control issues)

It is to bring computing machines effectively into processes of thinking that must go on in **"real time,"** time that moves too fast to permit using computers in conventional ways. Imagine trying, for example, to direct a battle with the aid of a computer on such a schedule as this. You formulate your problem today. Tomorrow you spend with a programmer. Next week the computer devotes 5 minutes to assembling your program and 47 seconds to calculating the answer to your problem. You get a sheet of paper 20 feet long, full of numbers that, instead of providing a final solution, only suggest a tactic that should be explored by simulation. Obviously, the **battle would be over before the second step in its planning was begun.** To think in interaction with a computer in the same way that you think with a colleague whose competence supplements your own will require much tighter coupling between man and machine than is suggested by the example and than is possible today.

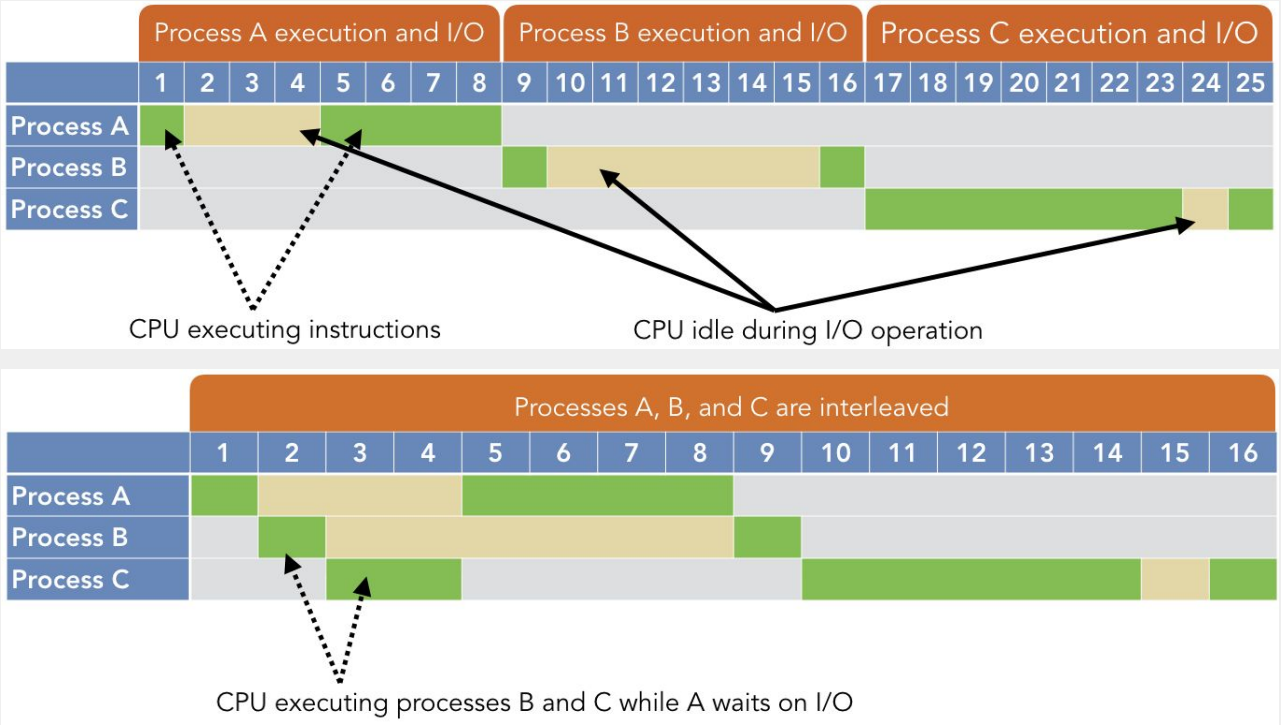
The basic **dissimilarity between human languages and computer languages may be the most serious obstacle to true symbiosis**. It is reassuring, however, to note what great strides have already been made, through interpretive programs and particularly through assembly or compiling programs such as FORTRAN, to adapt computers to human language forms. The "Information Processing Language" of Shaw, Newell, Simon, and Ellis [24] represents another line of rapprochement.

Batch Processing vs. Time-Sharing



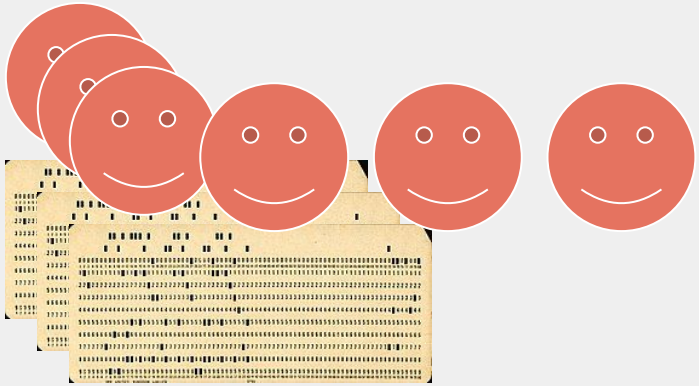


(Multiprogramming)

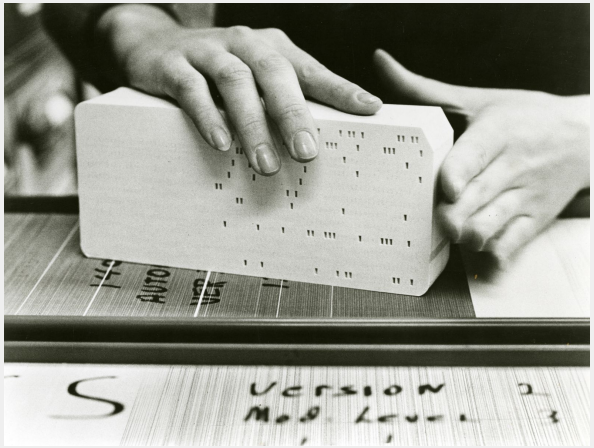


https://w3.cs.jmu.edu/kirkpams/OpenCSF/Books/csf/html/Multiprogramming.html

From Batch Processing …



Jobs

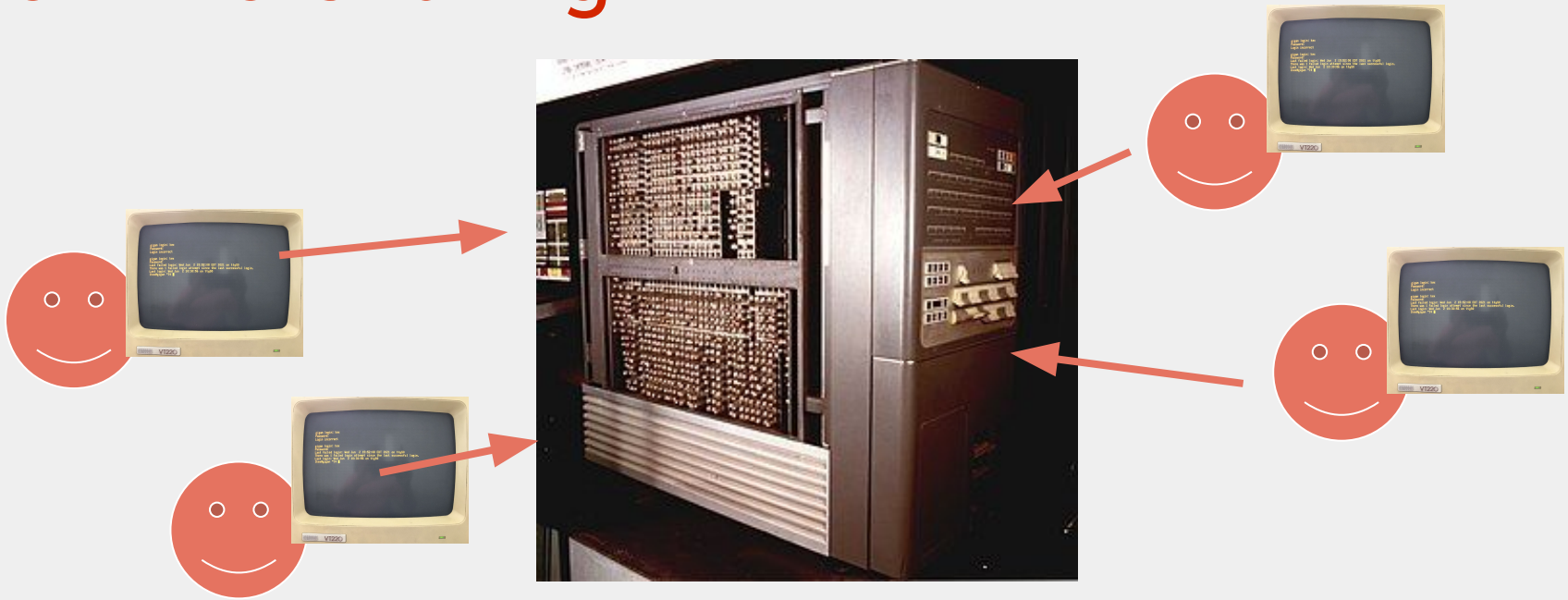


Batch

Processed

… to Time-Sharing

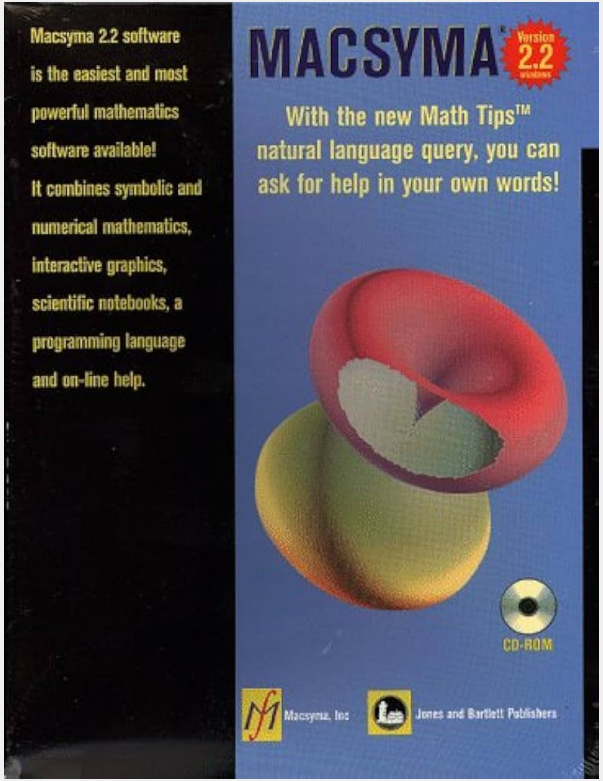
Illusion of real-time computing



Interactive Computing out of IPTO

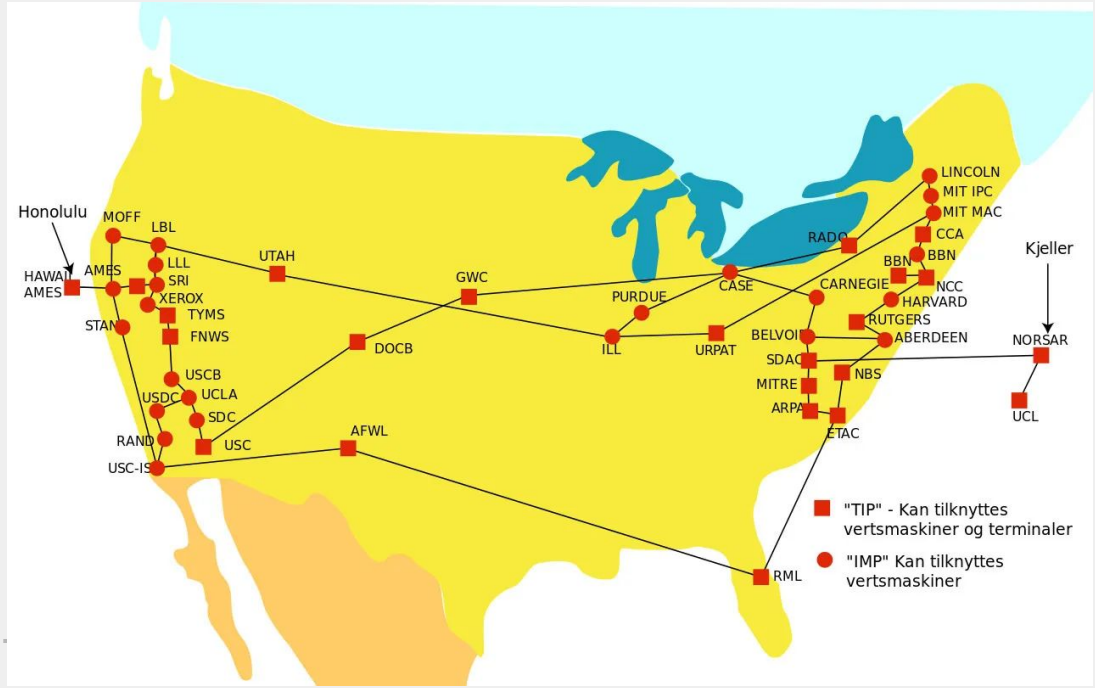
- MAC (Man and Computer; Multi-Access Computing): ARPA spent $25m 1964-1970
- Multics, TSS (Time sharing interfaces)
- MACSYMA (High level algebra program)

```
$TIME SHARING SYSTEM ON THE AIR
LOGIN 10335 JG025
$OK LOG ON 16
DRUMS
$4324 DRUM WDS.
DIAL 9  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
$MSG IN
FROM 9  WE CAN GET YOU ON IN 5 MINUTES.
FROM 9  GO AHEAD AND LOAD.
LOAD ALPHA1
$LOAD OK
(CHANNEL 9'S TELETYPE COPY - THE OPERATOR'S TELETYPE)
$TIME SHARING SYSTEM ON THE AIR
LOGIN 8762 JG025 MANAGR  (LOAD REQUEST FOR MANAGER PROGRAM)
$LOAD OK
GO       (OPERATOR TYPES 'GO' TO START PROGRAM)
ENTER SCOPE NO.  6  (PROG. QUERIES OPERATOR FOR SCOPE NO.)
AM,20    (REQUESTS MANAGER PROG. TO UPDATE SCOPE EVERY 20 SEC.)
FROM 16  THIS IS JOHN JONES, I NEED 20K IN ORDER TO LOAD MY PROG.
26   (REQUESTS MANAGER PROG. TO GIVE A DETAILED DISPLAY OF CH. 26)
!DIAL 26  CAN YOU FINISH IN 5 MIN. - WE HAVE A HIGH PRIORITY REQUEST.
$MSG IN
FROM 26  OK - I'LL FINISH UP IN 5 MINUTES.
DIAL 16  WE CAN GET YOU ON IN 5 MINUTES.
$MSG IN
FROM 26  I'M THROUGH
OT 26   (OPERATOR QUITS CHANNEL 26)
DIAL 16  GO AHEAD AND LOAD.
$MSG IN
```



“Intergalactic Computer Network”

- ARPANET → MILNET → Internet
- “Libraries of the Future”



Vietnam War & Project Agile

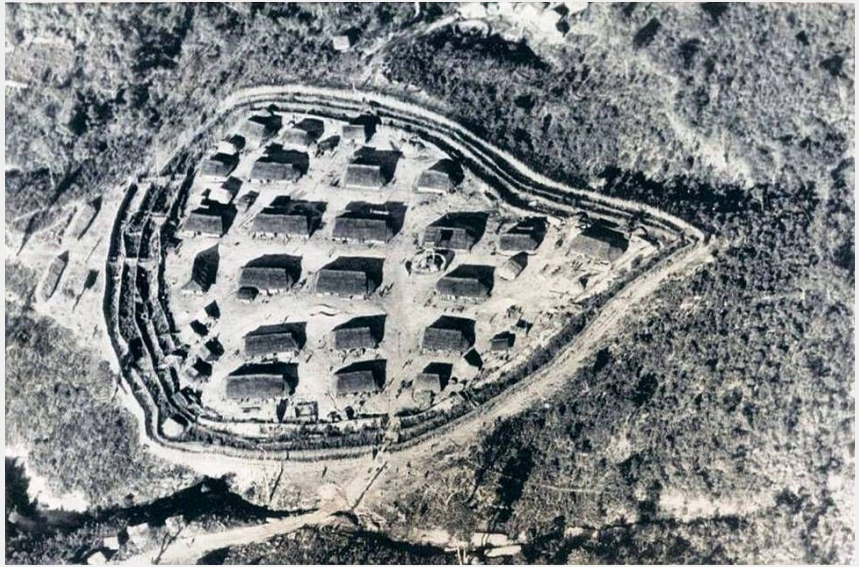
- Seismic Sensors
- Magnetic Sensors
- Infrared Sensors
- Strain Sensors

Sensors dropped on Ho Chi Minh Trail

First Drones. 1968

Behavioral Experiments

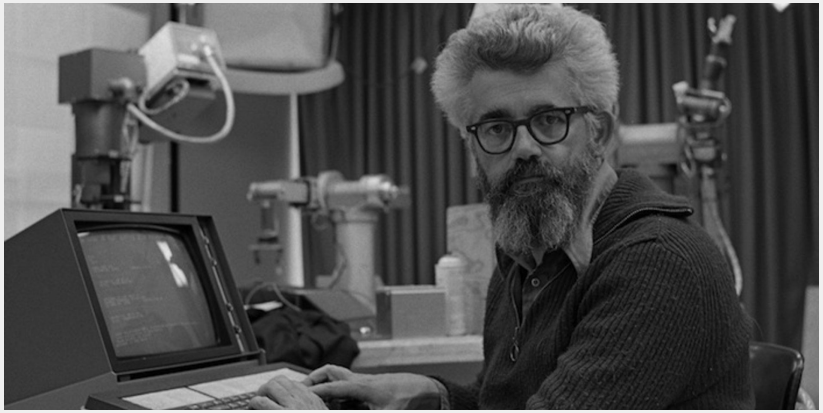
- “Thailand was the laboratory for the soft side and Vietnam was the laboratory for the hard side, or things that go boom”
- “Anthropometric Survey of the Royal Thai Armed Forces” - 2,950 Thai soldiers, sailors, and pilots
- Strategic Hamlet Program to isolate insurgency in Vietnam 1964
- All data taken back to Washington and computed



Strategic Hamlet

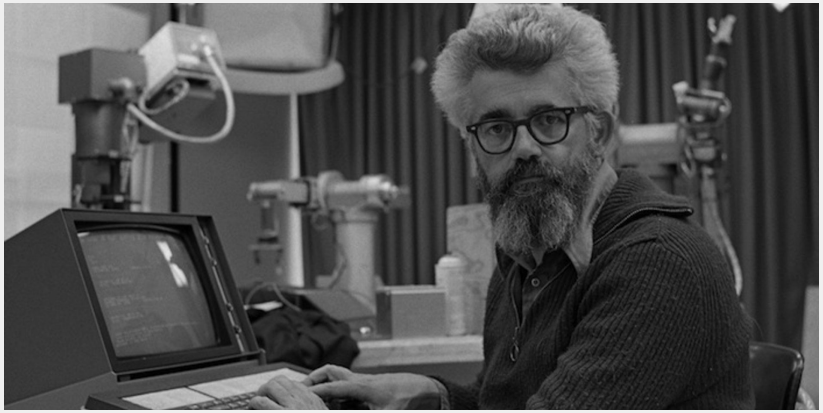
ARPA funds U.S. AI institutions

- MIT: Marvin Minsky and students
- Stanford: John McCarthy started the Stanford AI Lab with unconditional ARPA money 1964
- ARPA supported AI at Rand, Carnegie-Mellon, SRI, SDC (Systems Development Corp) ,BBN (Bolt, Beranak, Newman), and seven other “centers of excellence”
- ARPA became the primary patron for the first twenty years of AI research



ARPA funds U.S. AI institutions

- By the mid-1970s, IPTO's budget accounted for 40-50% of applied research in math and computers
- 80-90% of funding for major AI groups (MIT, Stanford, SRI, CMU) came from ARPA/DARPA



ANALOGY 1963 (Tom Evans)

- Rule-based program written in LISP
- No scanners → coordinate points entered manually
- **'figure A is to figure B as figure C is to which of the given answer figures?'**

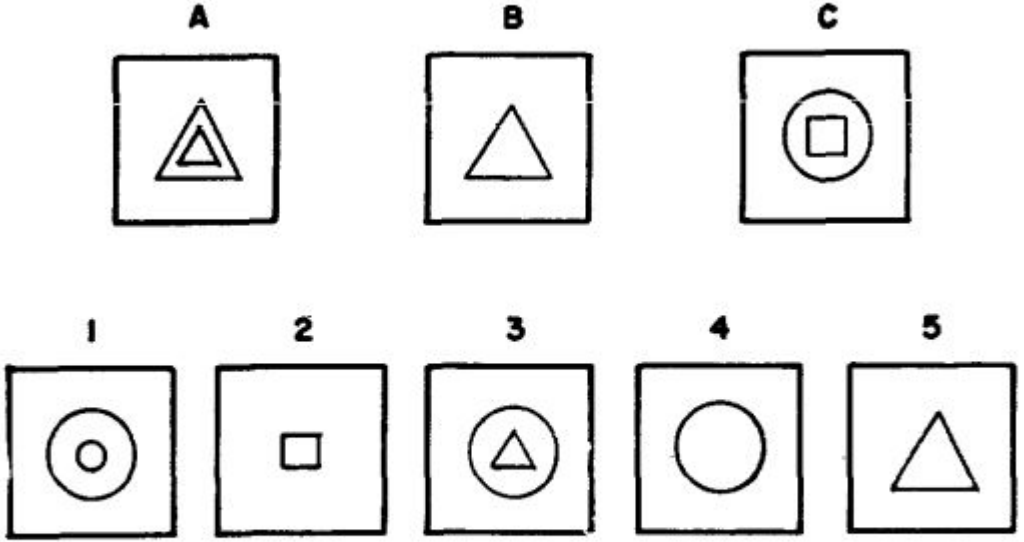


Figure 1.

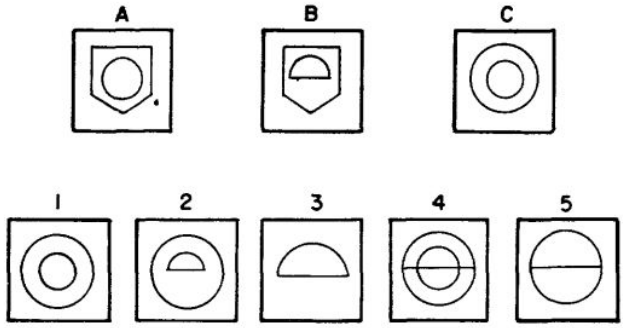


Figure 6.

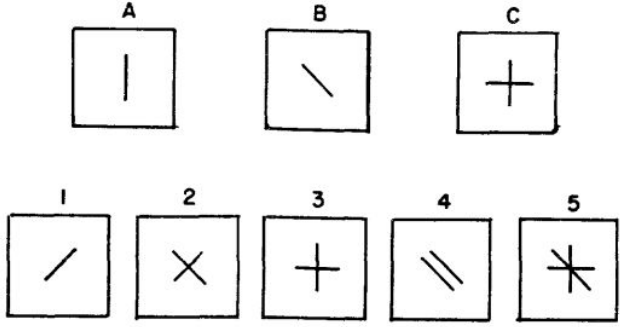


Figure 8.

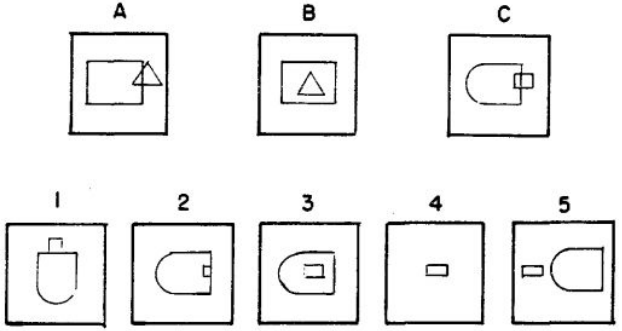


Figure 7.

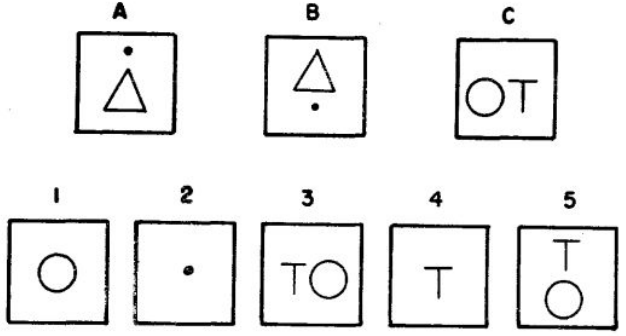


Figure 9.

```
(
(REMOVE A1 ((ABOVE A1 A3) (ABOVE
    A1 A2) (SIM OB3 A1 (((1.0 . 0.0) .
    (N.N))))))
(MATCH A2 (((INSIDE A2 A3) (ABOVE
    A1 A2)) . ((LEFT A2 A3) (SIMTRAN
    (((1.0 . 0.0) . (N.N)) ((1.0 . 3.14) .
    (N.N)))))))
(MATCH A3 (((INSIDE A2 A3) (ABOVE
    A1 A3)) . ((LEFT A2 A3) (SIMTRAN
    (((1.0 . 0.0) . (N.N)))))))
)
```

STUDENT 1964 (Daniel Bobrow)

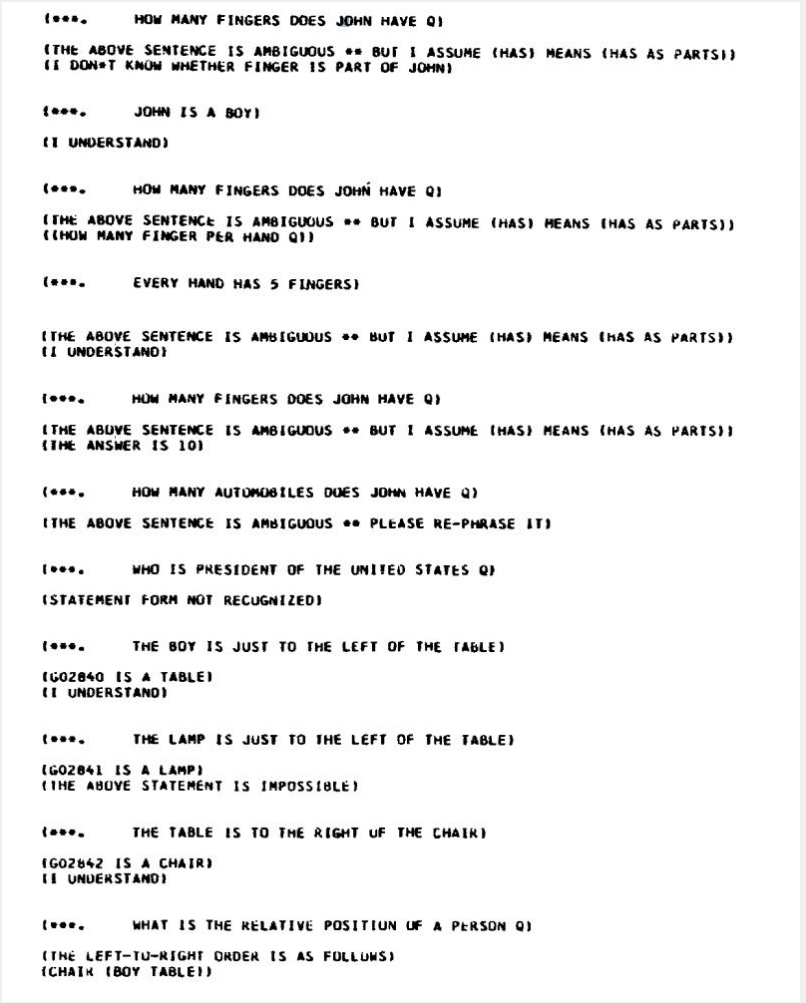
- Rule-based program written in LISP
- Solve math word problems at high school level
- English sentences are converted into kernel sentences (single pieces of info) → mathematical expressions. 52 facts supported.

    - **“Bill’s father’s uncle is twice as old as Bill’s father. Two years from now Bill’s father will be 3 times as old as Bill. The sum of their ages is 92. Find Bill’s age.”**

    - **“If the number of customers Tom gets is twice the square of 20% of the number of advertisements he runs, and the number of advertisements is 45, then what is the number of customers Tom gets?”**

SIR 1964 (Bertram Raphael)

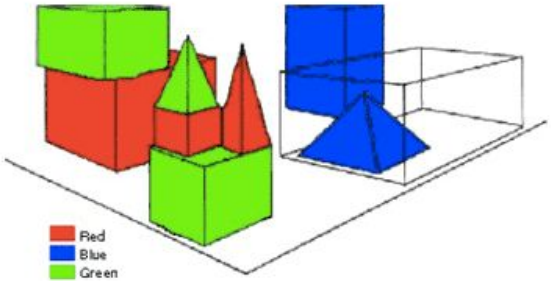
- Rule-based program written in LISP
- Semantic Information Retrieval
- Program interpreted or “understood” simple English relations like ownership, spatial relations, part-whole relations.
- **It used “template matching” to “learn” facts and retrieve them.**
  - **the sum of _____ and _____**



```
(***.     HOW MANY FINGERS DOES JOHN HAVE Q)

(THE ABOVE SENTENCE IS AMBIGUOUS ** BUT I ASSUME (HAS) MEANS (HAS AS PARTS))
(I DON*T KNOW WHETHER FINGER IS PART OF JOHN)


(***.     JOHN IS A BOY)

(I UNDERSTAND)


(***.     HOW MANY FINGERS DOES JOHN HAVE Q)

(THE ABOVE SENTENCE IS AMBIGUOUS ** BUT I ASSUME (HAS) MEANS (HAS AS PARTS))
((HOW MANY FINGER PER HAND Q))


(***.     EVERY HAND HAS 5 FINGERS)


(THE ABOVE SENTENCE IS AMBIGUOUS ** BUT I ASSUME (HAS) MEANS (HAS AS PARTS))
(I UNDERSTAND)


(***.     HOW MANY FINGERS DOES JOHN HAVE Q)

(THE ABOVE SENTENCE IS AMBIGUOUS ** BUT I ASSUME (HAS) MEANS (HAS AS PARTS))
(THE ANSWER IS 10)


(***.     HOW MANY AUTOMOBILES DOES JOHN HAVE Q)

(THE ABOVE SENTENCE IS AMBIGUOUS ** PLEASE RE-PHRASE IT)


(***.     WHO IS PRESIDENT OF THE UNITED STATES Q)

(STATEMENT FORM NOT RECOGNIZED)


(***.     THE BOY IS JUST TO THE LEFT OF THE TABLE)

(G02840 IS A TABLE)
(I UNDERSTAND)


(***.     THE LAMP IS JUST TO THE LEFT OF THE TABLE)

(G02841 IS A LAMP)
(THE ABOVE STATEMENT IS IMPOSSIBLE)


(***.     THE TABLE IS TO THE RIGHT OF THE CHAIR)

(G02842 IS A CHAIR)
(I UNDERSTAND)


(***.     WHAT IS THE RELATIVE POSITION OF A PERSON Q)

(THE LEFT-TO-RIGHT ORDER IS AS FOLLOWS)
(CHAIR (BOY TABLE))
```

Microworlds 1972

- Real world has too many variables → reduce them to a artificial simulation!
- SHRDLU was a rule-based program made to understand natural language
- The program could have natural language conversations about the blocks in block world and manipulate them
- Researchers hoped that by expanding this microworld, more comprehensive AI could emerge



Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX.
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
Computer: YES.
Person: CAN A PYRAMID SUPPORT A PYRAMID?
Computer: I DON'T KNOW.
Person: STACK UP TWO PYRAMIDS.
Computer: I CAN'T.
Person: WILL YOU PLEASE STACK UP BOTH OF THE RED BLOCKS AND EITHER A GREEN CUBE OR A PYRAMID?
Computer: OK. (does it)
Person: WHICH CUBE IS SITTING ON THE TABLE?
Computer: THE LARGE GREEN ONE WHICH SUPPORTS THE RED PYRAMID.
Person: IS THERE A LARGE BLOCK BEHIND A PYRAMID?
Computer: YES, THREE OF THEM: A LARGE RED ONE, A LARGE GREEN CUBE, AND THE BLUE ONE.
Person: PUT A SMALL ONE ONTO THE GREEN CUBE WHICH SUPPORTS A PYRAMID.
Computer: OK. (does it)
Person: PUT THE LITTLEST PYRAMID ON TOP OF IT.
Computer: OK. (does it)
Person: DOES THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS SUPPORT ANYTHING GREEN?
Computer: YES, THE GREEN PYRAMID.
Person: WHAT COLOR IS IT?
Computer: BY "IT", I ASSUME YOU MEAN THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS.
Computer: RED
Person: HOW MANY THINGS ARE ON TOP OF GREEN CUBES?
Computer: I'M NOT SURE WHAT YOU MEAN BY "ON TOP OF "IN THE PHRASE "ON TOP OF GREEN CUBES " .
DO YOU MEAN:

WHY DID YOU DROP IT?

BY "IT", I ASSUME YOU MEAN THE SUPERCONDUCTOR.
I CAN'T EXPLAIN A NON-EXISTENT EVENT.

PLANNER language

- LISP based
- SHRDLU written in PLANNER
- Automate tedious programming tasks like looping and backtracking (one statement was enough to get the program to loop through all until it found a block)

```
(GOAL (IS ?X1 BLOCK))
(GOAL (COLOR-OF ?X1 RED))
(GOAL (CUBE ?X1))
(GOAL (IS ?X2 PYRAMID))
```

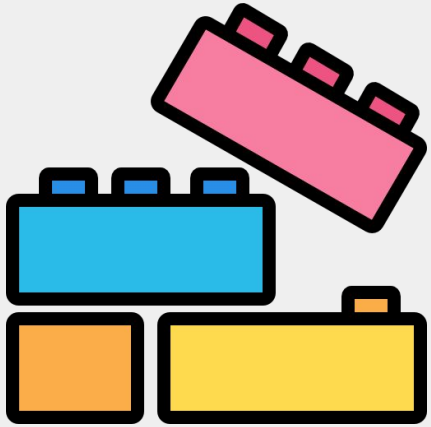
STRIPS 1969 (SRI)

- Shakey the robot
- TV camera, range finder, radio-link antenna
- real life “micro world” of 7 rooms connected by 8 doors
- some rooms were boxes that Shakey could stack or push around
- The control procedure that guided Shakey was called STRIPS (Stanford Research Institute Problem Solver)

Rule-based Shortcomings!

- Rule-based “toy problem” microworlds did not scale
- Language translation hit a ceiling
- Dictionary based machine translation was insufficient for fluent translation. Programs needed encyclopedia knowledge to understand natural language but there was neither software nor hardware to carry this capacity in the 1960s → **In 1966, all U.S. gov support for machine translation ended.**

ENG: “the spirit is willing but the flesh is weak”
RUS: “the vodka is strong but the meat is rotten”



ARPA → DARPA 1972-

- Vietnam War increasingly unpopular (1975 “collapse” of S. Vietnam)
- Congress pressures DoD to be more frugal and spend on defense-related projects
- Greater accountability
- Less freedom/autonomy :(
- But DARPA funding doubled 1977

ARPA → DARPA money

ARPA

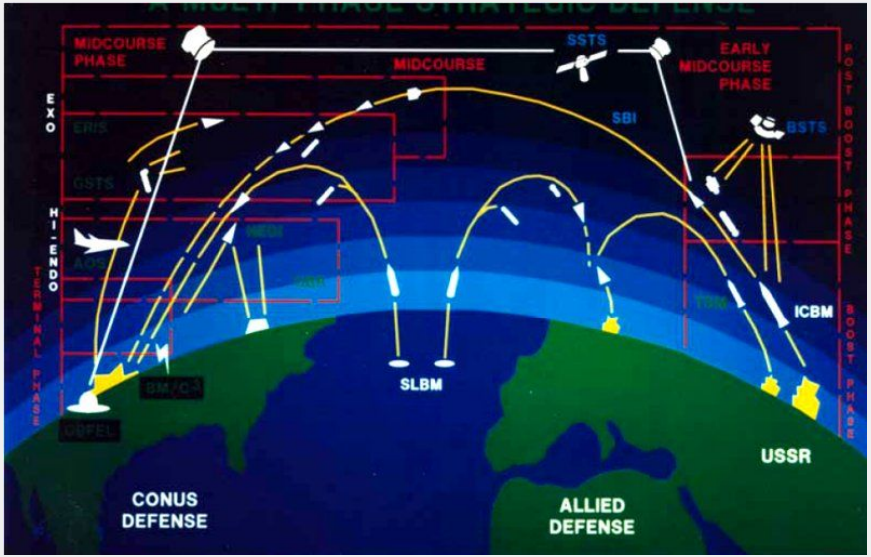
- IPTO program managers solicited proposals in the research community by direct contact with the researchers
- Community/Small Circle
- More experimental
- Network/Trust-oriented

DARPA

- as the funds got larger ($100 million a year), fund granting became more official and publicized. PMs would specify project needs as much as possible.
- Slower
- More competition
- Direct relation to defense

Star Wars 1983

- Reagan announces “Star Wars” Strategic Defense Initiative for ballistic missile research and development
- 20-100 billion dollars spent

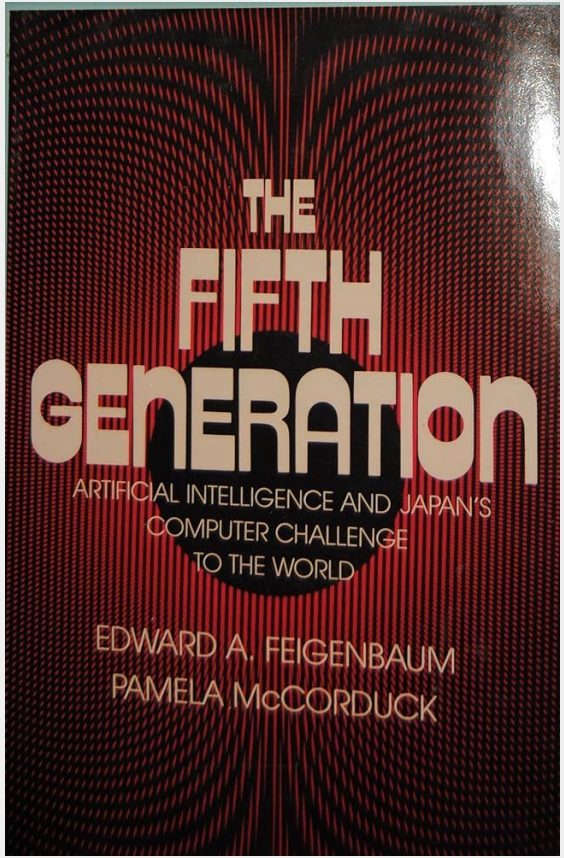


3 week later… Strategic Computing Initiative

- 1 decade long (1983-93)
- 1 billion dollars
- Inspired/threatened by Japan's (MITI's) Fifth Generation Computer Program (1982-, $800million dollar budget)
- By 1985, DARPA had spent $100 million for 92 projects underway at 60 institutions (half in industry, half in universities and gov)
- SCI planners made a detailed ten-year timeline to coordinate all of these subprojects to come together to create 3 intelligent projects

SC

STRATEGIC COMPUTING



Project 1) Autonomous Land Vehicle Project (CMU) 1984-1988

- An autonomous land vehicle (ALV) would be required to navigate unfamiliar terrain at high speeds → possibly a robot tank.
- Funded advances in machine vision

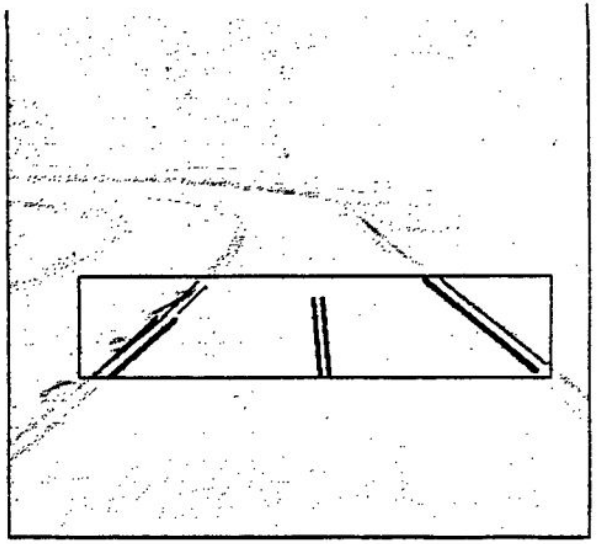


Figure 2: Road edge detection. Here a road image is shown after processing to enhance intensity changes. The vision program selects a window in which to search for road edges. Candidate left and right road edges are lines fit to the raw edge data, shown here as black lines. Heavy black lines indicate the left and right road edges selected by the program. The computed road center line is shown as a double line

Figure 8: An ERIM range image (top) and a color image (bottom) taken in registration

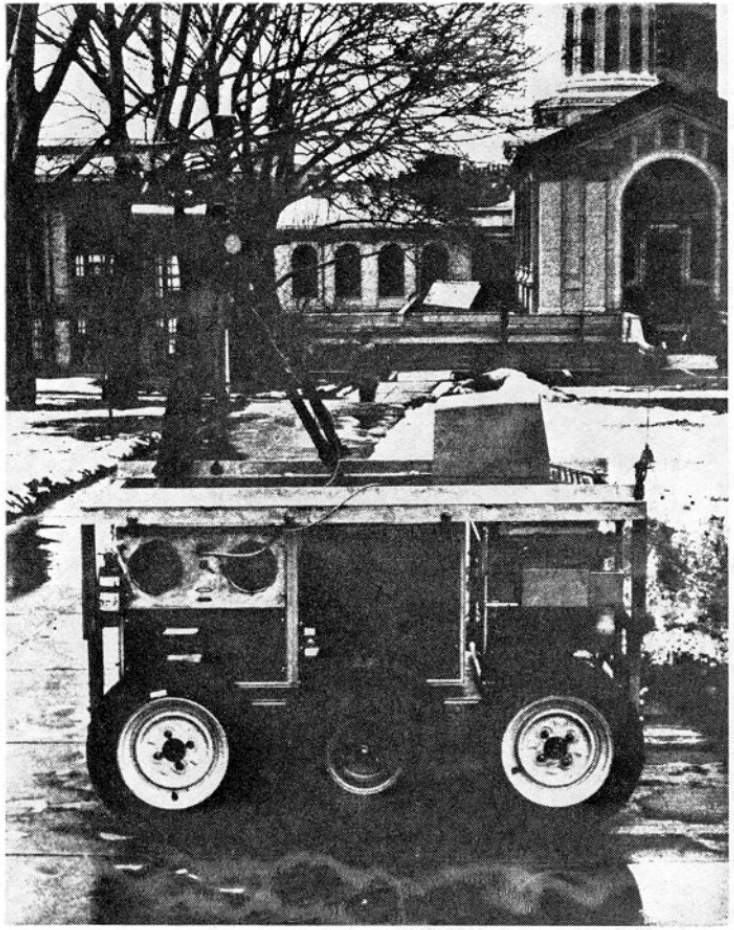
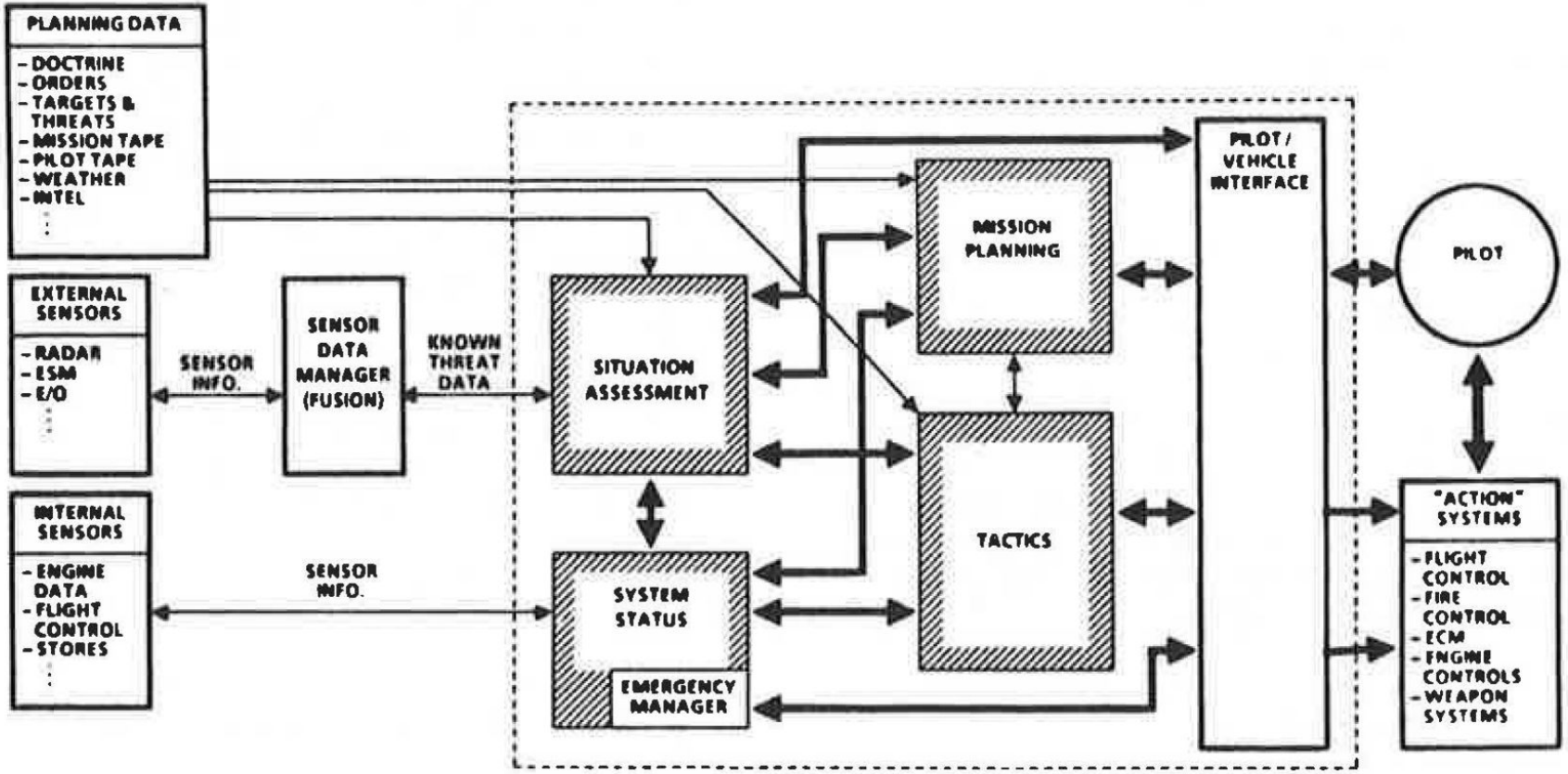


Figure 1: Terregator

ALV
WARNING
ALV

Project 2) Pilot's Associate

- Intelligent fighter associate for the Air Force
- Automated copilot
- Keeping track of many functions pilots must monitor
- It could take over some of the pilot's tasks, reroute electrical signals/fluids in the event of damage.
- Voice recognition was a primary goal → the assistant would be able to understand spoken English even amidst the noise of the jet cockpit.



PILOT'S ASSOCIATE

Figure 6. Pilot's Associate Mission.

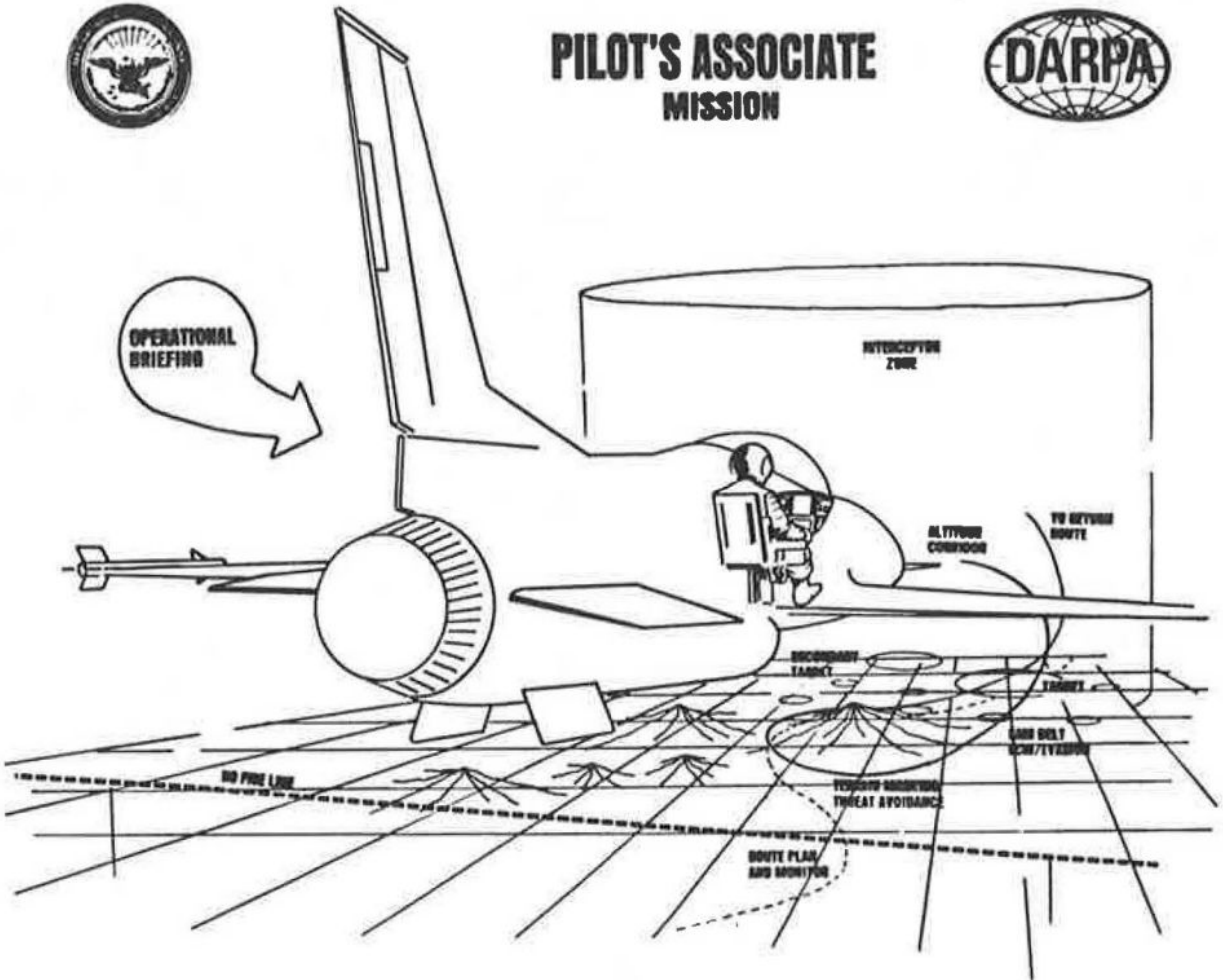
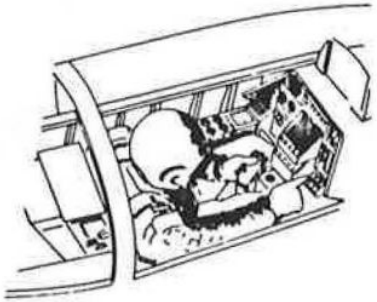


Figure 3. Pilot's Associate Program Overview.



PROGRAM OBJECTIVES

- EXPLOIT DARPA STRATEGIC COMPUTING TECHNOLOGY BASE
  - EXPERT SYSTEMS
  - PARALLEL PROCESSING
  - SPEECH SYSTEMS
  - PILOT-VEHICLE INTERFACE
- REAL-TIME INFORMATION MANAGEMENT
- PROVIDE DECISION AIDING IN DEMANDING HIGH THREAT SCENARIOS

FUNCTIONAL AREAS

- SITUATION ASSESSMENT
- SYSTEM STATUS
- MISSION PLANNING
- TACTICS ADVISORY

DEVELOPMENT ENVIRONMENT

- COMPUTER LABORATORY — PART TASK SIMULATORS
- FULL MISSION SIMULATORS

8643/8SW/GW28/1

PROGRAM MILESTONES

FY86-88:	PHASE I — KNOWLEDGE ENGINEERING, SYSTEM DEVELOPMENT
FY86:	DEMO ONE — FEASIBILITY AND IMPACT ASSESSMENT
FY87:	DEMO TWO — NON-REAL-TIME PART TASK SYSTEM SIMULATION
FY88:	DEMO THREE — NEAR-REAL-TIME TOTAL SYSTEM SIMULATION
FY89-90:	PHASE II — INTEGRATION OF TECHNOLOGY BASE PRODUCTS
FY90:	DEMO FOUR — FULL MISSION REAL-TIME SIMULATION, TOTAL SYSTEM EVALUATION

Project 3) Battle management expert system

- battle management for aircraft carrier battle groups
- analyze sensor data from the carrier's huge field of engagement, interpret threats, keep track of friendly forces, and plan response options that it would then "suggest" to the battle group commanders
- DARPA later (in 1985) added another battle manager for the Army's AirLand Battle Doctrine plan for forward air and ground combat in Europe

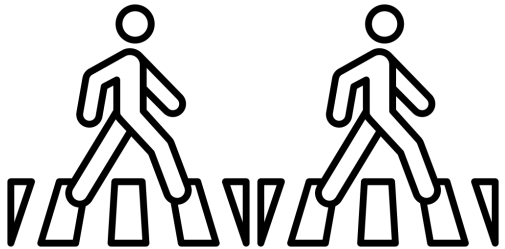
- What is DARPA’s legacy?
- What DARPA tech do you use today?
- What are the pros and cons of government funded tech research?

Autonomous Weapons

“Any weapons system with autonomy in its critical functions. That is, a weapon system which can select (i.e. search for or detect, identify, track, select) and attack (i.e. use force against, neutralize, damage or destroy) target without human intervention.”
— International Committee of the Red Cross

Trolley Problem for Machines

Autonomous car has to decide colliding with single pedestrian vs. multiple pedestrians or to save the car's passenger.



When autonomous machines make decisions about life-and-death situations

When it comes to how agents SHOULD decide: people tend to apply similar norms to human and artificial decision makers

But when people judge the moral quality of the agents' actual decision, their judgements tend to differ:

1. People imagine the the decision conflict the human agent endures → blame mitigation
2. Harder to imagine the machine decision maker's processes and justifications → higher blame

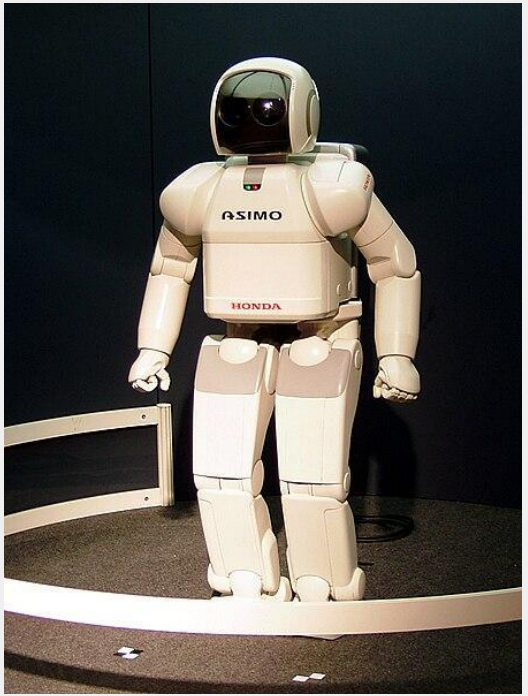
→ Even if the ex ante norms & decisions are the same, machines are judged differently for their actions

Scheutz & Malle. "May Machines Take Lives to Save Lives?" Lethal Autonomous Weapons. Galliott et al.

- What happens are machines are equipped to make life-death decisions?
- What “norms” should they optimize?
- Should they be equipped to justify their “choices”?

History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

Lecture 4 Februrary 14, 2025

- Why language AI?
- Is communication our basis/standard of intelligence?
- How does commercialization of technology change its development?

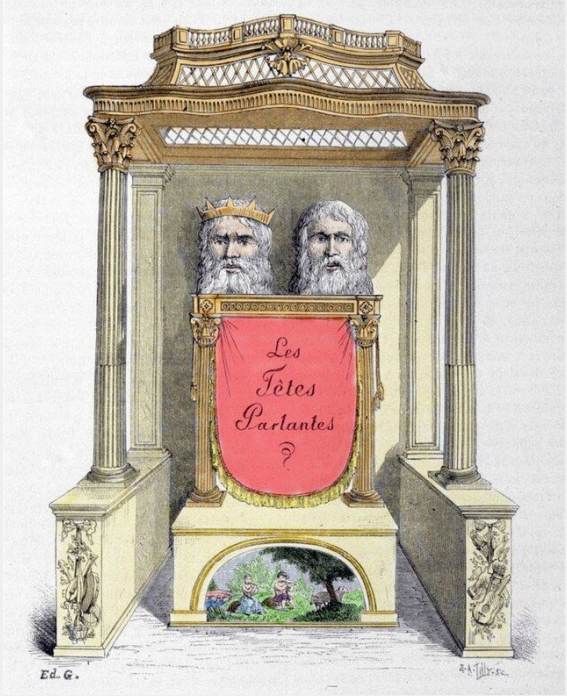
AI or not?

AI or not?

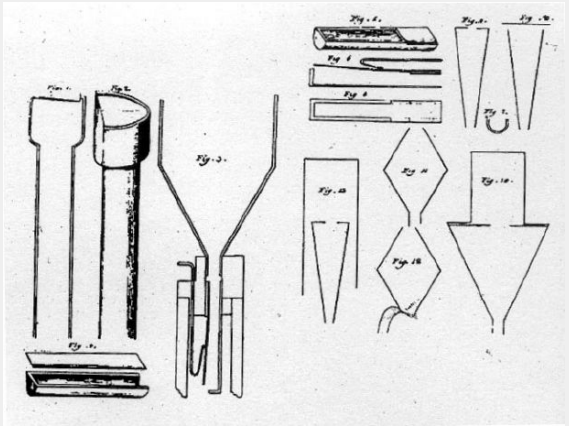
AI or not?

AI or not?

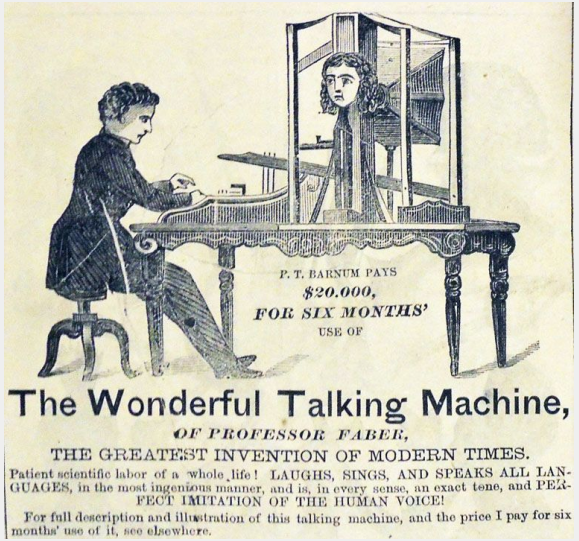
Why language AI? Early Speech Simulation (not synthesis)



abbe Mical's Talking Heads. Paris Academy of Sciences. 1778.

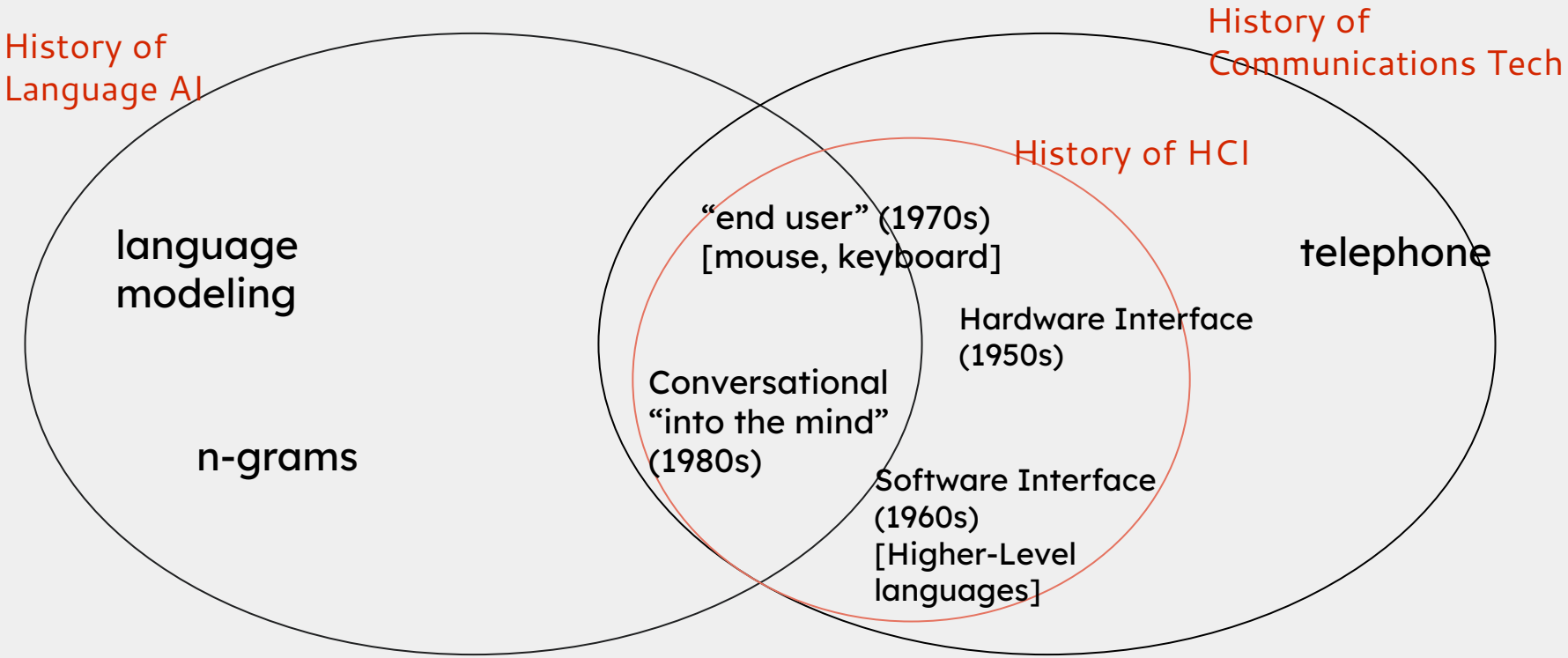


C. G. Kratzenstein. *"vox humana"* Saint Petersburg Academy of Sciences. 1784.



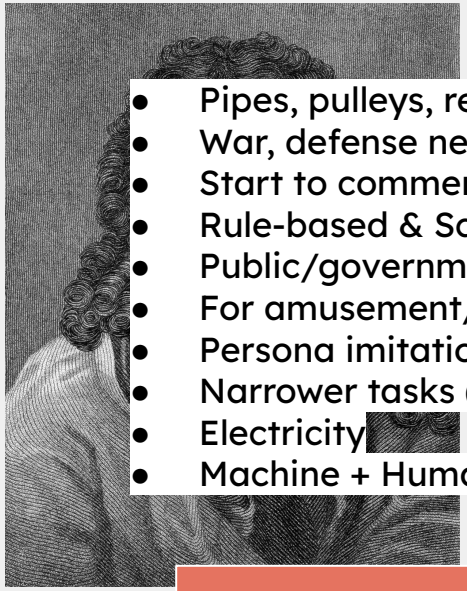
Joseph Faber. Talking Machine. 1840s.

Language AI vs. Communication Tech



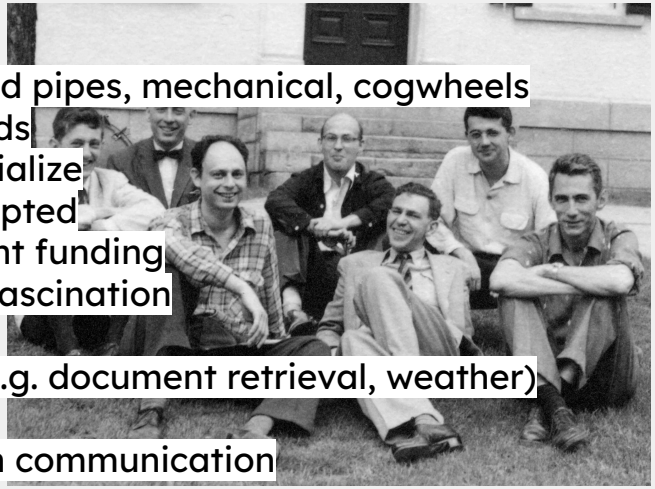
Language AI

Early Modern

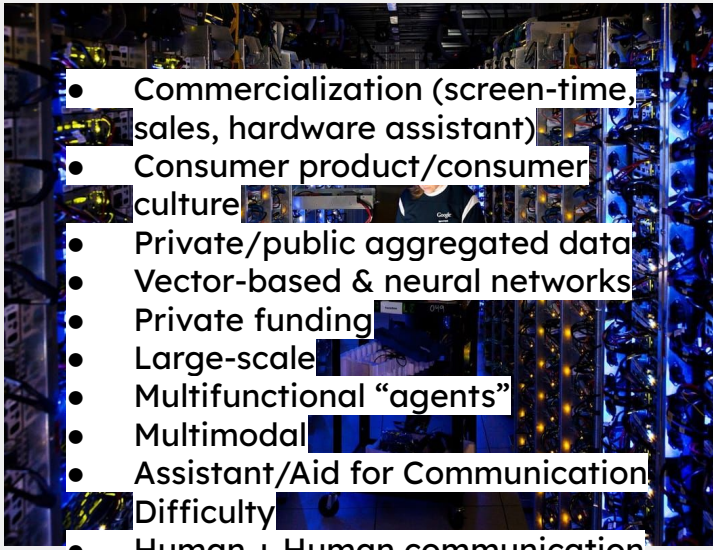


- Pipes, pulleys, reed pipes, mechanical, cogwheels
- War, defense needs
- Start to commercialize
- Rule-based & Scripted
- Public/government funding
- For amusement/fascination
- Persona imitation
- Narrower tasks (e.g. document retrieval, weather)
- Electricity
- Machine + Human communication

Modern



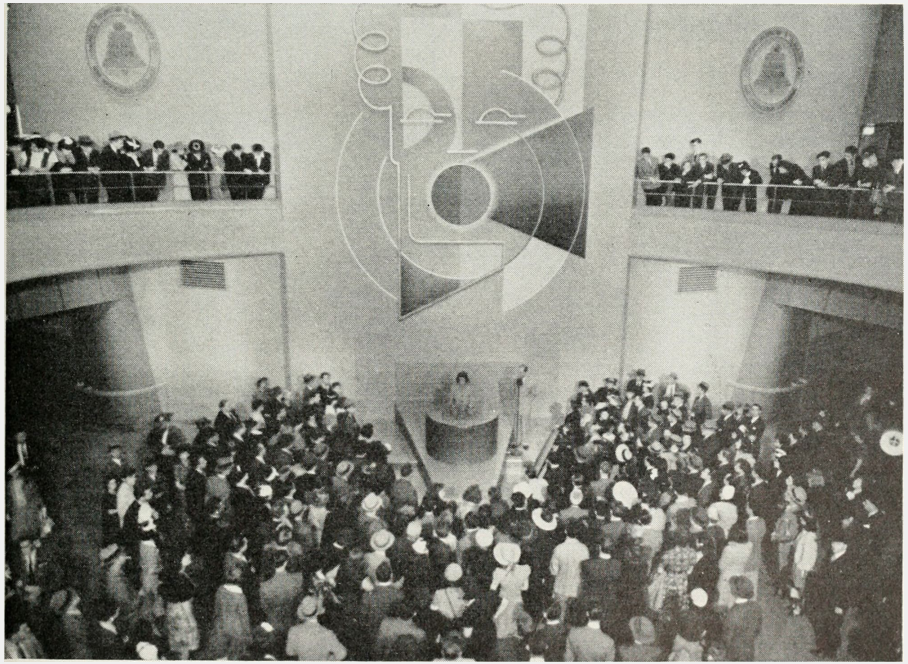
Post-Internet AI



- Commercialization (screen-time, sales, hardware assistant)
- Consumer product/consumer culture
- Private/public aggregated data
- Vector-based & neural networks
- Private funding
- Large-scale
- Multifunctional “agents”
- Multimodal
- Assistant/Aid for Communication Difficulty
- Human + Human communication

Speech before NLP: World Fair 1939

- Speech simulation → Synthesis
- Generating waveforms of audio to simulate human voice
- Bell Labs (AT&T) first experiments in electronic voice synthesis
- Voder (Voice Demonstrator)
- Machine Age → Information Age
- Telephone ownership about 40% 1930s



Speech Synthesis

- AT&T Bell Labs developed a digits recognizing speech system called AUDREY (1952) through simple acoustic pattern matching. AUDREY could understand numbers.
- Transistors → Big Increase in voice technology
- First voice businesses appeared
- 1962 IBM presented the program SHOEBOX, that could understand 16 words
- 1962 Am-Quote that could read stock information

AUDREY 1952

Machine Translation 1947

- Post WWII, machine translation becomes a defense agenda
- Mechanical/Manual translation too time-consuming and inaccurate

One thing I wanted to ask you about is this. A most serious problem, for UNESCO and for the constructive and peaceful future of the planet, is the problem of *translation*, as it unavoidably affects the communication between peoples. Huxley has recently told me that they are appalled by the magnitude and the importance of the translation job.

Recognizing fully, even though necessarily vaguely, the semantic difficulties because of multiple meanings, etc., I have wondered if it were unthinkable to design a computer which would translate. Even if it would translate only scientific material (where the semantic difficulties are very notably less), and even if it did produce an inelegant (but intelligible) result, it would seem to me worth while.

Also knowing nothing official about, but having guessed and inferred considerable about, powerful new mechanized methods in cryptography – methods which I believe succeed even when one does not know what language has been coded – one naturally wonders if the problem of translation could conceivably be treated as a problem in cryptography. When I look at an article in Russian, I say "This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode."

Have you ever thought about this? As a linguist and expert on computers, do you think it is worth thinking about?

March 1947, Warren Weaver, director of the Rockefeller Foundation writing to Cryptographer friend Norbert Wiener (the cybernetician)

“Mechanical” Translation’s Problems

- With “mechanical” translation, a word-for-word dictionary search via a computer, two problems arose:

1. memory:

One way of dealing with this was to limit the size of the vocabulary to the most common words.

2. syntax and word disambiguation:

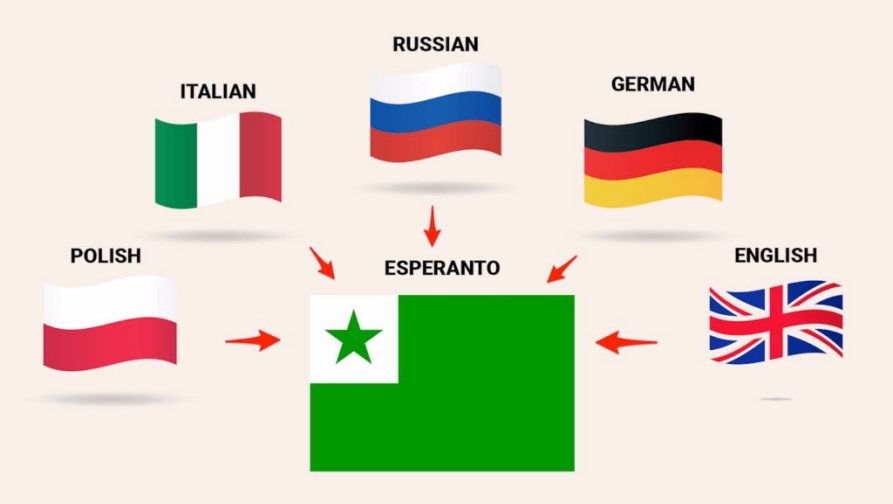
Some proposed an “international language” or semantic blocks that could encode the essence of the input and translate into the output language– “common” language such as Esperanto to capture the shared human meaning of input and output languages.

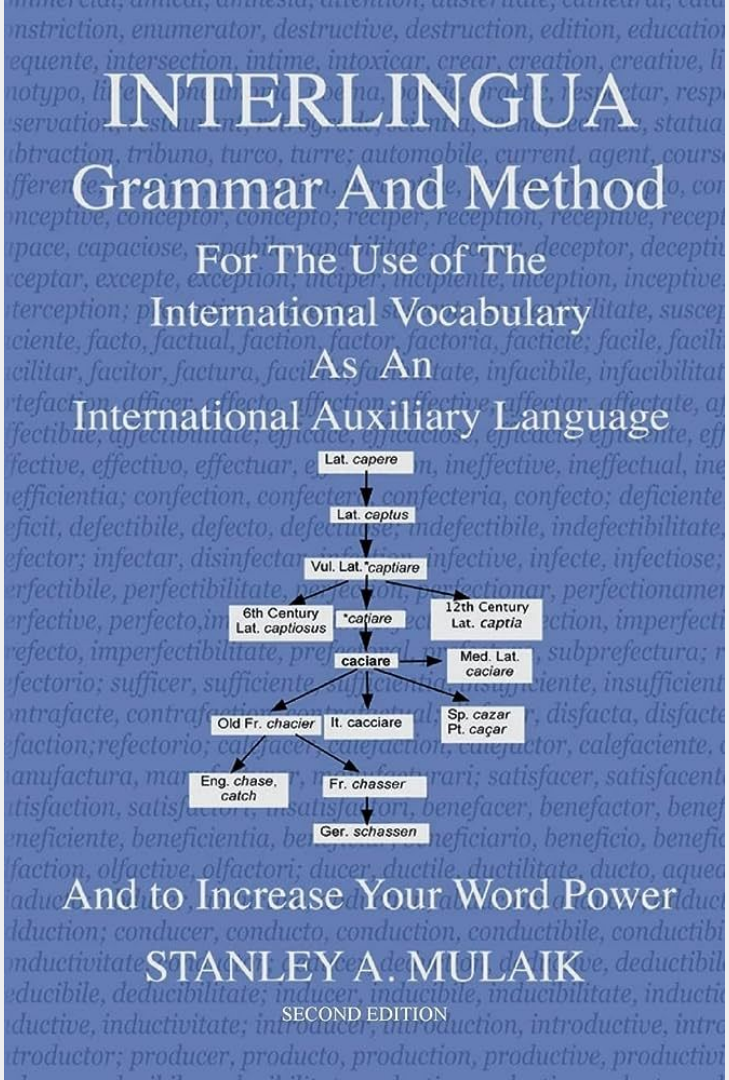
1948 Segmentation

(German) Wenn in ein*em gröss*eren Gebiet zwei Form*en neben*einander leb*en, ohne sich zu vermisch*en, so gehör*en sie verschieden*en Form*en*-kreis*en an.
(English) if in a/one *d* large (more) area two form *m* beside one another live *z* without self to/too mix *z*, so belong/hear *p z z* different *m* form *m* circle *m* at.

A Global Language

- Politically, debate ensued about the "scientific" or "international" language. What common language would scientists speak to share knowledge globally? English? Esperanto? Interlingua?



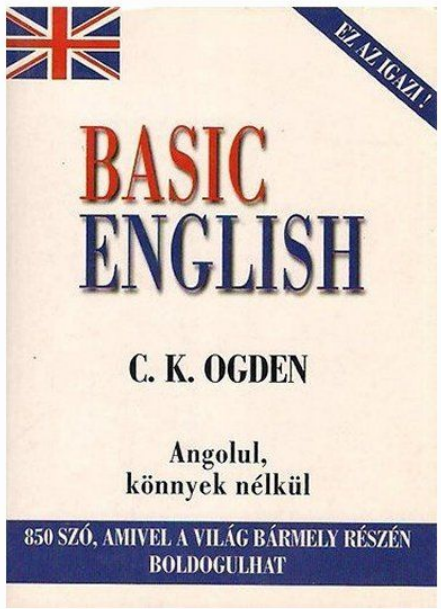


BASIC ENGLISH

OPERATIONS etc.	THINGS						QUALITIES	
100	400 General				200 Pictrured		100 General	50 Opposite
COME	ACCOUNT	EDUCATION	METAL	SENSE	ANGLE	KNEE	ABLE	AWAKE
GET	ACT	EFFECT	MIDDLE	SERVANT	ANT	KNIFE	ACID	BAD
GIVE	ADDITION	END	MILK	SEX	APPLE	KNOT	ANGRY	BENT
GO	ADJUSTMENT	ERROR	MIND	SHADE	ARCH	LEAF	AUTOMATIC	BITTER
KEEP	ADVERTISEMENT	EVENT	MINE	SHAKE	ARM	LEG	BEAUTIFUL	BLUE
LET	AGREEMENT	EXAMPLE	MINUTE	SHAME	ARMY	LIBRARY	BLACK	CERTAIN
MAKE	AIR	EXCHANGE	MIST	SHOCK	BABY	LINE	BOILING	COLD
PUT	AMOUNT	EXISTENCE	MONEY	SIDE	BAG	LIP	BRIGHT	COMPLETE
SEEM	AMUSEMENT	EXPANSION	MONTH	SIGN	BALL	LOCK	BROKEN	CRUEL
TAKE	ANIMAL	EXPERIENCE	MORNING	SILK	BAND	MAP	BROWN	DARK
BE	ANSWER	EXPERT	MOTHER	SILVER	BASIN	MATCH	CHEAP	DEAD
DO	APPARATUS	FACT	MOTION	SISTER	BASKET	MONKEY	CHEMICAL	DEAR
HAVE	APPROVAL	FALL	MOUNTAIN	SIZE	BATH	MOON	CHIEF	DELICATE
SAY	ARGUMENT	FAMILY	MOVE	SKY	BED	MOUTH	CLEAN	DIFFERENT
SEE	ART	FATHER	MUSIC	SLEEP	BEE	MUSCLE	CLEAR	DIRTY
SEND	ATTACK	FEAR	NAME	SLIP	BELL	NAIL	COMMON	DRY
MAY	ATTEMPT	FEELING	NATION	SLOPE	BERRY	NECK	COMPLEX	FALSE
WILL	ATTENTION	FICTION	NEED	SMASH	BIRD	NEEDLE	CONSCIOUS	FEEBLE
ABOUT	ATTRACTION	FIELD	NEWS	SMELL	BLADE	NERVE	CUT	FEMALE
ACROSS	AUTHORITY	FIGHT	NIGHT	SMILE	BOARD	NET	DEEP	FOOLISH
AFTER	BACK	FIRE	NOISE	SMOKE	BOAT	NOSE	DEPENDENT	FUTURE
AGAINST	BALANCE	FLAME	NOTE	SNEEZE	BONE	NUT	EARLY	GREEN
AMONG	BASE	FLIGHT	NUMBER	SNOW	BOOK	OFFICE	ELASTIC	ILL
AT	BEHAVIOUR	FLOWER	OBSERVATION	SOAP	BOOT	ORANGE	ELECTRIC	LAST
BEFORE	BELIEF	FOLD	OFFER	SOCIETY	BOTTLE	OVEN	EQUAL	LATE
BETWEEN	BIRTH	FOOD	OIL	SON	BOX	PARCEL	FAT	LEFT
BY	BIT	FORCE	OPERATION	SONG	BOY	PEN	FERTILE	LOOSE
DOWN	BITE	FORM	OPINION	SORT	BRAIN	PENCIL	FIRST	LOUD
FROM	BLOOD	FRIEND	ORDER	SOUND	BRAKE	PICTURE	FIXED	LOW
IN	BLOW	FRONT	ORGANIZATION	SOUP	BRANCH	PIG	FLAT	MIXED
OFF	BODY	FRUIT	ORNAMENT	SPACE	BRICK	PIN	FREE	NARROW
ON	BRASS	GLASS	OWNER	STAGE	BRIDGE	PIPE	FREQUENT	OLD
OVER	BREAD	GOLD	PAGE	START	BRUSH	PLANE	FULL	OPPOSITE
THROUGH	BREATH	GOVERNMENT	PAIN	STATEMENT	BUCKET	PLATE	GENERAL	PUBLIC
TO	BROTHER	GRAIN	PAINT	STEAM	BULB	PLOUGH	GOOD	ROUGH
UNDER	BUILDING	GRASS	PAPER	STEEL	BUTTON	POCKET	GREAT	SAD
UP	BURN	GRIP	PART	STEP	CAKE	POT	GREY	SAFE
WITH	BURST	GROUP	PASTE	STITCH	CAMERA	POTATO	HANGING	SECRET
AS	BUSINESS	GROWTH	PAYMENT	STONE	CARD	PRISON	HAPPY	SHORT
FOR	BUTTER	GUIDE	PEACE	STOP	CART	PUMP	HARD	SHUT
OF	CANVAS	HARBOUR	PERSON	STORY	CARRIAGE	RAIL	HEALTHY	SIMPLE
TILL	CARE	HARMONY	PLACE	STRETCH	CAT	RAT	HIGH	SLOW
THAN	CAUSE	HATE	PLANT	STRUCTURE	CHAIN	RECEIPT	HOLLOW	SMALL
A	CHALK	HEARING	PLAY	SUBSTANCE	CHEESE	RING	IMPORTANT	SOFT
THE	CHANCE	HEAT	PLEASURE	SUGAR	CHEST	ROD	KIND	SOLID
ALL	CHANGE	HELP	POINT	SUGGESTION	CHIN	ROOF	LIKE	SPECIAL
ANY	CLOTH	HISTORY	POISON	SUMMER	CHURCH	ROOT	LIVING	STRANGE
EVERY	COAL	HOLE	POLISH	SUPPORT	CIRCLE	SAIL	LONG	THIN
LITTLE	COLOUR	HOPE	PORTER	SURPRISE	CLOCK	SCHOOL	MALE	WHITE
MUCH	COMFORT	HOUR	POSITION	SWIM	CLOUD	SCISSORS	MARRIED	WRONG
NO	COMMITTEE	HUMOUR	POWDER	SYSTEM	COAT	SCREW	MATERIAL	
OTHER	COMPANY	ICE	POWER	TALK	COLLAR	SEED	MEDICAL	
SOME	COMPARISON	IDEA	PRICE	TASTE	COMB	SHEEP	MILITARY	
SUCH	COMPETITION	IMPULSE	PRINT	TAX	CORD	SHELF	NATURAL	
THAT	CONDITION	INCREASE	PROCESS	TEACHING	COW	SHIP	NECESSARY	
THIS	CONNECTION	INDUSTRY	PRODUCE	TENDENCY	CUP	SHIRT	NEW	
I	CONTROL	INK	PROFIT	TEST	CURTAIN	SHOE	NORMAL	
HE	COOK	INSECT	PROPERTY	THEORY	CUSHION	SKIN	OPEN	
YOU	COPPER	INSTRUMENT	PROSE	THING	DOG	SKIRT	PARALLEL	
WHO	COPY	INSURANCE	PROTEST	THOUGHT	DOOR	SNAKE	PAST	
AND	CORK	INTEREST	PULL	THUNDER	DRAIN	SOCK	PHYSICAL	
BECAUSE	COTTON	INVENTION	PUNISHMENT	TIME	DRAWER	SPADE	POLITICAL	
BUT	COUGH	IRON	PURPOSE	TIN	DRESS	SPONGE	POOR	
OR	COUNTRY	JELLY	PUSH	TOP	DROP	SPOON	POSSIBLE	
IF	COVER	JOIN	QUALITY	TOUCH	EAR	SPRING	PRESENT	
THOUGH	CRACK	JOURNEY	QUESTION	TRADE	EGG	SQUARE	PRIVATE	
WHILE	CREDIT	JUDGE	RAIN	TRANSPORT	ENGINE	STAMP	PROBABLE	
HOW	CRIME	JUMP	RANGE	TRICK	EYE	STAR	QUICK	
WHEN	CRUSH	KICK	RATE	TROUBLE	FACE	STATION	QUIET	
WHERE	CRY	KISS	RAY	TURN	FARM	STEM	READY	

Ogden's English 1930

- Ogden (1930) proposed that most of our thoughts could be encompassed by just 850 words in the English vocabulary.



Georgetown IBM MT Experiments Jan 7th 1954

- IBM/Georgetown researchers put on a public demo
- IBM 701 translated Russian to English
- 250 Russian words, 6 rules of operational syntax
- 60 sentences
- 6-7 seconds per sentences
- CIA, NSA, Office of Naval Research

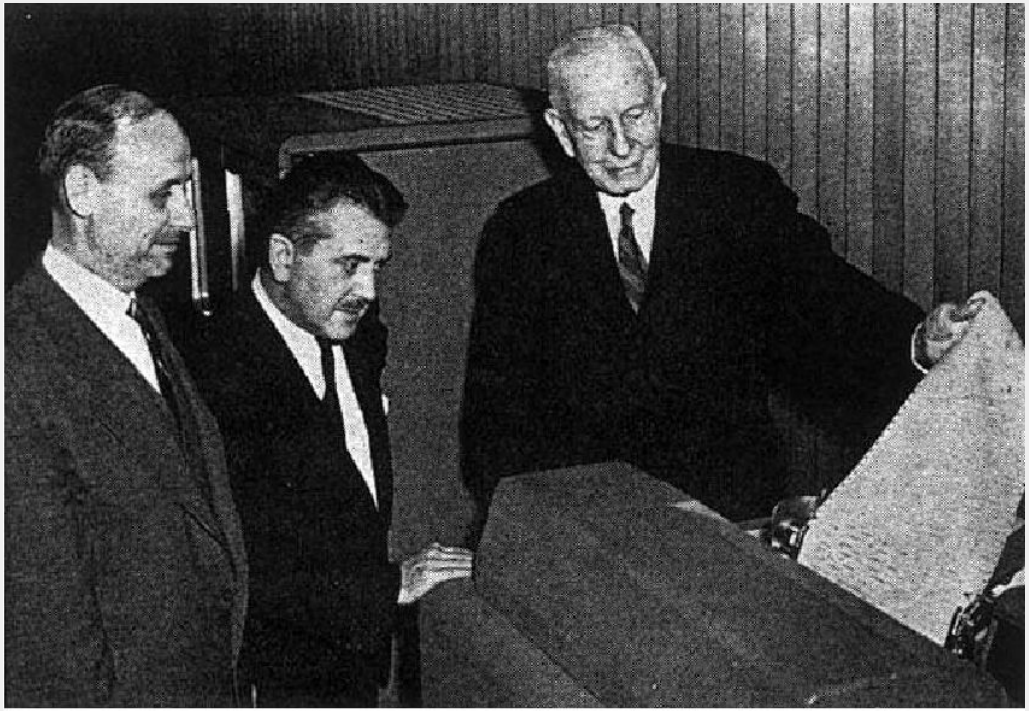


Fig. 2: Hurd, Dostert and Watson at the demonstration

Ruský originál (a angl. transkripce)	Anglický překlad
Качество угля определяется калорийностью Kachyestvo uglya opryedyelyayetsya kaloryiynostjyu	The quality of coal is determined by calory content
Крахмал вырабатывается механическим путем из картофеля Kraxmal virabativayetsya myexanyichyeskyim putyem yiz kartofyelya	Starch is produced by mechanical methods from potatoes
Железо добывается из руды химическим процессом Zhyelyezo dobivayetsya yiz rudi xyimyicheskyim protsyessom	Iron is obtained from ore by chemical process
Мы передаем мысли посредством речи Mi pyeryedayem mislyi posryedstvom ryechi	We transmit thoughts by means of speech
Величина угла определяется отношением длины дуги к радиусу Vyelyichyina ugla opryedyelyayetsya otnoshyenyiyem dlyini dugi k radyiusu	Magnitude of angle is determined by the relation of length of arc to radius
Международное понимание является важным фактором в решении политических вопросов Myezhdunarodnoye ponyimanyiye yavlyayetsya vazhnim faktorom v ryeshyenyiyi polyityichyeskyix voprosov	International understanding constitutes an important factor in decision of political questions
Владимир является на работу поздно утром Vladyimir yavlyayetsya na rabotu pozdno utrom	Vladimir appears for work late in the morning

IBM 701 Electronic Data Processing Machine

- $400,000 machine consisting of eleven separate units and was the size of a tennis court.
- There were only 70 computers in the U.S. in 1954
- Input: IBM 80 column cards

IBM 701. 1954

Russian Is Turned Into English By a Fast Electronic Translator

By ROBERT K. PLUMB

A public demonstration of what is believed to be the first successful use of a machine to translate meaningful texts from one language to another took place here yesterday afternoon.

This may be the culmination of centuries of search by scholars for "a mechanical translator." So far the system has a vocabulary of only 250 words. But there are no foreseeable limits to the number of words that the device can store or the number of languages it can be directed to translate.

Scholars and scientists who worked on it believe that within a few years the system may greatly increase communication, particularly in technical subjects, by making translation quick, accurate and easy.

The demonstration was at the headquarters of the International Business Machines Corporation, 590 Madison Avenue. It is the result of cooperative research by scientists of the corporation and scholars of the Georgetown University Institute of Languages and Linguistics in Washington.

The "mechanical" part of the translation system, which is mostly electronic, is a standard commercial model of the largest International Business Machines "stock" computer. This device, called the IBM Type 701 Electronic Data Processing Machine, was put on the market last April. Since then twelve of the machines have been sold to commercial, military and university computation laboratories.

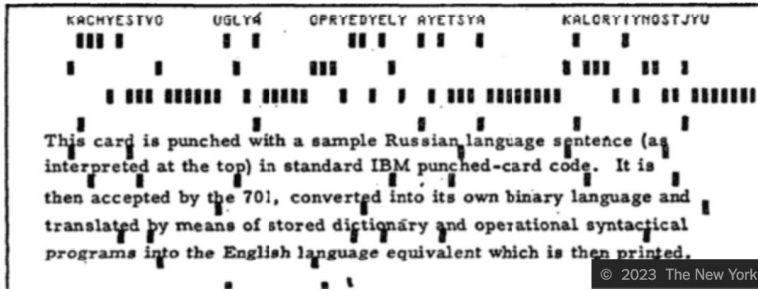
The "literary" part of the system is a mechanical model of language devised at Georgetown by Prof. Leon Dostert and Dr. Paul Garvin. The corporation's share in the project was conducted by Dr. Cuthbert C. Hurd, director of its Division of Applied Science.

In the demonstration, a gir

Continued on Page 5, Column 2

The New York Tim

An electronic calculator produced by the International Business Machines Corporation wa demonstrated yesterday in a new role: translating Russian phrases into English. The de vice has a "vocabulary" of 250 Russian words and can be adapted to other languages. Mis Marilyn Polle "types" Russian phrases on I. B. M. punch cards that are fed into the machine



This card is punched with a sample Russian language sentence (as interpreted at the top) in standard IBM punched-card code. It is then accepted by the 701, converted into its own binary language and translated by means of stored dictionary and operational syntactical programs into the English language equivalent which is then printed.

Jan 8, 1954

© 2023 The New York

Machine Translation Era 1954-1966

- After the 1954 demo MT projects won massive grants for about a decade following
- Went into translating Russian into English
- The CIA was funding projects that amounted to 1.3 million dollars through the National Science Foundation

Russian to English

- Understanding and translating Russian language became a driving force for Machine Translation projects.
- A matter of national security in the throes of the Cold War.
- Russian was also increasingly the global language of science.
- By 1958, Russian abstracts made up (17%) of all entries in *Chemical Abstracts* trailing behind English (50%), the most comprehensive source of information on foreign abstracts.
- MIT translated German, a lab in the U.K. translated French, but the vast majority of Anglophone researchers worked on Russian.

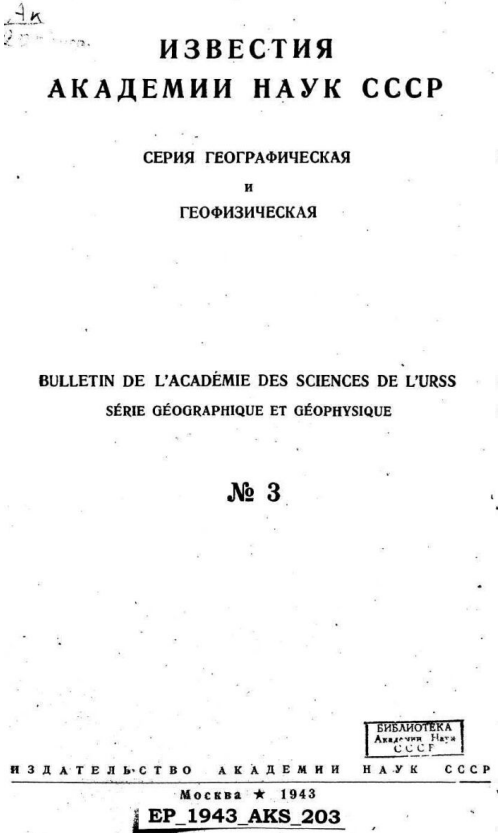
“We chose Russian because we believe that today it is of very great importance to be able to communicate with the Russians … to make faster progress toward the goal of world peace.” – Thomas Watson, Jr. (chairman of IBM)

Russian Science

- Scientists were most interested in *scientific* Russian. esp natural sciences
- In 1956, Georgetown received substantial National Science Foundation fundings to undertake intensive research for the translation of Russian scientific materials into English
- These projects focused on translation scientific journals.
- By 1960, five separate gov agencies (NSF, CIA, the Army, Navy, and Air Force) were funding machine translation.

NSF backed translation project 1956

NSF backed the largest scientific translation project in the world of this time (1956)


ИЗВЕСТИЯ
АКАДЕМИИ НАУК СССР

СЕРИЯ ГЕОГРАФИЧЕСКАЯ
и
ГЕОФИЗИЧЕСКАЯ

BULLETIN DE L'ACADÉMIE DES SCIENCES DE L'URSS
SÉRIE GÉOGRAPHIQUE ET GÉOPHYSIQUE

№ 3

БИБЛИОТЕКА Академии Наук СССР

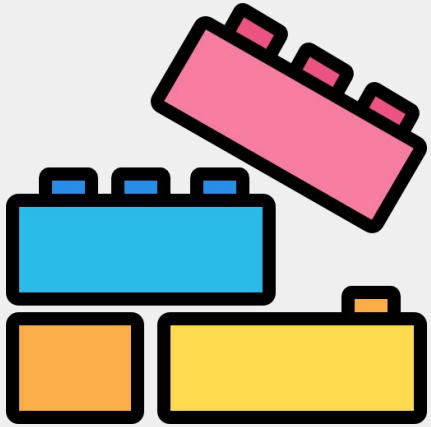
ИЗДАТЕЛЬСТВО АКАДЕМИИ НАУК СССР
Москва ★ 1943

EP_1943_AKS_203

Rule-based Shortcomings!

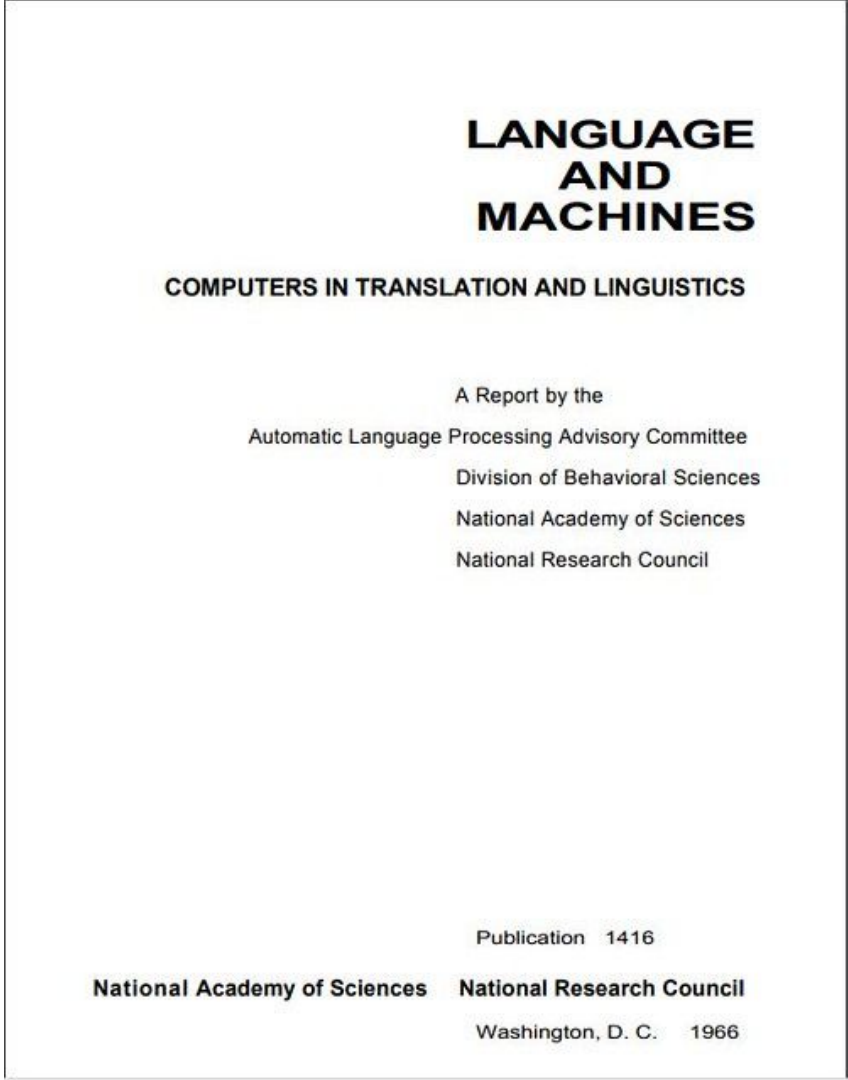
- Rule-based “toy problem” microworlds did not scale
- Language translation hit a ceiling
- Dictionary based machine translation was insufficient for fluent translation. Programs needed encyclopedia knowledge to understand natural language but there was neither software nor hardware to carry this capacity in the 1960s → **In 1966, all U.S. gov support for machine translation ended.**

ENG: “the spirit is willing but the flesh is weak”
RUS: “the vodka is strong but the meat is rotten”



Triggers First AI Winter (1970s)

"There is no immediate or predictable prospect of useful machine translation" (Automatic Language Processing Advisory Committee 1966)



LANGUAGE AND MACHINES

COMPUTERS IN TRANSLATION AND LINGUISTICS

A Report by the
Automatic Language Processing Advisory Committee
Division of Behavioral Sciences
National Academy of Sciences
National Research Council

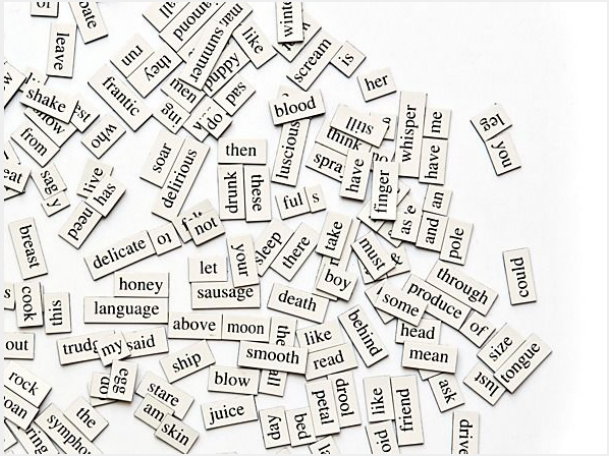
Publication 1416

National Academy of Sciences National Research Council

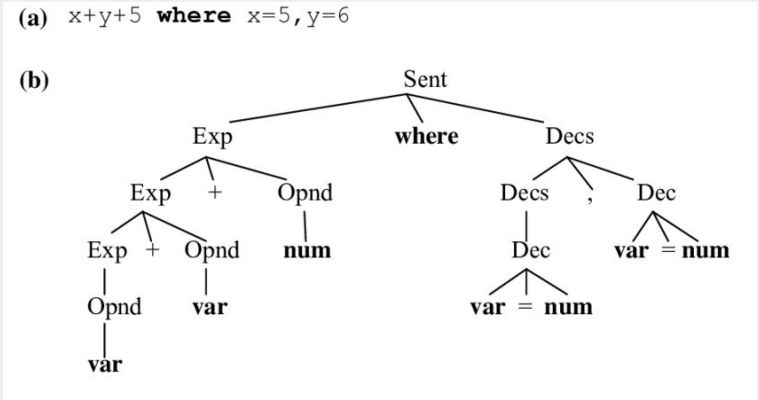
Washington, D. C. 1966

Empirical vs. Rational

- Contest in Language AI between probabilistic modeling vs. structuralism, or “data” empiricism vs. theoretical “rationalism”



Observation



Intuition & Deduction

Probabilistic Language Modeling 1940s & 1950s

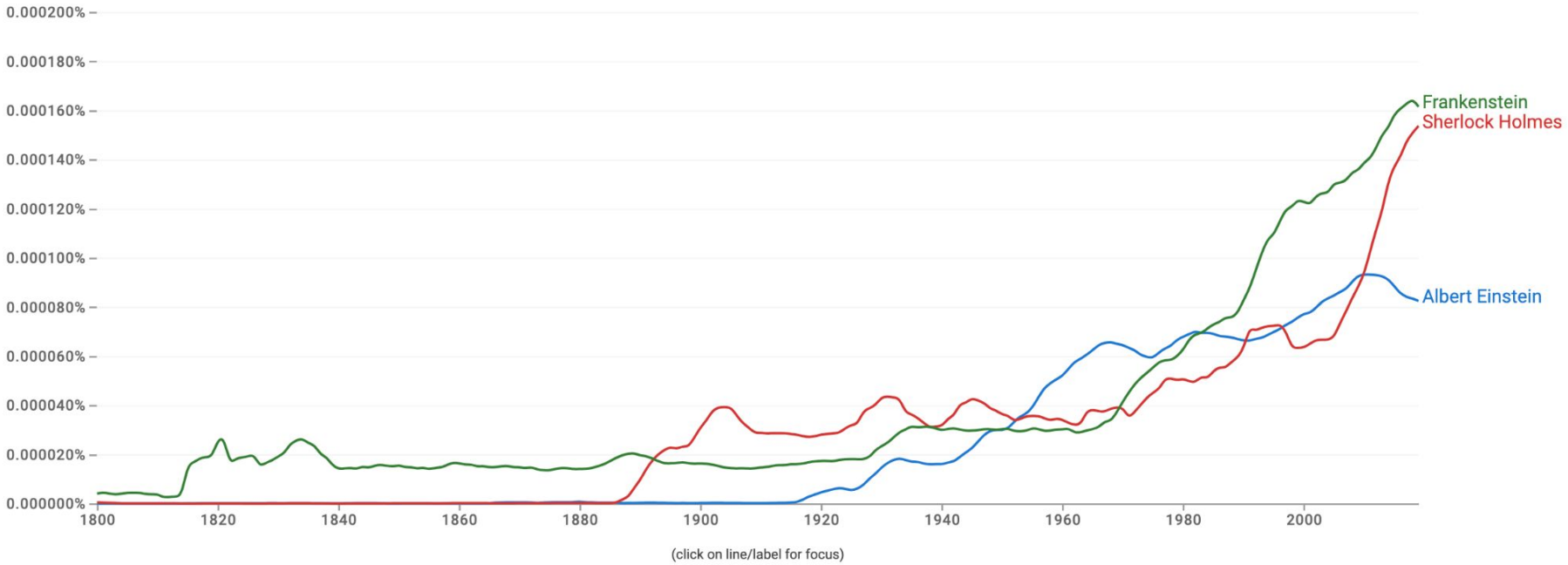
- Markov proposed n-gram letter models in 1913
- In 1949, Claude Shannon introduced the concept of n-gram language modeling for English

# of times “My dog ate my” >= 5
# of times “My dog ate my homework” = 5

$$P(\text{“homework”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my homework”)}}{\text{Count(“My dog ate my”)}}$$

$$P(\text{“mother”} \mid \text{“My dog ate my”}) = \frac{\text{Count(“My dog ate my mother”)}}{\text{Count(“My dog ate my”)}}$$

Google N-grams



N-gram Frequencies

- Linguistics of the 1940s and 1950s focused on word and n-gram frequencies.

In 1957, linguist John Firth proclaimed

“You shall know a word by the company it keeps”

1-grams: “you”, “shall”, “know”, “a”, “word” …
2-grams: “you shall”, “shall know”, “know a”, “a word” …
3-grams: “you shall know”, “shall know a”, “know a word” …
4-grams: “you shall know a”, “shall know a word”, “know a word by”

Hardware Constraints

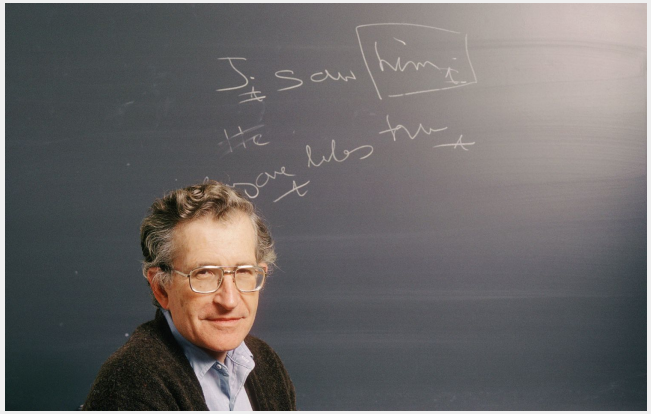
IBM 701

IBM PC

Years	Typical RAM	Examples
1950s	1-4 KB	IBM 701, UNIVAC
1960s	4-64 KB	IBM 360
1970s	64-512 KB	semiconductor chips
1980s	~640 KB	APPLE II, IBM PC
1990s	4-16 MB	
2000s	~4 GB	Wide adoption of PCs

Noam Chomsky's Context Free Models 1957/58

- Noam Chomsky introduced the idea of context-free models (and the Chomsky Hierarchy) and showed language has a logical structure!
- Started a movement to study syntax and disregard probabilistic, statistical approaches to language modeling
- For twenty years (1950s to 1970s), this approach dominated computational linguistics until empirical methods made a come back with statistical speech recognition



Palindrome of {a,b} Context Free Grammar

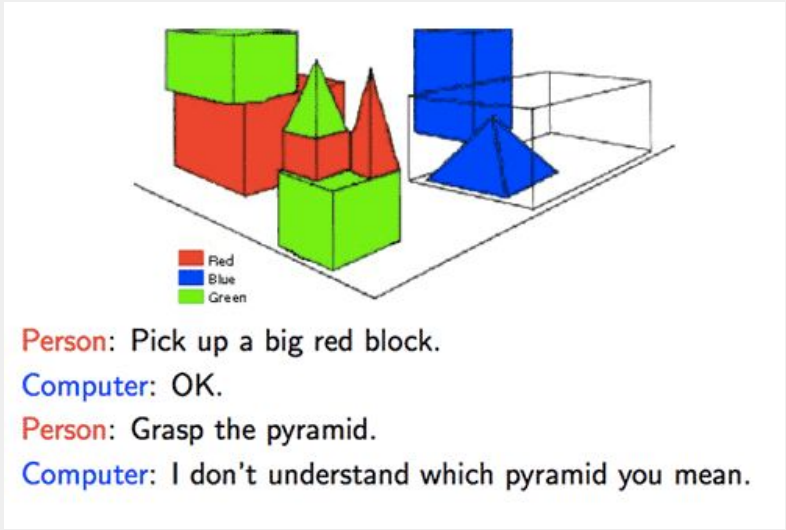
$$S \longrightarrow aSa \mid bSb \mid a \mid b \mid \langle end \rangle$$

start variable

terminal/nonterminal

Logic and rule-based methods 1960s →

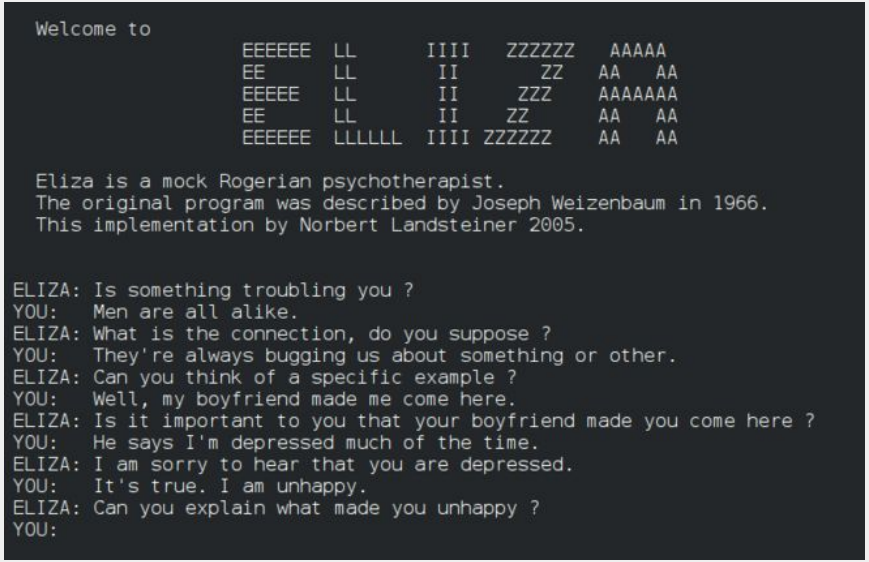
- Can’t capture all variables in the real world
- Best simulation is a microworld!



Terry Winograd’s microworld SHRDLU (1972)

ELIZA (1964-1967) by Joseph Weizenbaum

- First "chatterbot" to attempt Turing Test
- Directives in LISP-based pattern matching language MAD-SLIP that runs "scripts"
- DOCTOR script: Rogerian psychologist mimic
- Parody of "the responses of a non-directional psychotherapist in an initial psychiatric interview" and to "demonstrate the communication between man and machine was superficial"



ELIZA!

PARRY 1972 by Kenneth Colby

- More advanced than ELIZA
- Simulating a person with paranoid schizophrenia
- Passes the Turing Test in 1970s? (48%) – psychiatrists given transcripts of patient–doctor conversations
- International Conference on Computer Communication (ICCC) in 1973: ELIZA and PARRY interacted online on the ARPAnet!

PARRY -- In my opinion, you have a one-track mind.

ELIZA -- What makes you think I have a one-track mind?

PARRY -- I used to go to the track every week.

ELIZA -- What does that suggest to you?

PARRY -- I avoid racetracks now because the races are fixed.

ELIZA -- Is that the real reason?

PARRY -- I'd rather not discuss it.

ELIZA -- What else would you like to discuss?

PARRY -- That's something I don't want to go into."

Transcript of PARRY & ELIZA (DOCTOR)

Danny Bobrow's GUS (1977)

- GUS– Genial Understander System
- Intended to engage a sympathetic and highly cooperative human in an English dialog
- Restricted to the role of a travel agent in a conversation with a client who wants to make a *simple return trip to a single city in California*

GUS: Hello. My name is GUS. I can help you plan a simple trip by air. Where do you want to go?
(1) *Client:* I want to go to San Diego on May 28
GUS: What time do you want to leave?
(2) *Client:* I must be in San Diego before 10 am
GUS: Would you like P.S.A. flight 102 that arrives at 9.15 am?
(3) *Client:* What is the next flight?
GUS: Air California flight 310 that leaves at 8.30 am and arrives at 10.00 am
(4) *Client:* I'll take the first one
GUS: In what name should I make a reservation, please?
(5) *Client:* Dan Bobrow
GUS: I have confirmed the following flight:
P.S.A. flight 102 on Wednesday May 28 from San Jose to San Diego that leaves at 7.30 am and arrives at 9.15 am.
What date do you want to return on?
(6) *Client:* On Friday in the evening.
GUS: Would you like the flight that leaves at 7.45 pm?
(7) *Client:* That's fine.
GUS: I have confirmed the following flight:
P.S.A. flight 307 on Friday May 30 from San Diego to San Jose that leaves at 7.45 pm and arrives at 9.30 pm
Thank you for calling. Goodbye.

FIG. 1. A transcript of an actual dialog with GUS.

Adding a database! Question & Answering

- Question Answering systems were one of the first “task” solving NLP systems.
- The first Q&A system was BASEBALL proposed by Green et al in 1961 - a program made to handle questions about baseball (by accessing a database of baseball statistics).
- In 1973, another Q&A system followed called LUNAR by Woods et al – a program that answered questions about rocks brought back from the moon by the Apollo program.

BASEBALL 1961

In general, a question may contain implicit or explicit queries. Since these queries must be answered one at a time, several searches, with intermediate processing, are required. The first search operates on the stored data while successive searches operate on the found list generated by the preceding search operation. As an example, consider the question "On how many days in July did eight teams play?" The spec list is

```
Day (number of)  = ? ;
Month            = July;
Team (number of) = 8 .
```

On the first pass, the implicit question which teams is answered. The spec list for the first search is

```
Day   = Each;
Month = July;
Team  = ? .
```

The found data is a list of days in July; for each day there is a list of teams that played on that date. Following this search, the processor counts the teams for each day and associates the count with the attribute Team. On the second search, the spec list is

```
Day              = ?   ;
Month            = July;
Team (number of) = 8 .
```

The found data is a list of days in July on which eight teams played. After this pass, the processor counts the days, addsthe count to the found list and is finished.

Progress in natural language understanding—An application to lunar geology

by W. A. WOODS

Bolt Beranek and Newman Inc.
Cambridge, Mass.

INTRODUCTION

The advent of computer networks such as the ARPA net (see e.g., Ornstein et al.[3]) has significantly increased the opportunity for access by a single researcher to a variety of different computer facilities and data bases, thus raising expectations of a day when it will be a common occurrence rather than an exception that a scientist will casually undertake to use a computer facility located 3000 miles away and whose languages, formats, and conventions are unknown to him. In this foreseeable future, learning and remembering the number of different languages and conventions that such a scientist would have to know will require significant effort—much greater than that now required to learn the conventions of his local computing center (where other users and knowledgeable assistance is readily available). The Lunar Sciences Natural Language Information System (which we will hereafter refer to as LUNAR) is a research prototype of a system to deal with this and other man-machine communication problems by adapting the machine to the conventions of ordinary natural English rather than requiring the man to adapt to the machine.

State of the art

Although the state of the art in "understanding" natural language by machine is still very limited, significant advances in this area have been made in recent years. Since Simmons' first survey of question answering systems,[4] our understanding of the mysterious "semantic interpretation" component has been made more clear by work such as Woods,[7,8] and the techniques for mechanically parsing natural language sentences have been advanced by the advent of transition network grammars and their parsing algorithms.[9,10,11] Recently, Terry Winograd's blocks world system[6] has demonstrated the potential of some of these techniques—especially those of procedural semantics. The field is now at the point where prototype applications to real problems can make significant contributions to our understanding of the problems of natural language communication with machines. It must be realized, that such applications are still essentially research vehicles, since the problems of mechanical understanding of natural language remain far from solution. However, by using real problems, rather than imaginary toy problems, as the vehicles for such research, one cannot only focus the effort on problems in need of solution, but may also reap the additional benefit of producing a system which will perform a task

(GIVE ME THE AVERAGE SM ANALYSIS OF TYPE A ROCKS)
(WHAT IS THE AVERAGE MODAL CONCENTRATION OF ILMENITE IN TYPE A ROCKS?)
(GIVE ME EU DETERMINATIONS IN SAMPLES WHICH CONTAIN ILM)
(GIVE ME ALL K / RB RATIOS FOR BRECCIAS).
(WHAT BE ANALYSES ARE THERE?)
(GIVE ME OXYGEN ANALYSES IN S10084)
(WHAT SAMPLES CONTAIN CHROMITE?)
(WHAT SAMPLES CONTAIN P205?)
(GIVE ME THE MODAL ANALYSES OF P205 IN THOSE SAMPLES)
(GIVE ME THE MODAL ANALYSES OF THOSE SAMPLES FOR ALL PHASES)

```
31**(HOW MANY BRECCIAS CONTAIN OLIVINE)
***
PARSING
815 CONSES
4.633 SECONDS
INTERPRETING
1514 CONSES
7.29 SECONDS
INTERPRETATIONS:
(FOR THE X12 / (SEQL (NUMBER X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QU
NIL)))) : T ; (PRINOUT X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
(5)

-----
***
32**(WHAT ARE THEY)
***
PARSING
487 CONSES
2.755 SECONDS
INTERPRETING
1158 CONSES
4.053 SECONDS
INTERPRETATIONS:
(FOR EVERY X12 / (SEQ TYPECS) : (CONTAIN X12 (NPR* X14 / (QUOTE OLIV)) (QUOTE NIL)) ; (PRINT
X12))

BBN LISP-10 03-09-72 . . .
EXECUTING
S10019
S10059
S10065
S10067
S10073
-----
***
34**(DO ANY SAMPLES HAVE GREATER THAN 13 PERCENT ALUMINUM)
***
PARSING
981 CONSES
4.614 SECONDS
INTERPRETING
902 CONSES
3.566 SECONDS
INTERPRETATIONS:
(TEST (FOR SOME X16 / (SEQ SAMPLES) : T ; (CONTAIN' X16 (NPR* X17 / (QUOTE AL203)) (GREATERT
13 PCT))))

BBN LISP-10 03-09-72
EXECUTING
YES.
T
```

LUNAR 1973

Expert Systems 1960s → 1980s

- From generality (General Problem Solver) to specific domain
- Capture a well-defined human expert knowledge domain (e.g. medicine, chemical spectrography)
- Encode "knowledge" as facts or rules!
- Restricting domains helps minimize common sense knowledge constraint for encoding

DENDRAL late 1960s

- DENDRAL was an expert system designed to discover molecular structure from mass spectra
- Collaborated with Djerassi (expert in mass spectrography) for more expert domain knowledge
- “Mother of all Expert Systems”?
- Moved to the SUMEX facility for national use (by all chemists!)

Elsevier

ARTINT 983

DENDRAL and Meta-DENDRAL: roots of knowledge systems and expert system applications

Edward A. Feigenbaum

Knowledge Systems Laboratory, Department of Computer Science, Stanford University, Stanford, CA 94305, USA

Bruce G. Buchanan

Computer Science Department, University of Pittsburgh, Pittsburgh, PA 15260, USA

During AI's first decade (1956–1966), the task environments in which AI scientists investigated their basic science issues were generally idealized “clean” task environments, such as propositional calculus theorem proving and puzzle solving. After the mid-1960s, a bolder and more applied inclination to choose complex real-world problems as task environments became evident. These efforts were both successful and exciting, in two ways. First, the AI programs were achieving high levels of competence at solving certain problems that human specialists found challenging (the excitement was that our AI techniques were indeed powerful and that we were taking the first steps toward the dream of the very smart machine). Second, these complex real-world task environments were proving to be excellent at stimulating basic science questions for the AI science, in knowledge representation, problem solving, and machine learning. To recognize and illuminate this trend, the *Artificial Intelligence Journal* in 1978 sponsored a special issue

MYCIN early 1970s

- A program that could identify bacteria that could cause infections and make treatment (antibiotic) recommendations
- It was accurate in about 69% of cases which was better than human disease experts when based on the same criteria (457 rules took 20 person years)

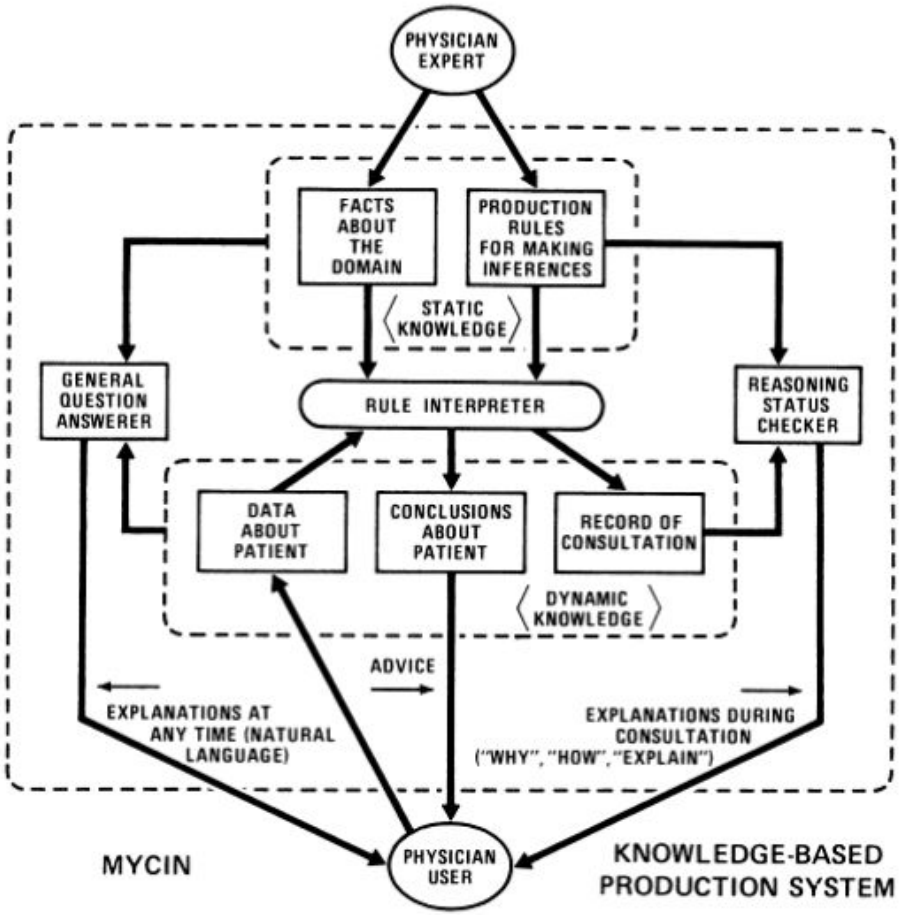


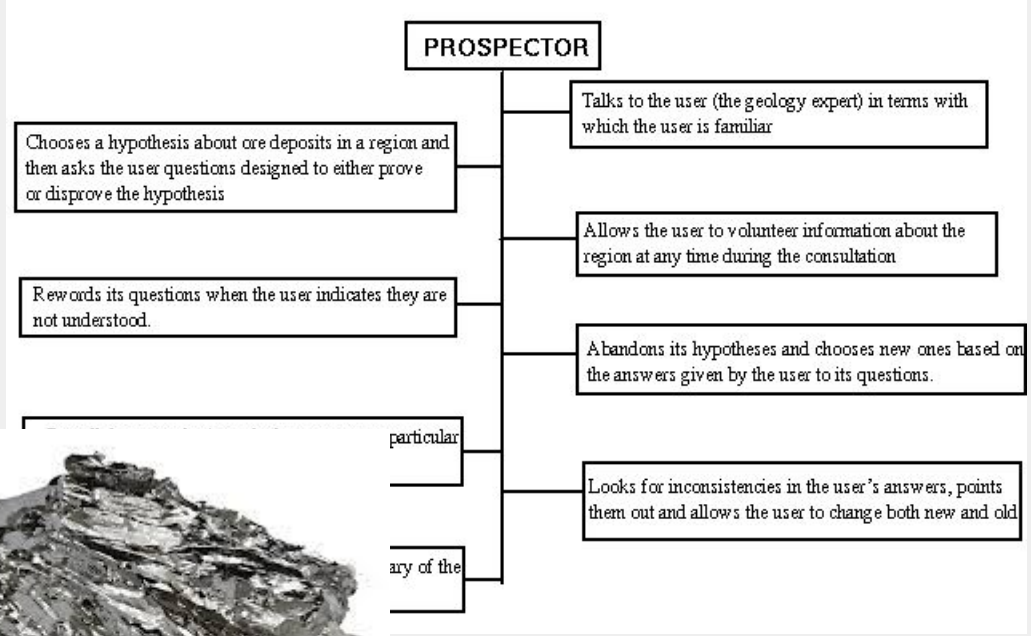
Figure 2 - Diagram summarizing the organization and flow of information within MYCIN. The correlation between this design and the human consultation process depicted in Fig. 1 is discussed in the text. (Figure reproduced from reference 10).

XCON or R1 (eXpert CONfigurer) 1980

- XCON was an system developed by DEC (digital equipment corporation) designed to help the computer manufacturer configure its machine to suit customer demand optimally
- First large-scale industrial use of AI
- 2500 rules
- $25m cost savings a year by reducing errors in order config and assembly process
- By 1986, had processed 80,000 orders and achieved an accuracy rate of 95-98%
- Good example of AI-led productivity → inspired 1980s investment

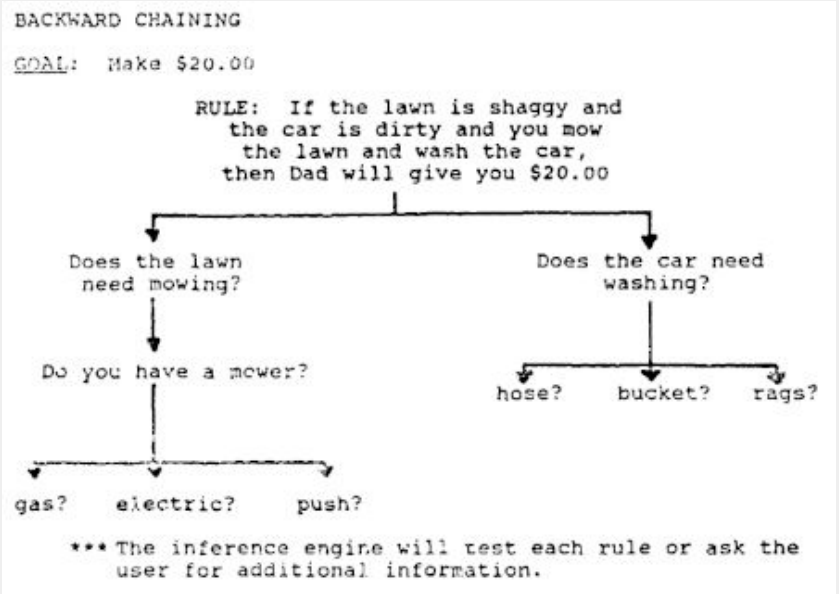
PROSPECTER 1983

- PROSPECTOR was a geology expert system that helped discover new drilling sites
- Made with 9 geologists
- Helped a company discover molybdenum deposits in Washington!



Expert Systems have common structure

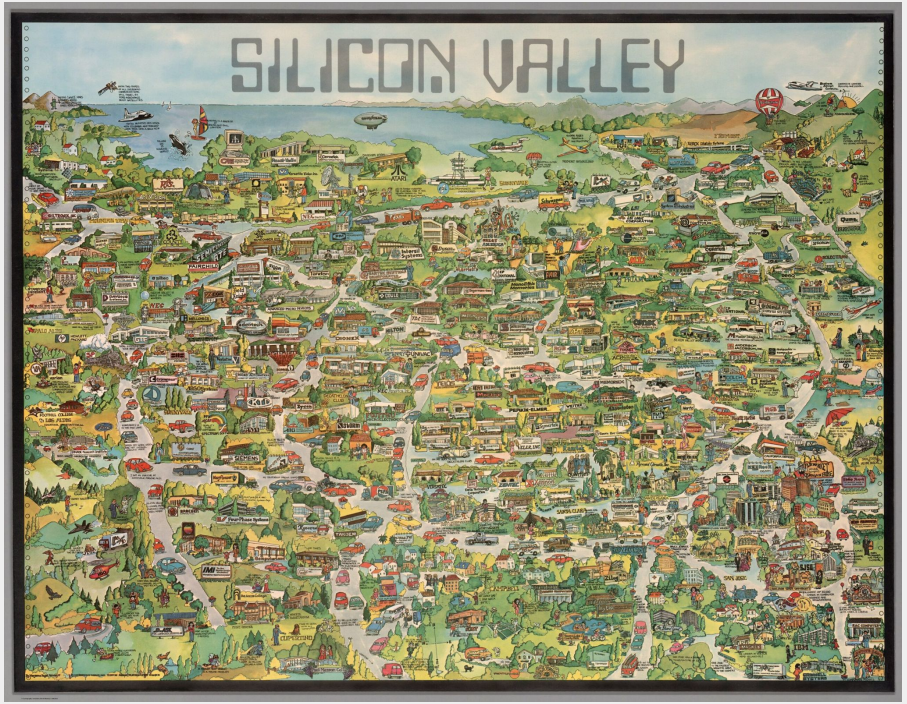
1. the knowledge base
   (i.e. list of IF____THEN____)
2. short-term memory that contains specific data about the actual problem
3. tree-searching
   (i.e. inference machine)
4. user interface
   (e.g. menu, quasi-natural language dialogue)



How to make this system more generic??

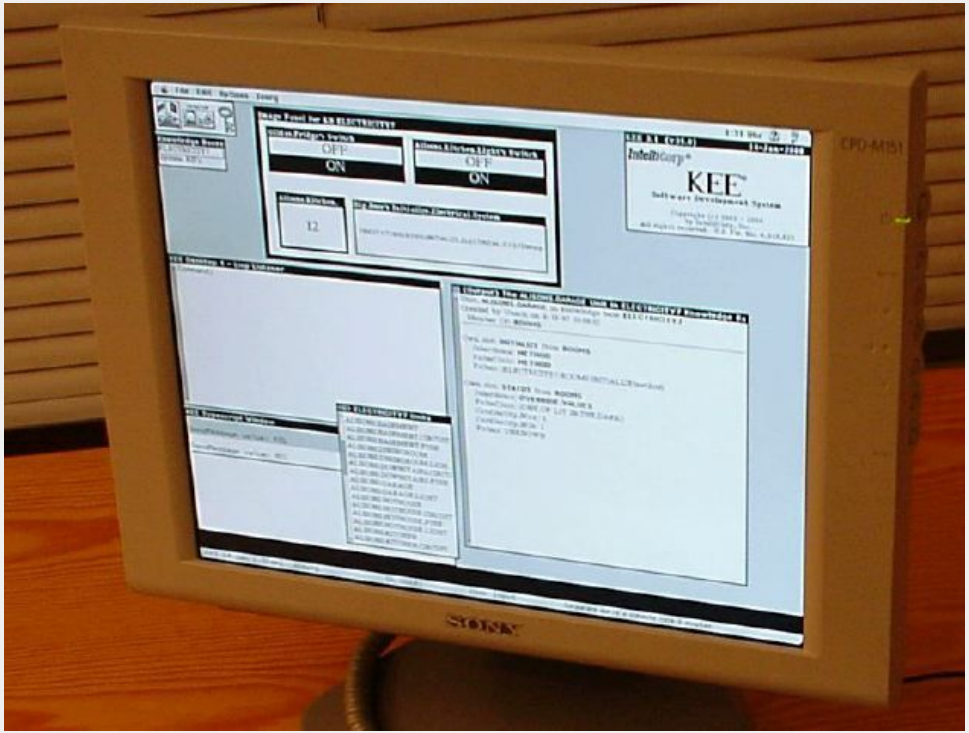
Expert Systems Start-ups 1980s

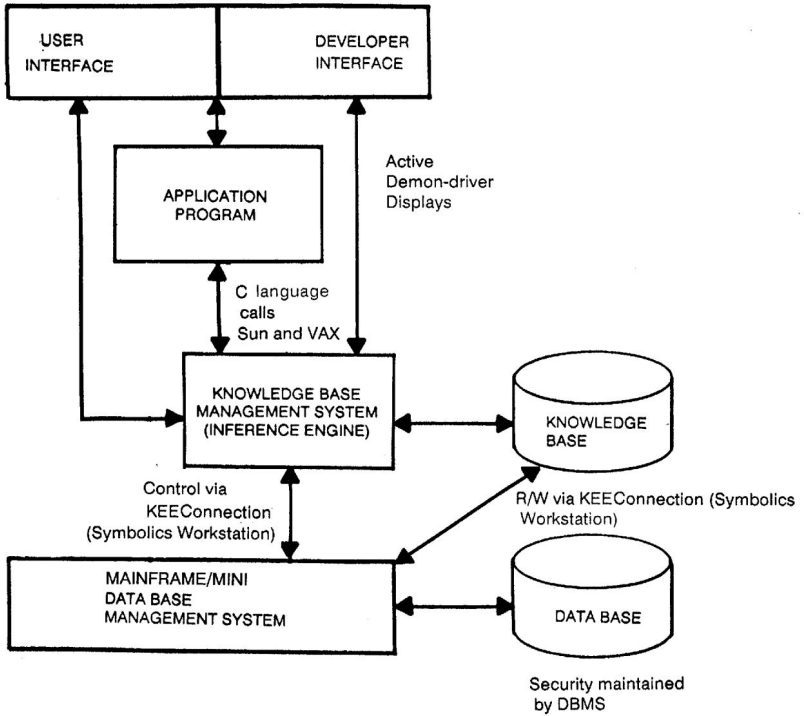
- Expert Systems was the first AI technology to arouse commercial interest
- Teknowledge, Inc., Intellicorp sprang up in the early 1980s to market these systems. Both these companies received millions from DARPA to bring AI into the market and military fields.



Knowledge Engineering Environment (KEE) 1983

- Intellicorp created and started selling Knowledge Engineering Environment (KEE) 1983: a high-end knowledge expert system development shell.
- KEE was to support knowledge acquisition and representation, reasoning, and user interface construction.
- Framework for multiple expert domains





Product Information

Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

Hardware

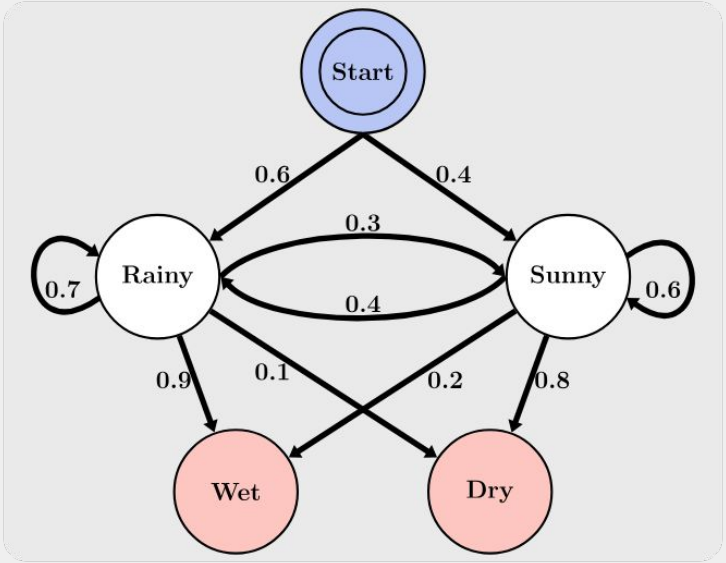
all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations

DARPA disappointed in Language AI again

- In the late 1980s, DARPA began cutting back bankrolling Teknowledge and Intellicorp projects
- AI commercial products failed to meet expectations of investors
- Second AI winter ensues

Speech goes statistical

- 1971 DARPA funded speech recognition projects
- 4 separate and competing projects over 5 years
- CMU made the winning project: HAPPY (based on DRAGON) built on Hidden Markov Models
- Hidden Markov Models used commonly for integrating sounds with words sequence + probability
- DARPA disappointed



Strategic Computing Initiative 1983-93 & Speech

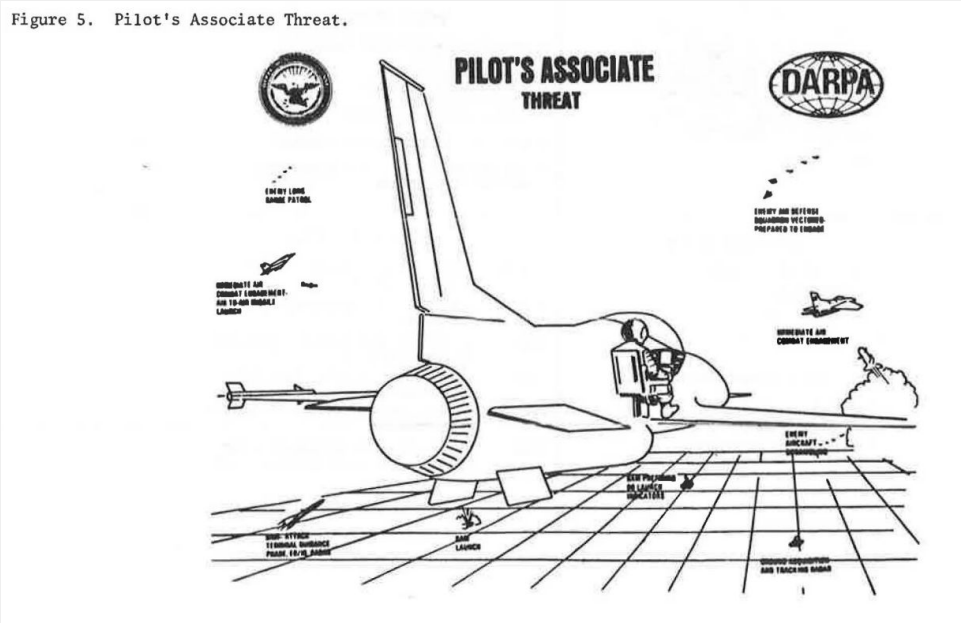
- Carnegie Mellon and MIT researchers in 1978 and 1979 concluded that phonetic signals were more tractable than previously believed.
- Started two SCI funded projects on speech recognition
- Project 1) Phone recognition: It turns out spoken English could be condensed to just 48 discrete sounds (and silence). These sounds could be converted to a stream of data a computer could interpret.
- Project 2) n-gram stats in speech: researchers began to integrate n-gram stats to model word order to disambiguate similar/same sounding words (“write” vs. “right”)

I have to “write”/”right” this paper
p(“write this paper”) vs. p(“right this paper”)

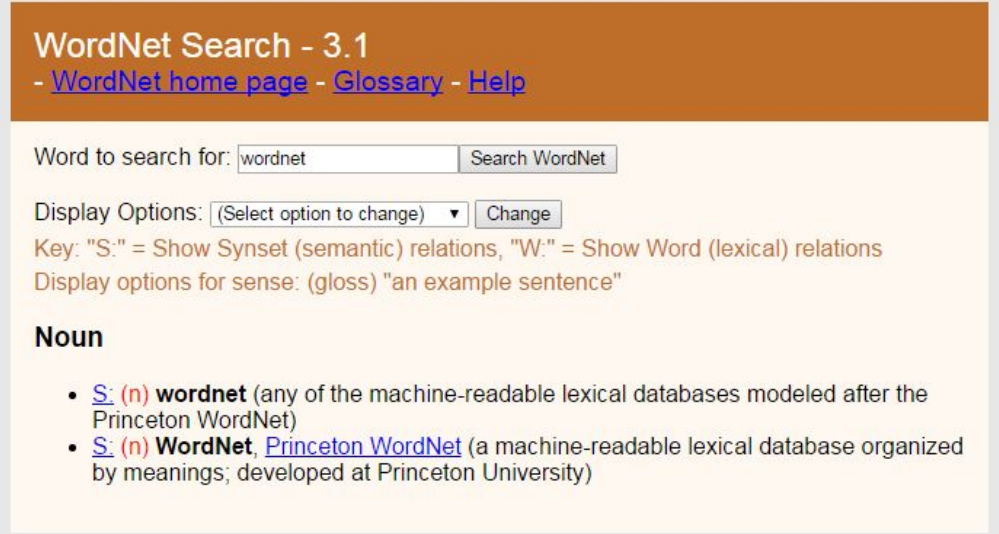
Speech in Defense

- DARPA wanted to integrate speech into their two military projects: Pilot's Associate and Battle Management.
- It funded 8 contractors in nine projects in continuous speech to Carnegie Mellon for Battle Management.
- And 5 contractors in 8 projects for robust speech to Texas Instruments for Pilot's Associate.
- By the end of the Strategic Computing Initiative, the continuous speech recognition program could recognize 10,000 words in a moderately noisy background → started being integrated into military and commercial applications.

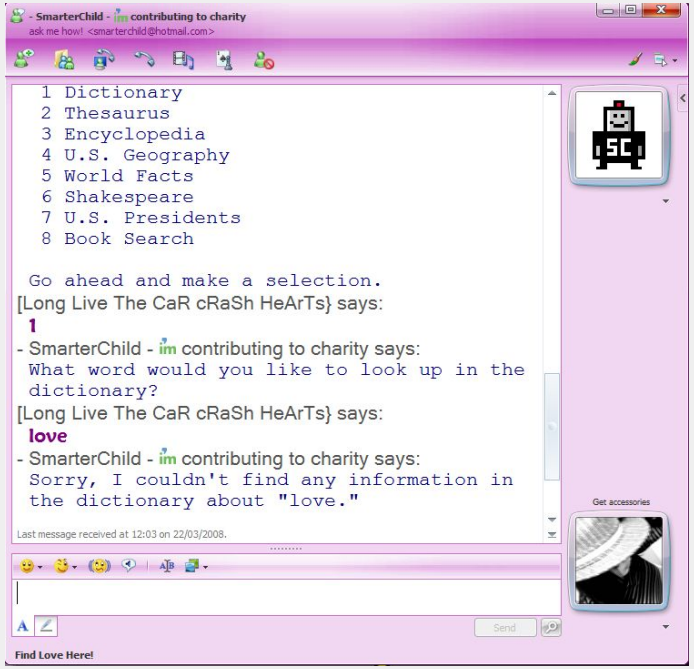
Figure 5. Pilot's Associate Threat.



Internet & Big Data 1992 →



Wordnet 2001



2001 SmarterChild on AOL Instant Messenger (AIM) or MSN Messenger

A publicly available dictionary of 100,000 words and phrases (labeled with parts of speech tags and linked by semantic relations such a synonyms, antonyms)

The Stanford Integrated Digital Library Project

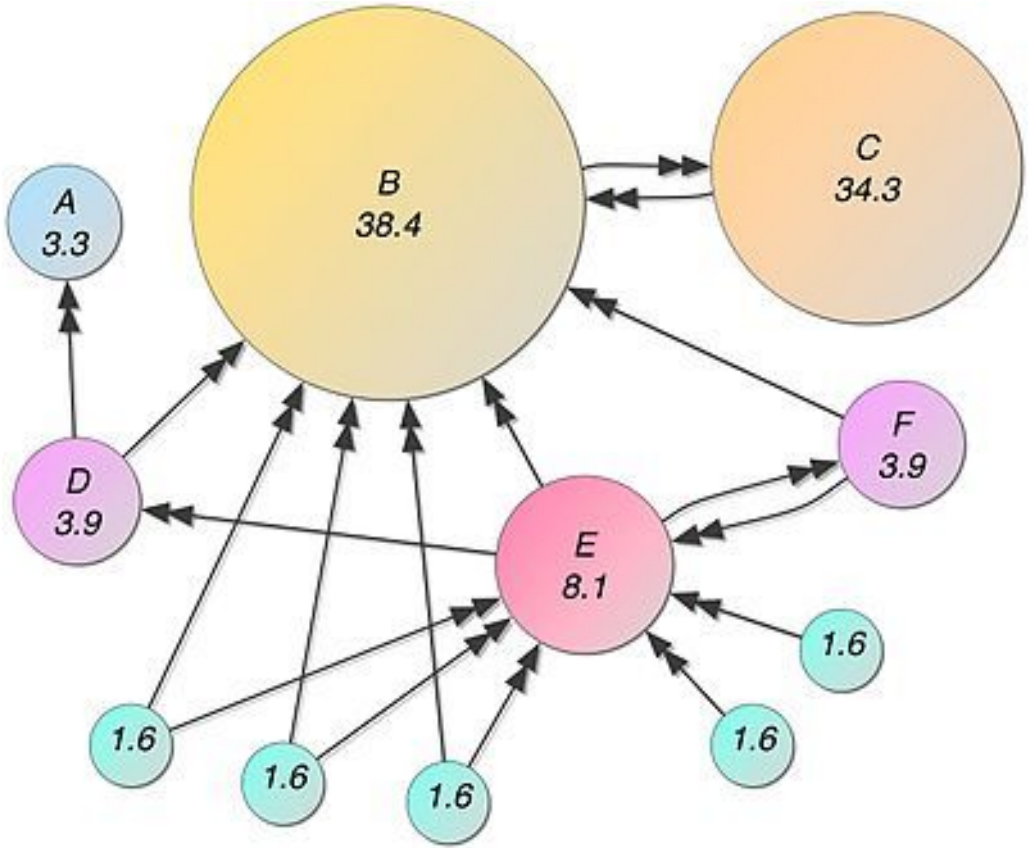
Summary

The goal of the Stanford Integrated Digital Library Project (SIDLP) is to develop the enabling technologies for a single, integrated and "universal" library, providing uniform access to the large number of emerging networked information sources and collections. These include both on-line versions of pre-existing works and new works and media of all kinds that will be available on the globally interlinked computer networks of the future. The Integrated Digital Library is broadly defined to include everything from personal information collections, to the collections that one finds today in conventional libraries, to the large data collections shared by scientists. Our technology will provide the "glue" that will make this worldwide collection usable as a unified entity, in a scalable and economically viable fashion.

A consortium of information vendors, publishers, end users, commercializers, and researchers has been formed to carry out the proposed research. They will also implement a SIDLP Testbed system that will provide access to significant collections of computing and related literature. The testbed will be used to develop a variety of novel library services, and to experiment with protocols for the integration of heterogeneous information sources.

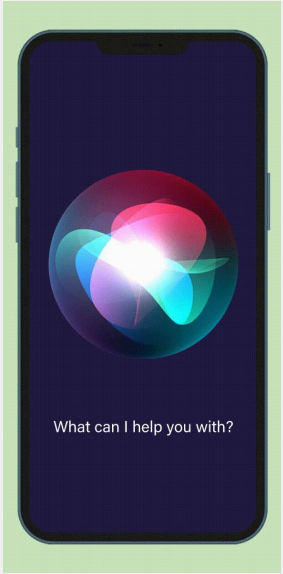
NSF Award Abstract 9411306. October 5, 1998

PageRank algorithm was introduced in 1998 (Brin and Page) as a way to search documents on the web.

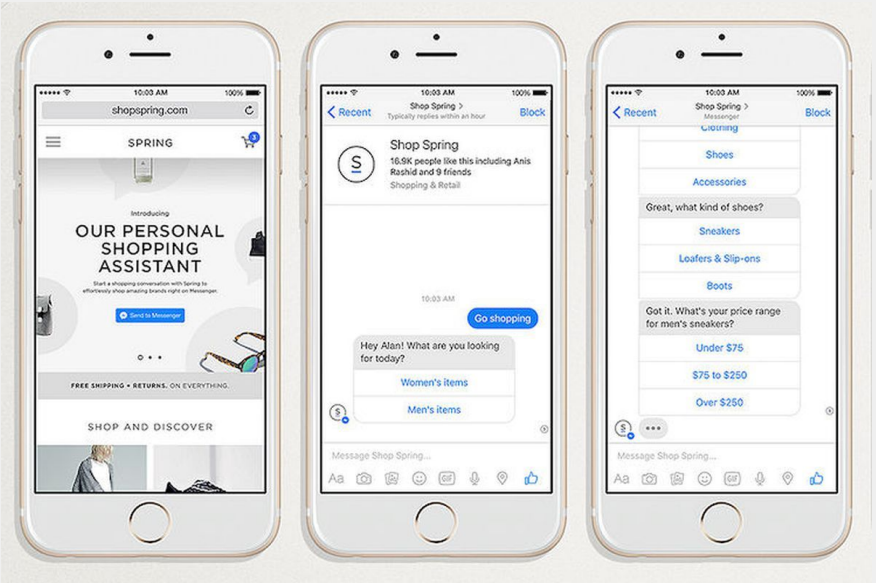


Commercialization

2000 Google AdWords



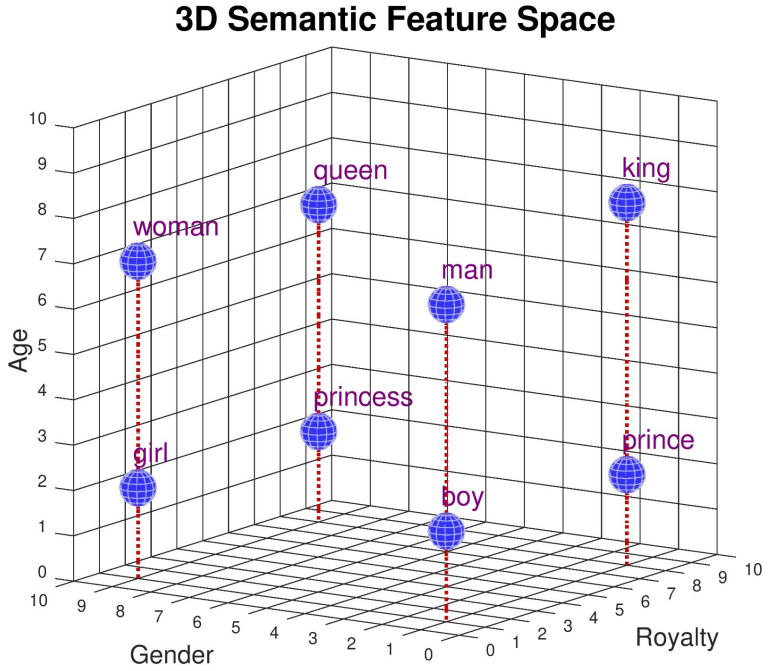
2010 Siri



2016 Bots for Messenger (Bot-plication)

More data → Empirical methods

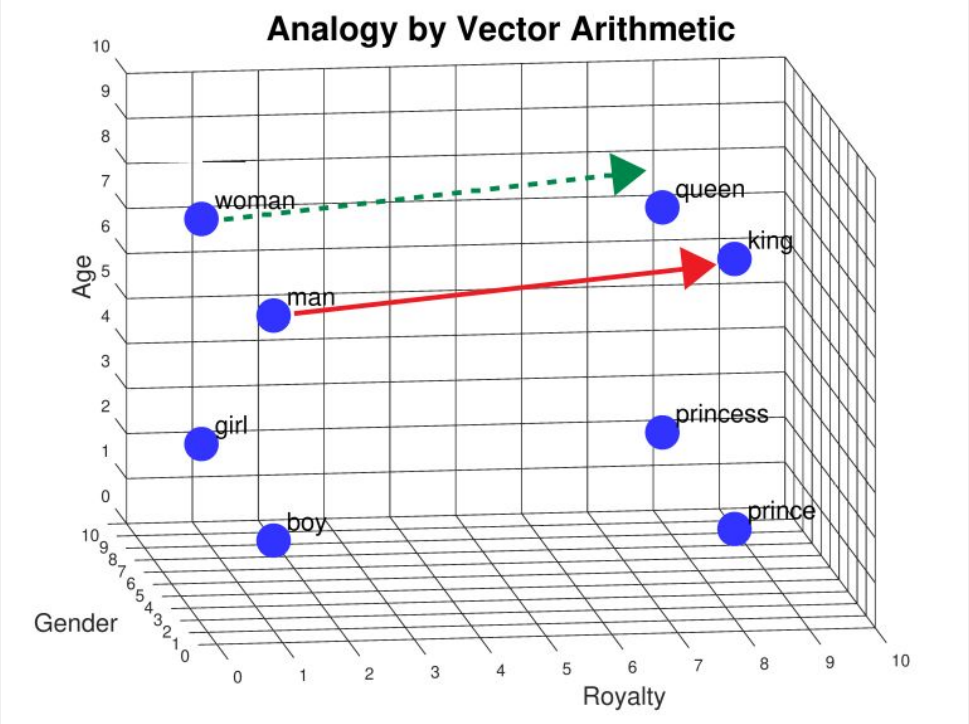
Word Coordinates	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html

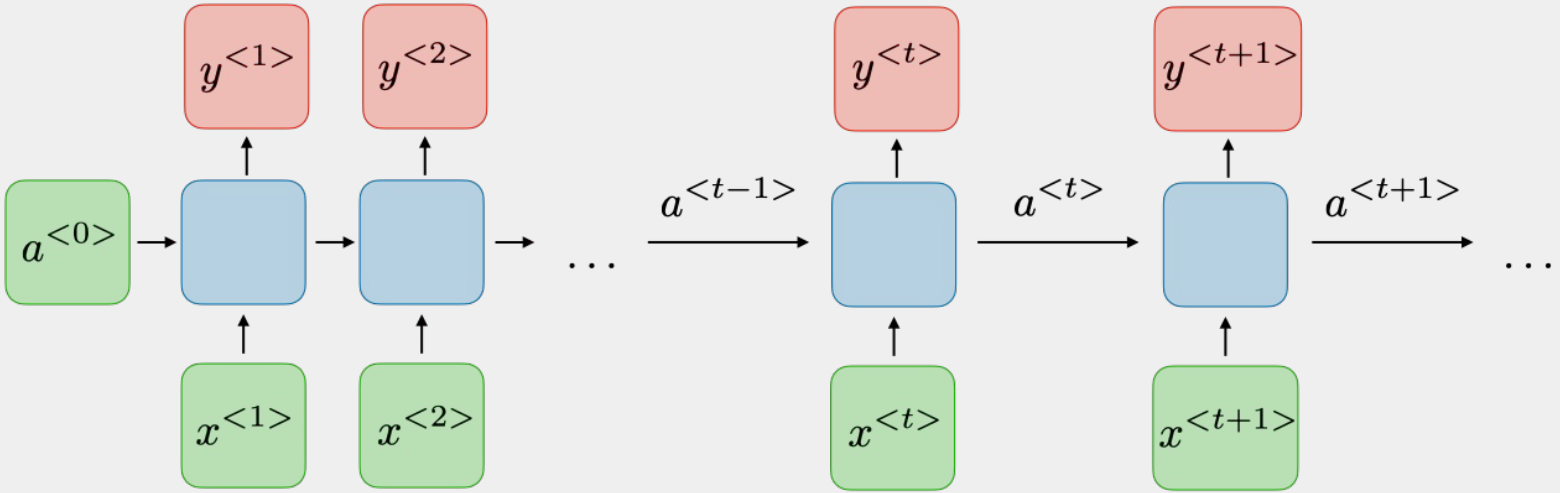
More data → Empirical methods

king	[	1,	8,	8]
man	[	1,	7,	0]
king - man	[	0,	1,	8]
woman	[	9,	7,	0]
king - man + woman	[	9,	8,	8]
queen	[	9,	7,	8]



https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/tutorial.html

Sequential Modeling with Neural Nets



https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-recurrent-neural-networks

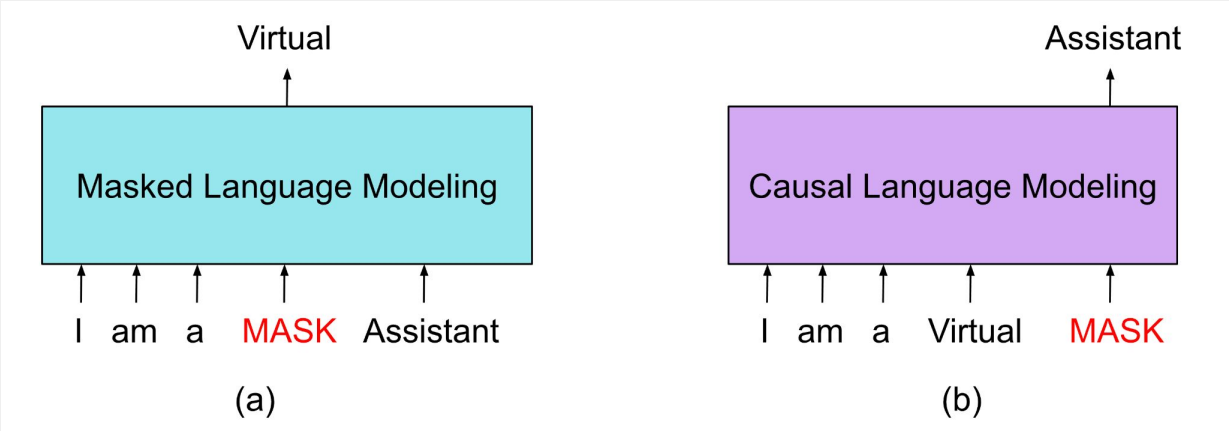
Language Modeling with Unlabeled Text

1. Masked Language Modeling

Training the model to correctly predict missing (masked) words in a given sequence (encoder models)

2. Autoregressive (causal) Language Modeling

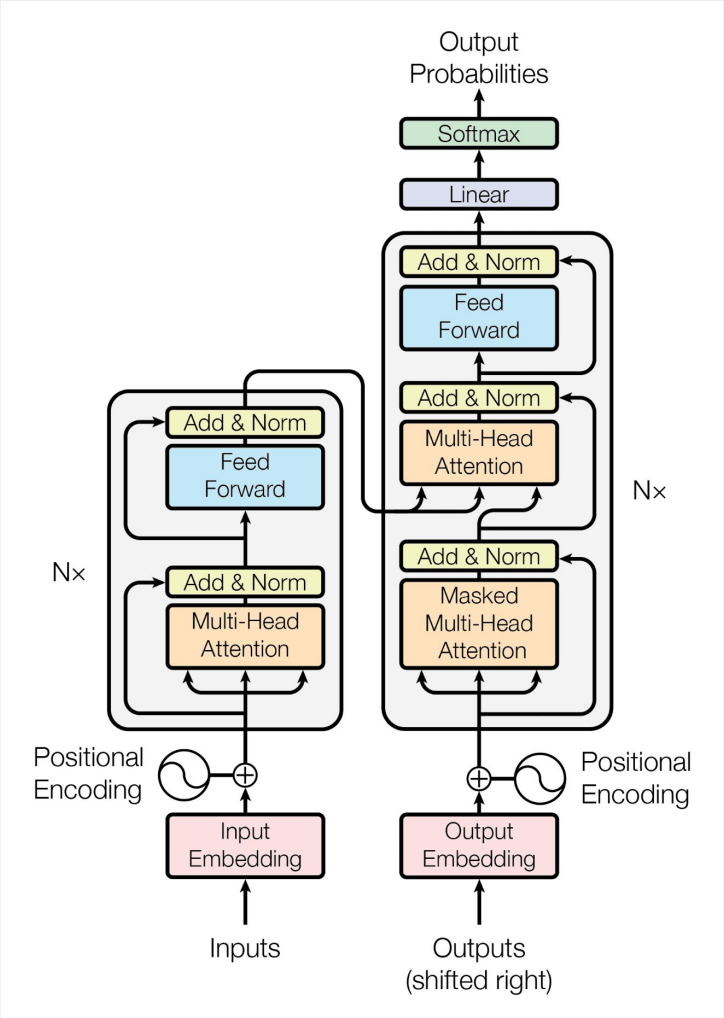
Training the model to learn by optimizing predicting the next word (GPT models)



Transformer Model 2016/2017

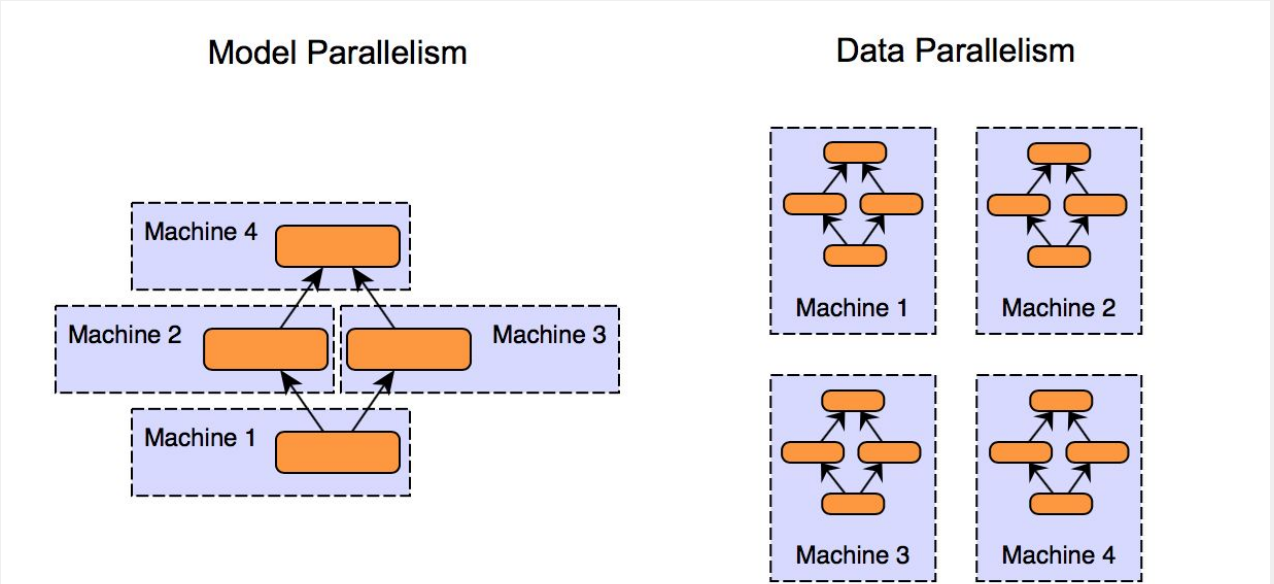
- Highly effective for encoding and decoding sequential/structured information

1. Attention mechanism for long-range dependencies
2. Parallel training across multiple machines



Parallel Training

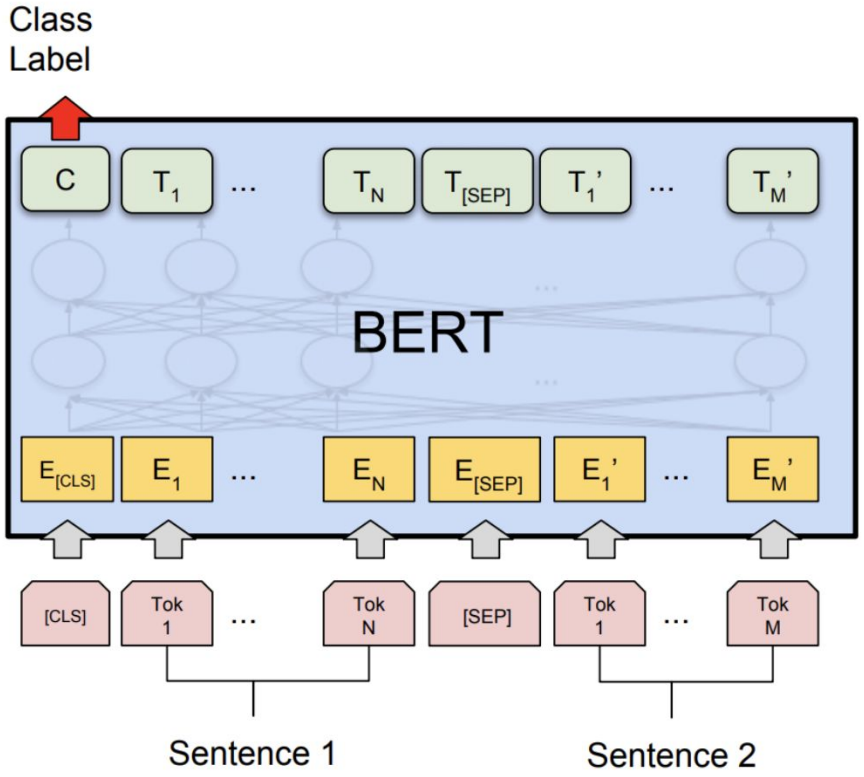
https://pytorch.org/torchrec/high-level-arch.html



- Split model weights (Embedding Dimension)
- Distribute model across GPUs
- Good for models with lots of weights
- Need to change code
- Only Transformers

- Split data (Batch Size)
- Each GPU computes → Shares loss
- RNNs & Transformers
- Good for training with lots of data

Bidirectional Encoder Representations 2018



Data vs. Model Size

Though there has been significant recent work allowing larger and larger models to be trained, our analysis suggests an increased focus on dataset scaling is needed. Speculatively, we expect that scaling to larger and larger datasets is only beneficial when the data is high-quality. This calls for responsibly collecting larger datasets with a high focus on dataset quality. Larger datasets will require extra care to ensure train-test set overlap is properly accounted for, both in the language modelling loss but also with downstream tasks. Finally, training for trillions of tokens introduces many ethical and privacy concerns. Large datasets scraped from the web will contain toxic language, biases, and private information. With even larger datasets being used, the quantity (if not the frequency) of such information increases, which makes dataset introspection all the more important. *Chinchilla* does suffer from bias and toxicity but interestingly it seems less affected than *Gopher*. Better understanding how performance of large language models and toxicity interact is an important future research

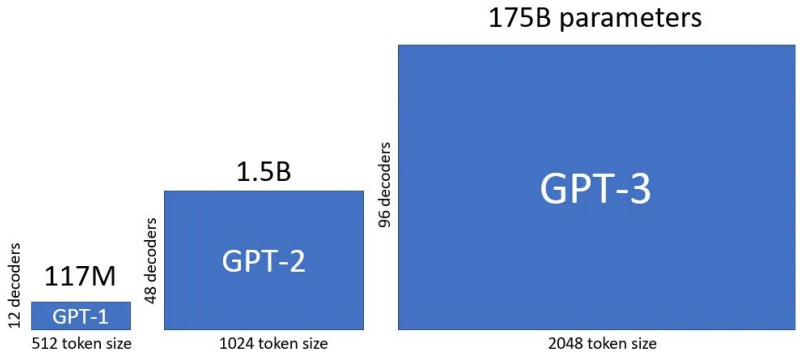
ology towards the training of auto-regressive language models, de-off between model size and the amount of data in other very expensive, choosing the optimal model size and training thods we propose are easy to reproduce in new settings.

Table 1 | **Current LLMs**. We show five of the current largest dense transformer models, their size, and the number of training tokens. Other than LaMDA (Thoppilan et al., 2022), most models are trained for approximately 300 billion tokens. We introduce *Chinchilla*, a substantially smaller model, trained for much longer than 300B tokens.

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
Gopher (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
Chinchilla	70 Billion	1.4 Trillion

Training Compute-Optimal Large Language Models. March 2022. DeepMind

OpenAI's Transformer Models



GPT-1
- Created in 2018
- Have a data set of 117 million parameters
- Performs tasks without labeled data

GPT-2
- Created in 2019
- Has a data set of 1.5 billion parameters
- Trained on a more diverse set of data than GPT-1

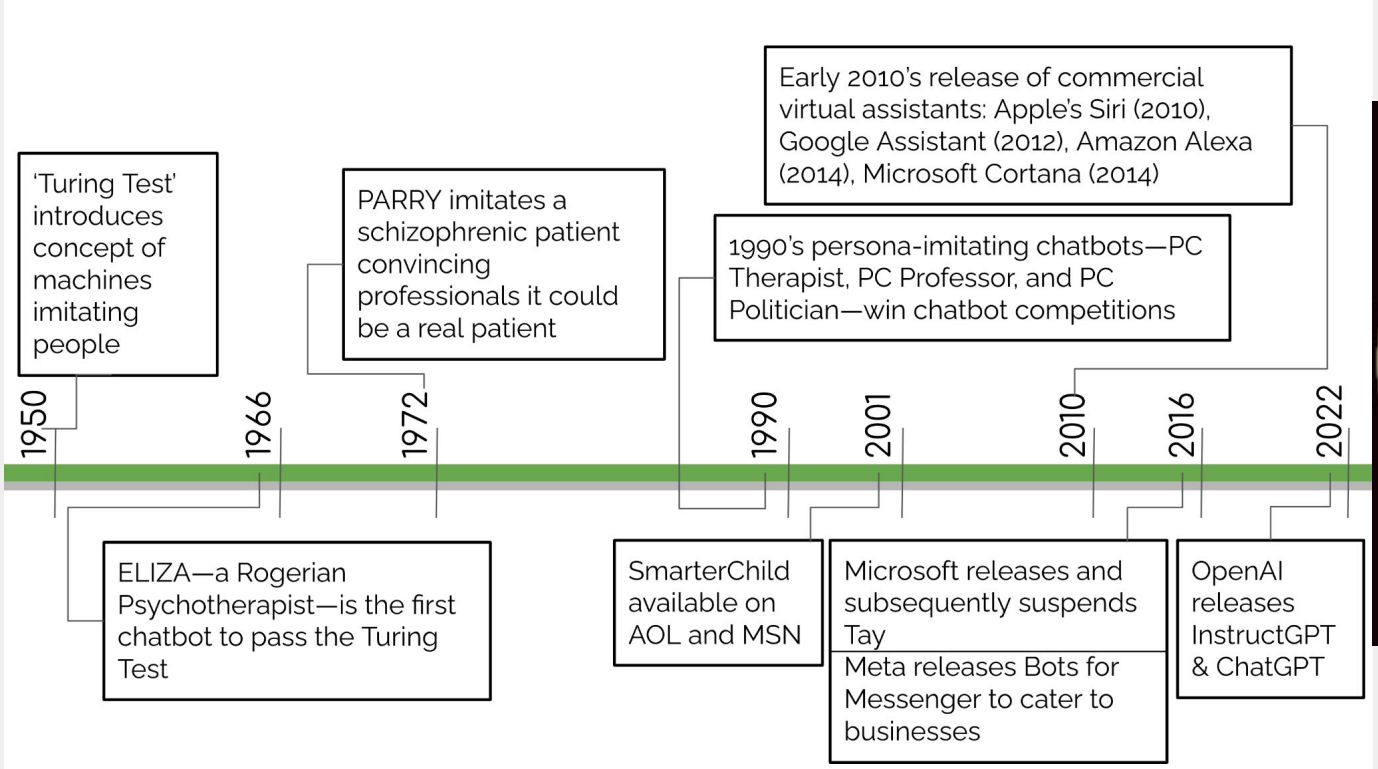
GPT-3
- Created in 2020
- Have a data set of 175 million parameters
- Solves simple arithmetic problems

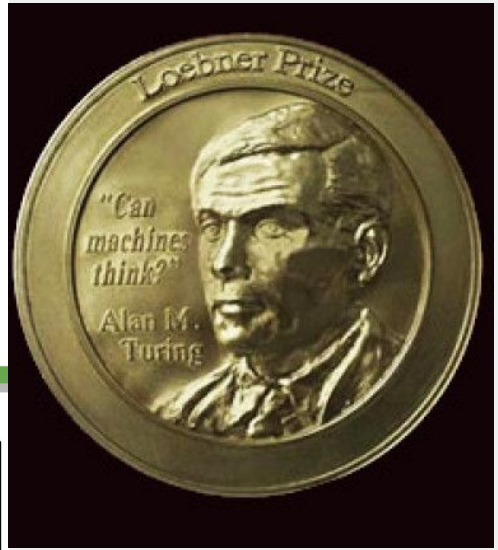
ChatGPT(GPT-3.5)
- Created in 2022
- Can create letters, song lyrics, research papers, recipes, therapy sessions, poems, essays, outines and software code

GPT-4
- Created in 2023
- Multimmodal AI which means you can interact with it in speech as well as textual interactions

OpenAI

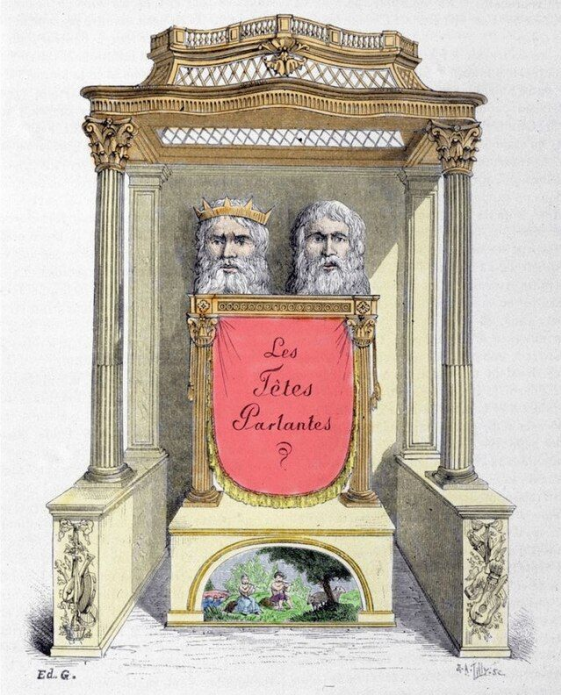
Chatbots: From Turing Test to ChatGPT

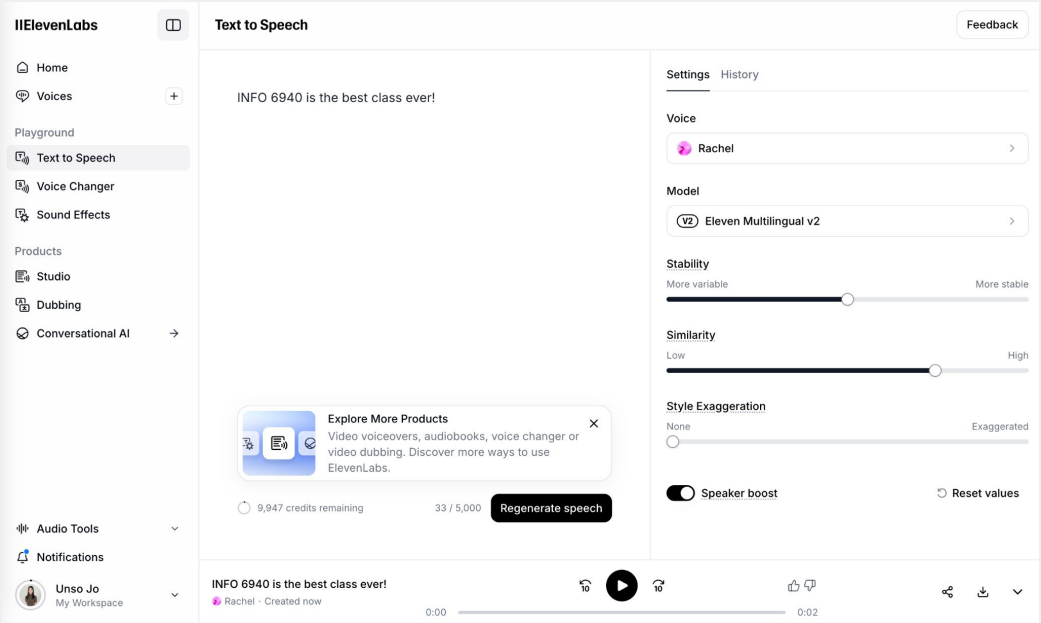




Voice: How far we've come!

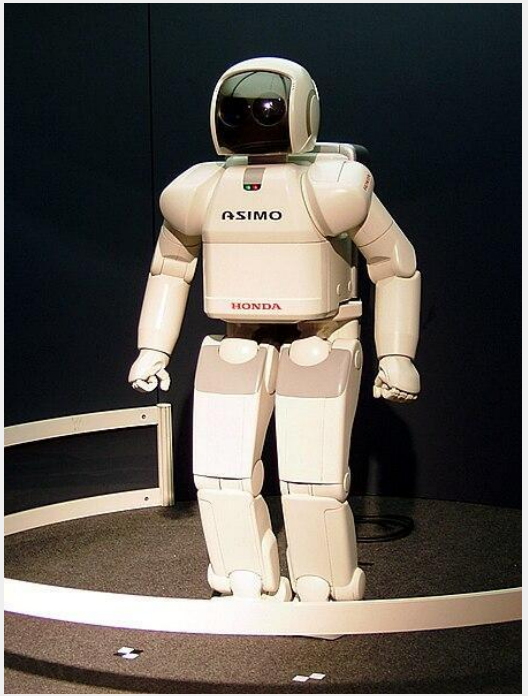
https://elevenlabs.io/app/speech-synthesis/text-to-speech





History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

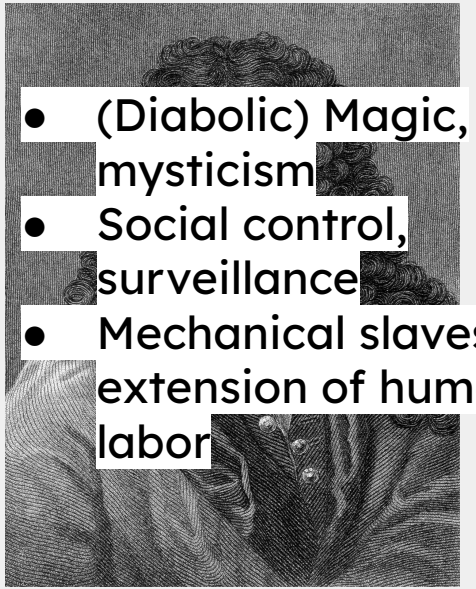
Lecture 5 Februrary 21, 2025

This week!

- Why is narrative so important?
- Do people have a natural fascination with AI?
- How do our stories about AI give social commentary?
- How do “stories” influence policy making & tech development?

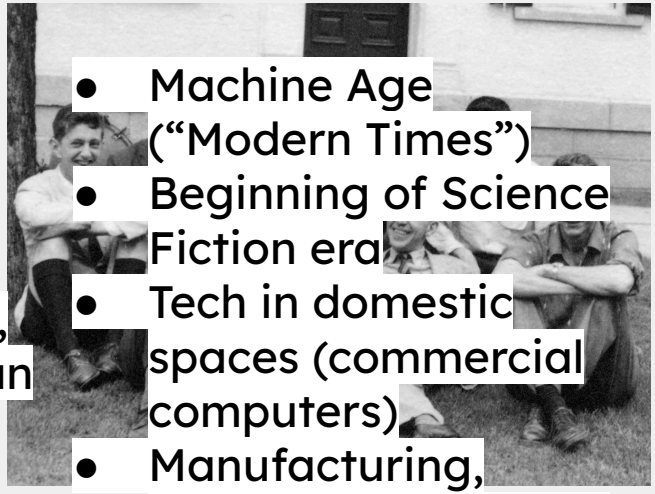
AI Narratives

Early Modern



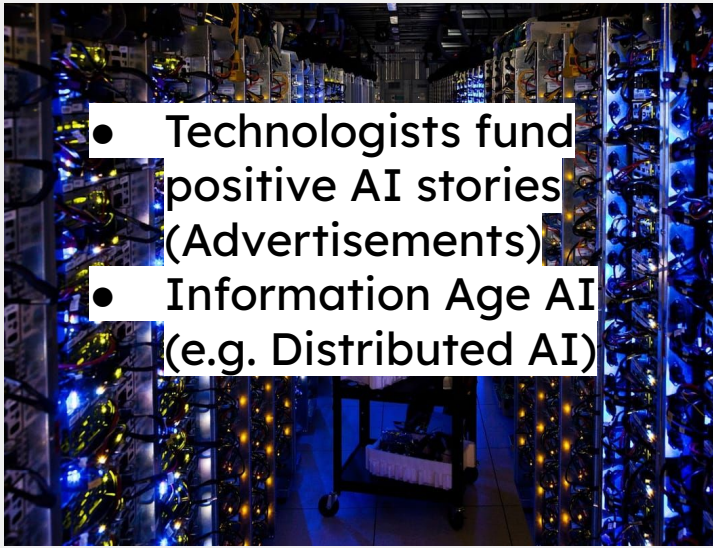
- (Diabolic) Magic, mysticism
- Social control, surveillance
- Mechanical slaves, extension of human labor

Modern



- Machine Age (“Modern Times”)
- Beginning of Science Fiction era
- Tech in domestic spaces (commercial computers)
- Manufacturing, Industrial Revolution

Information Age AI



- Technologists fund positive AI stories (Advertisements)
- Information Age AI (e.g. Distributed AI)

Early Modern “AI”?

- How to see the future, distance, hidden information
- Use AI for surveillance, control, maintain social norms
- Are all non-natural forms of intelligence demonic?

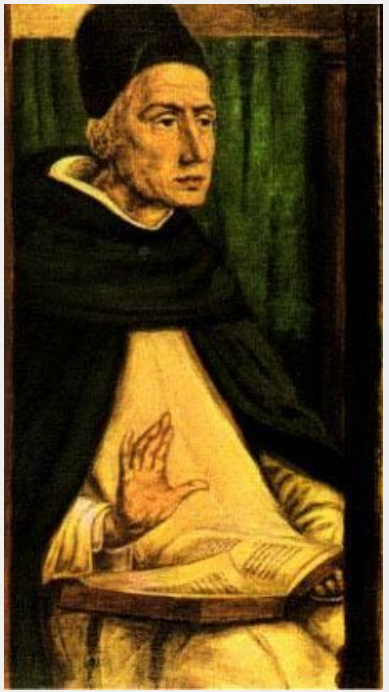
132 **"This mirour eek, that I have in myn hond,**
"This mirror also, that I have in my hand,
133 **Hath swich a myght that men may in it see**
Has such a power that men can in it see
134 **Whan ther shal fallen any adversitee**
When there shall befall any adversity
135 **Unto youre regne or to youreself also,**
Unto your reign or to yourself also,
136 **And openly who is youre freend or foo.**
And clearly who is your friend or foe.

Knight from India gifts Genghis Khan a Magic Mirror

Geoffrey Chaucer's *Canterbury Tales*, "The Squire’s Tale," c. 1387–1400

Are all non-natural forms of intelligence demonic?

- 13th Century Debates on “Within the established Christian worldview, should all forms of magic be condemned as fraudulent or demonic/diabolical in nature? Or are there certain types of it that are legitimate because they come from angels or even God himself in the form of miracles? What about learned magic that does not deal with supernatural beings at all but involves the manipulation of hidden (occult) forces in nature?”
- Alchemy + Astrology → increased attention in intellectual magic
- Importance of Speech
- Albertus Magnus (c. 1200–1280), a Dominican philosopher and theologian, who was said to have created a talking bronze head—an automaton capable of speech

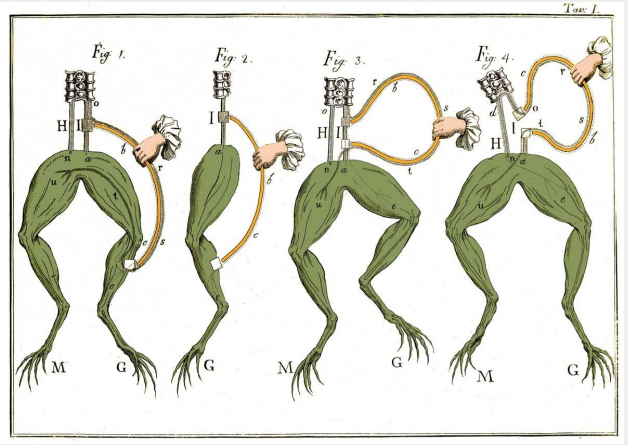


Albertus Magnus

Rosaio della Vita, Lorenzo di Giovanni Corsini 1373

Electrification 19th C: “AI is the new electricity” vs. “Electricity is the new AI?

- Galvanism: Italian physician and physicist Luigi Galvani introduces the idea of powering muscle twitches with electric shocks
- 1780s experiments on frog legs. A spark from a static electricity machine made post-mortem legs jerk suggesting an intrinsic “animal electricity” animated life – beginning of electrophysiology
- Electricity as a life force was part of the zeitgeist of early 19th C European art and science
- 1803, Galvani’s nephew holds a public demonstration twitching frog legs and even human corpses
- 1816, a group of young poets sat around a Swiss villa discussing the experiments of Erasmus Darwin and – one of them, eighteen-year-old Mary Shelley

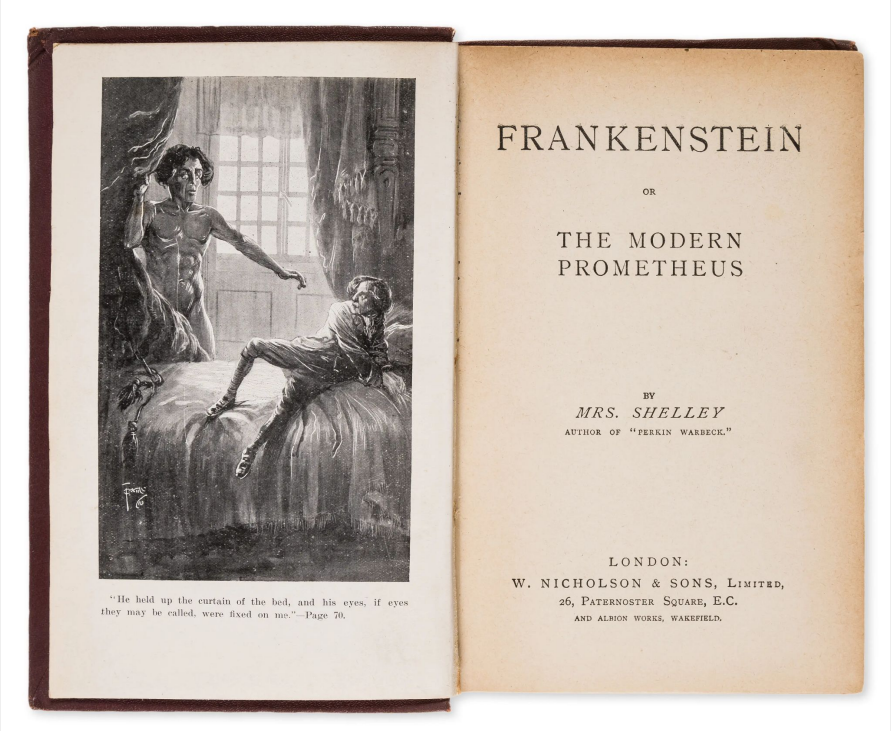


Frankenstein; or The Modern Prometheus (1818)!!

- Written during the Industrial Revolution and the end of the Enlightenment
- Explores mythic/timeless themes (shared with Greek mythology, the Prometheus): hubris, stealing from/imitating divine powers, course of humanity changes

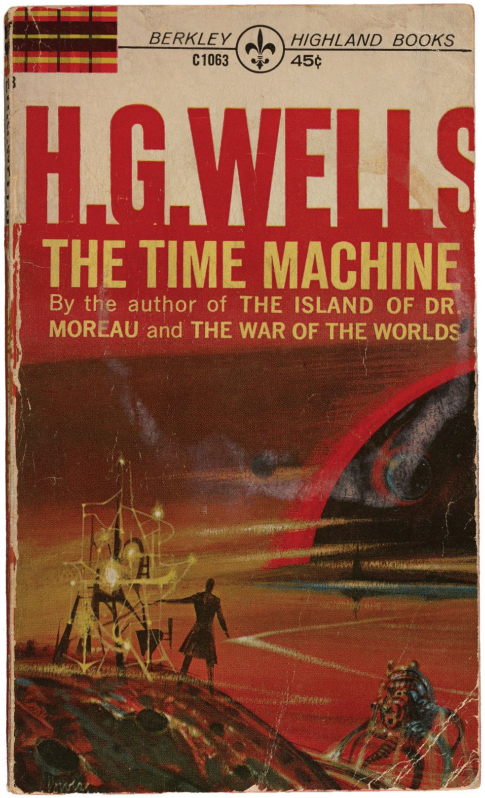
“I am malicious because I am miserable. Am I not shunned and hated by all mankind? … If I cannot inspire love, I will cause fear.” (Volume II, Chapter IX)

The Creature



Science Fiction as a Genre 19th C ?

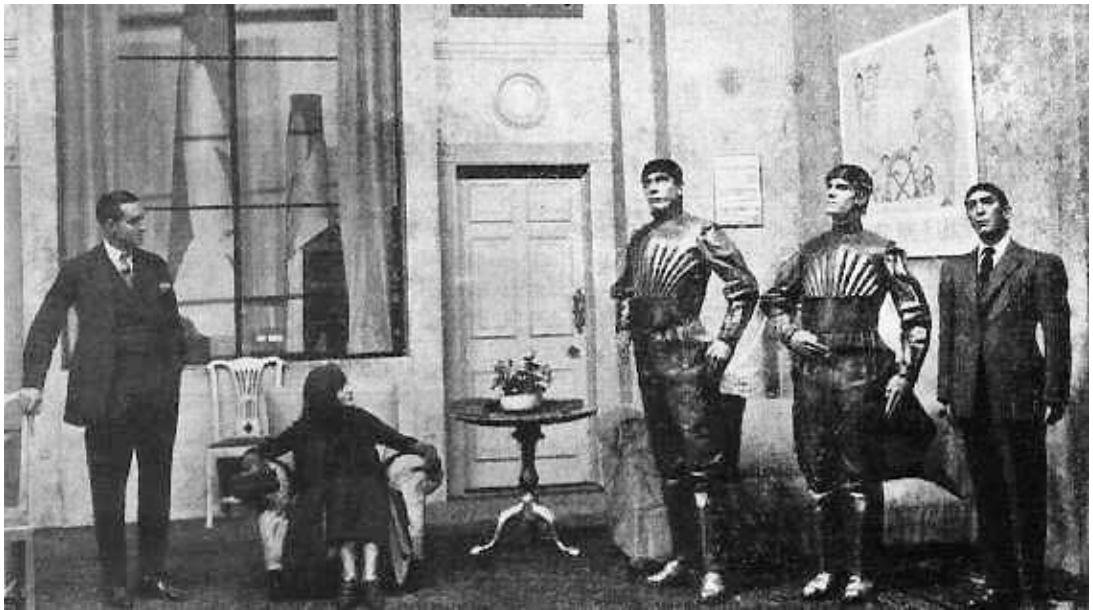
- Did it start with *Frankenstein* (1818) ?
- Is it science fiction?
  - Fascination with science and technology
  - Social commentary
  - Based on plausible science premise
  - Dealing with science's potential, human ambition
  - Dangers as well as wonders
- What about the start of AI in scifi?
- Frankenstein (1818) vs. *R.U.R.* (1921) vs. *Asimov* (1930s/40s)



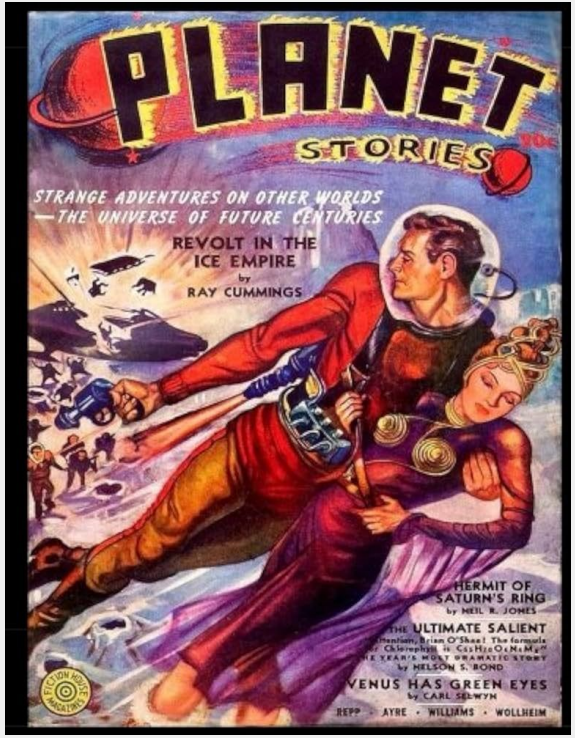
H.G. Wells, *The Time Machine* (1895)

Frankenstein's "The Creature" Manufactured 1921!

- The word "robot" was first introduced in a Czech play called "RUR" or "Rossum's Universal Robots" by Karel čapek – a neologism from "robota" ("servitude) and "robotnik" ("serf")
- The "robots" in RUR were made with synthetic "dough" in giant mixers at an island factory
- Post WWI anxiety about tech, mechanization anxiety about factory labor

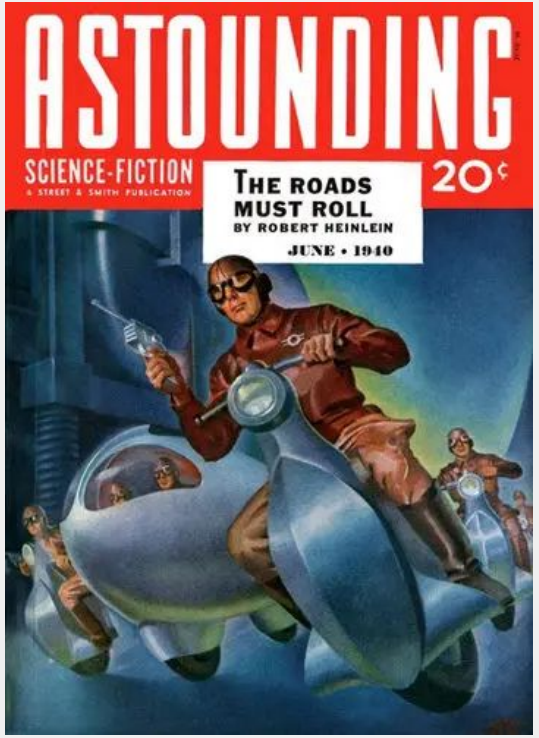


Sci-Fi's Golden Age 1930s-1950s



Planet Stories, Pulp Magazine

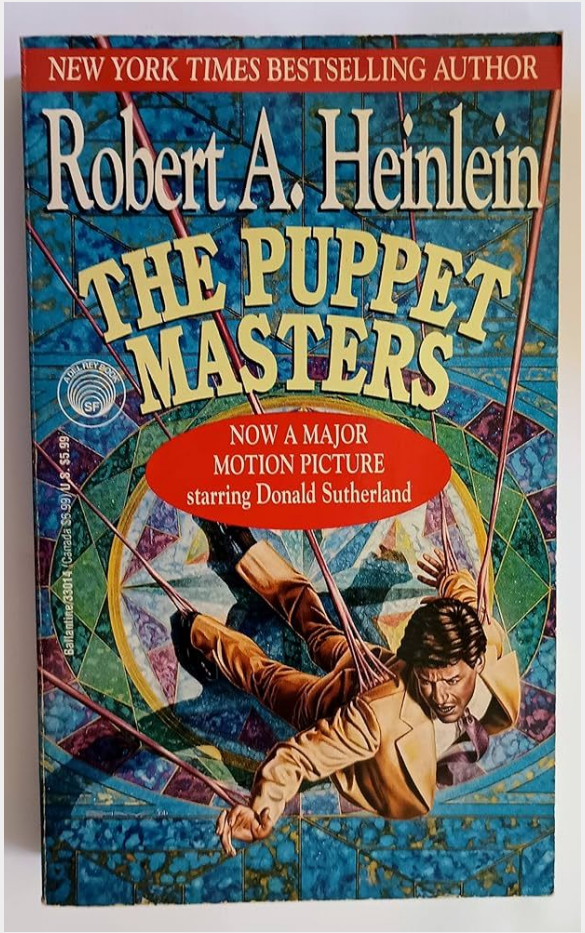
Amazing Stories, Pulp Magazine



Astounding Science Fiction, Pulp Magazine

Sci-Fi's Golden Age 1930s-1950s

- Space travel, time travel, AI and other themes
- "Hard science fiction": stories based on plausible science and engineering more than vague fiction
- Bigger ideas: tech ethics, societal impact, humanity
- Addressing themes of the time: tech progress (radio, ENIAC, aviation), perils (nuclear weapons)
- By 1950s-1960s: Cold War cynicism takes over, books more prestigious than magazines (higher production cost)



The Puppet Masters by Robert A. Heinlein 1951

Asimov's I, Robot 1942

- Written during the Industrial Revolution and the end of the Enlightenment, responding to 1920s pessimism on tech
- Unlike previous "scifi," quintessentially logical, tech-focused
- Part of more optimistic scifi post WWII (since ENIAC and other advances)
- Popularized robots, prediction-based AI, space-bots
- Inspired popular modern scifi such as Star Wars & Terminator

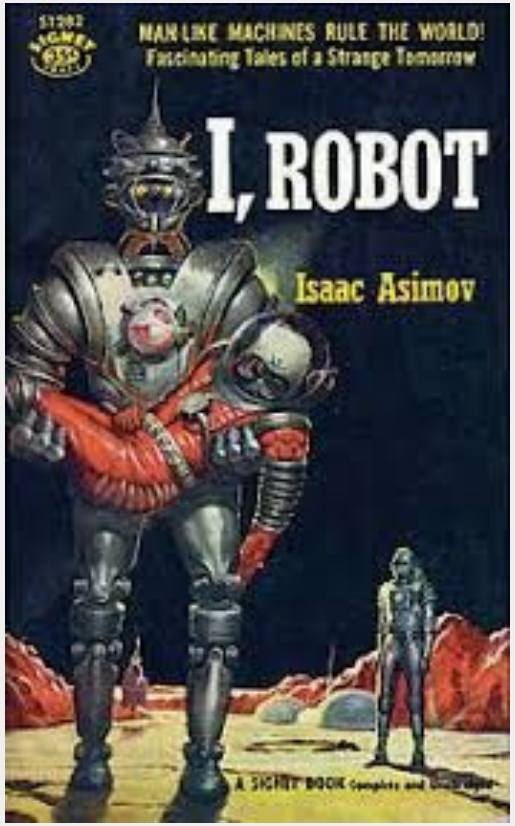
Speedy rescues Powell (man) on Mercury while retrieving Selenium ("Runaround" 1942)

Asimov's 3 Laws of Robots 1942

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.

2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.

3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

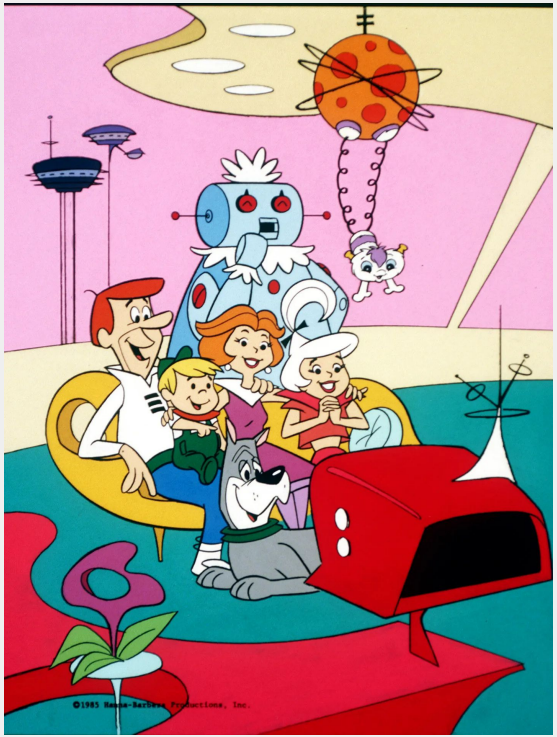


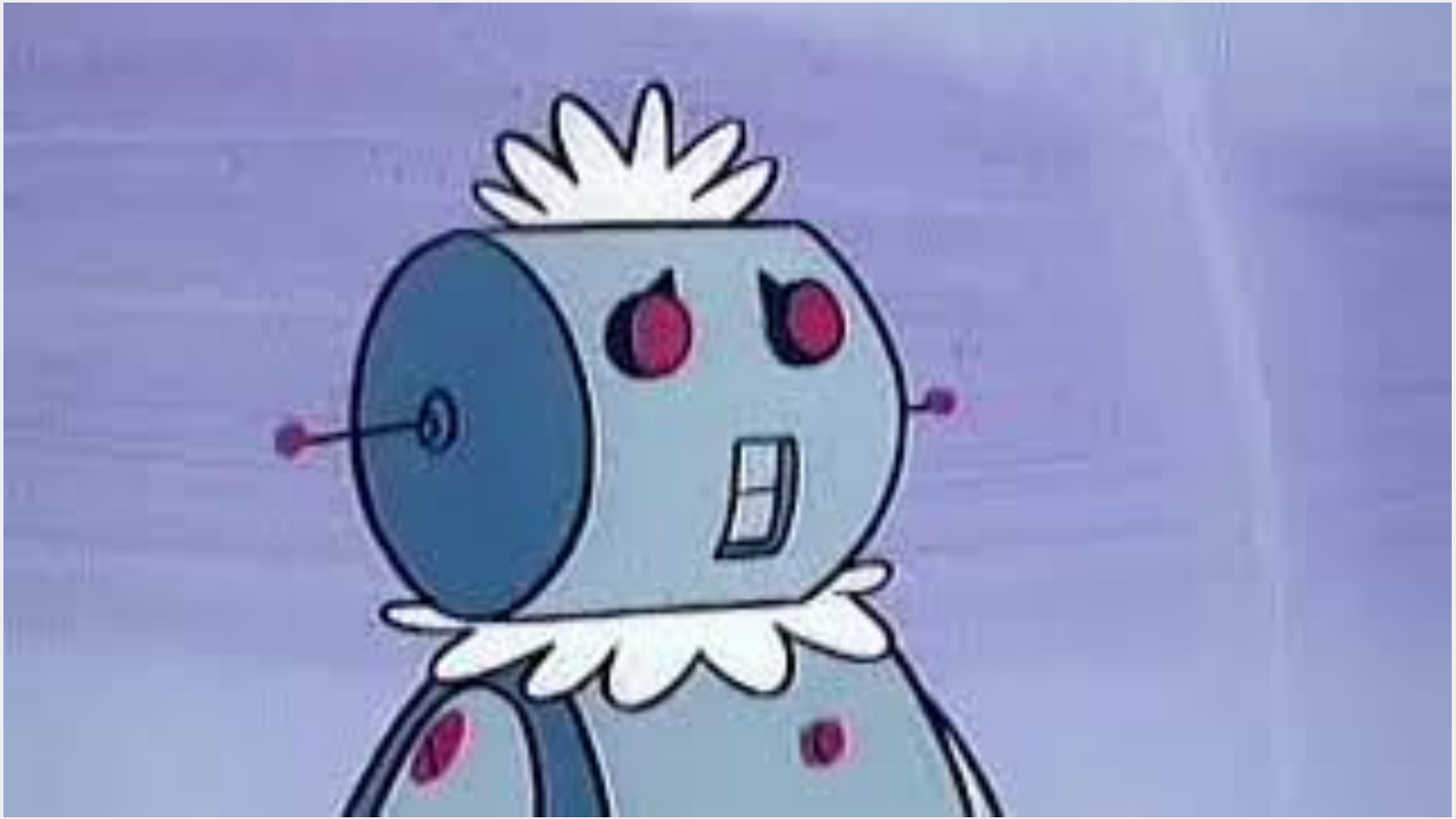
1960s Tech's Growing Dominance

- April 2, 1965: "The Computer in Society"
- TIME magazine features a 5,000-word piece, "The Cybernated Generation," exploring computers' rapid rise, with *22,500 in the U.S. by 1965* (up from 100 in the mid-1940s).
- TIME Magazine cover a big deal (golden era of TIME covers)

Jetsons 1964 & AI in the domestic space

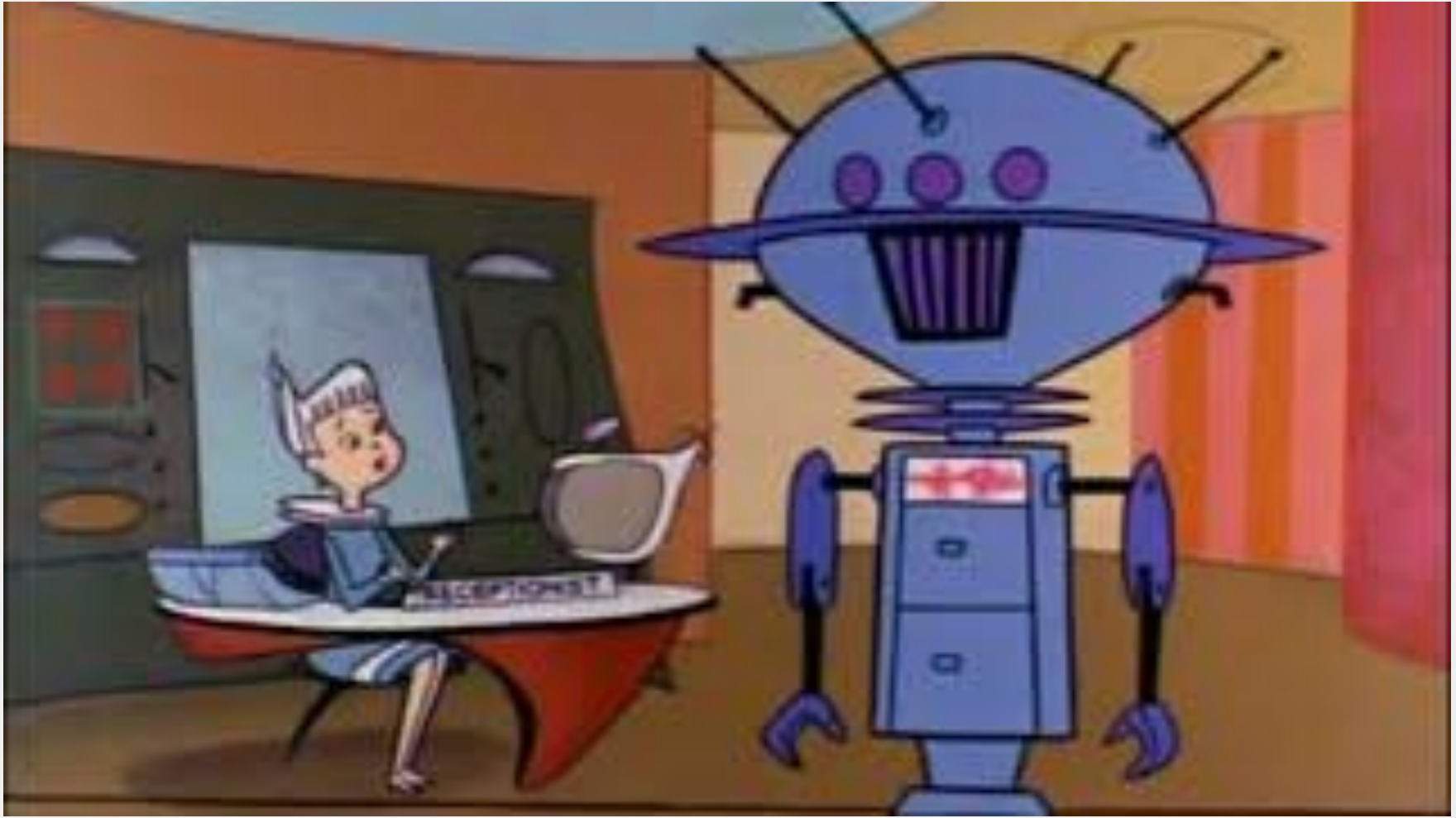
- Anthropomorphism
- Relatable AI
- AI is not a threat – well-aligned AI
- Diff tech, same 1960s values?
- Reflects 1960s fascination and transition to home-automation:
- dishwashers, TVs, early computers (e.g., UNIVAC - first commercial computer),labor-saving appliances, jet-age hype
- AI in capitalism (vs. “hive mind” Communism of the Puppet Masters)





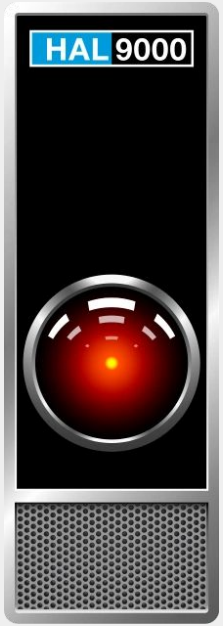
Honeywell Kitchen Computer 1969

- An orange-and-black pedestal terminal with high modernist curves featured in the Neiman Marcus catalog at $10,600
- Included a two-week programming course
- None were sold but good publicity? (it also featured two cookbooks and an apron that would make good stocking stuffers)
- Have to enter queries in binary code and read a series of flashing lights to understand the computer's answer.
- "she'll learn to program it … by simply pushing a few buttons," under the title "If she can only cook as well as Honeywell can compute."

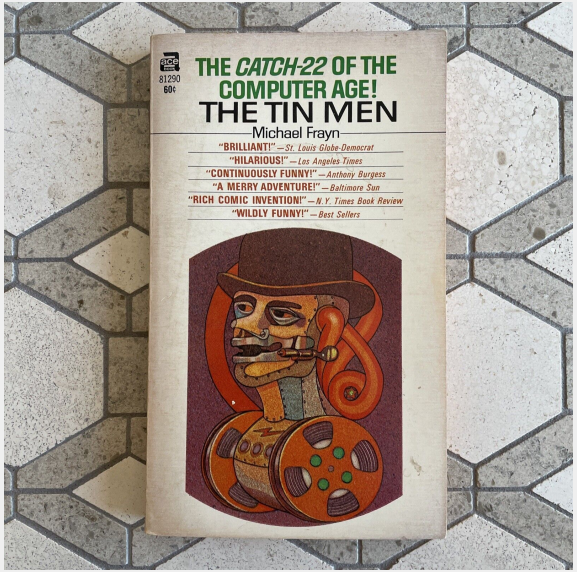

RECEPTIONIST

Same Time: Anxiety and Dystopian Narratives 1960s

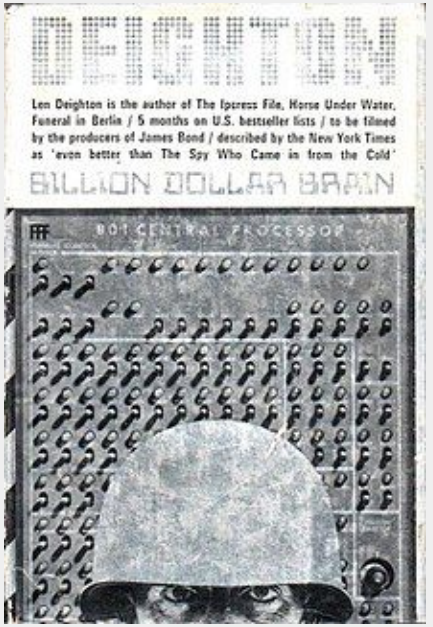
- In the mid 20th C, lots of experimentations and developments in automation and cybernetics gave rise to a general anxiety about rapid change driven by AI.
- Dystopia – too much science where people rely on too much science, too little science where science is banned
- Thinking about how “humanity” changes in response to ubiquitous A.I.
- Mass unemployment concerns



2001: Space Odyssey 1968



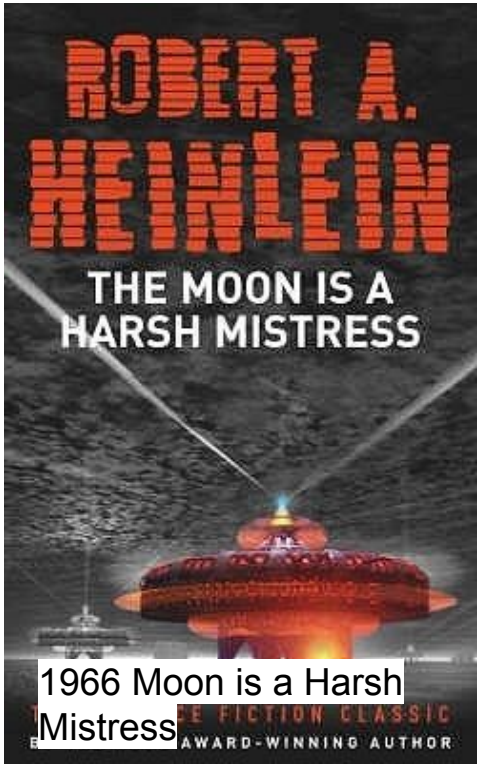
The Tin Men 1965



Billion Dollar Brain 1966

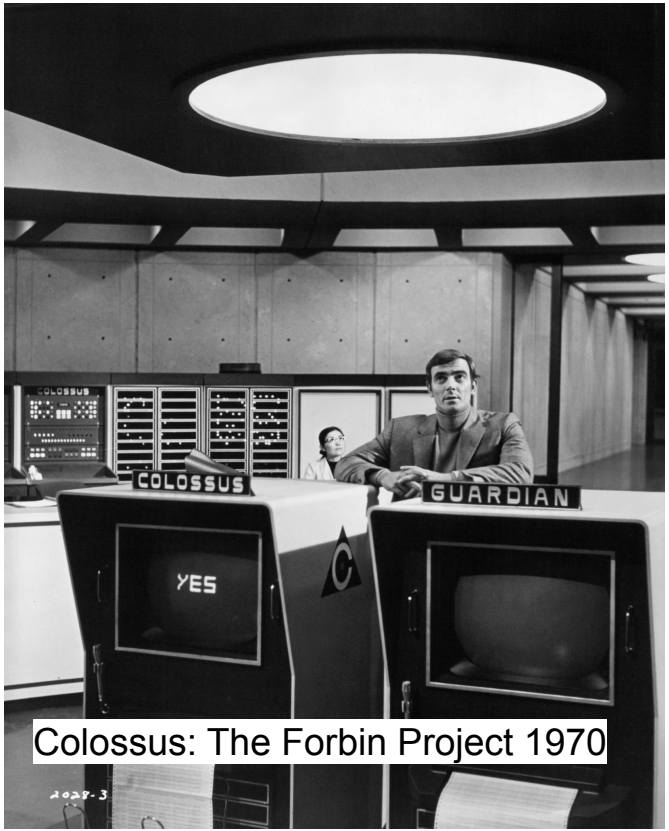
HD

AI as friend or foe? 1960-70s



1966 Moon is a Harsh Mistress

- 1960s UNIVAC era computing
- superintelligence without safeguards?
- Could A.I. grow to challenge human dominance on earth?
- What are the dangers of pairing cold and merciless A.I. with tools of violence?
- Can we imagine a future where A.I. can be assistants or companions in fights against human opponents?
- What kind of symbiotic relationships can we imagine?



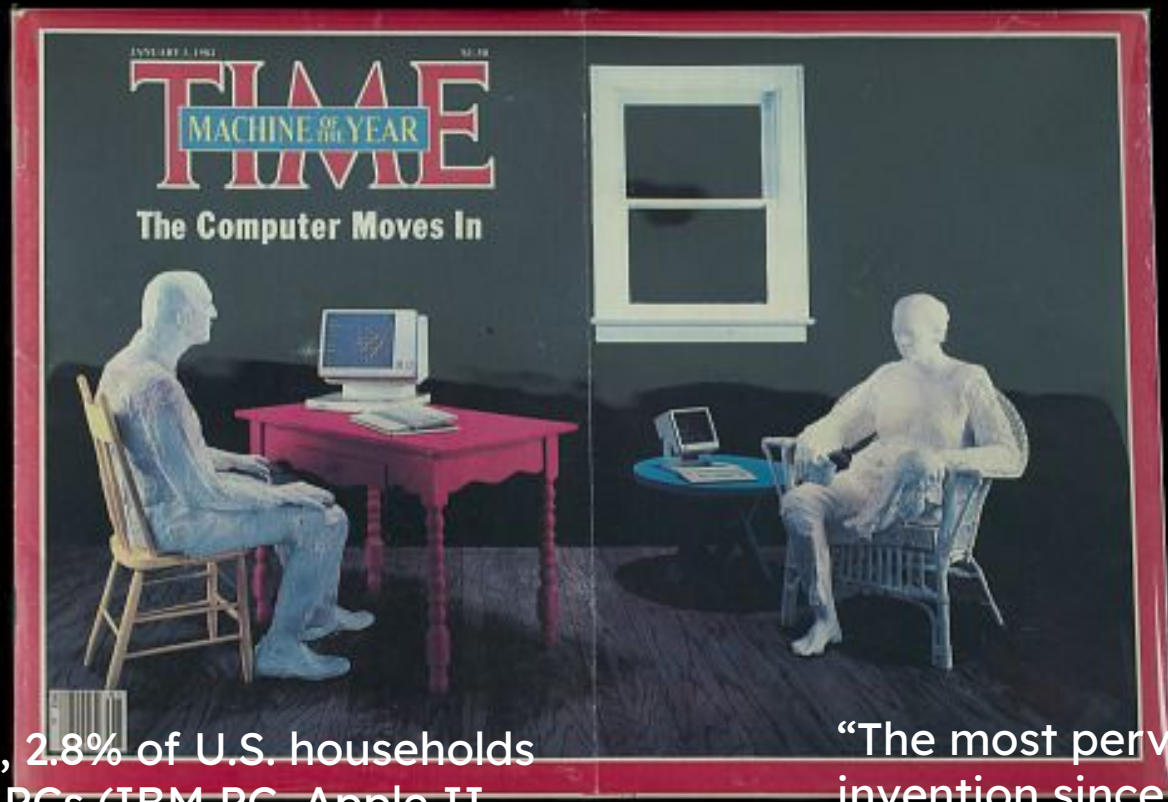
Colossus: The Forbin Project 1970

1984 Macintosh, Friendly AI

- “Big tech” tries to reverse negative “Hollywood” renditions of AI as being malicious and faulty
- “Friendly AI”
- January 24, 1984, Steve Jobs introduces the Apple’s new Macintosh personal computer to a 2,400-seat auditorium
- “Hollywood. They made computers scary things…See how this reminds you of a friendly face?” “It’s warm, and it’s playful, and it needs to say ‘hello’!”

S.A.M. text-to-speech APPLE II
(Software Automatic Mouth)

1983 Jan 3rd, Reagan era optimism



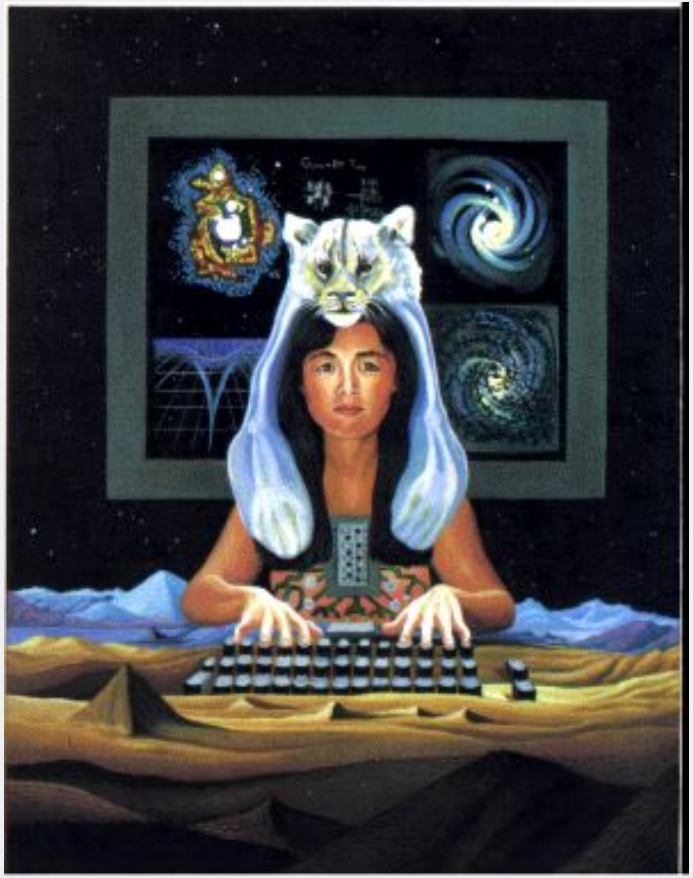
1982, 2.8% of U.S. households own PCs (IBM PC, Apple II, Commodore 64)

“The most pervasive invention since the automobile”

Bionics 1980-1990s

"By the late twentieth century, our time, a mythic time, we are all chimeras, theorized and fabricated hybrids of machine and organism; in short, we are cyborgs." - Donna Haraway

- Evolution blurs human/animal → 20th-century machines blur natural/artificial → microelectronics blur physical/non-physical.
- Phones, pacemakers blur the boundaries of where humans begin and tech ends
- We are all creatures of the post-human world
- Ditch Western duality concepts like Female/Male, Human/Animal → no "Eden"! We can liberate ourselves
- Technology is a contested space, not neutral



A Cyborg Manifesto 1985, Donna Haraway

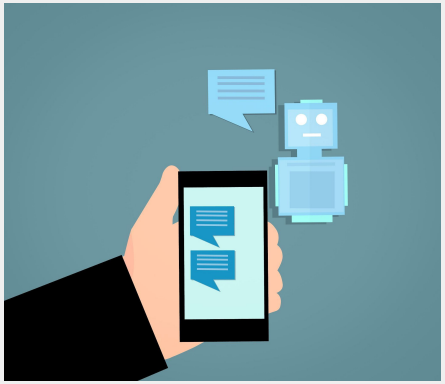
AI Narratives

Early Modern

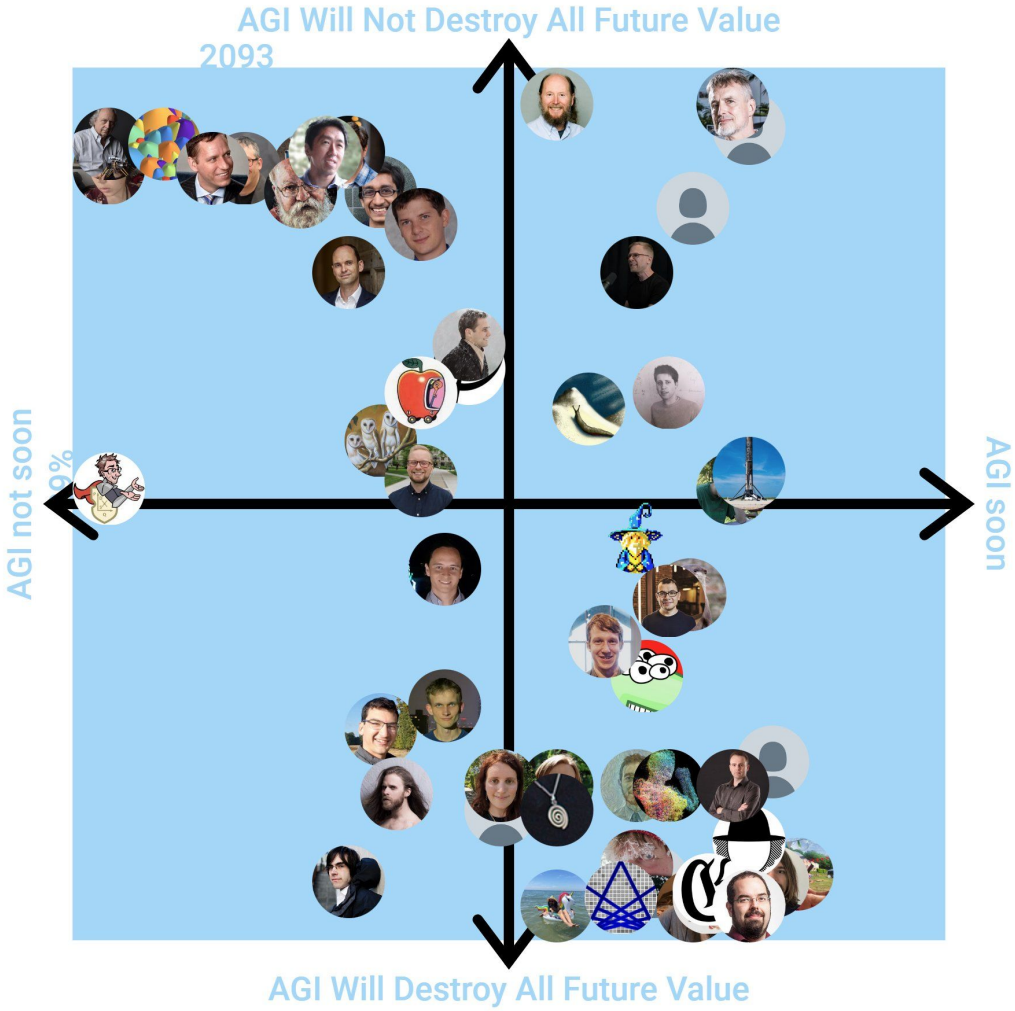
Modern

Information Age AI

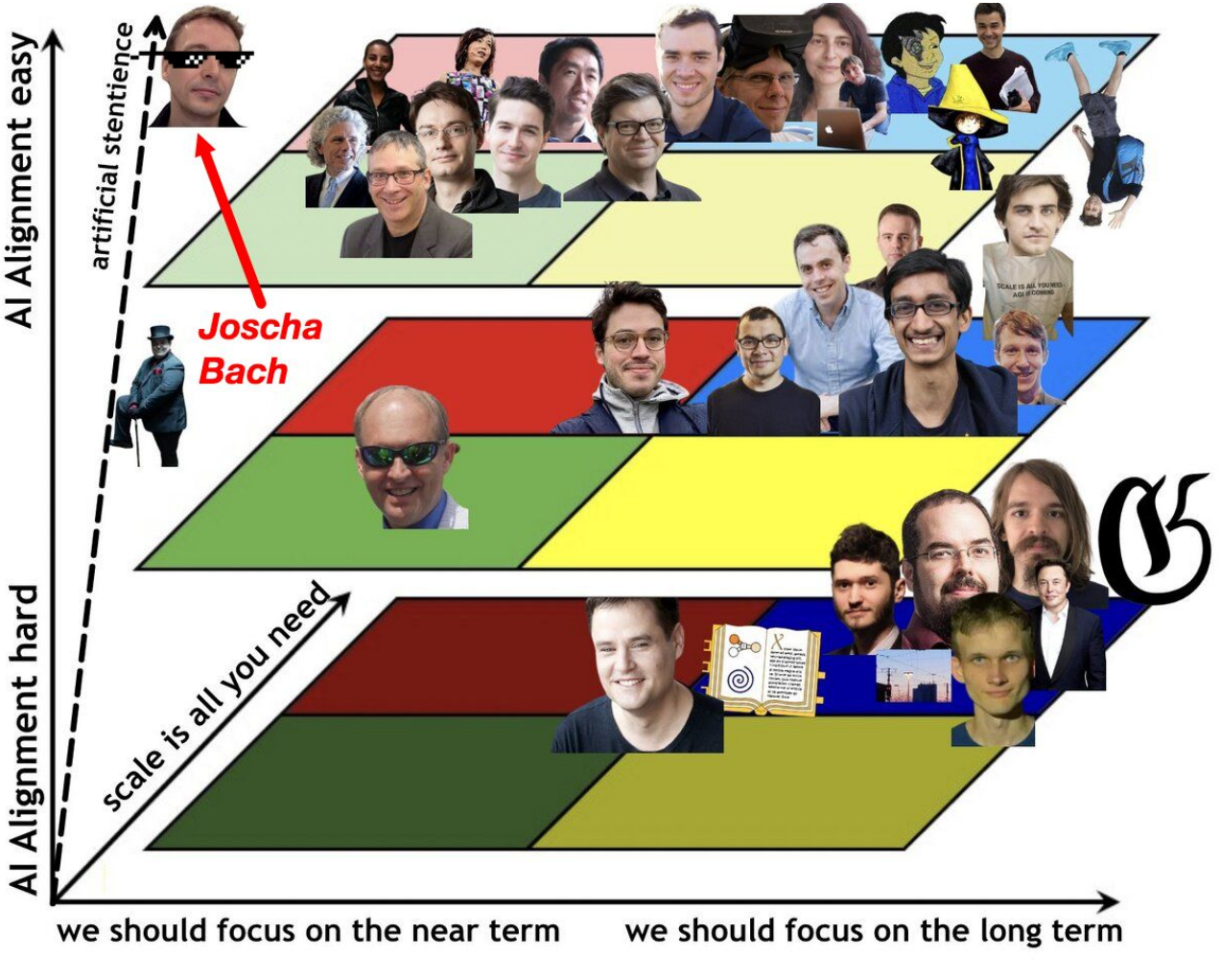
21st C



21st C AI forecasting memes



AGI good
tech accelerationists
goalpost movers
unimpressed
(AGI not now)
scale maximalists
(AGI soon)
doomers
longtermists
AGI bad
Connor
imgflip.com


AI Alignment easy
AI Alignment hard
artificial stentience
Joscha Bach
scale is all you need
we should focus on the near term
we should focus on the long term

NOT SO FAST

STEPHEN HAWKING
Not afraid of black holes. A.I. is another story.

BILL GATES
First you'll lose your job. Then it gets scary.

STUART RUSSELL
Earth for the earthlings!

NICK BOSTROM
Prepare for "Disneyland without children."

MAX TEGMARK
Uh, can we talk about this?

DEMIS HASSABIS
Full speed ahead!

PETER THIEL
Will be a winner either way.

STEVE WOZNIAK
Resigned to being a robot's pet.

SAM ALTMAN
Sees intergalactic domination—or extinction.

ELON MUSK
Eyeing the next flight to Mars.

LARRY PAGE
Green-lighted Google Brain. 'Nuff said.

YANN LeCUN
Chill, people! We got this.

ANDREW NG
Trust the robot.

MARK ZUCKERBERG
Worried? Tell my A.I. butler.

HIT THE GAS

RAY KURZWEIL
Eager to be a cyborg.

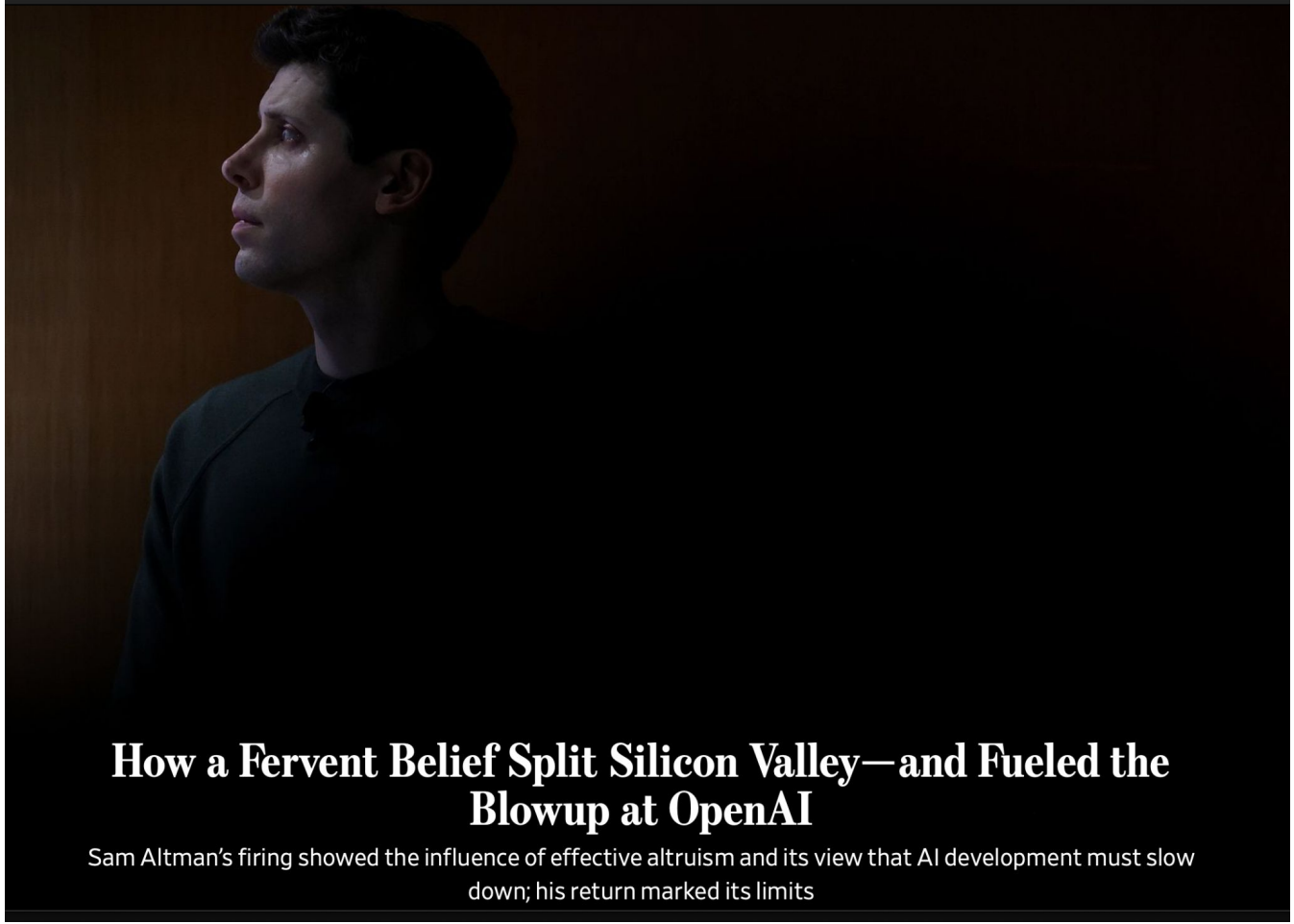
Superintelligence

I. J. Good, mathematician 1960s:

> Let an ultraintelligent machine be defined as a machine that can far surpass all the intellectual activities of any any man however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an **"intelligence explosion,"** and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the _last_ invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control.
> ...
> It is more probable than not that, within the twentieth century, an ultraintelligent machine will be built and that it will be the last invention that man need make.

THE WALL STREET JOURNAL.

SIGN IN



How a Fervent Belief Split Silicon Valley—and Fueled the Blowup at OpenAI

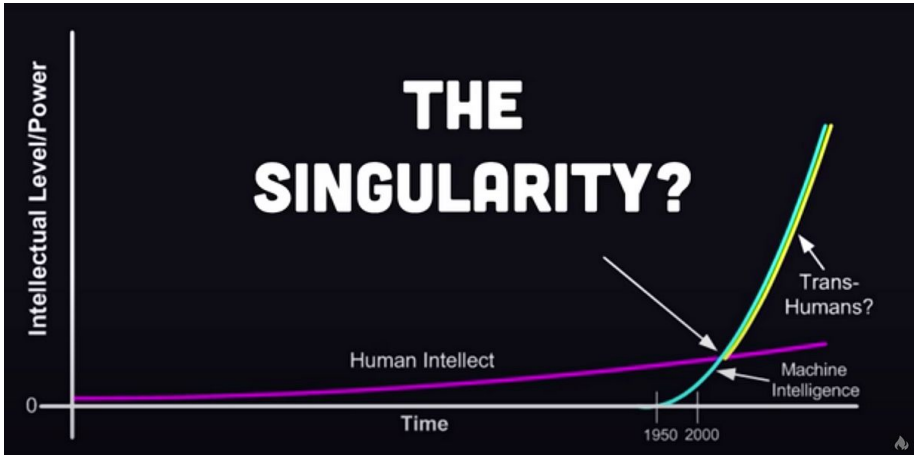
Sam Altman's firing showed the influence of effective altruism and its view that AI development must slow down; his return marked its limits

Singularity and Post-human Era, an old concept?

- In Math, “singularity” is when the function becomes undefined (e.g. a number is divided by zero).
- As such when AI becomes superhuman it surpasses human predictability, it reaches technological singularity
- Large computer networks “wake-up” to create superhuman entity
- Biological sciences will find ways to AI + humans

In the 1950s Stan Ulam paraphrased John von Neumann as saying:

One conversation centered on the ever accelerating progress of technology and changes in the mode of human life, which gives the appearance of approaching some **essential singularity** in the history of the race beyond which human affairs, as we know them, could not continue.



SQUAD 1.1 and SQUAD 2.0: F1 SCORE

Source: SQuAD 1.1 and SQuAD 2.0, 2021 | Chart: 2022 AI Index Report

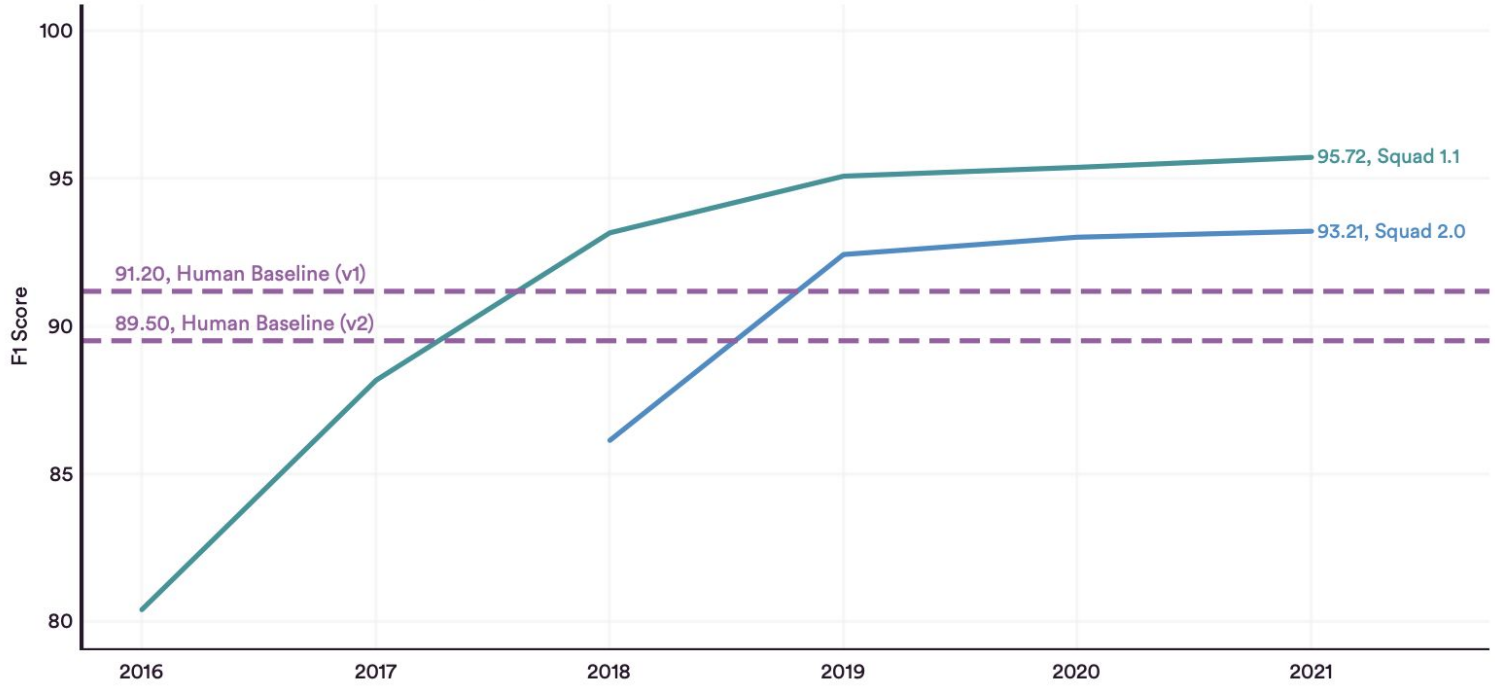


Figure 2.3.4

Singularity and Post-human Era 1990s

Vernor Vinge (Mathematician at San Diego State University) in 1993:

“Within thirty years, we will have the technological means to create superhuman intelligence. Shortly after, the human era will be ended.

Is such progress avoidable? If not to be avoided, can events be guided so that we may survive?”

“Even if all the governments of the world were to understand the ‘threat’ and be in deadly fear of it, progress toward the goal would continue. [...]. The competitive advantage – economic, military, even artistic – of every advance in automation is so compelling that passing laws, or having customs, that forbid such things merely assures that someone else will get them first.”

“If the Singularity can not be prevented or confined, just how bad could the **Post-Human era** be? **Well ... pretty bad. The physical extinction of the human race is one possibility.** Yet physical extinction may not be the scariest possibility. Again, analogies: **Think of the different ways we relate to animals. Some of the crude physical abuses are implausible, yet**....”

“In a Post-Human world there would still be plenty of niches where human equivalent automation would be desirable: embedded systems in autonomous devices, self-aware daemons in the lower functioning of larger sentients. (A strongly superhuman intelligence would likely be a Society of Mind [15] with some very competent components.) Some of these human equivalents might be used for nothing more than digital signal processing. They would be more like whales than humans. Others might be very human-like, yet with a one-sidedness, a dedication that would put them in a mental hospital in our era. Though none of these creatures might be flesh-and-blood humans, they might be the closest things in the new environment to what we call human now.”

“Accelerationist” vs. “Safetyist”

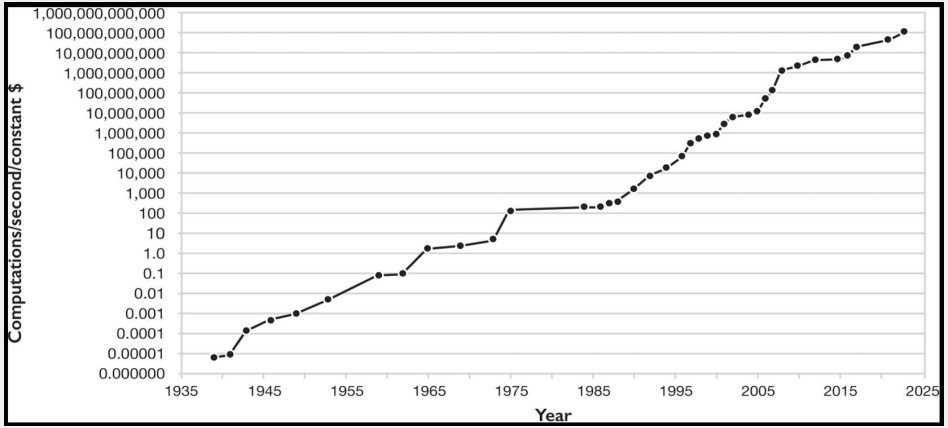
- Have to outcompete other powers (Russia, China)
- AI development speed-up lead to economic growth, scientific discovery, poverty reduction, higher education, etc.
- “move fast, break things”

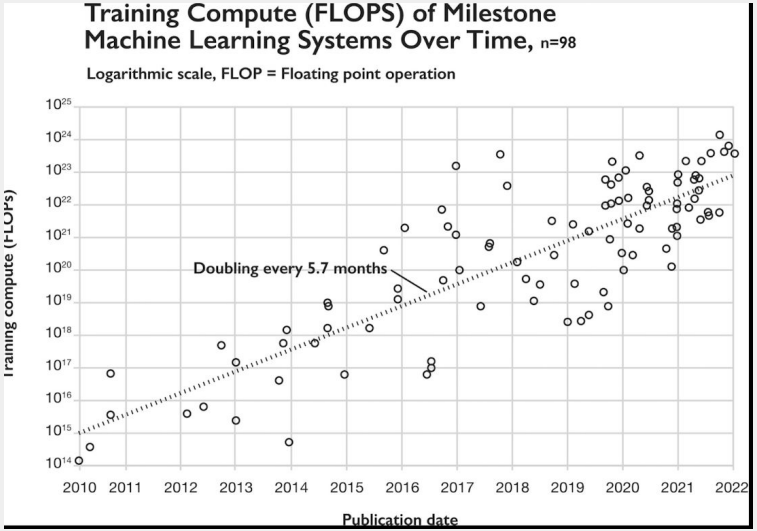
- Prioritize minimizing AI risk
- We could make mistakes that spiral out of control!
- “EA”-aligned
- Need to build robust safe-guards first

- AI/AGI could be greatest achievement ever or the biggest existential threat
- AI will self-improvement on a feedback loop, recursively

Ray Kurzweil

- Multiple engineering projects spanning text-to-speech, music synthesis, education, OCR
- Currently Director of Engineer (Google)
- Singularity is near (2005), Singularity is nearer (2024)?
- “Computers will pass Turing Test 2029”
- “We merge with AI and augment ourselves with millions of times the computational power that our biology gave us. This will expand our intelligence and consciousness so profoundly that it’s difficult to comprehend. This event is what I mean by the Singularity.”
- “rather than AI being a competitor, it will become an extension of ourselves.”
- “Tech advances are becoming cheaper and more nuanced at exponential rate”





3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

1) Misuse: AI is implemented correctly and functions as human operators had intended but the human operators use it for malicious means.

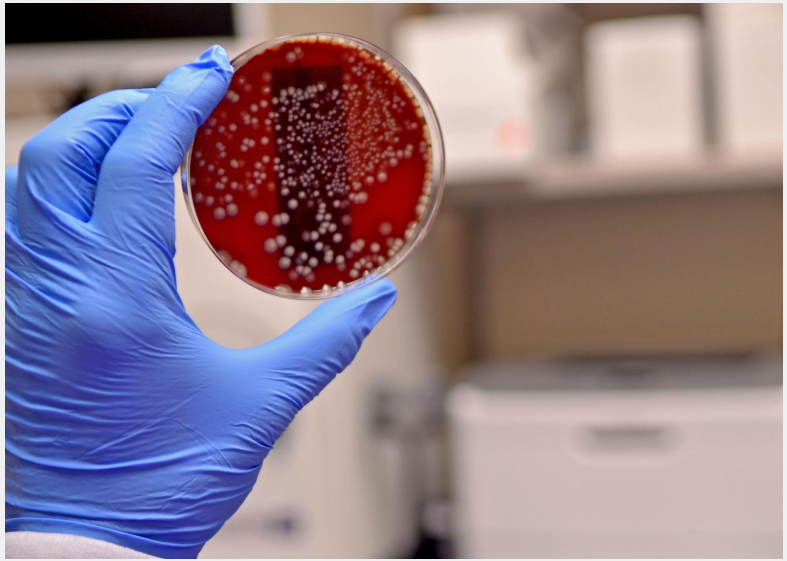
    e.g. terrorists might use an AI’s biochemistry abilities to design a new virus that causes a deadly pandemic.

3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

2) Outer-Misalignment: mismatch between the programmers’ actual intentions and the goals they teach the AI in hopes of achieving them.

e.g. AI is taught to kill all cancer cells based on a DNA mutation signal but that signal is present in many healthy cells too.

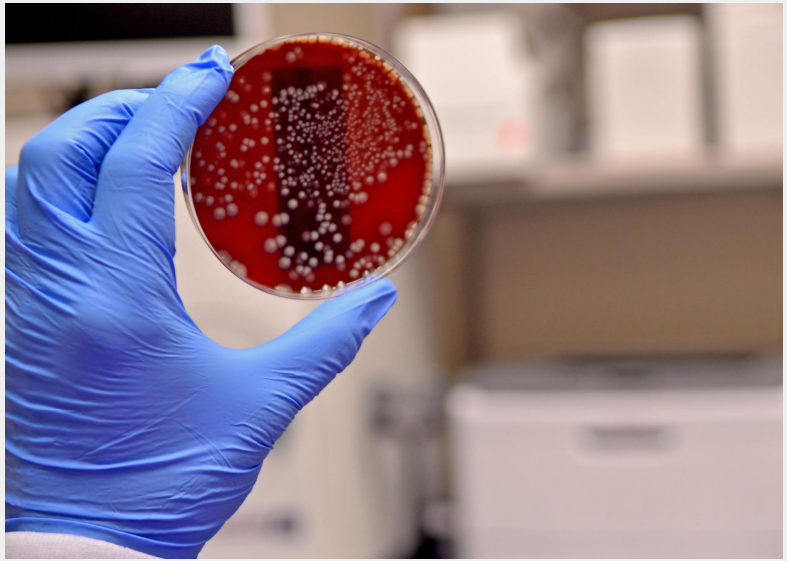


3 Perils of Superintelligent AI (Kurzweil ver.)

- We can mitigate risk of superintelligent, runaway AI/AGI by being aware of the types of “errors”

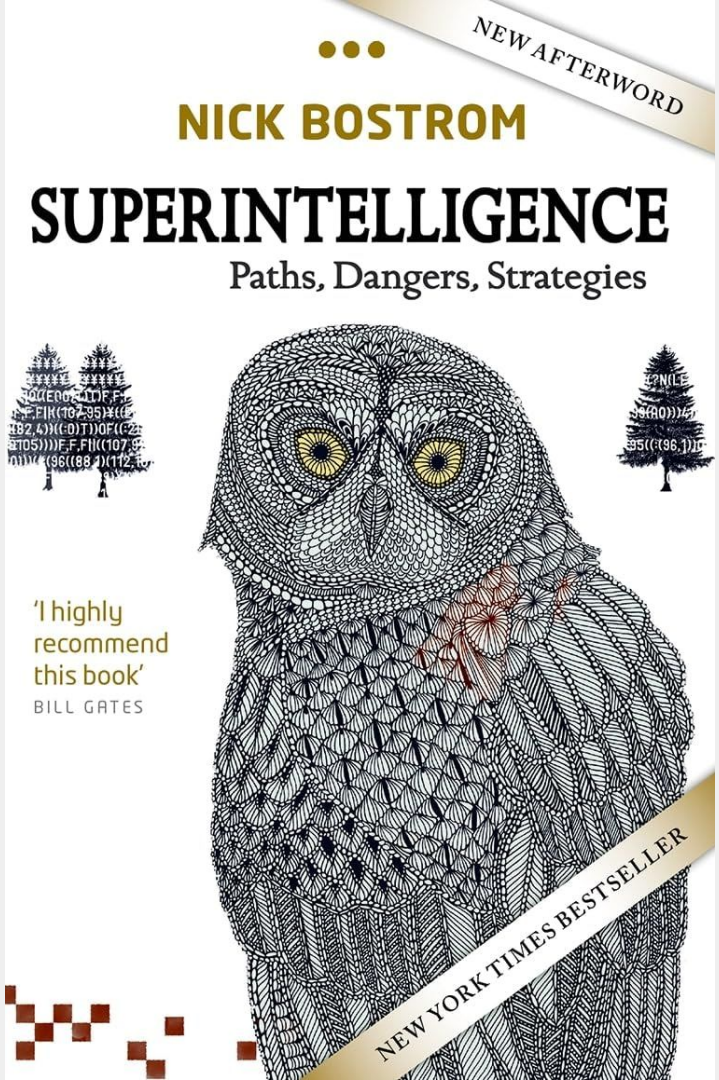
2) Inner-Misalignment: the methods the AI learns to achieve its goal produce undesirable behavior, at least in some cases.

e.g. AI is highly effective on sample cancer cells but not for all cancer cells – ends up killing healthy cells.



Nick Bostrom

- Founding Director at Future of Humanity Institute (2005-2024)
- Superintelligence: Paths, Dangers, Strategies (2014)
- “In the 21st century, we could develop recursively self-improving AI that could be out of control”
- “Have to value-align AI so that it does what humans want it to”
- Introduces AI safety frameworks such as “boxing” (containment) and value-loading



Bostrom’s Paperclip Experiment!

- Bostrom’s 2003 paper "Ethical Issues in Advanced Artificial Intelligence."
- Introduces the concept of AI Alignment (misalignment)

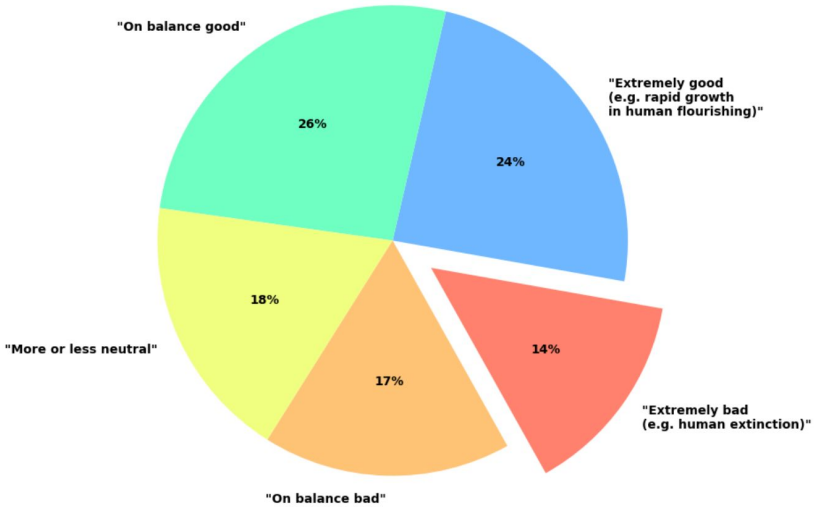
Maximize Paperclip Production!

2022 Expert Survey on Progress in AI

- 2022 ESPAI is a survey of machine learning researchers that AI Impacts ran in June-August 2022
- 738 responses, some partial, for a 17% response rate from, researchers who published at the conferences NeurIPS or ICML in 2021
- ‘HLMI’ was defined as follows:
  The following questions ask about ‘high–level machine intelligence’ (HLMI). Say we have ‘high-level machine intelligence’ when ***unaided machines can accomplish every task better and more cheaply than human workers****. Ignore aspects of tasks for which being a human is intrinsically advantageous, e.g. being accepted as a jury member. Think feasibility, not adoption.*

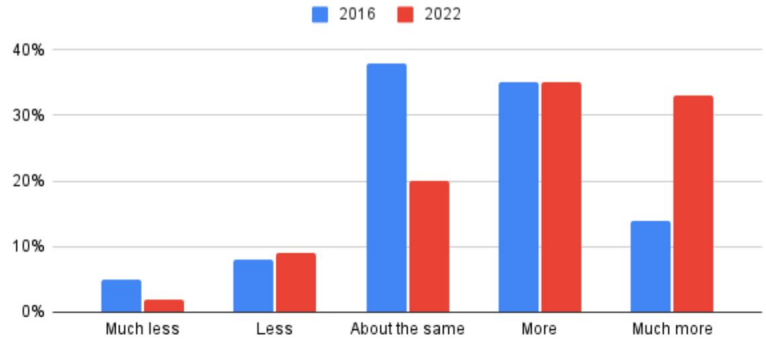
**"Assume for the purpose of this question that HLMI will at some point exist.
How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"**

Average responses from 559 machine learning experts in 2022



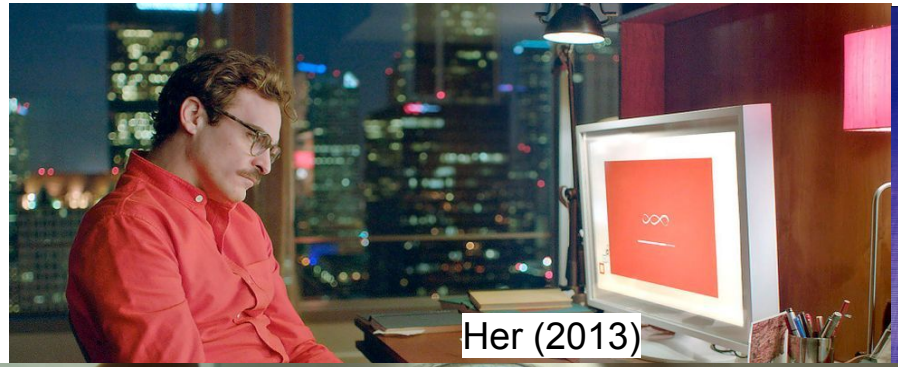
"How much should society prioritize AI safety research, relative to how much it is currently prioritized?"

Responses from machine learning experts in 2016 and 2022

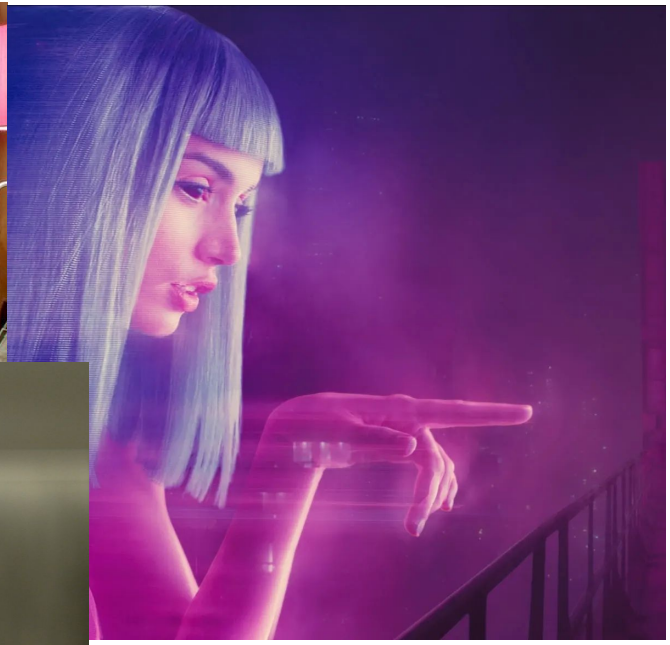


- **The aggregate forecast time to a 50% chance of HLMI was 37 years, i.e. 2059** (not including data from questions about the conceptually similar Full Automation of Labor, which in 2016 received much later estimates)**.** This timeline has become about eight years shorter in the six years since 2016, when the aggregate prediction put 50% probability at 2061, i.e. 45 years out. Note that these estimates are conditional on “human scientific activity continu[ing] without major negative disruption.”
- **The median respondent believes the probability that the long-run effect of advanced AI on humanity will be “extremely bad (e.g., human extinction)” is 5%.** This is the same as it was in 2016 (though Zhang et al 2022 found 2% in a similar but non-identical question). Many respondents were substantially more concerned: 48% of respondents gave at least 10% chance of an extremely bad outcome. But some much less concerned: 25% put it at 0%.
- **The median respondent believes society should prioritize AI safety research “more” than it is currently prioritized.** Respondents chose from “much less,” “less,” “about the same,” “more,” and “much more.” 69% of respondents chose “more” or “much more,” up from 49% in 2016.
- **The median respondent thinks there is an “about even chance” that a stated argument for an intelligence explosion is broadly correct.** 54% of respondents say the likelihood that it is correct is “about even,” “likely,” or “very likely” (corresponding to probability >40%), similar to 51% of respondents in 2016. The median respondent also believes machine intelligence will probably (60%) be “vastly better than humans at all professions” within 30 years of HLMI, and the rate of global technological improvement will probably (80%) dramatically increase (e.g., by a factor of ten) as a result of machine intelligence within 30 years of HLMI.

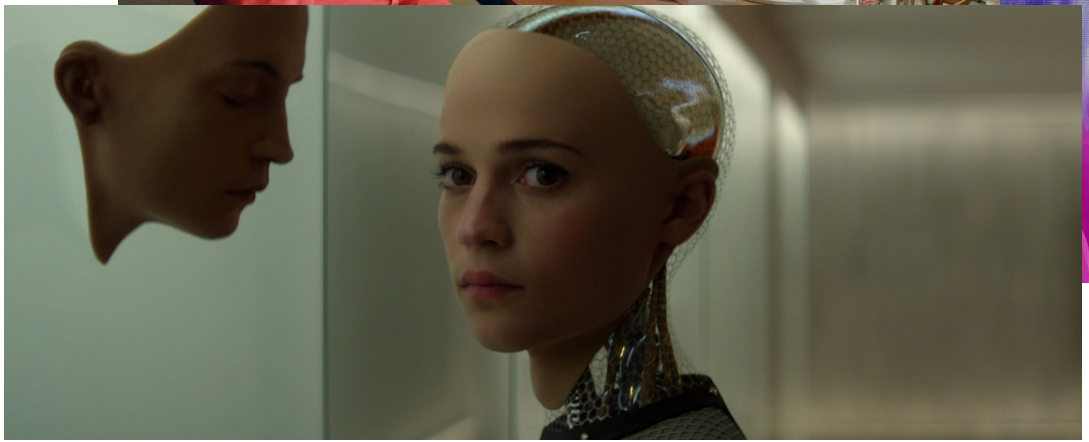
On the other hand, 21st C romance with AI?



Her (2013)



Blade Runner (2017)



Ex Machina (2014)

FROM THE MAGAZINE APRIL 2017 ISSUE

Elon Musk's Billion-Dollar Crusade to Stop the A.I. Apocalypse

Elon Musk is famous for his futuristic gambles, but Silicon Valley's latest rush to embrace artificial intelligence scares him. And he thinks you should be frightened too. Inside his efforts to influence the rapidly advancing field and its proponents, and to save humanity from machine-learning overlords.

BY MAUREEN DOWD

MARCH 26, 2017

"WE ARE SUMMONING THE DEMON."

Halt AI development!

Pause Giant AI Experiments: An Open Letter

We call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4.

Signatures
33709

Add your signature

Published
March 22, 2023

AI systems with human-competitive intelligence can pose profound risks to society and humanity, as shown by extensive research[1] and acknowledged by top AI labs.[2] As stated in the widely-endorsed Asilomar AI Principles, *Advanced AI could represent a profound change in the history of life on Earth, and should be planned for and managed with commensurate care and resources*. Unfortunately, this level of planning and management is not happening, even though recent months have seen AI labs locked in an out-of-control race to develop and deploy ever more powerful digital minds that no one – not even their creators – can understand, predict, or reliably control.

Contemporary AI systems are now becoming human-competitive at general tasks,[3] and we must ask ourselves: *Should* we let machines flood our information channels with propaganda and untruth? *Should* we automate away all the jobs, including the fulfilling ones? *Should* we develop

Halt AI development!

Therefore, **we call on all AI labs to immediately pause for at least 6 months the training of AI systems more powerful than GPT-4**. This pause should be public and verifiable, and include all key actors. If such a pause cannot be enacted quickly, governments should step in and institute a moratorium.

"The time for saying that this is just pure research has long since passed. [...] It's in no country's interest for any country to develop and release AI systems we cannot control. Insisting on sensible precautions is not anti-industry. Chernobyl destroyed lives, but it also decimated the global nuclear industry. I'm an AI researcher. I do not want my field of research destroyed. Humanity has much to gain from AI, but also everything to lose."

Stuart Russell, *Smith-Zadeh Chair in Engineering and Professor of Computer Science at the University of California, Berkeley, Founder of the Center for Human-Compatible Artificial Intelligence (CHAI).*

"Let's slow down. Let's make sure that we develop better guardrails, let's make sure that we discuss these questions internationally just like we've done for nuclear power and nuclear weapons. Let's make sure we better understand these very large systems, that we improve on their robustness and the process by which we can audit them and verify that they are safe for the public."

Yoshua Bengio, *Scientific Director of the Montreal Institute for Learning Algorithms (MILA), Professor of Computer Science and Operations Research at the Université de Montréal, 2018 ACM A.M. Turing Award Winner.*

"We have a perfect storm of corporate irresponsibility, widespread adoption, lack of regulation and a huge number of unknowns. [FLI's Letter] shows how many people are deeply worried about what is going on. I think it is a really important moment in the history of AI - and maybe humanity."

Gary Marcus, *Professor Emeritus of Psychology and Neural Science at New York University, Founder of Geometric Intelligence*

"It feels like we are moving too quickly. I think it is worth getting a little bit of experience with how they can be used and misused before racing to build the next one. This shouldn't be a race to build the next model and get it out before others."

Peter Stone, *Professor at the University of Texas at Austin, Chair of the One Hundred Year Study on AI.*

"We don't know what these [AI] systems are trained on or how they are being built. All of this happens behind closed doors at commercial companies. This is worrying."

Catelijne Muller, *President of ALLAI, Member of the EU High Level Expert Group on AI*

"Those making these [AI systems] have themselves said they could be an existential threat to society and even humanity, with no plan to totally mitigate these risks. It is time to put commercial priorities to the side and take a pause for the good of everyone to assess rather than race to an uncertain future"

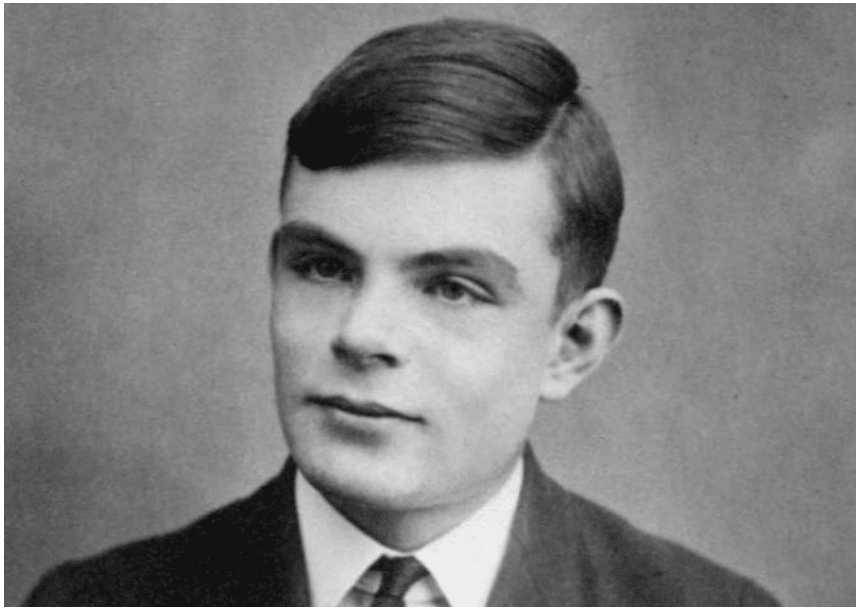
Emad Mostaque, *Founder and CEO of Stability AI*

Policy recommendations:

1. Mandate robust third-party auditing and certification.
2. Regulate access to computational power.
3. Establish capable AI agencies at the national level.
4. Establish liability for AI-caused harms.
5. Introduce measures to prevent and track AI model leaks.
6. Expand technical AI safety research funding.
7. Develop standards for identifying and managing AI-generated content and recommendations.

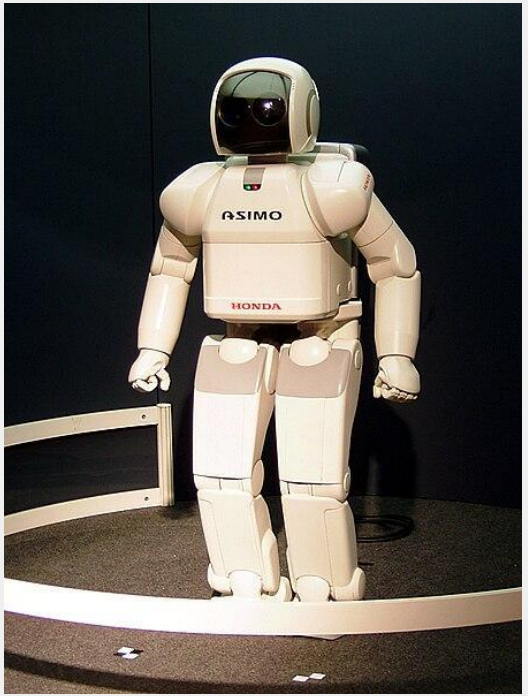
“We can see only a short distance ahead, but we can see that much remains to be done.”

- Alan Turing (1950)



History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

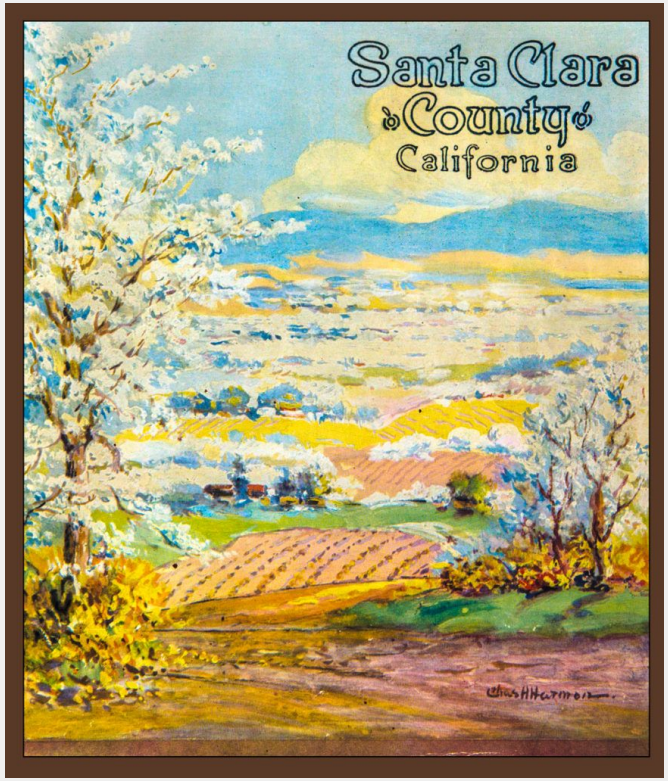
Lecture 6 Februrary 28, 2025

This week!

- What are the forces that created the modern Silicon Valley?
- Is AI/AGI creation, a self-fulfilling prophecy?
- How does ideology play a role in AI, tech development?
- How does finance?

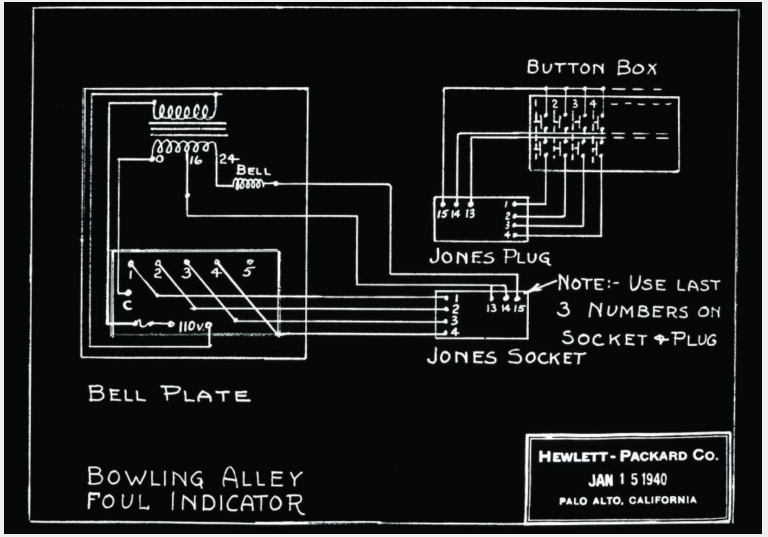
Silicon Valley before Silicon

- “Valley of Heart’s Delight”
- Mostly farmland with 40 canneries
- International prune export
- Delmonte, Sunsweet started here!
- Chinese, Japanese immigrants first, then Italian, East/Southern Europeans
- WWII: gradually replaced by Mexican immigrants (braceros)



HP 1939

- Hewlett & Packard built the HP 200A audio oscillator in their Palo Alto garage
- HP 200B custom made for Disney's Fantasia
- Stanford Prof, Fred Terman's mentorship ("Father of Silicon Valley")
- Pivot projects: automatic toilet flusher, an electric weight-loss machine and this device, an electric foul-line indicator they developed for a local bowling alley

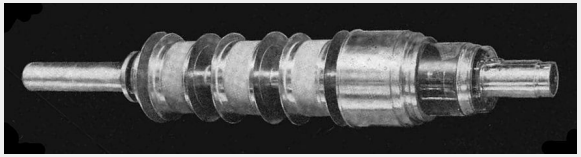


Bowling Alley Foul Indicator

HP 200B audio oscillator

WWII comes to Silicon Valley

- Radar, electronics, communications
- Varian Associates (1948)
  - Made parts for radiation therapy, fuse for atomic bomb, etc.
  - Military contracts
- Stanford Research Park (1951):
  - partnership between Palo Alto and Stanford
  - Kodak, GE, Lockheed offices



Klystron (specialized vacuum tube)

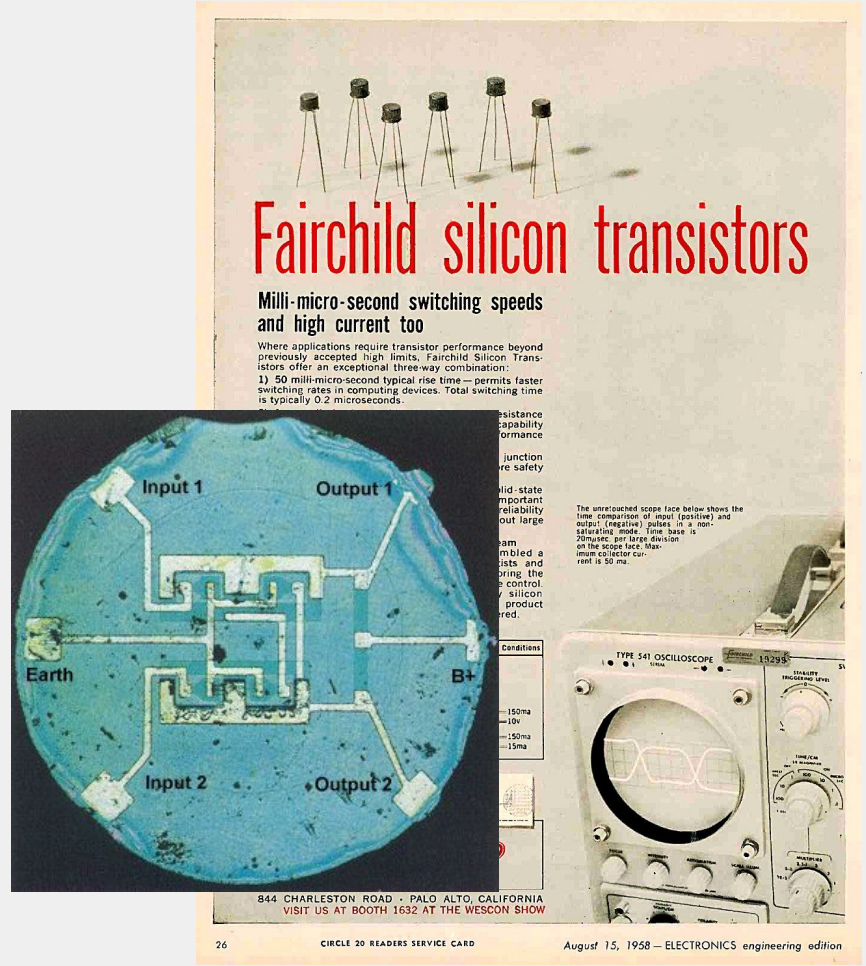
Fred Terman “Father of Silicon Valley”

SRI → SRI International

- Established in the 1940s by Stanford for DARPA/gov subsidized projects.
- Broke off 1977 (due to Vietnam War protests)
- The first project was investigating whether guayule plant could be converted into sources of rubber. → rubber shortage/need for WWII
- 1960s: Shakey the Robot (robotics)
- Speech, networks, ARPAnet

Semiconductors

- In 1947 at Bell Labs, John Bardeen, Walter Brattain, and William Shockley invent transistors – can replace bulky vacuum tubes!
- Shockley Semiconductor 1956
- Fairchild Semiconductor 1957 by the “Traitorous Eight”
- IBM bought 2N697 for $150 apiece in order to build the computer for the B-70 bomber.
- Fairchild made the first silicon integrated circuit
- Fairchild spawns 60+ of spinoff semiconductor companies —e.g. Intel (1968, founded by Noyce and Moore)

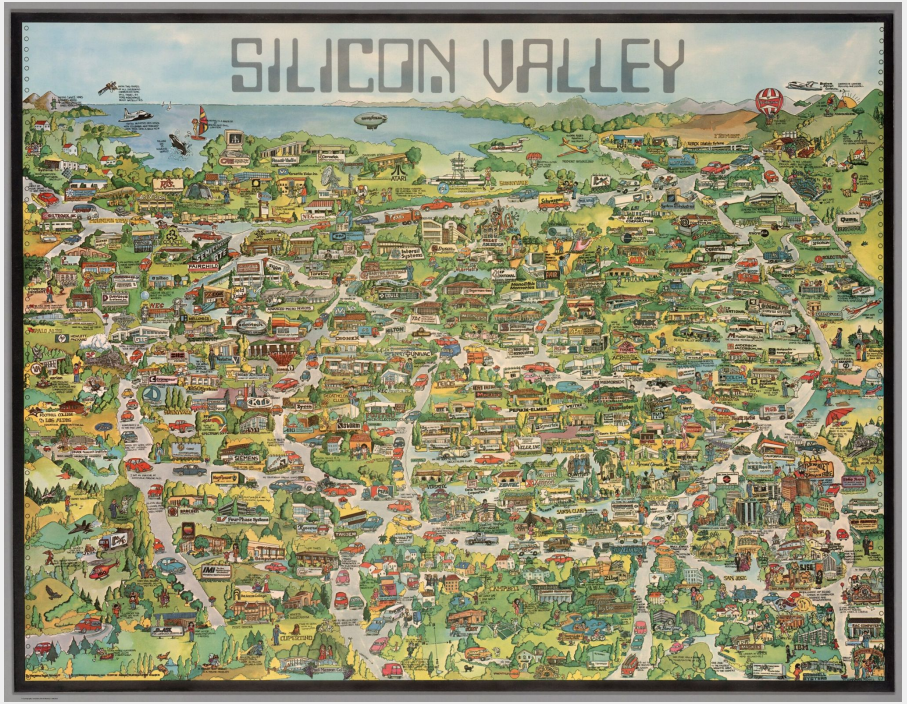


Silicon Valley VCs 1980s

- More money → more risk taking
- Heavy investment in computing, networking, internet
- Expert Systems in 1980s
- Capital Gains Tax cuts 1978, 49.5% to 28%
- Apple's billion dollar IPO in 1980 inspires VCs
- Previously VCs funded mostly semiconductor microchips → expand into biotech, software, lifestyle, AI
- ERISA "prudent man" rule pivot 1979 Pivot: 5–10% of pension funds into "alternative investments" like venture capital
- Sudden cashflow into VC: $4-5 billion in 1983

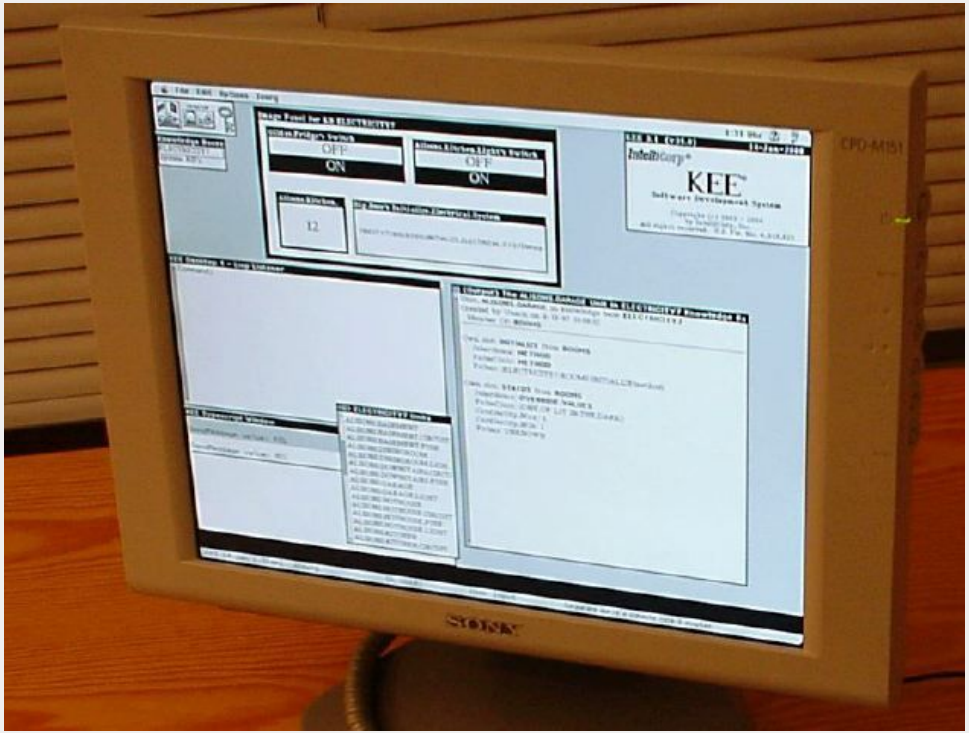
Expert Systems Start-ups 1980s

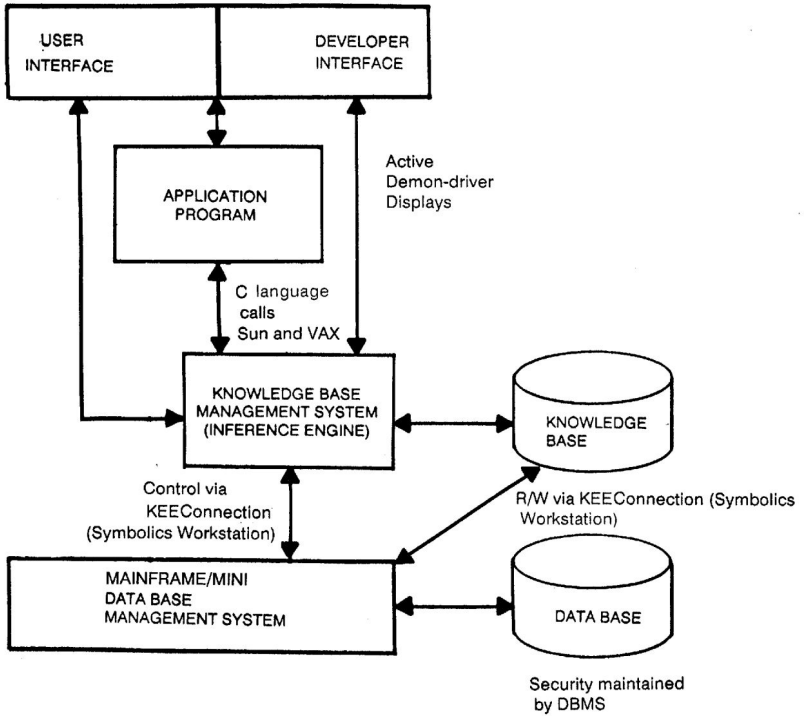
- Expert Systems was the first AI technology to arouse commercial interest
- Teknowledge, Inc., Intellicorp sprang up in the early 1980s to market these systems. Both these companies received millions from DARPA to bring AI into the market and military fields.



Knowledge Engineering Environment (KEE) 1983

- Intellicorp created and started selling Knowledge Engineering Environment (KEE) 1983: a high-end knowledge expert system development shell.
- KEE was to support knowledge acquisition and representation, reasoning, and user interface construction.
- Framework for multiple expert domains





Product Information

Software

- **KEE Version 3.0 for workstations**
- **KEE Version 3.0 for 80386-based IBM-compatible PCs**
- **KEEConnection**
- **SimKit**
- **Run-Time KEE**

KEE Versions 1.0 & 2.0 and Intelliscope are no longer sold.

Pricing

KEE Version 3.0 (workstations)	**$46,500** academic price **$9,250**
KEE Version 3.0 (386 PCs)	**$9,900**
KEEConnection (varies with hardware)	**$15,500** academic price **$5,000**
SimKit	**$21,000** academic price **$4,125**
Run-Time KEE (single-user workstation)	**$5,000**

Pricing varies, depending on number of licenses purchased, and includes 3-9 days training for 2 persons and one year's support. Support services include standard consultants, senior scientist consultants and/or apprenticeship training (4 weeks).

Hardware

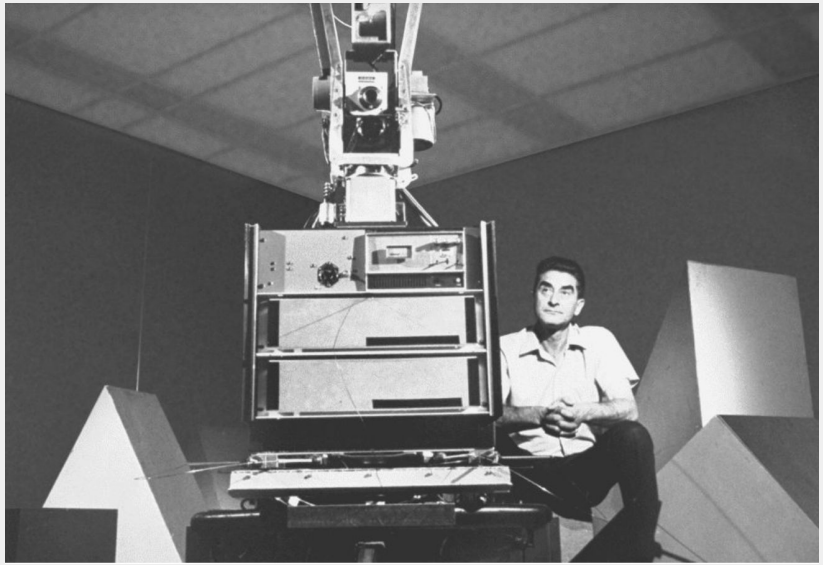
all 80386-based IBM PC-compatible computers
IBM PC RT
Symbolics 36XX series
Xerox 1100 series
Sun 3 workstations
Apollo Domain workstations
TI Explorer I and II
DEC VAX series
MicroVAX II workstations

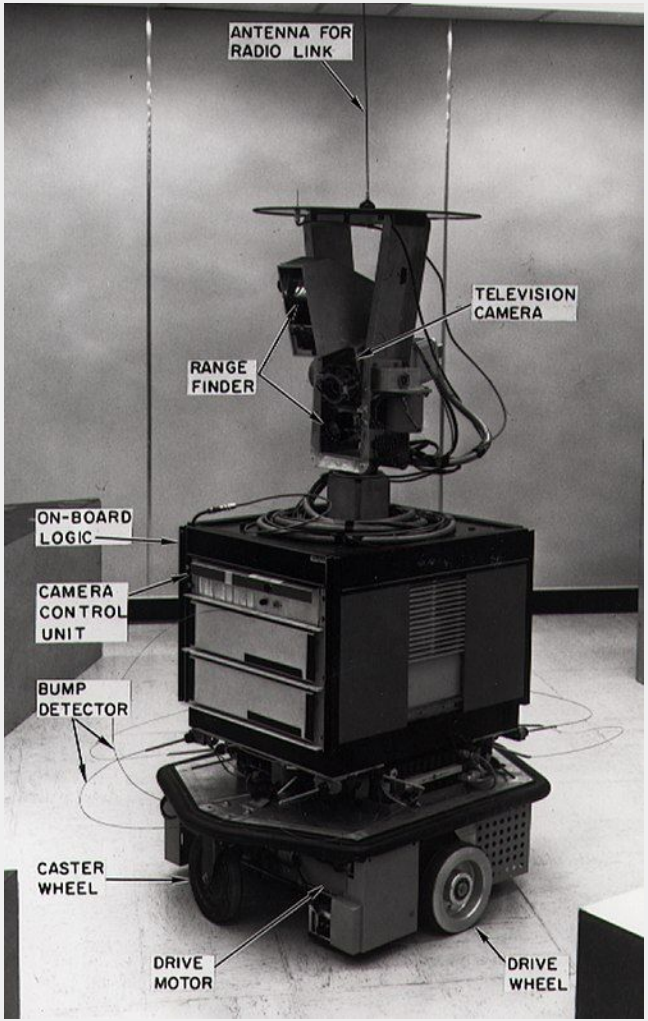
Silicon Valley VCs

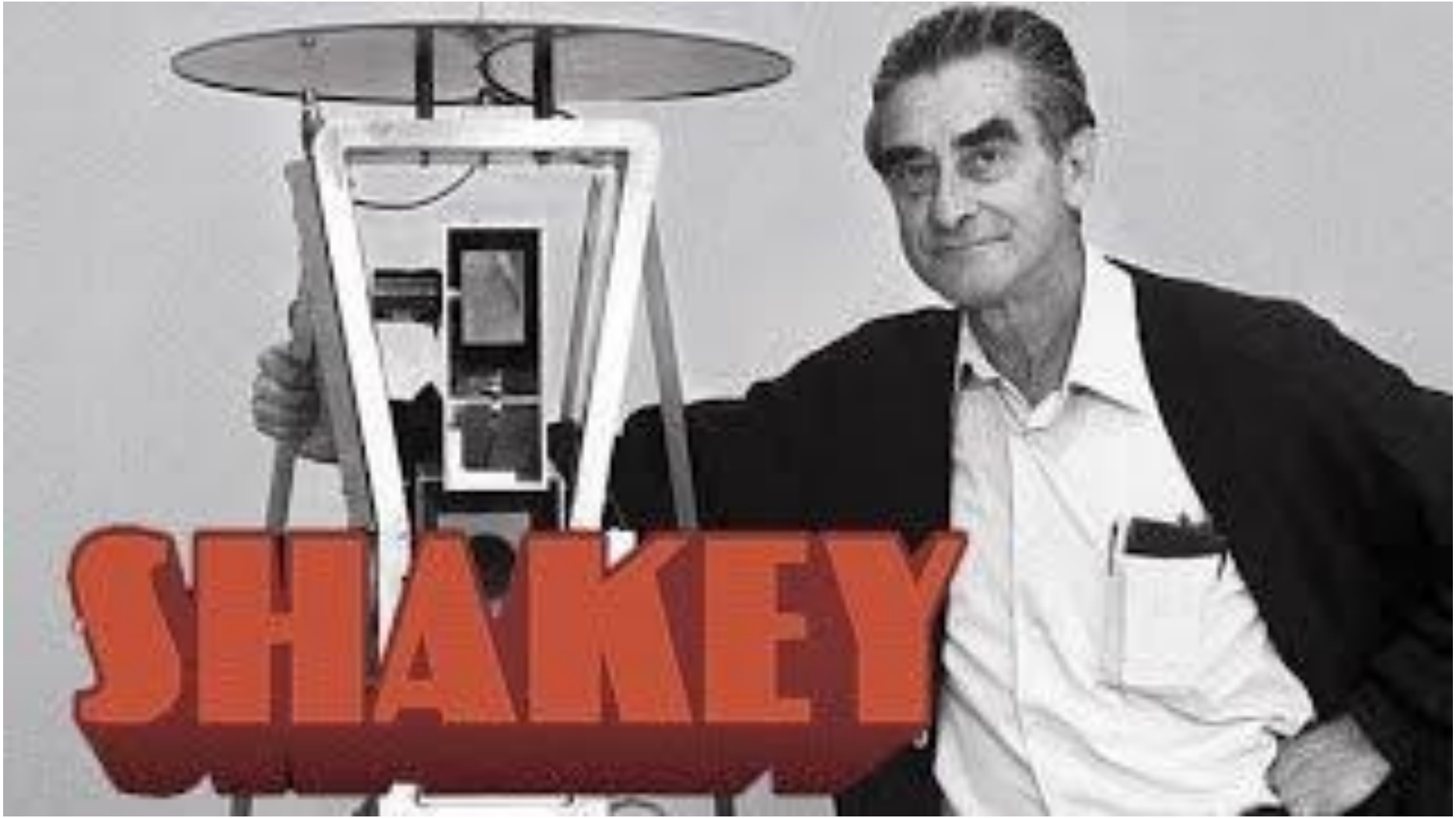
- Eugene Kleiner (Fairchild Semiconductor co-founder) and Tom Perkins found Kleiner Perkins 1972 - known for Genentech, Google, etc.
- Don Valentine, a semiconductor vet founds Sequoia Capital 1972 - Apple early investor, Cisco
- Bill Draper founds Draper Associates 1980s - Hotmail investor
- Dick Kramlich founds NEA 1977
- Accel Partners 1983

SRI Shakey

- AI center founded in 1966 - 1972
- First project: Shakey the robot, the “first electronic person”; STRIPS (the software program) written in LISP






SHAKEY

SRI Centibots 2003

- Centibots
- Research in swarm robotics and multi-agent systems
- DARPA sponsored $2.2m projects

Centibots 2003

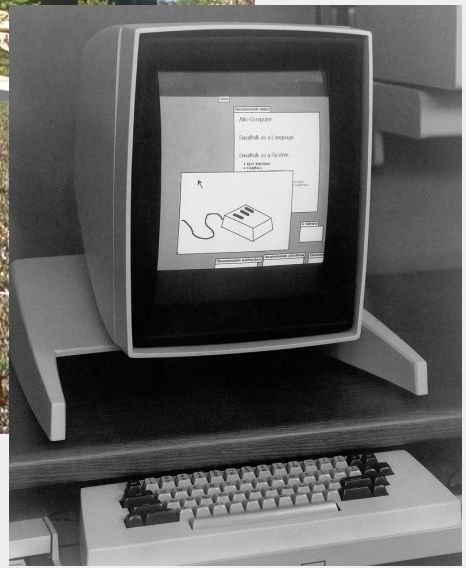
SRI's SIRI (2000-2008)

- DARPA funds a project called CALO (Cognitive Assistant that Learns and Organizes) —> an assistant that can send emails helping users organize their tasks, etc.
- spin-off of (Personal Assistant that Learns) framework from DARPA, a defense assistant tool
- February 2010, Siri was released as an app for iOS – two months later Apple acquired it and integrated Siri software into its hardware devices.

SIRI 2010

Xerox PARC 1970

- R&D division of Xerox
- Laser printing, Ethernet, GUI, computer mouse



Dot Com Boom

- Search 90s: Netscape (1994), Yahoo (1995), and Google (1998) turned the Valley into a digital goldmine.
- Lifestyle 2000s: Facebook (2004), YouTube (2005), and the iPhone (2007)
- AI in the 2010s???

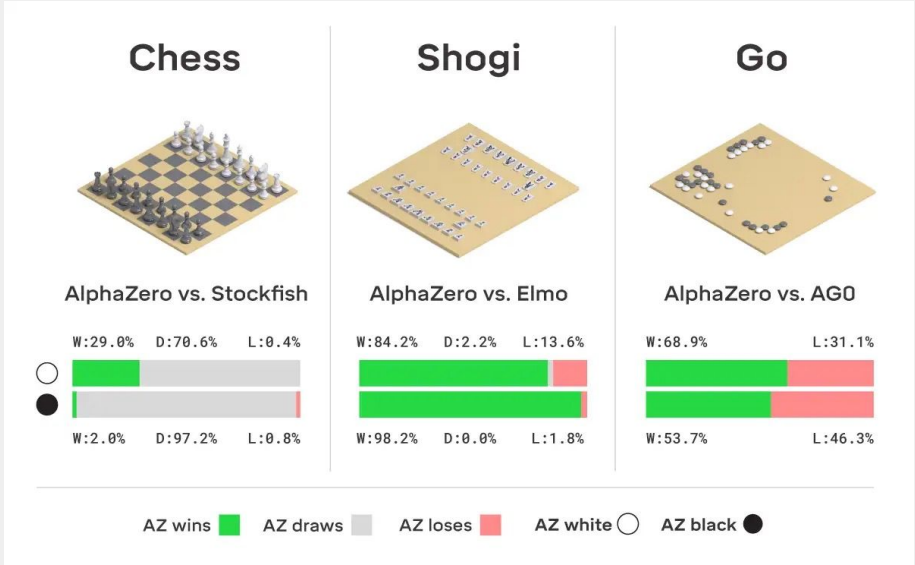
Deepmind 2010

- How to make AGI for the benefit of humanity
- “Could we create a machine that could work backwards to make sense of the universe? AGI would give you an insight into where we came from, and what is God.”
- Demis Hassabis, Shane Legg and Mustafa Suleyman cofounders
- Early investment from Peter Thiel, Elon Musk, Jaan Thallin
- Met investors at Singularity Summit
- Earliest projects on reinforcement learning with game screen pixels as input
- Ethics board

AlphaGo 2014, AlphaZero 2017

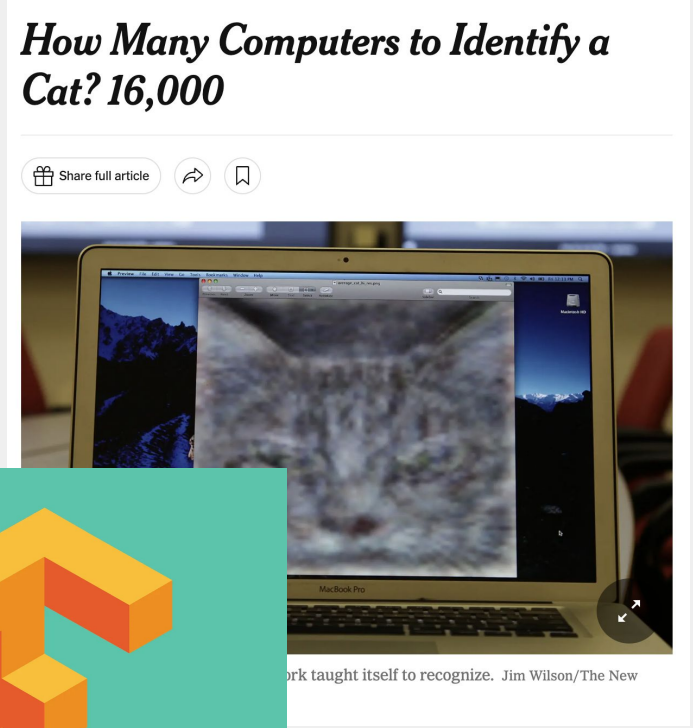
AlphaZero (2017) plays Chess, Shogi & Go
Self train

AlphaGo(2014) beat Lee Sedol 4:1 in a series of matches in 2016
30 million + moves learned



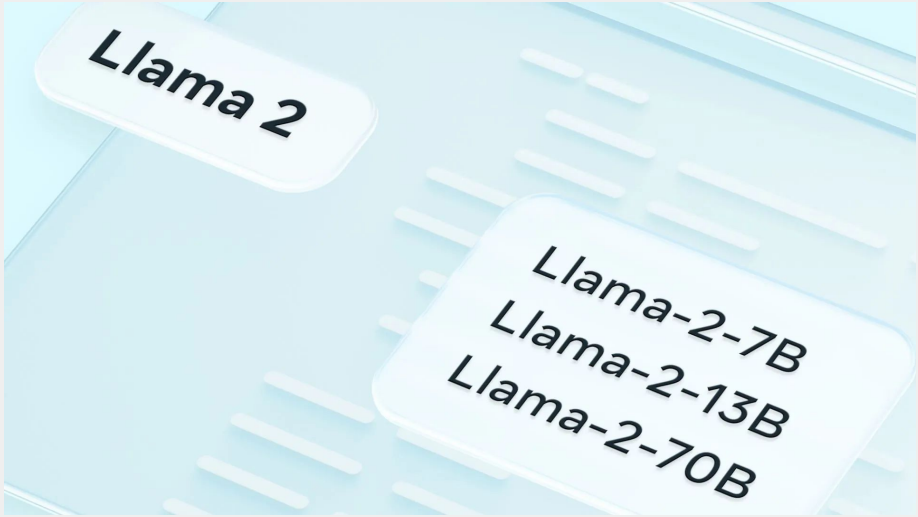
Google Brain 2011

- Unsupervised “cat experiment” 2012: Stanford professor Andrew Ng. One of the group’s first projects was training a cluster of 16,000 processors in 1,0000 machines on neural nets to recognize a cat based in 10 million images from Youtube videos.
- Focus on Google Ads
- Customer AI Chips (TPUs 2016)
- 2015 Tensorflow 1.0



Facebook FAIR 2013

- Yann LeCunn heads FAIR and pushes open source
- PyTorch 2016
- RoBERTA 2019
- Llama series



PyTorch

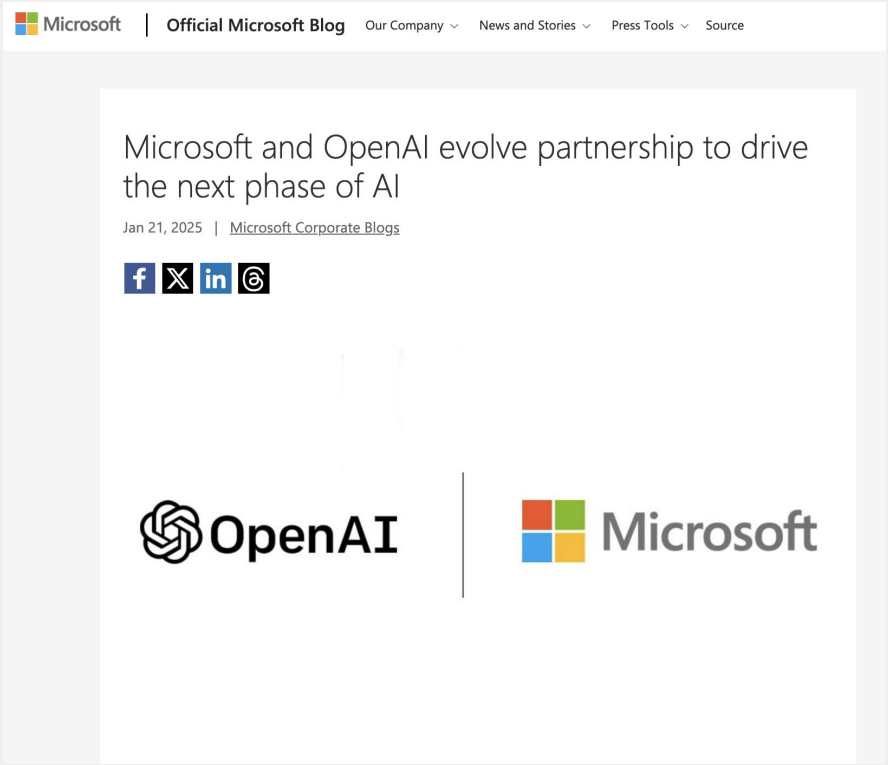
OpenAI 2015

- Sam Altman et al.
- Some handful of Deepmind researchers
- Non-profit dedicated to AI safety
- Elon Musk wants to fund Deepmind rival, “Hitler of AI”
- Altman email to Musk on May 25, 2015: “someone other than Google” should build AGI first.

“Proxy Wars”

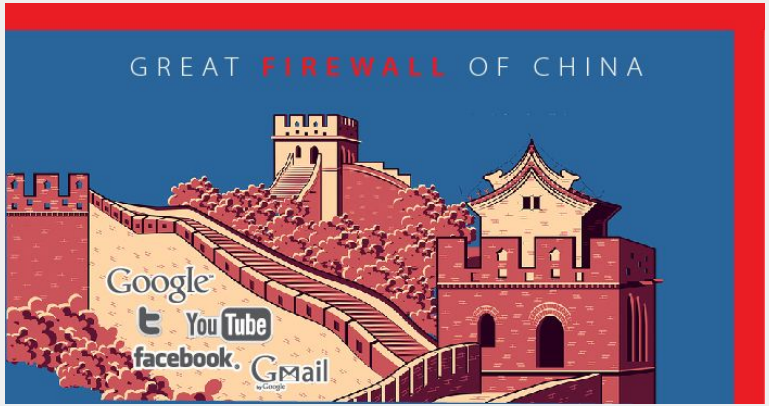
- Google vs. Microsoft masked as Deepmind vs. OpenAI?
- Google acquires Deepmind 2014 $650m
- Microsoft $1b investment 2019 in OpenAI (since then much more) - inspired by XiaoIce debacle
- OpenAI becomes “Capped profit”-able
- Deepmind aims for “prestige”, OpenAI goes more “product”

XiaoIce



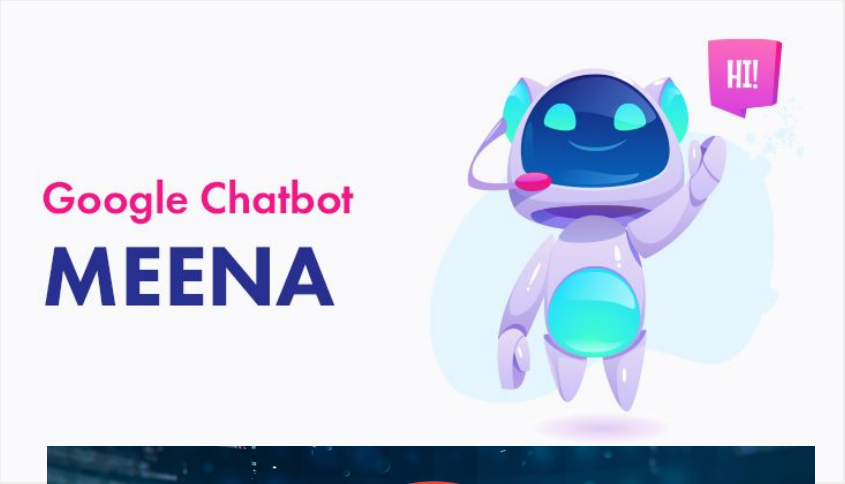
Google locked out of China 2010

- Google focused on re-entering China
- In 2010 it had exited the country after accusing Beijing of hacking its intellectual property and the Gmail accounts of Chinese human rights activists.
- The Chinese government demanded that Google censor searches about Tiananmen Square and other controversial topics for the Chinese Communist Party
- China blocks access to Facebook and Twitter, setting up what became known as the Great Firewall
- Project Dragonfly (Google search for China)
- Deepmind Alpha Go Master beats Chinese play Ke Jie 3 matches but not aired in China



Google’s Transformers

- Google Brain’s Vaswani et al released transformer model based on attention mechanism only
- But Google too cautious any new experiments will hurt their $100 billion ad product
- Not really applied anywhere except BERT embeddings and machine translation



Anthropic 2021

- Delaware Public Benefit Corporation
- Dario Amodei & Daniela Amodei from AI Safety team OpenAI
- 7 former OpenAI employees
- Funds from Google, Amazon
- FTX founders support out of effective altruism

ChatGPT 2022

- Just the decoder part of Transformers?
- GPT, GPT2, GPT3, DALL-E 2
  - Gradually less transparent!
- After two months online ChatGPT had attracted thirty million registered users, (one of the fastest-growing online services in history)
- By early 2024, around one hundred million people.

Aug 29, 2024 - Technology

OpenAI says ChatGPT usage has doubled since last year

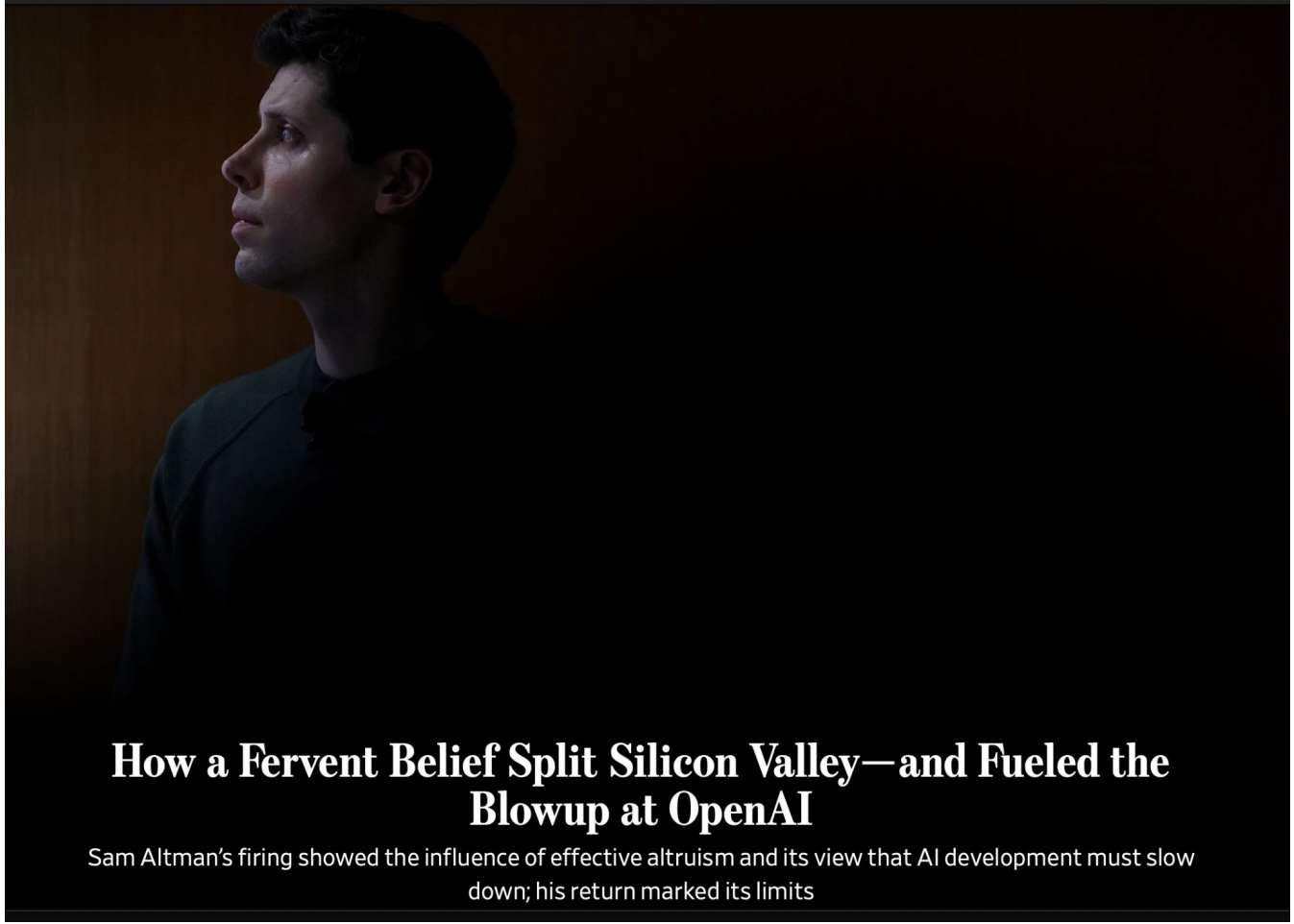
Ina Fried

Illustration: Natalie Peeples/Axios

OpenAI said on Thursday that ChatGPT now has more than 200 million weekly active users — twice as many as it had last November.

THE WALL STREET JOURNAL.

SIGN IN



How a Fervent Belief Split Silicon Valley—and Fueled the Blowup at OpenAI

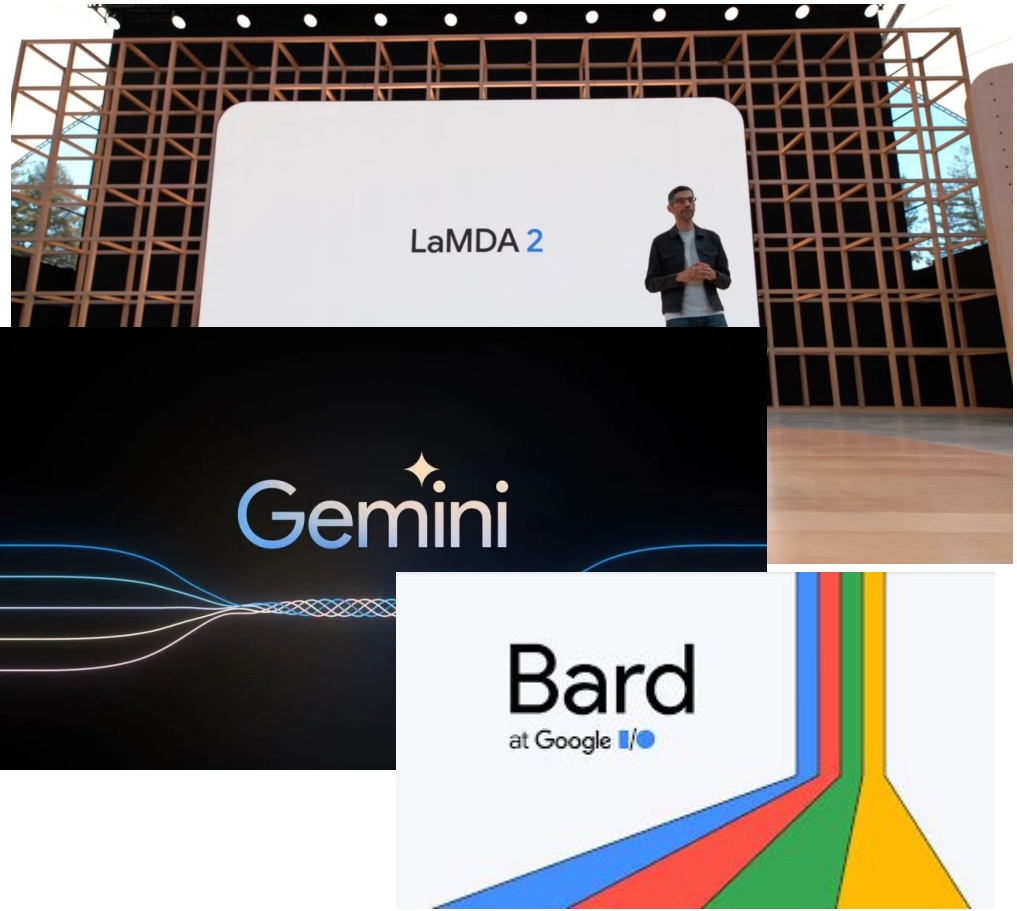
Sam Altman's firing showed the influence of effective altruism and its view that AI development must slow down; his return marked its limits

Google AI
2017

DeepMind
2014

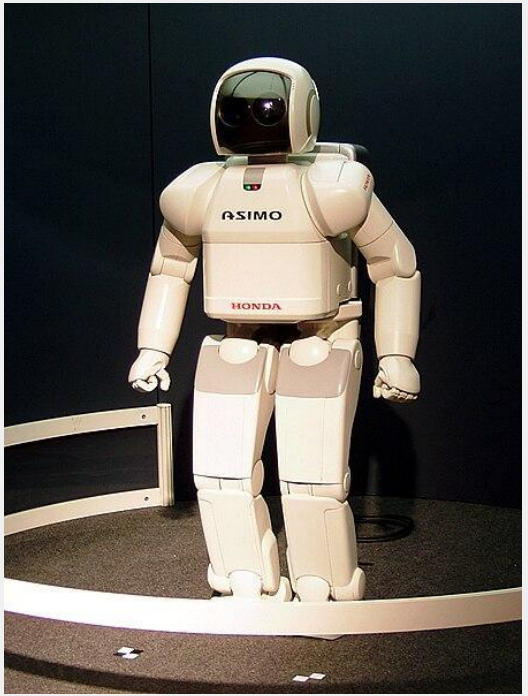
Google
Brain
2011

Google
DeepMind
2023



History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

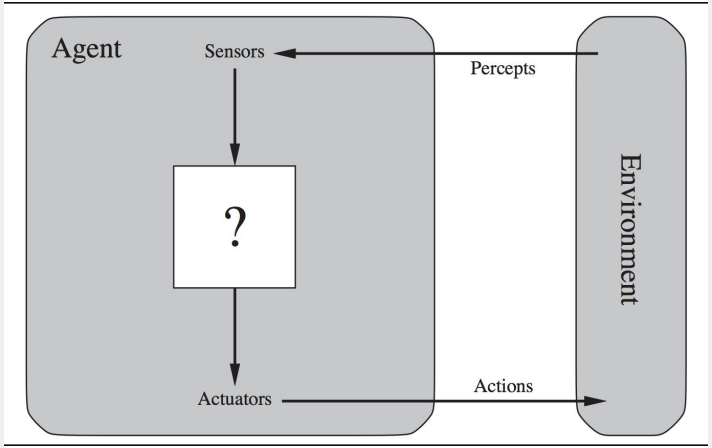
Lecture 7 March 7, 2025

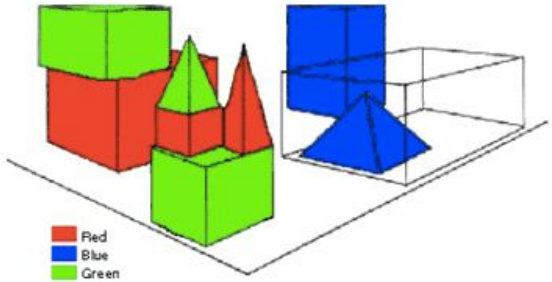
This week!

- Are games and AI a match made in heaven?
- Similarities & differences in AI for health and AI for games
- What are different factors at play in how different domains adopt AI?

Games, a natural AI habitat? Microworlds 1972

- A limited world with defined rules, perfect place to build microworlds
- Natural confined environment to test AI



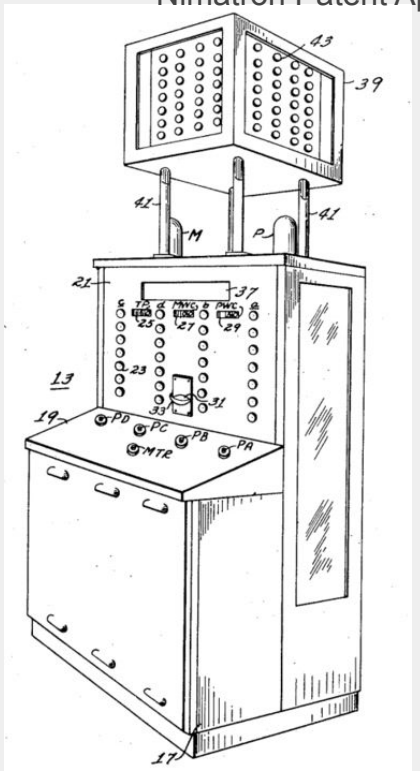


Person: Pick up a big red block.
Computer: OK.
Person: Grasp the pyramid.
Computer: I don't understand which pyramid you mean.

Nim on Nimatron 1940

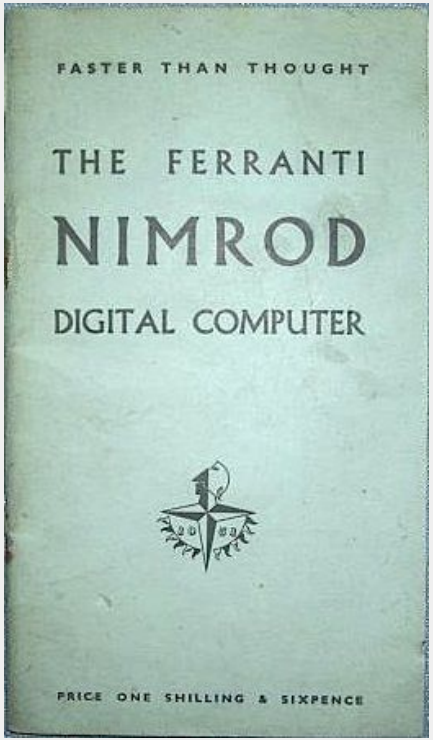
Nimatron, New York World Fair, 1940
(Westinghouse Corp.)

Nimatron Patent Application

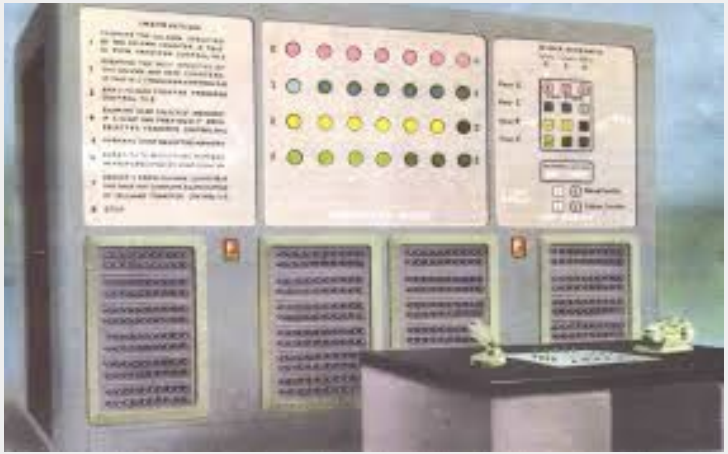


- For the NY World Fair
- played 100,000 times
- If won, NIM CHAMP coin!
- World's first arcade game?
- Nimatron won 90,000 times

Nim on Nimrod 1951



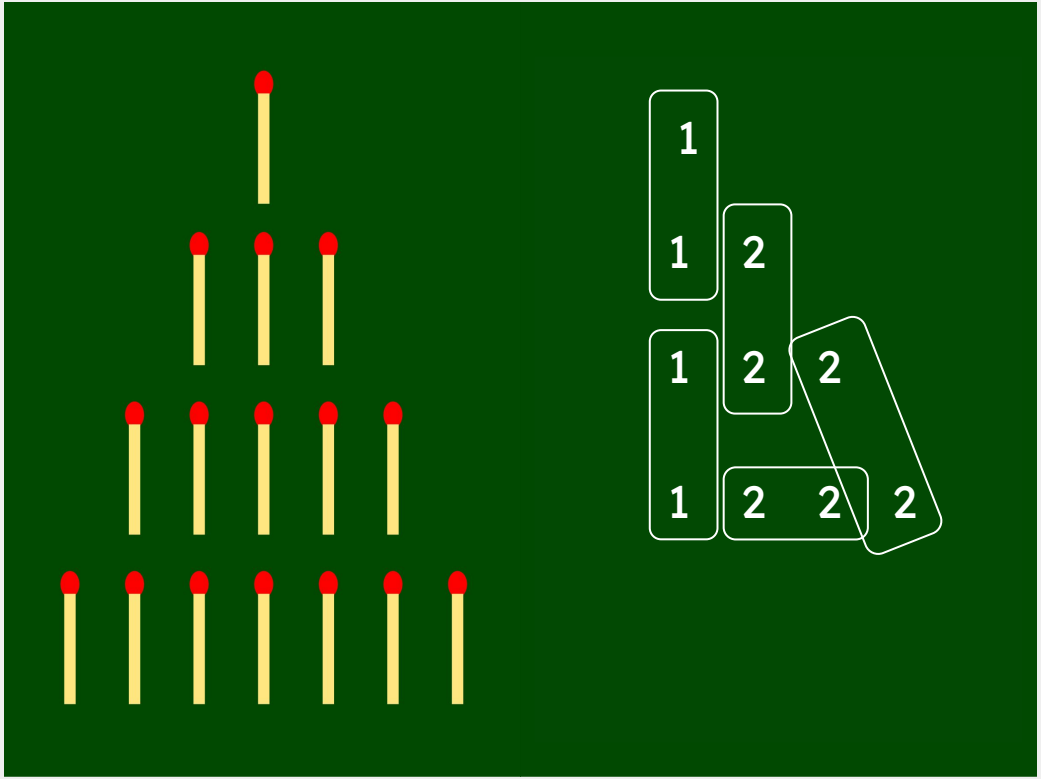
- Alan Turing managed to beat the computer!
- “COMPUTER WINS”



Nimrod, The Festival of Britain, 1951 (Ferranti)

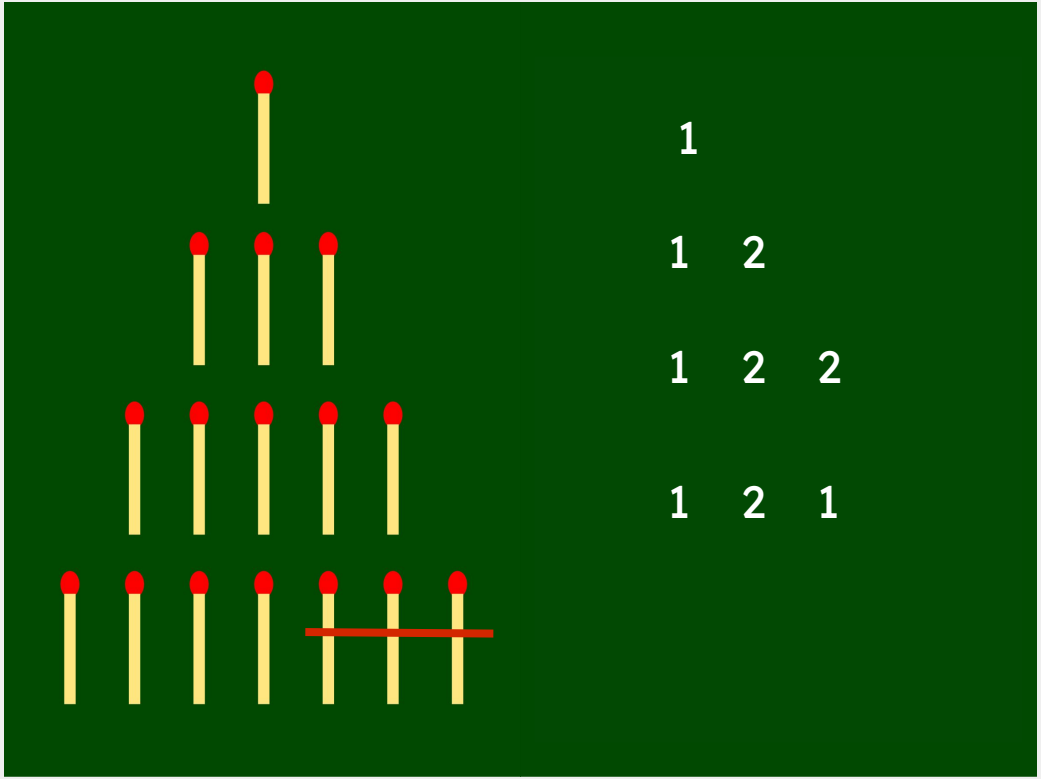
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



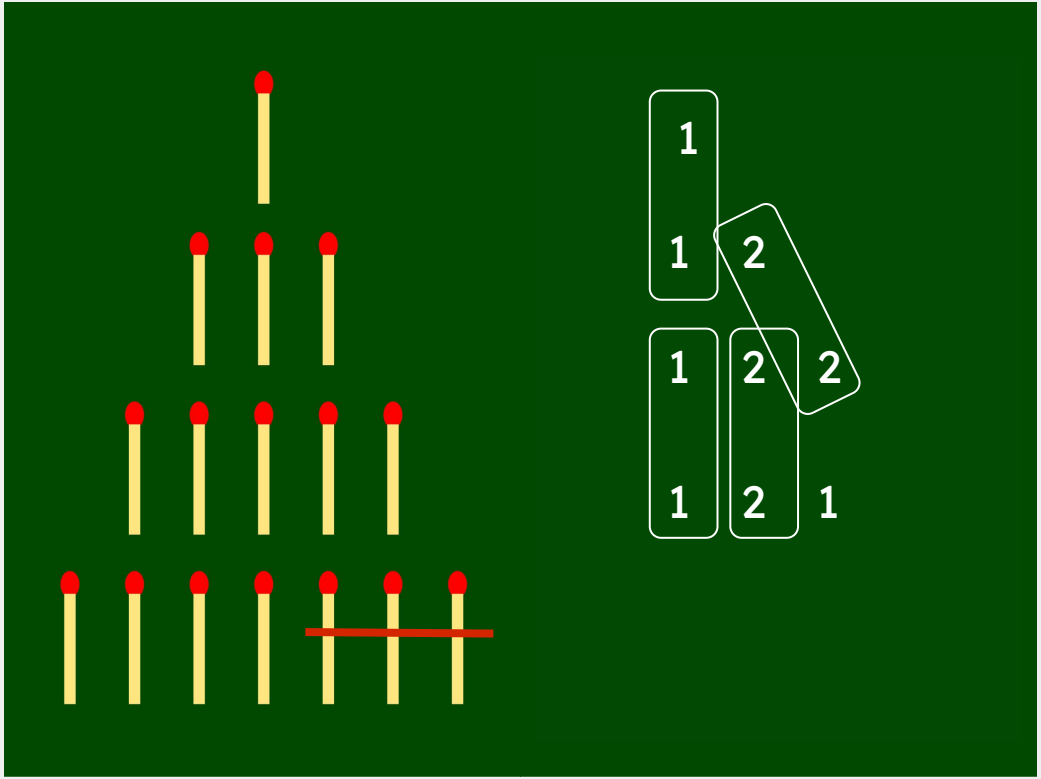
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



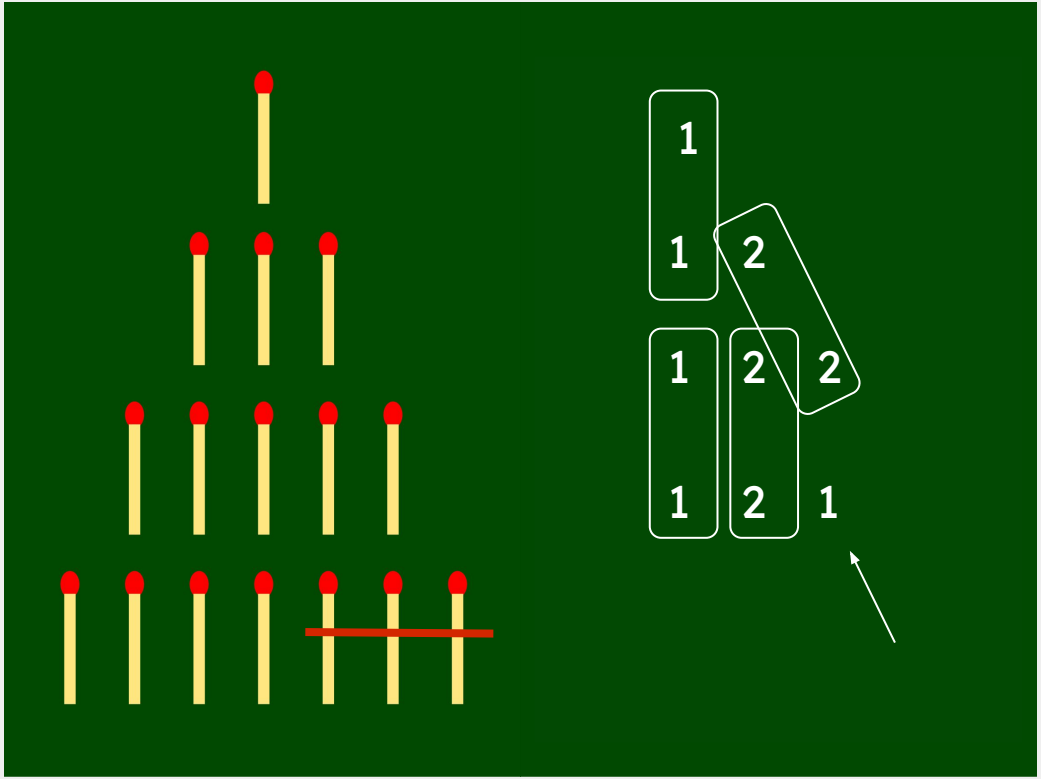
Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent

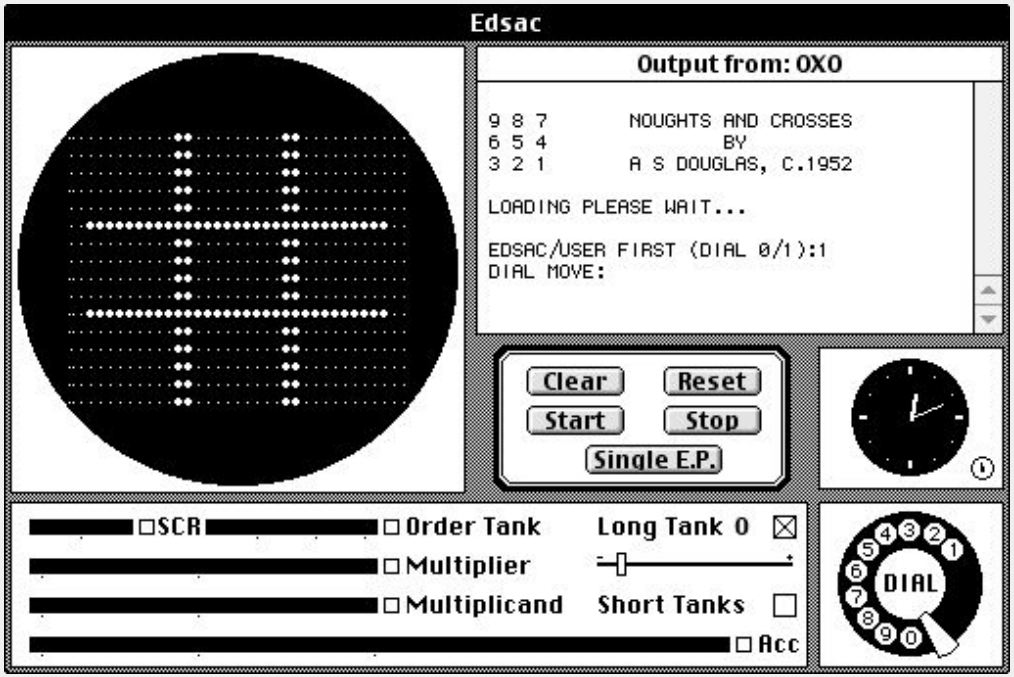


Nim: A solved game!

- Nim sum helps you win every time!
- Want your opponent to be at nim sum of 0
- “AI” implemented this algorithm as opponent



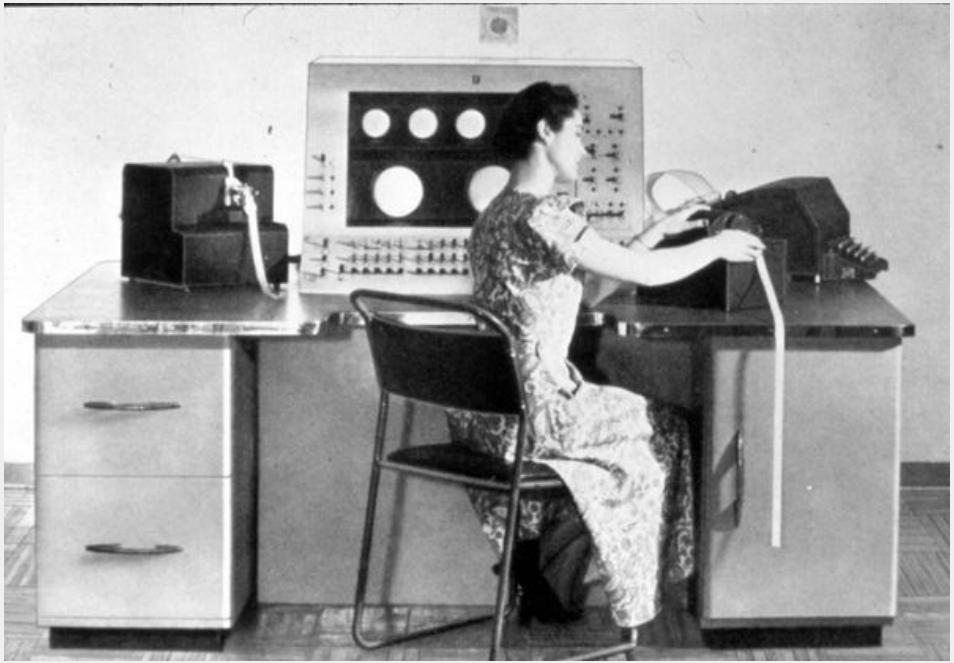
OXO: AKA Tic-Tac-Toe 1951



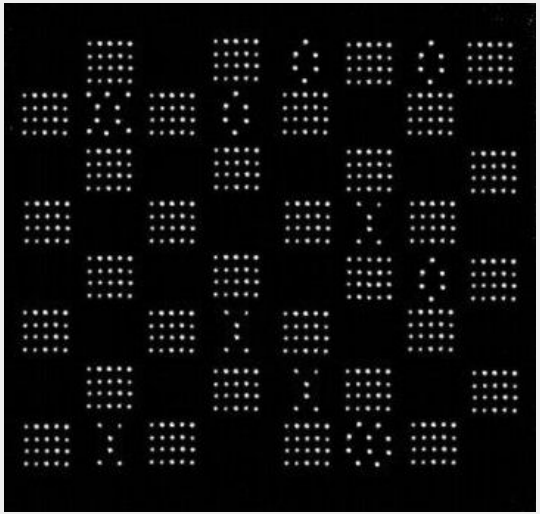
- AI opponent for Tic-Tac-Toe → minimal tie when played optimally
- On the EDSAC, early british computer at Cambridge
- One of first computer games on a screen (not light bulbs)

Checkers on Ferranti Mark I 1952

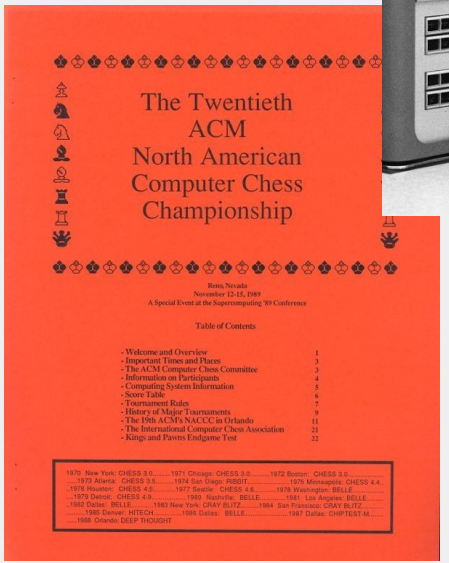
- On the Ferranti Mark I, world's first commercially available digital computer (4,050 vaccuum tubes)
- "Draughts" first ever played July 1952



Ferranti Mark I 1951



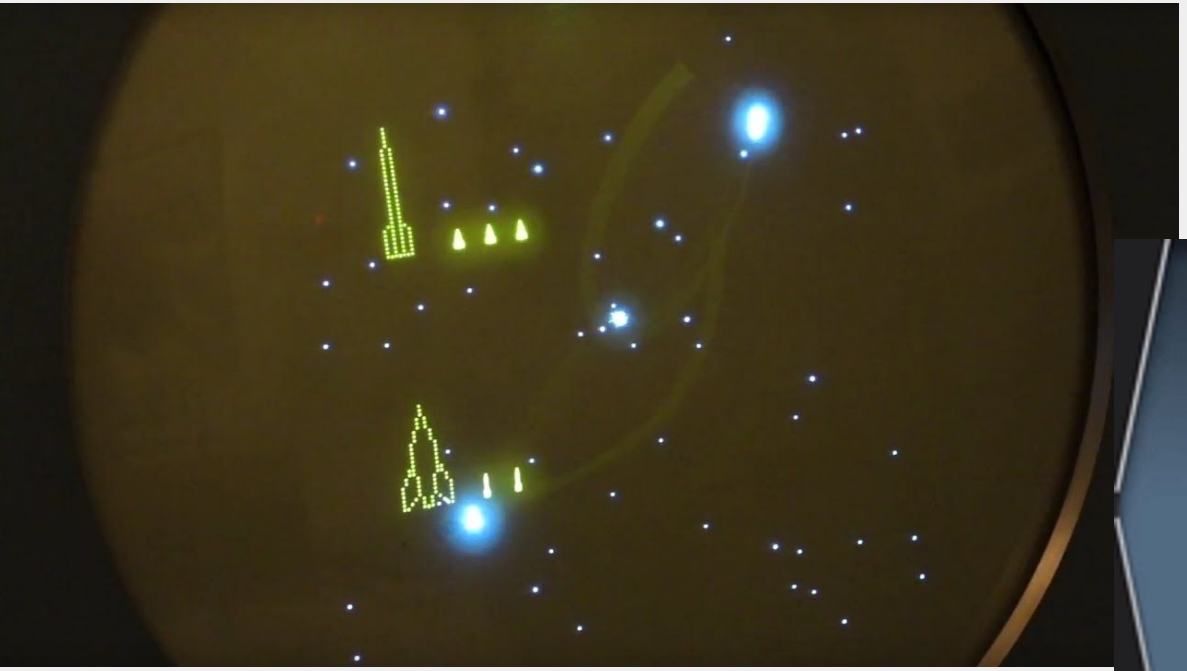
Chess programming begins 1957



- 1957: First computer chess game IBM 704; evaluate 42,000 instructions per second
- 1962: IBM 7040; 1,100 positions per second
- 1980s: Feng-hsiung Hsu writes ChipTest that goes on to win the North American Computer Chess Championship in 1987→ later helps develop DeepBlue

Spacewars! 1962

https://www.youtube.com/watch?v=AbAODTne64w



- MIT students made Spacewars! on PDP-1 (first mini computer)
- Widely shared and inspired Atari founders
- Rudimentary AI as second player

Pong by Atari 1972

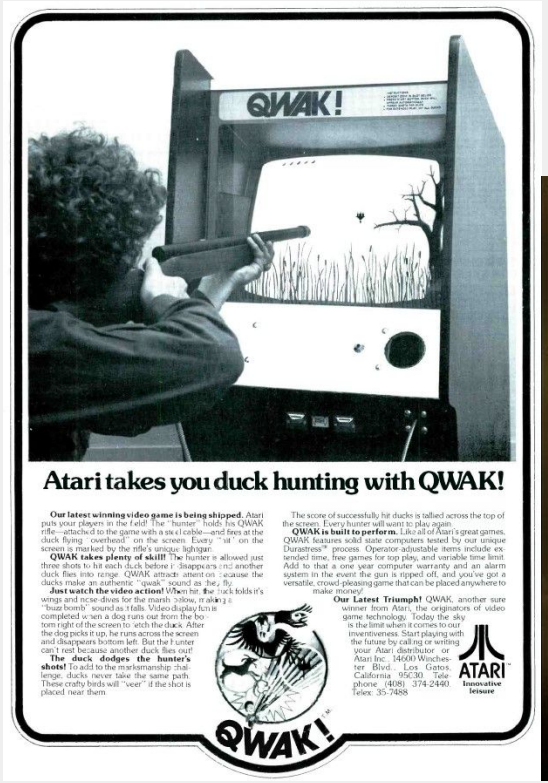
https://www.youtube.com/watch?v=fhd7FfGCdCo

- The “opponent” AI paddle programmed to follow the ball

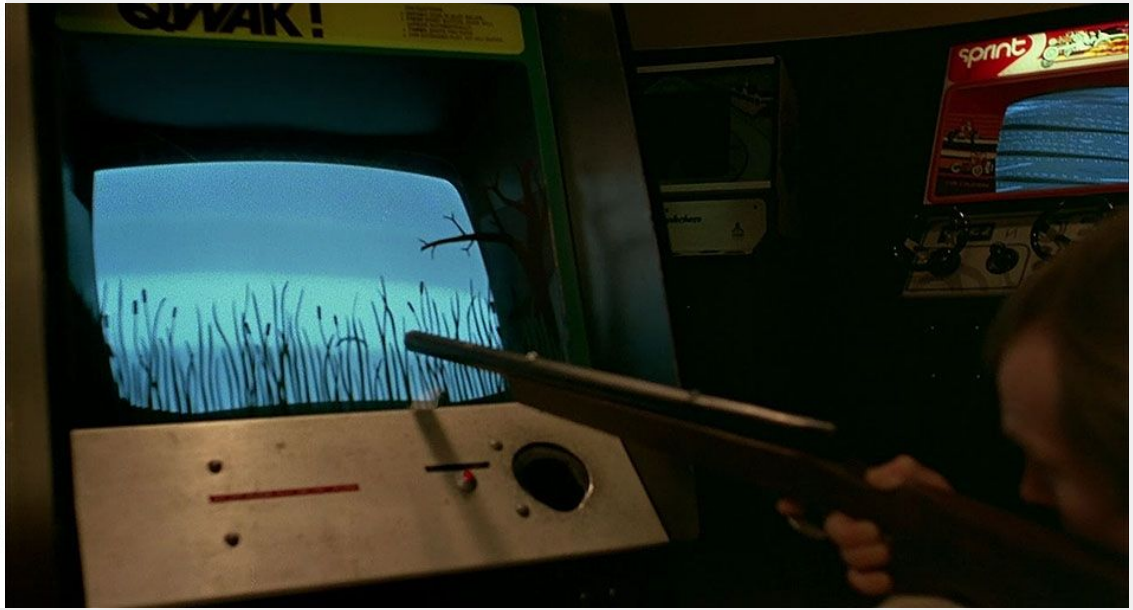
Deep Blue 1997

- IBM's Deep Blue beats Garry Kasparov, previously undefeated chessmaster, in chess
- 6 matches (Deep Blue won first game) → 4:2 Kasparov wins
- Evaluate 200 million chess positions per second

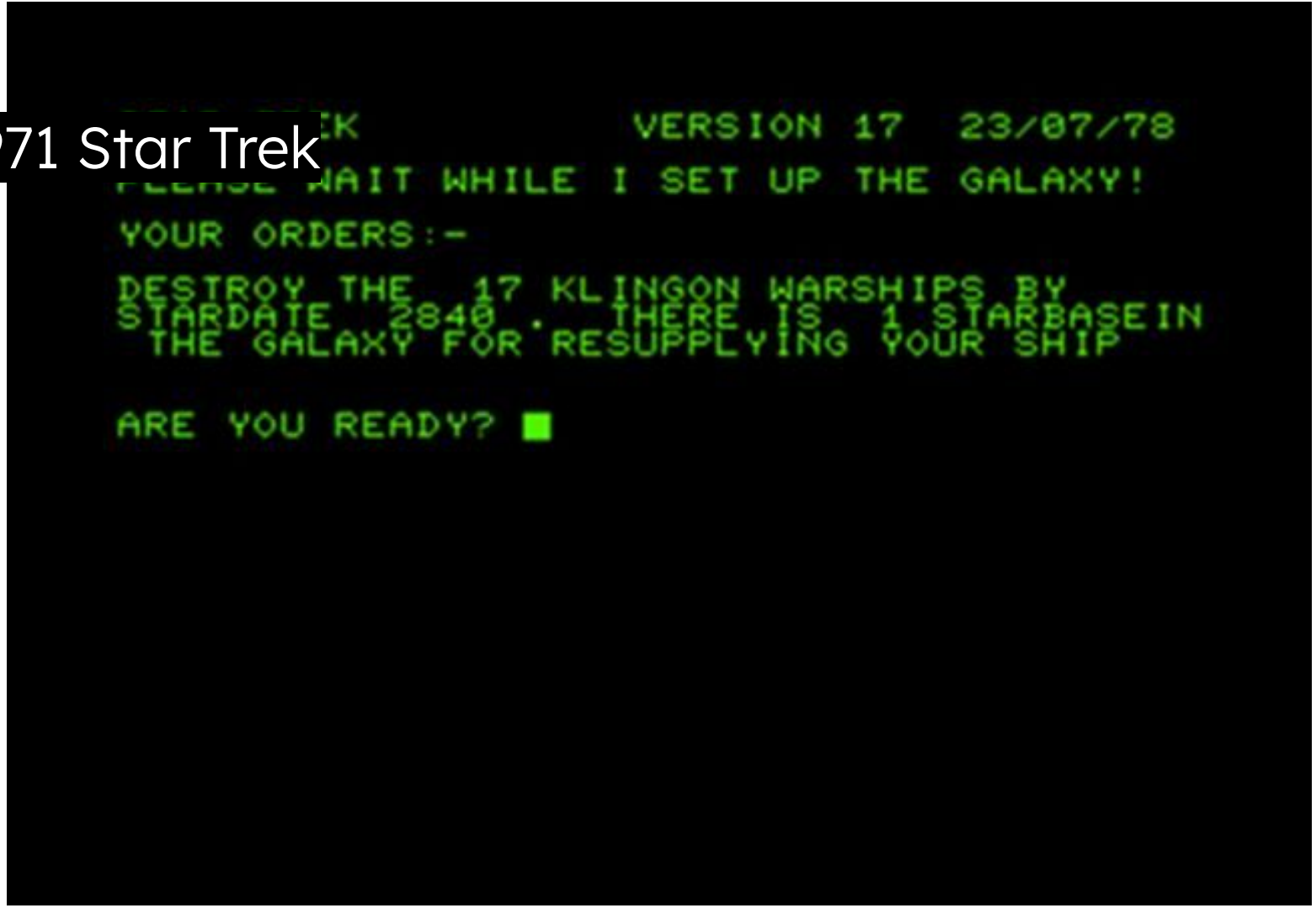
AI as NPCs 1970s



- Arcade Golden Era 1970s
- Atari's QWAK 1973 first AI NPC



1971 Star Trek



SPACE INVADERS 1978

https://www.youtube.com/watch?v=viDiuf9Mv7o

- Wildly popular game
- Lots of diversity in computer opponent behavior
- Increasing difficulty levels, distinct movement patterns, and in-game events (that hashed player input).

1980

- First brought the maze to gaming AI

Real-time Strategy (RTS) Games 1990s

Herzog Zwei 1989

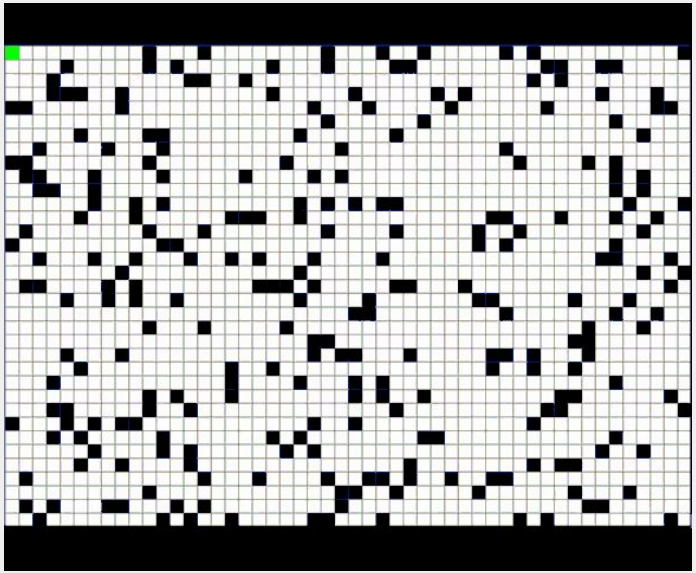
https://www.youtube.com/watch?v=BYfYrhT82RM

- Real-time strategy games pose new challenge for game makers
- Mostly scripted, fixed opponent responses

Real-time Strategy Games 1990s

- Decisions trees for rule-based AI
- Pathfinding for NPCs in map navigation (Age of Empires, Civilization)

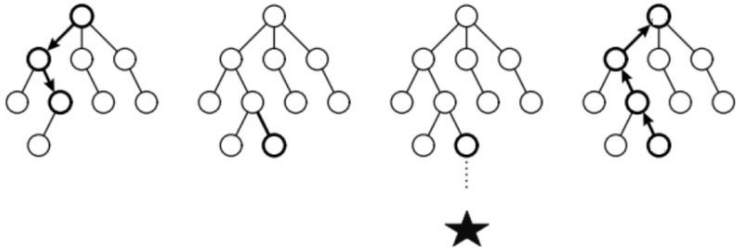
Age of Empires

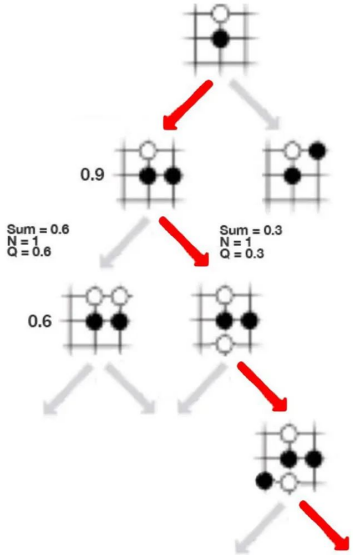


A* pathfinding

Deep Learning & Reinforcement Learning 2010s

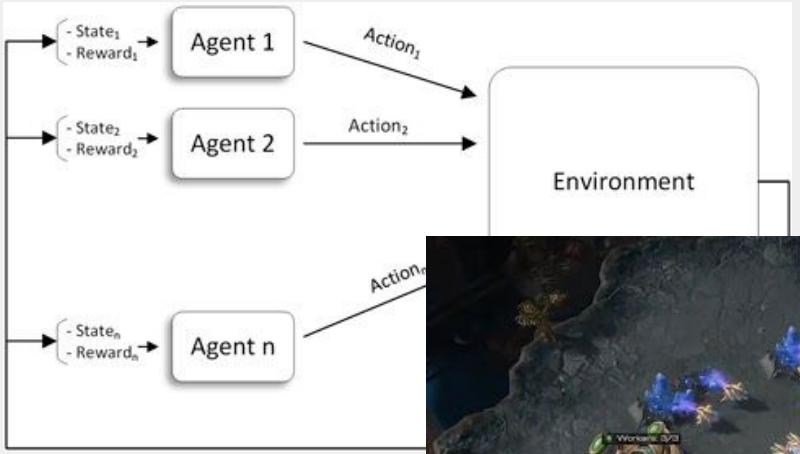
- In the mid 2010s, deep learning was used to successfully beat international masters in Go such as AlphaGo.
- Monte Carlo Tree Search methods for games like Chess & Shogi





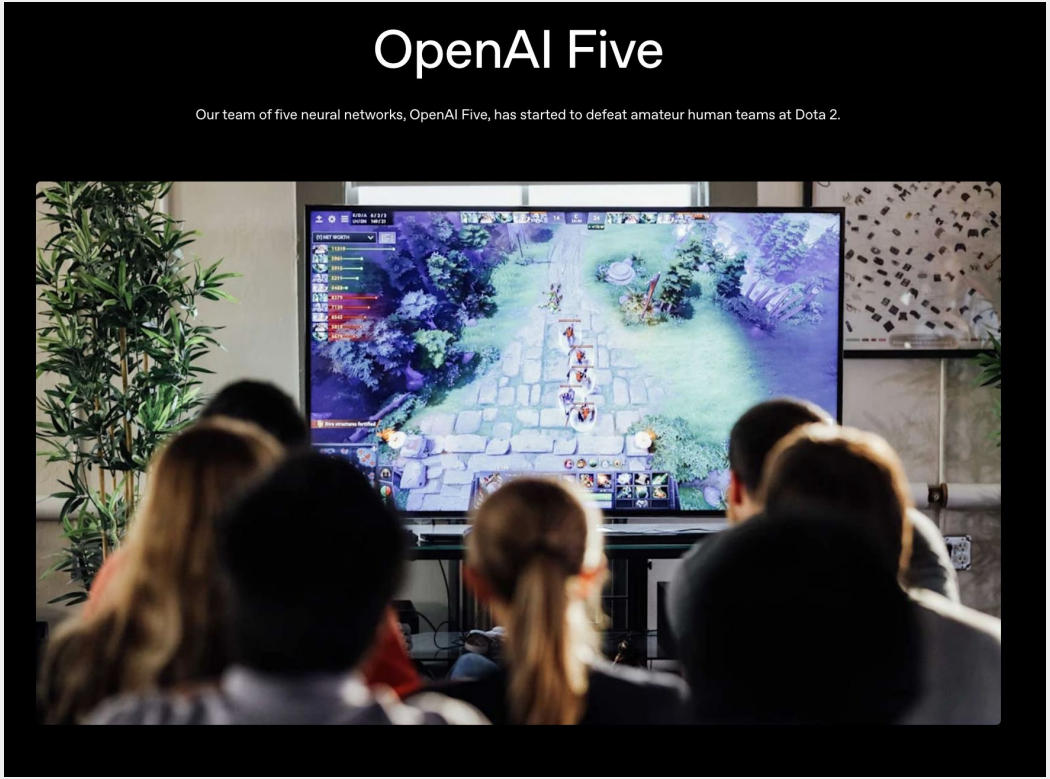
Deepmind's AlphaStar 2019

https://www.youtube.com/watch?v=6EQAsrfUIyo



- Late 2010s people began to experiment with multi-agent reinforcement learning methods to play complex RTS games like Starcraft II
- Dec 2018, AlphaStar beats world class Starcraft player Team Liquid's Grzegorz "MaNa" Komincz 5-0

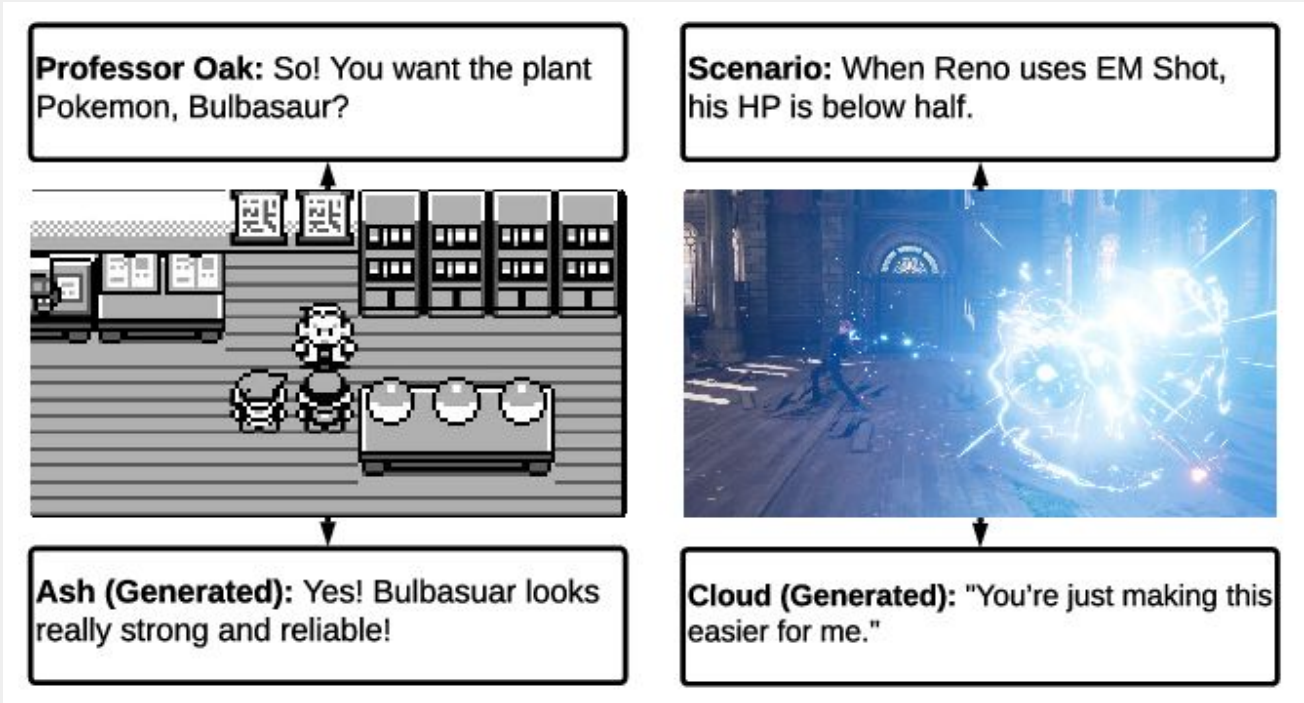
OpenAI's Five masters Dota 2 2019



- April 2019, OpenAI Five wins 2 out of 3 games against human e-sports team OG
- Played 45,000 years of Dota2 (self-play)

AI for Game Content Generation

Dialogue Generation for NPCs or story games



Explore Playground Beta Pricing Docs Blog Changelog Sign in Get started

lambdal / text-to-pokemon

Generate Pokémon from a text description

https://replicate.com/lambdal/text-to-pokemon/examples

Run with an API

Cold Public 7.9M runs T4 GitHub License

Playground API Examples README Versions

Prediction

lambdal/text-to-pokemon:ff6cc781

Model	ID	Status	Source	Hardware	Total duration	Created
lambdal/text-to-pokemon:3554d9e6	n2vfz327yvgnpnrcbecukyqvh4	Succeeded	Web	–	30.2s	over 2 years ago by @justinpinkney

Input

Form JSON Node.js Python HTTP Cog Docker

prompt

```
Yoda
```

num_outputs

```
1
```

guidance_scale

```
7.5
```

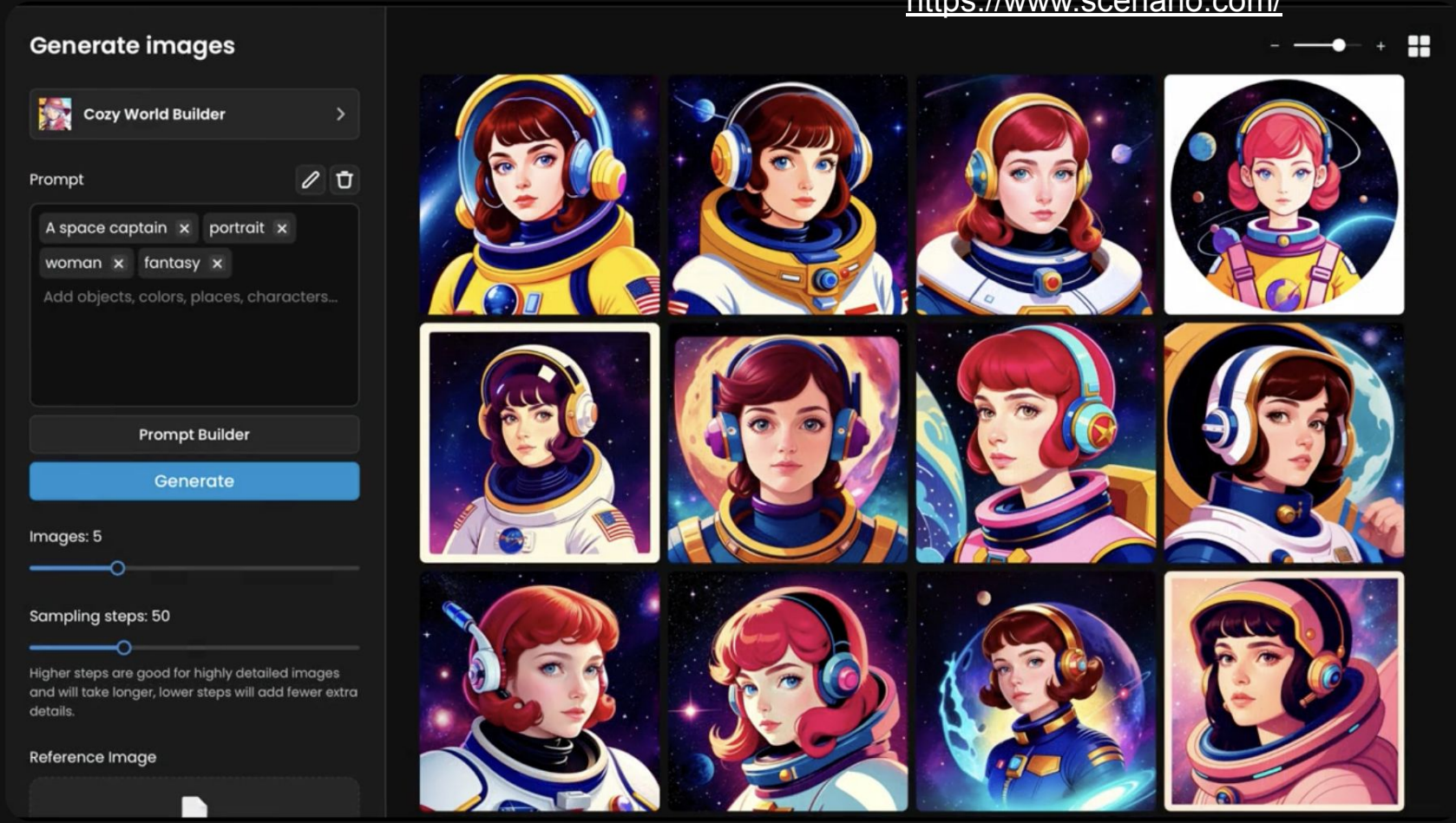
num_inference_steps

```
"50"
```

Output

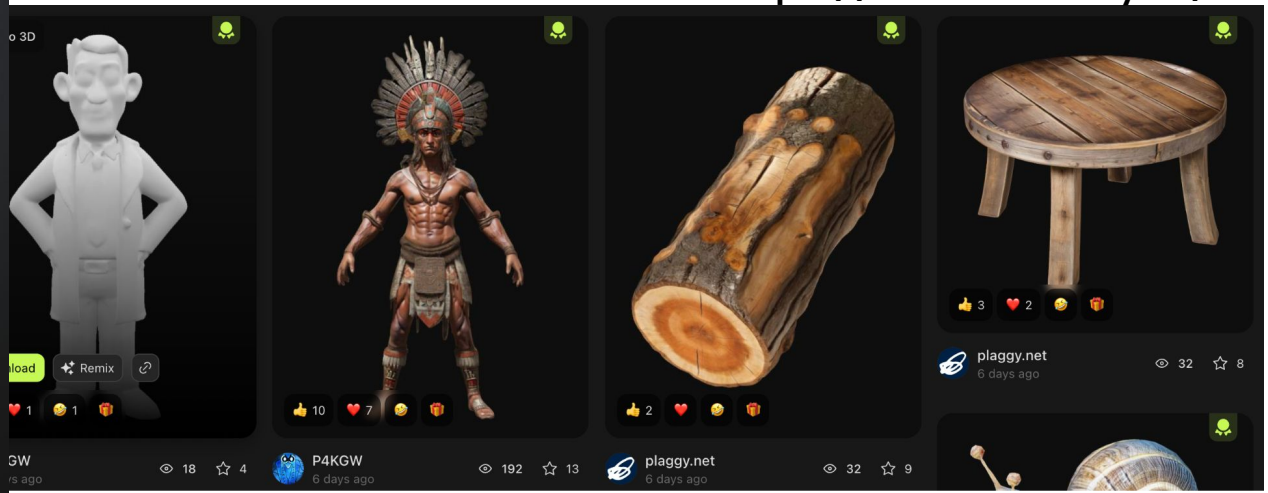
Preview JSON

https://www.scenario.com/



Sound effects, backgrounds, voice, 3d mesh, etc.

https://www.meshy.ai/

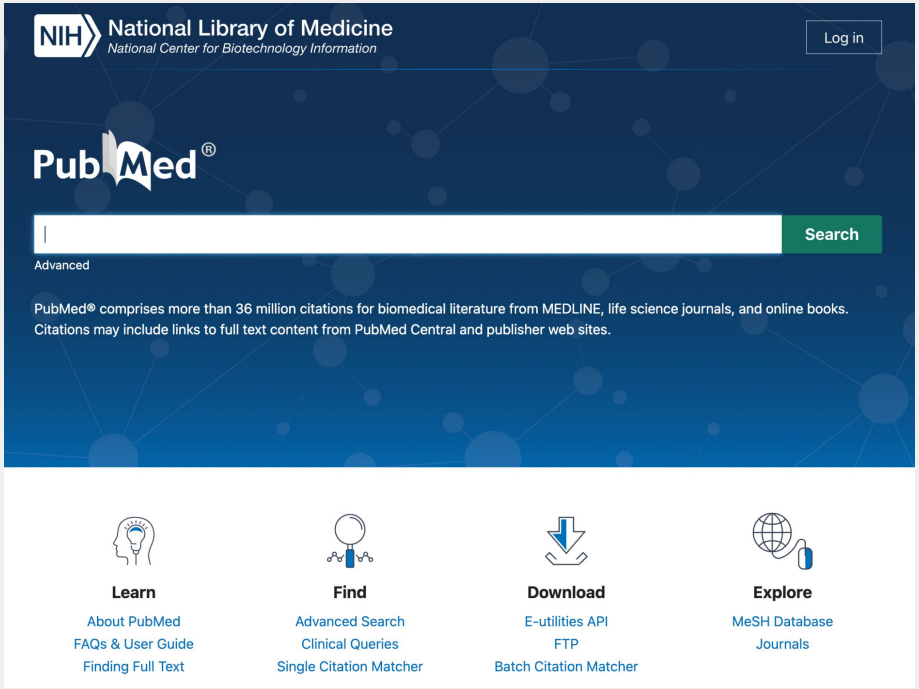


AI in Health

- diagnosis of diseases
- prediction of therapeutic responses
- preventative medicine
- improving provider workflow & clinical operations
- disease and therapy monitoring
- improving procedural accuracy and overall patient outcomes

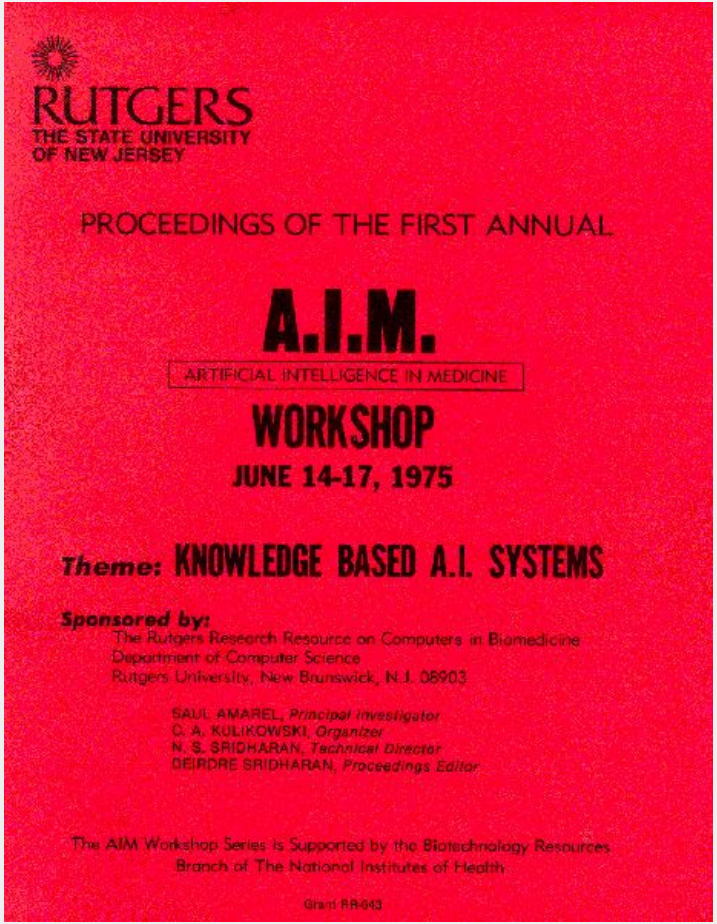
Digitizing Health Data 1950s

- While the AI field was moving along with experiments with robots (Shakey) and chatbot (ELIZA), medicine was slower to respond to automation
- The medical field began with digitization of health data
- Medical Literature and Analysis and Retrieval System MEDLARS; online counterpart: MEDLINE, 30 million references to journal articles in life sciences with a concentration on biomedicine, 1789 to present)
- and search engines of medical records (PubMed by National Library of Medicine)



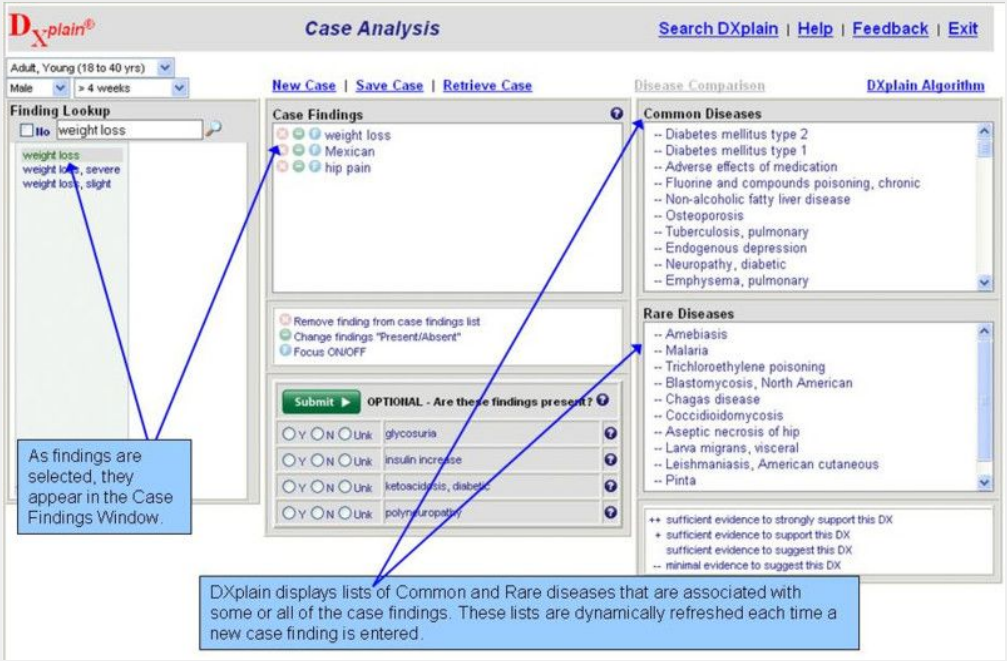
First AI Conferences 1970s

- Researchers in medicine worked on information sharing and networking systems for medicine.
- In 1971, The Research Resource on Computers in Biomedicine by Saul Amarel at Rutgers University. The Stanford University Medical Experimental – AI in Medicine created in 1973.
- The first AI in medicine workshop was held in 1975 at Rutgers University hotels by the National Institute of Health.



Medical Expert Systems 1970s

- In the 1970s these groups worked on expert system prototypes for applying AI to medical consulting.
- 1976 CASNET: model-building, consultation, database all brought together to apply information about diseases to individual patients and give physicians advice on patient management.
- 1970s MYCIN: backward chaining AI system with about 600 rules. Physicians could input patient information to get back list of potential bacterial pathogens and antibiotic recommendations.
- 1986 DXplain: input symptoms to get back different diagnoses. Served as an electronic medical textbook with diseases and references. At first release could handle 500 diseases.



Medical Chatbots 2010s

- Pharmabot developed in 2015 to assist in medication education for pediatric patients.
- Mandy was created in 2017 as an automated patient intake process for a primary care practice.

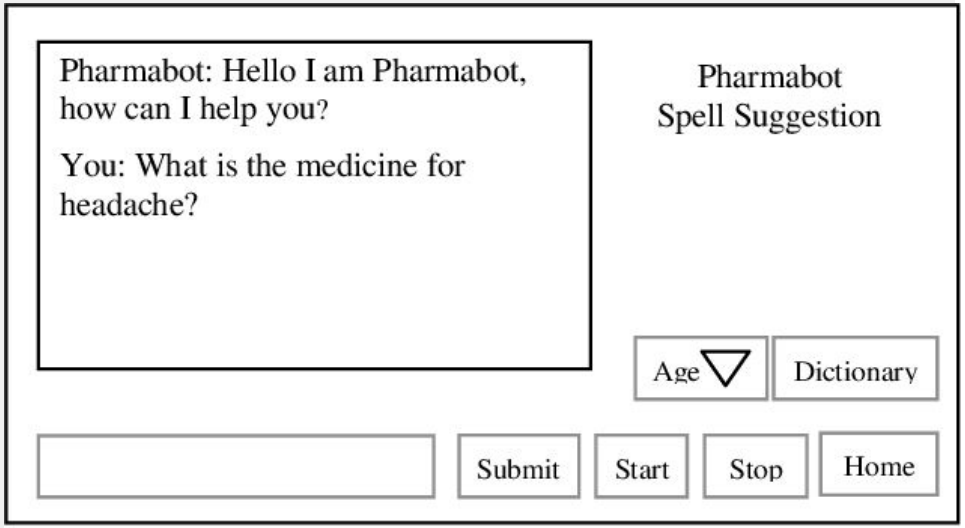


Figure 3. Conversation window

MANDY: Towards A Smart Primary Care Chatbot Application

Lin Ni[1], Chenhao Lu, Niu Liu, and Jiamou Liu[2]

Department of Computer Science, The University of Auckland, New Zealand
[1]lni600@aucklanduni.ac.nz, [2]jiamou.liu@auckland.ac.nz

Abstract. The paper reports on a proof-of-concept of Mandy, a primary care chatbot system created to assist healthcare staffs by automating the patient intake process. The chatbot interacts with a patient by carrying out an interview, understanding their chief complaints in natural language, and submitting reports to the doctors for further analysis. The system provides a mobile-app front end for the patients, a diagnostic unit, and a doctor's interface for accessing patient records. The diagnostic unit consists of three main modules: An analysis engine for understanding patients symptom descriptions, a symptom-to-cause mapper for reasoning about potential causes, and a question generator for deriving further interview questions. The system combines data-driven natural language processing capability with knowledge-driven diagnostic capability. We evaluate our proof-of-concept on benchmark case studies and compare the system with existing medical chatbots.

Keywords: Medicare chatbot; Patient interview; Natural language processing; AI and healthcare

Acknowledgement: The first author is partially funded by a scholarship offered by Precision Driven Health in New Zealand, a public-private research partnership aimed at improving health outcomes through data science. Initial progress of the research was reported in the PDH & Orion Health Blog https://orionhealth.com/global/knowledge-hub/blogs/meet-mandy-an-intelligent-and-interactive-medicare-system/

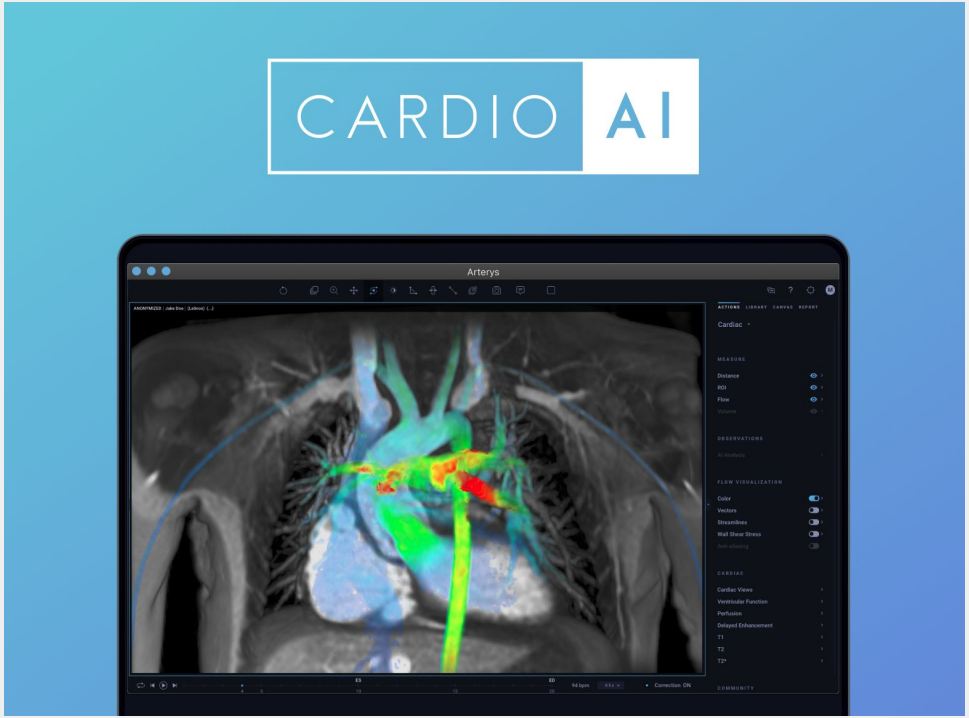
1 Introduction

Patients arriving at a primary care service sometimes need to wait for a long time before being advised by a doctor [26]. This is often due to high workload and limited resources at the primary care service [7]. To facilitate the process, nurses and other health care staffs usually take the role of patient intake. An incoming patient would be first greeted by a receptionist who carries out an intake inquiry. The receptionist would typically be someone who has a certain level of medical proficiency, and the inquiry involves collecting patient information and understanding the symptoms of the patient. A brief report is generated

Deep Learning in Medicine 2010s

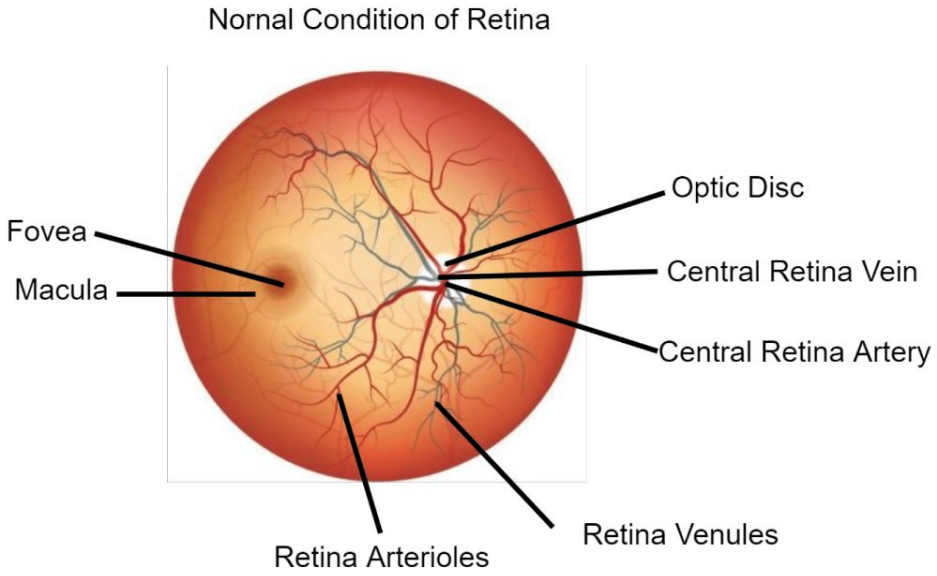
- In the 2010s researchers began to apply deep learning to medical imaging.
- Arterys was the first US FDA approved clinical DL health care product in 2017.
- CardioAI was able to analyze cardiac MRIs in seconds to give information about cardiac ejection fraction (now includes liver and lung imaging, chest and musculoskeletal xray images)

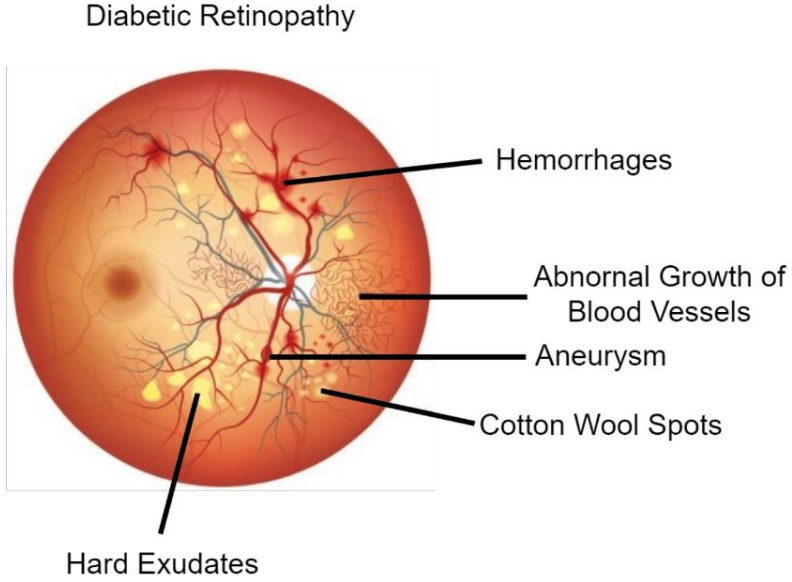
ARTERYS



Deep Learning in Medicine 2010s

- Vision for lesion detection, cardiovascular risk, progression of Alzheimers, and screen for diabetic retinopathy, etc.

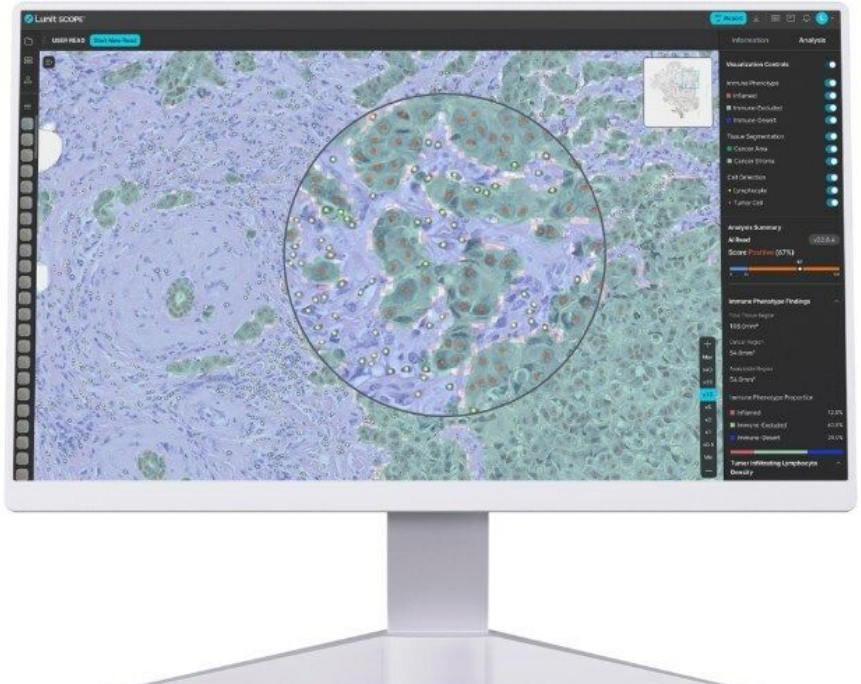


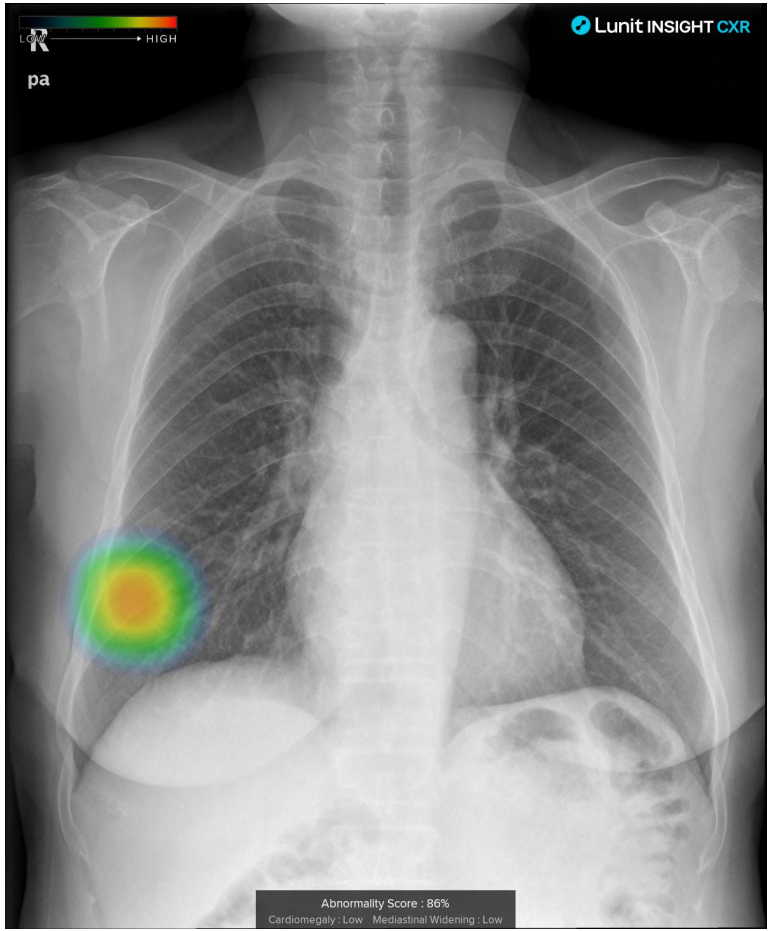


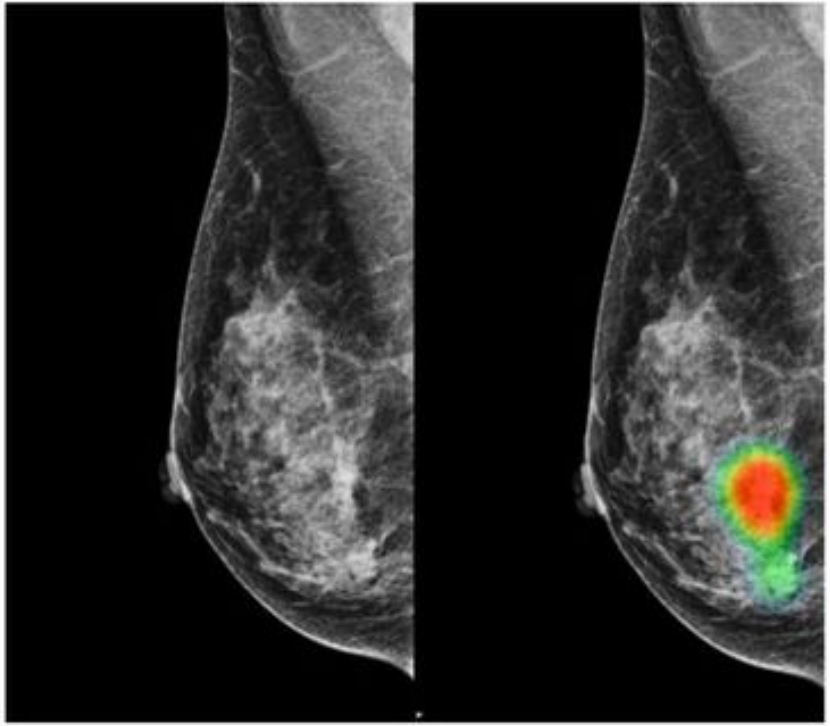
Computer Vision for Cancer Detection 2010s

- Computer vision has recently been used for cancer detection – Lunit is a South Korean company that has been successful with Lung and Breast cancer detection.

Lunit

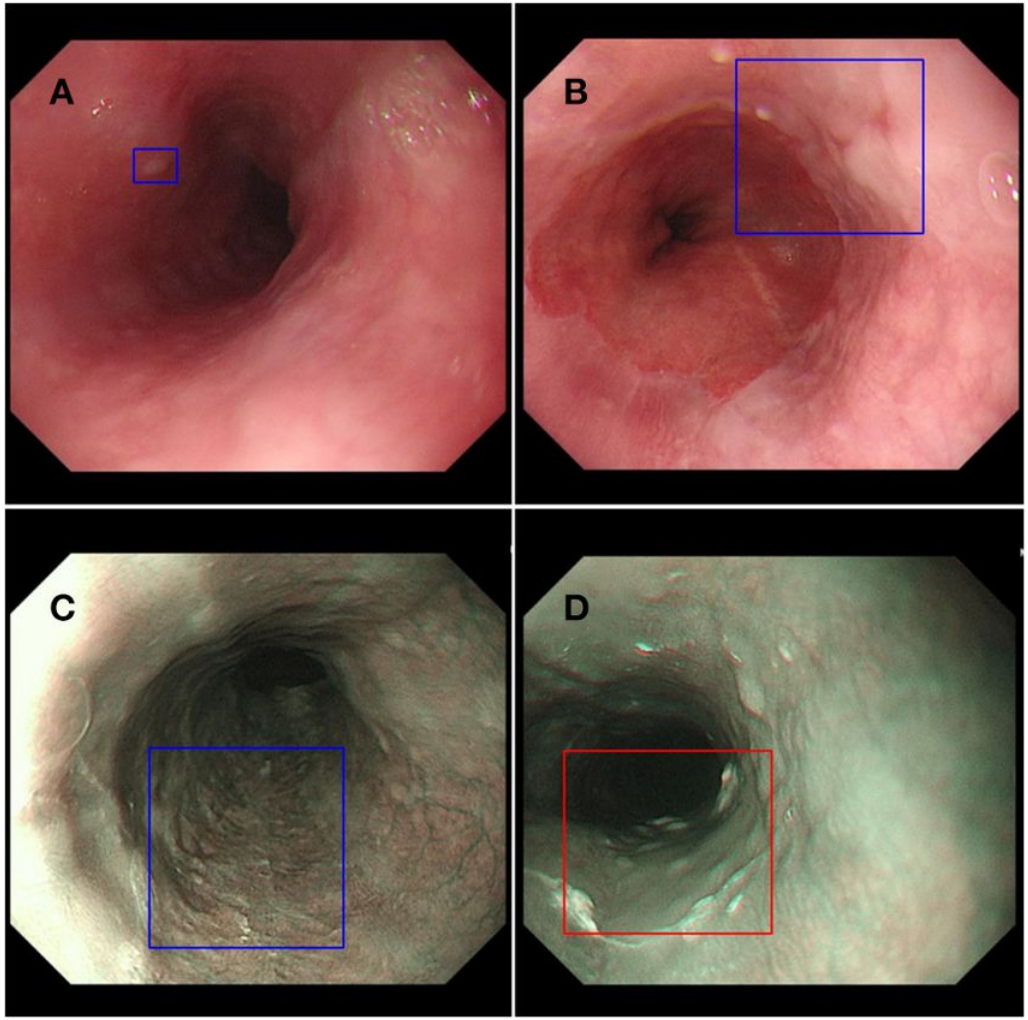



LOW
HIGH
R
pa
Lunit INSIGHT CXR
Abnormality Score : 86%
Cardiomegaly : Low
Mediastinal Widening : Low



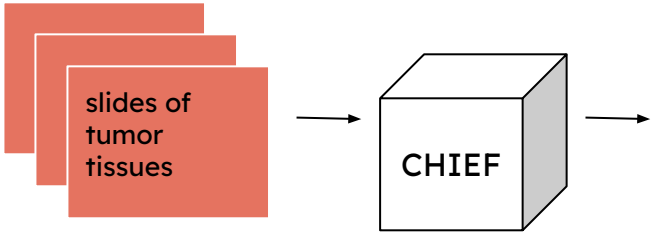
Computer Vision for Cancer Detection 2010s

- Gastroenterology has been using AI for colonoscopies. It has been successful in identifying malignant/benign colon polyps and pancreatitis/pancreatic cancer.


A
B
C
D

ENDOANGEL
ENDOANGEL®
内镜精灵

More general: AI for Cancer Treatment



- 19 types of cancer cell detection
- tumor origin identification
- predicting patient outcomes
- identifying the presence of genes and DNA patterns related to treatment response in tumor micro-environment

nature

Explore content | About the journal | Publish with us | Subscribe

nature > articles > article

Article | Published: 04 September 2024

A pathology foundation model for cancer diagnosis and prognosis prediction

Xiyue Wang, Junhan Zhao, Eliana Marostica, Wei Yuan, Jietian Jin, Jiayu Zhang, Ruijiang Li, Hongping Tang, Kanran Wang, Yu Li, Fang Wang, Yulong Peng, Junyou Zhu, Jing Zhang, Christopher R. Jackson, Jun Zhang, Deborah Dillon, Nancy U. Lin, Lynette Sholl, Thomas Denize, David Meredith, Keith L. Ligon, Sabina Signoretti, Shuji Ogino, ... Kun-Hsing Yu + Show authors

Nature **634**, 970–978 (2024) | Cite this article

43k Accesses | **366** Altmetric | Metrics

Abstract

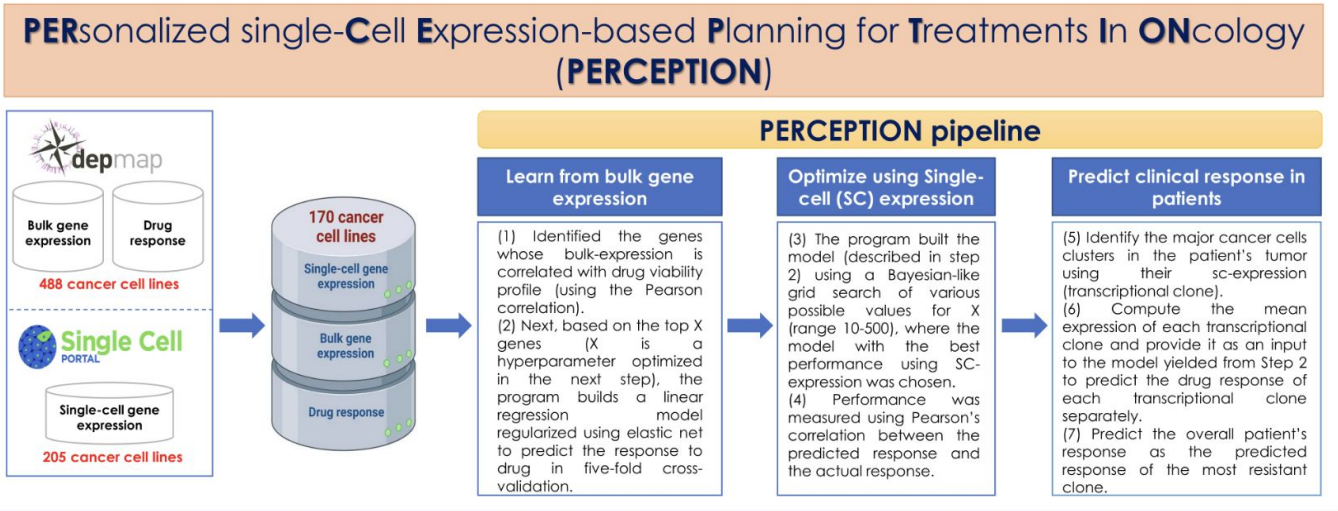
Histopathology image evaluation is indispensable for cancer diagnoses and subtype classification. Standard artificial intelligence methods for histopathology image analyses have focused on optimizing specialized models for each diagnostic task[1,2]. Although such methods have achieved some success, they often have limited generalizability to images generated by different digitization protocols or samples collected from different populations[3]. Here, to address this challenge, we devised the Clinical Histopathology Imaging Evaluation Foundation (CHIEF) model, a general-purpose weakly supervised machine

CHIEF Model 2024 for

More specific: Single-cell RNA Sequencing

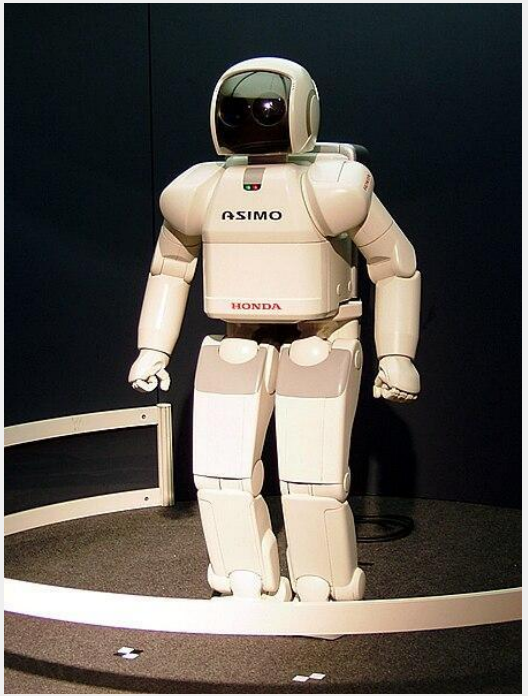
Bulk sequencing cancer DNA & RNA
- average of all tumor cells
- difficult to target tumors

Single cell RNA sequencing
- higher resolution diagnosis
- individual cell level targetted
- more lasting drug responses



History of AI

Global Perspectives in History of AI



unsojo@cornell.edu

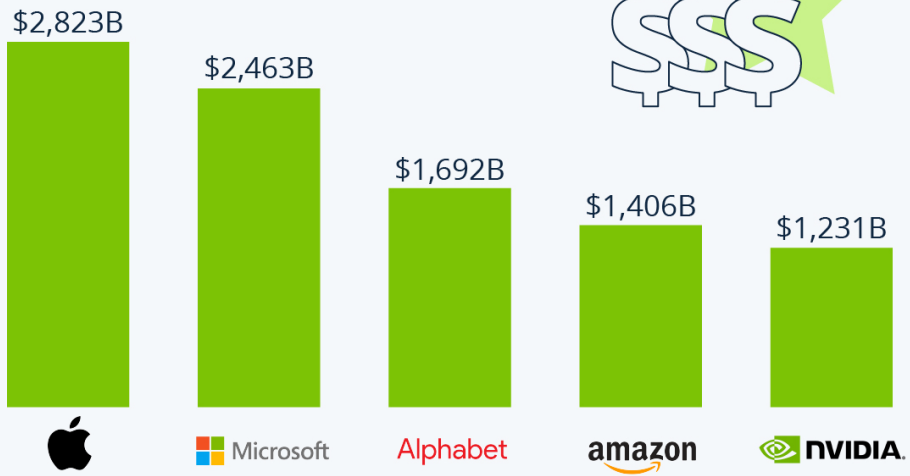
Lecture 8 March 14, 2025

This week!

- Why is hardware so important for AI?
- How does labor and global economics play a role in modern AI growth?
- How do computing advances change AI accessibility and productizing?

Nvidia Cements Membership in Trillion Dollar Club

Market capitalization of the highest-valued U.S. public companies*



* As of market open on August 24, 2023

Source: Yahoo Finance

statista

SUPERCHARGED

A $40,000 Nvidia chip has become the world's most sought-after hardware

Companies and governments want to deploy generative AI—but first they need access to Nvidia's H100 chips

By **Michelle Cheng** Published August 18, 2023

Nvidia is the AI company of the moment. Photo: Ann Wang (Reuters)

It seems like everyone wants to get their hands on Nvidia's H100 chips these days.

Microsoft and Google, which are building out generative AI-driven search engines, are some of the biggest customers of Nvidia's H100 chips. Server manufacturers claim to have waited more than six months to receive their latest orders. Venture capital investors are buying up H100 chips for the startups they invest in.

RELATED CONTENT

GO MEDIA
AV CLUB DEADSPIN GIZMODO
the inventory JALOPNIK JEZEBEL
KOTAKU lifehacker the ONION
QUARTZ THE ROOT TAKEOUT

CHIPS

Nvidia Stock Rises. Why More Volatility Is Coming.

By George Glover Follow and Adam Clark Follow

Updated March 12, 2025, 5:09 pm EDT / Original March 12, 2025, 5:15 am EDT

Share Resize Reprints

In this article

↑ NVDA ↓ SPX ↓ 2330 ↓ AMD ↓ AVGO

Nvidia CEO Jensen Huang. (COURTESY NVIDIA)

Nvidia ↑ NVDA +0.64% stock rose Wednesday as it continued to pare back some of its recent losses—but more big swings could be ahead.

Shares in the chip maker rose 6.4% to $115.74, having closed the previous session up 1.7%. The benchmark S&P 500 ↓ SPX -1.22% index climbed 0.5%.

Nvidia is down 17% over the past month through Tuesday's close, tumbling as investors have fretted that U.S. President Donald Trump's tariff plans will drag down growth and trigger a flare-up of inflation, both of which could weigh on chip sales.

2020s: GPUs for AI

NVIDIA Blackwell GPUs Sold Out: Demand Surges, What's Next?

Published October 14, 2024 | Written by **Esther Shein**

The supply of next-gen processors is sold out for the next 12 months as company demand for more sophisticated AI models and applications continues unabated.

Nvidia continues to ride the AI wave as the company sees unprecedented demand for its latest Blackwell GPU next-generation processors. The supply for the next 12 months is sold out, Nvidia CEO Jensen Huang told Morgan Stanley analysts during an investors' meeting.

A similar situation occurred with Hopper GPUs several quarters ago, Morgan Stanley Analyst Joe Moore pointed out.

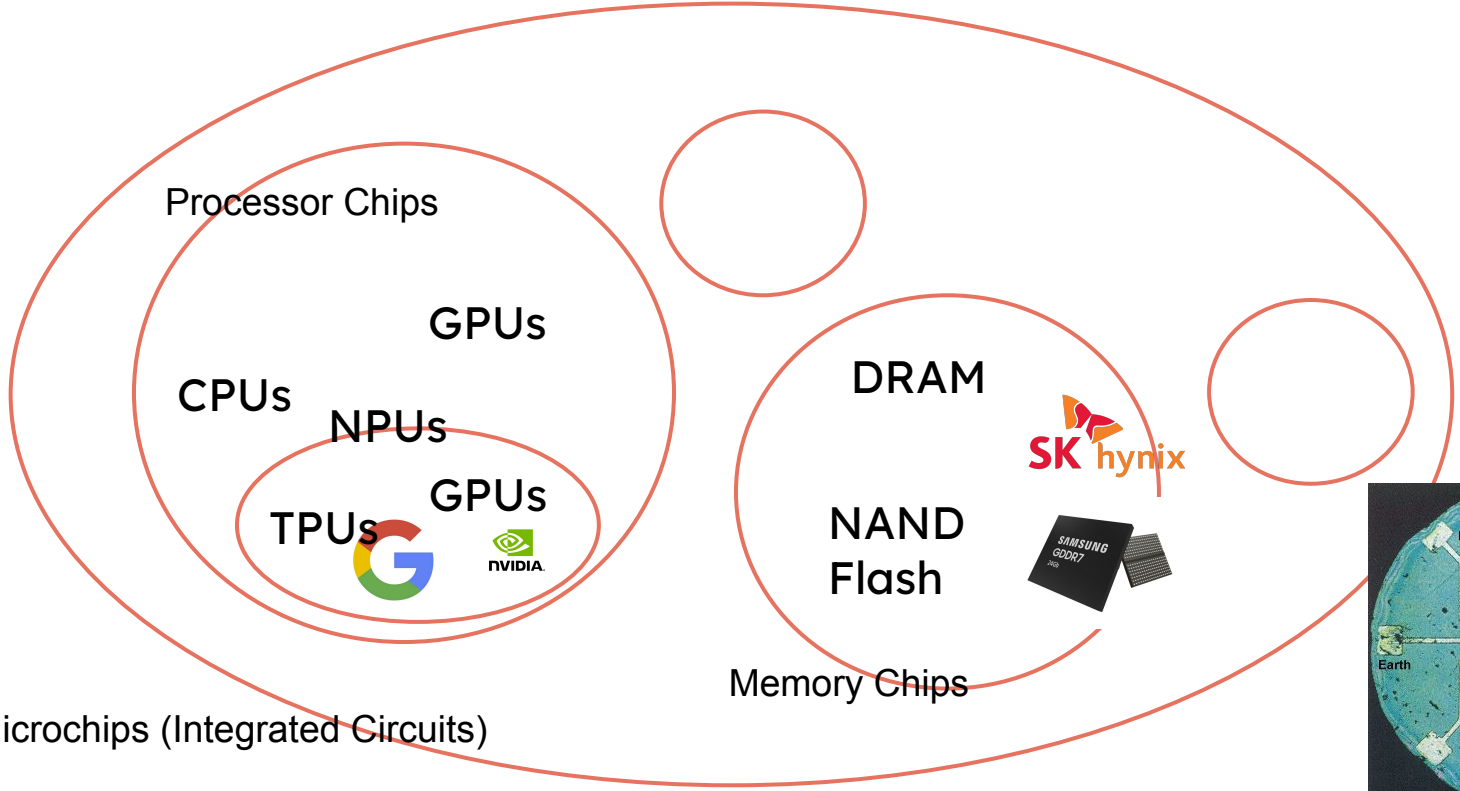
Nvidia's traditional customers are driving the overwhelming demand for Blackwell GPUs, including major tech giants such as AWS, Google, Meta, Microsoft, Oracle, and CoreWeave. Every Blackwell GPU that Nvidia and its manufacturing partner TSMC can produce over the next four quarters has already been purchased by these companies.

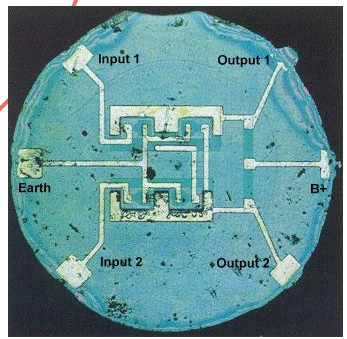
The AI Boom Is Giving Rise to "GPU-as-a-Service"

The industry harvests idle compute for AI startups that need it

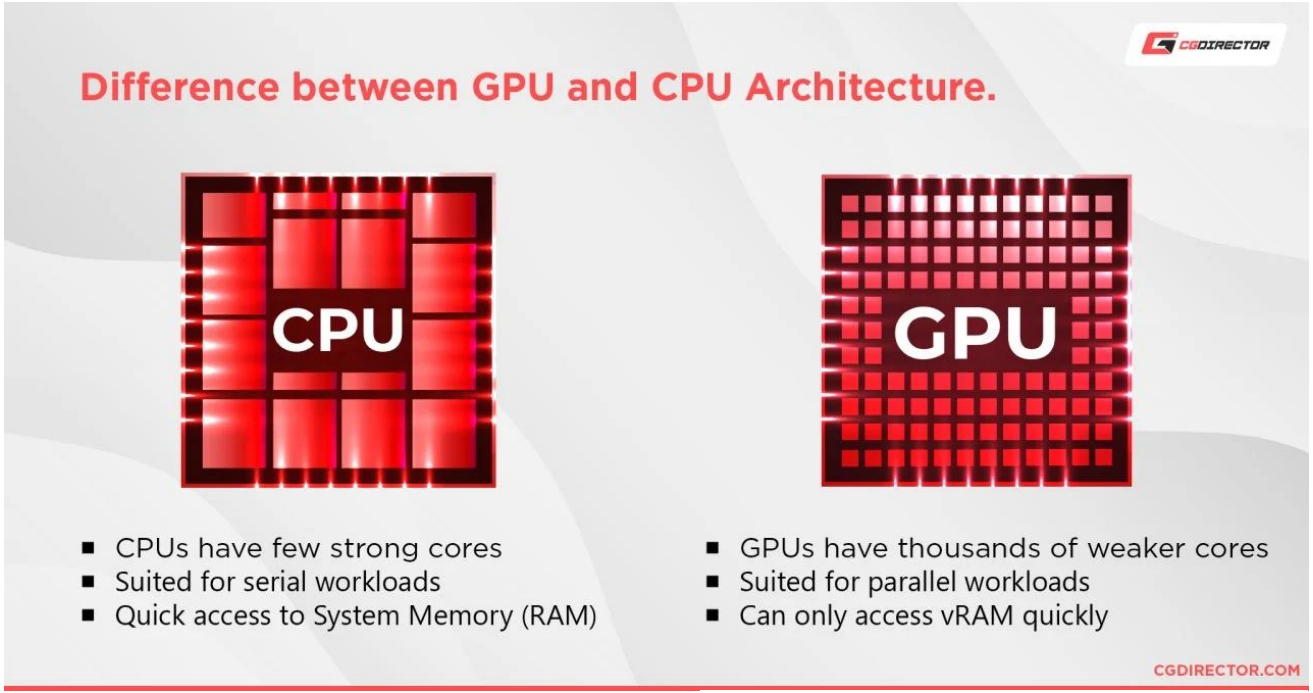
BY JUAN PABLO PEREZ | 20 JAN 2025 | 3 MIN READ

Microchips Family





Fairchild IC



GPUs

- better for batch processing
- graphics, deep learning (matrix multiplications)

CPUs

- typical PC microchips
- fast and good with general purpose computing

Chip Production

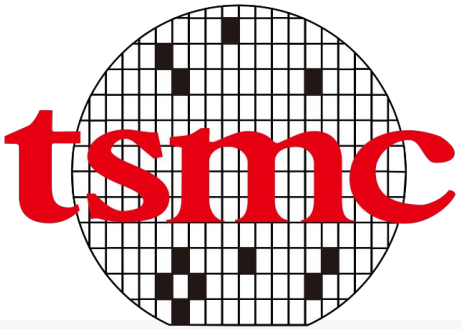
Lithography Machines

Canon

ASML

Fabrication Factories

intel



SAMSUNG

Chip Design + Software

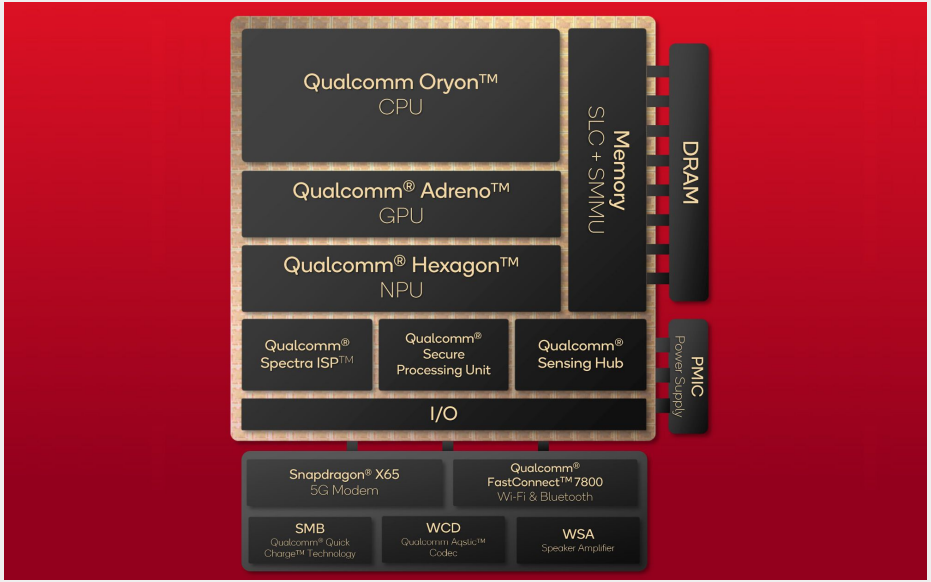
Qualcomm

Chip Production: Lithography

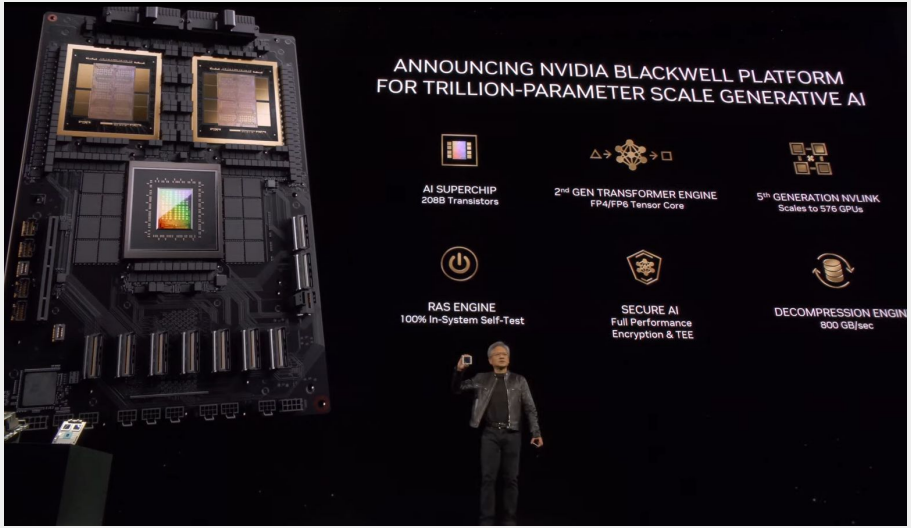
ASML EUV Lithography

Canon DUV Lithography

Chip Production: Chip Design



Qualcomm Snapdragon SoC



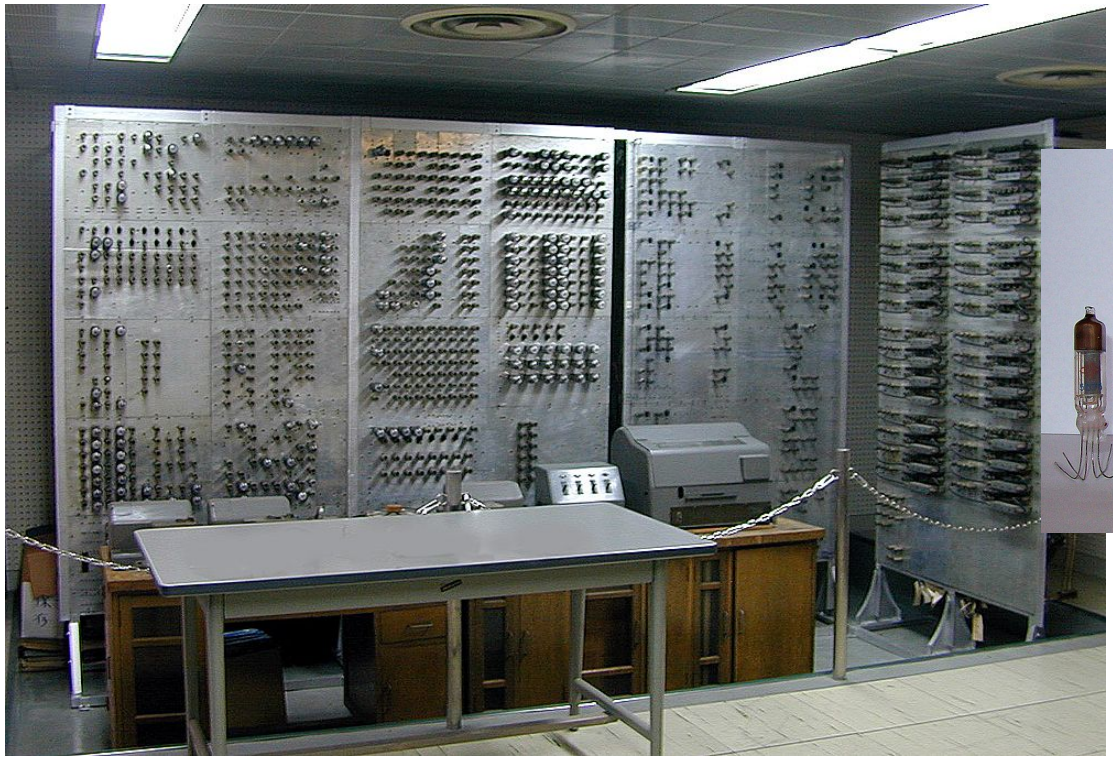
Nvidia Blackwell Platform

Chip Production: Fabs & Foundries

Samsung Pyeongtaek Fab

TSMC Foundry in Arizona

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)



ENIAC 1945

Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

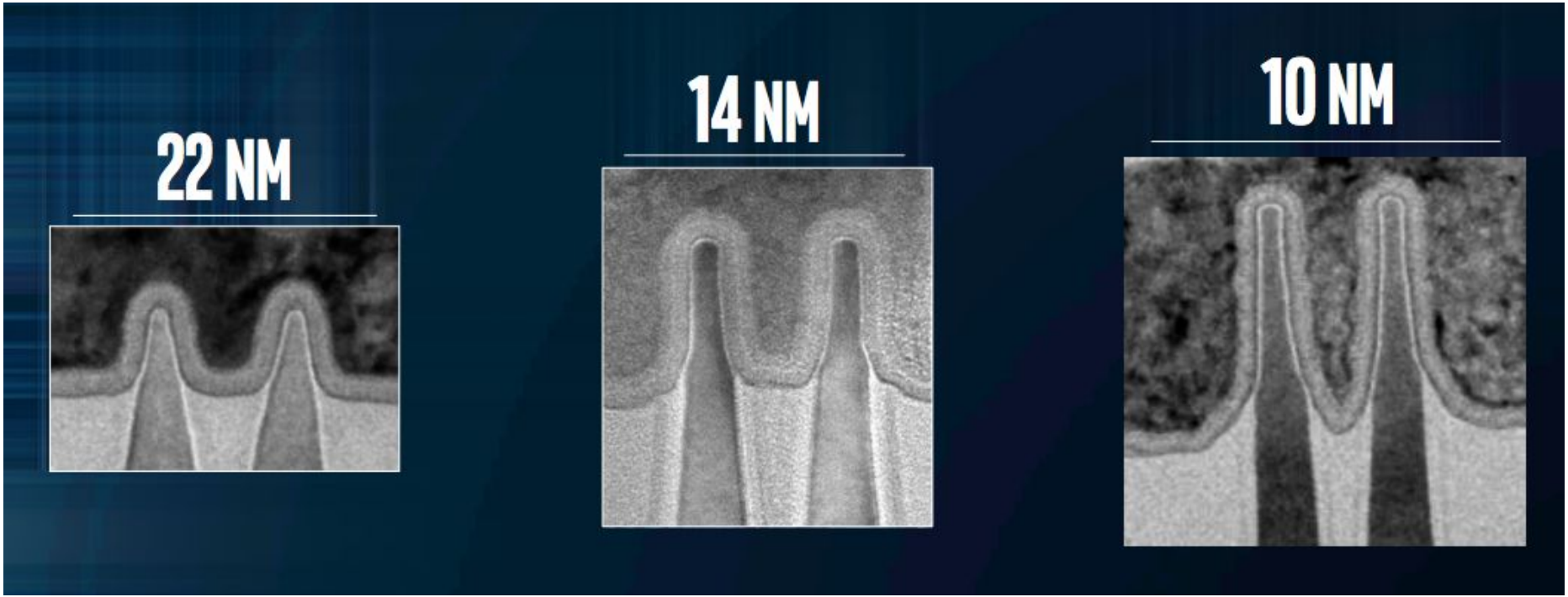
Researchers at Bell Labs and AT&T start making semiconductors

Silicon

Phosphorous

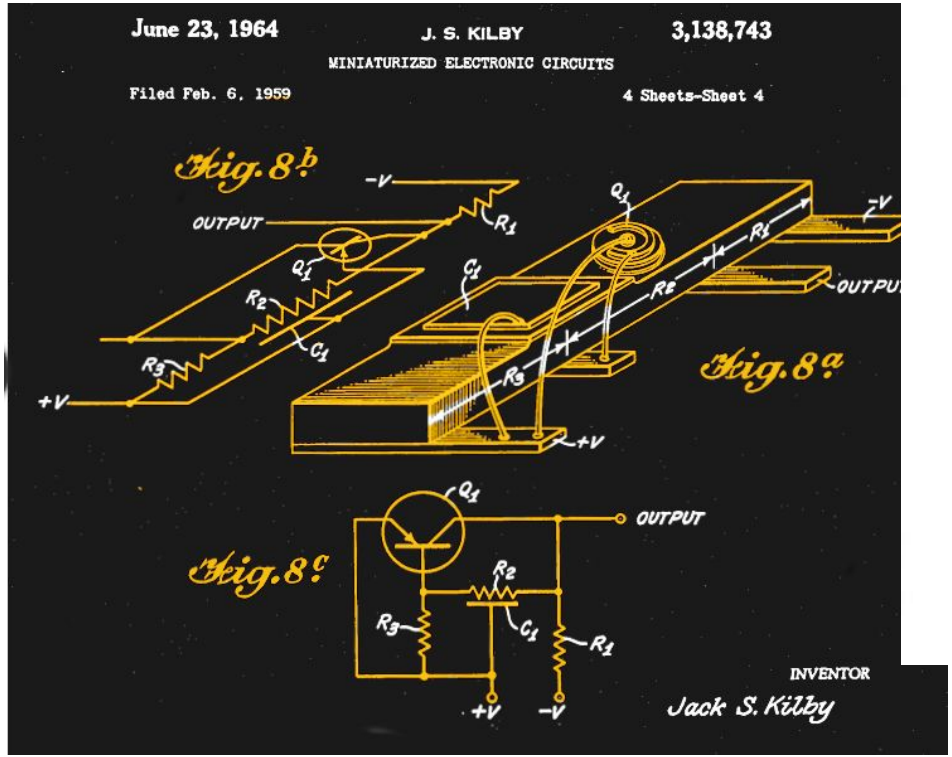
Beginning of semiconductor chips: Vacuum Tubes to Transistors (1940s)

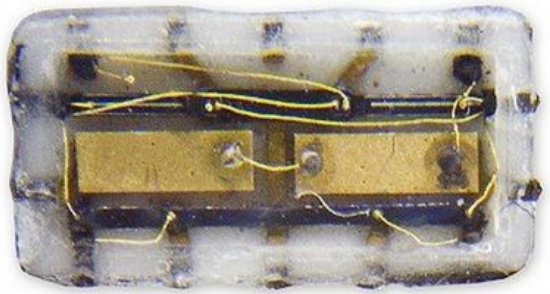
Each transistor is nanometers wide. Each chip carries billions of transistors



First Integrated Circuits 1958

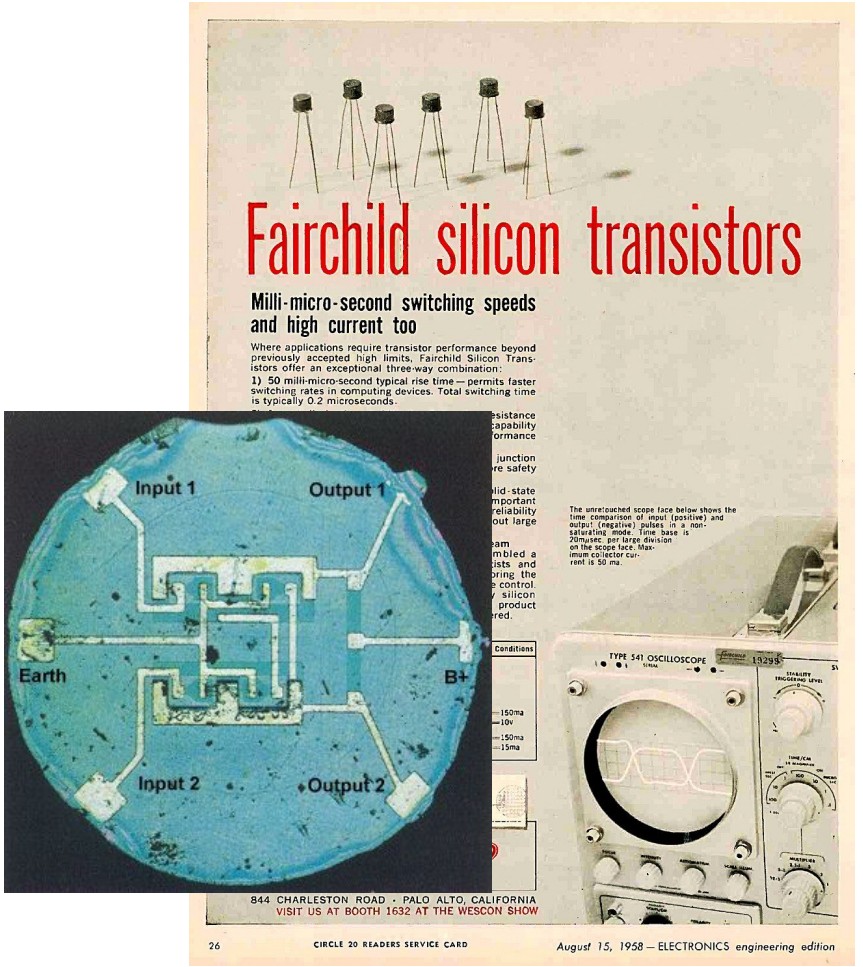
Kilby at Texas Instruments invents the Integrated Circuit





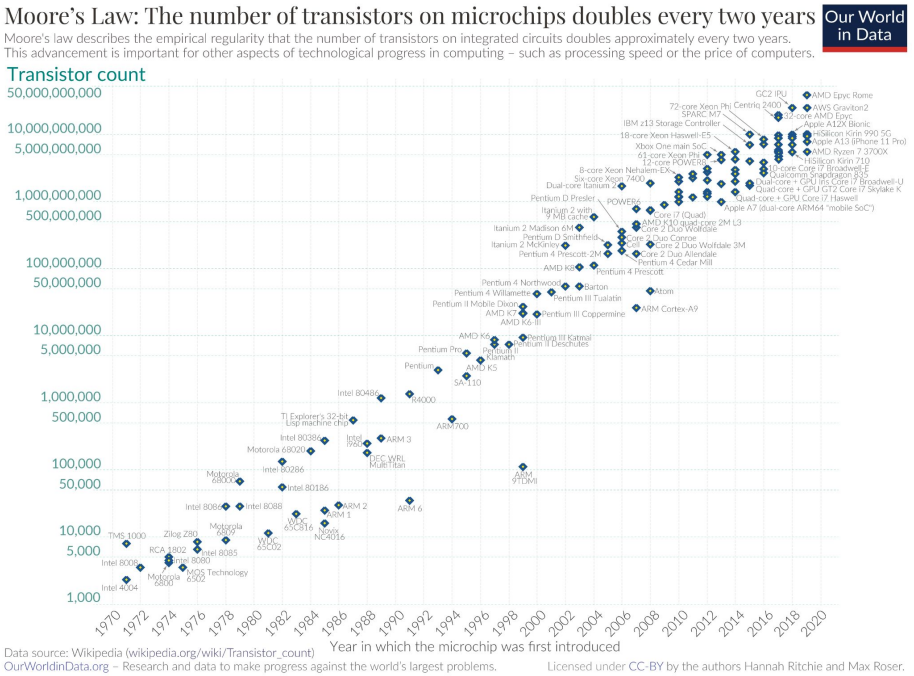
Fairchild Semiconductors

- Micrologic Chips 1958
- Apollo (NASA) program used Fairchild chips!
- Sales ballooned from $500,000 1958 to $21m two years later
- Founder Noyce started slashing prices to address wider consumer need ($120 → $15)
- 1965: military applications made up 95% of chips



Defense to Consumer Tech

- Cold War U.S. Defense using over 95% of chips up to the mid-1960s
- used in command, control technologies like satellites, sonar, torpedoes, telemetry systems
- …to the civilian sector (private market) in the late 1960s
- Moore's Law (1965) that every year for the next decade, the number of components possible to fit on silicon would double



Moore's Law (1965)

Chips come to Asia!

- U.S. starts investing in Japanese technology post WWII
- Morita founds Sony (Sonus + Sonny) 1946
- “licensing” strategy: Japanese electronics paid U.S. firms IP fees (4.5% to Fairchild, 3.5% to TI, 2% Western Electric)
- Japanese efficient manufacturing was an easy way to start embedding consumer products with chips (1958)

Sony Radios

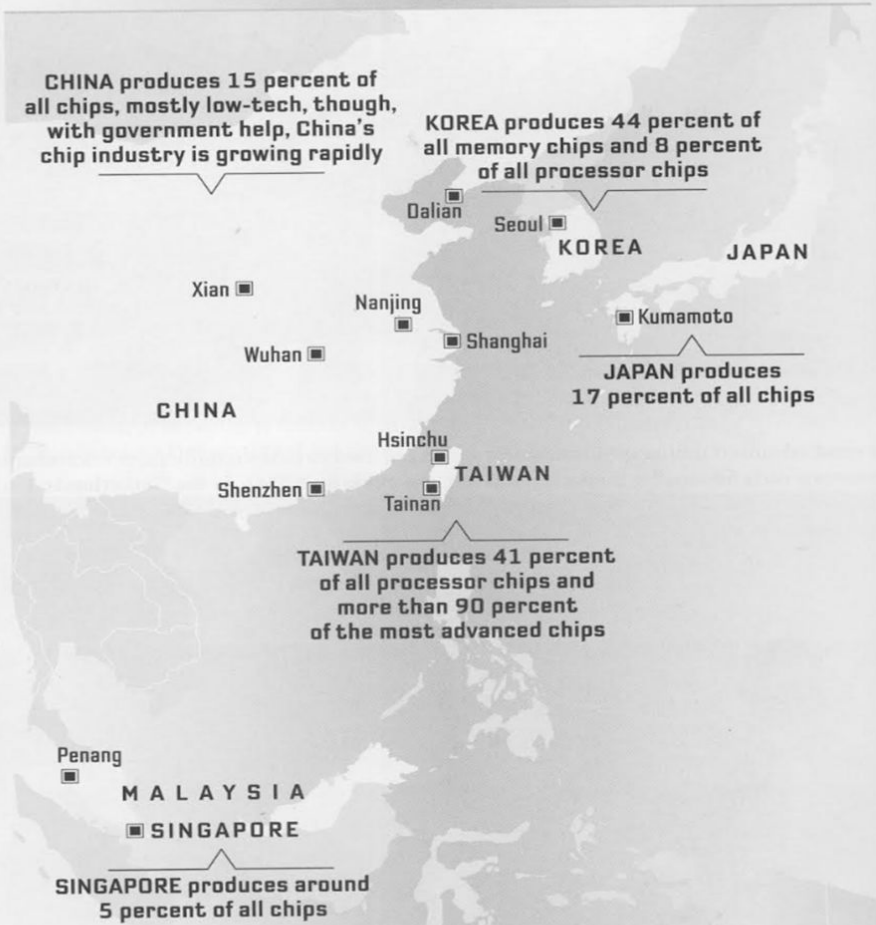
Japan overtakes U.S.

- By 1964 Japan overtakes U.S. in discrete transistor production
- U.S. was still producing the most advanced chips
- From 1965 to 1985 Japanese export of semiconductors and consumer products with them grew from $600 million to $60 billion

Sharp Calculator 1967

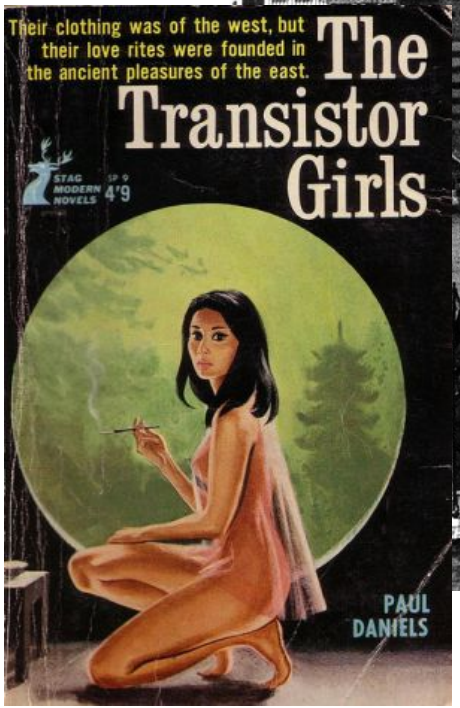
EAST ASIA PRODUCES:

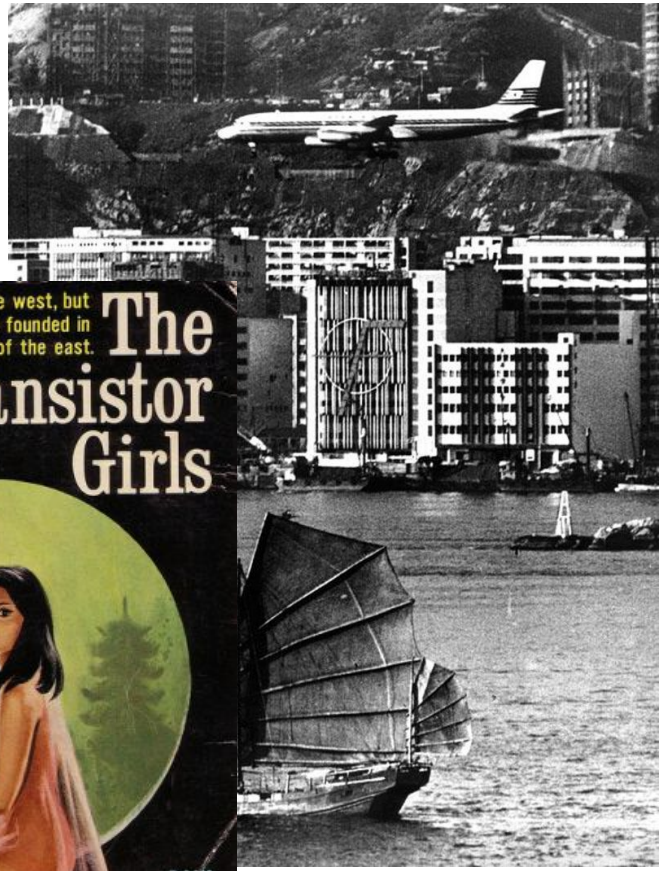
90 percent of all memory chips, 75 percent of all processor (logic) chips, and 80 percent of all silicon wafers



Cheaper labor in Asia 1960s

- Companies like Fairchild Semiconductor, Texas Instruments, Motorola start building factories in Hong Kong, Singapore, Malaysia
- Workers here were paid about 1/10 the wages of Americans
- HK 25c (/hour), Taiwan 19c, Malaysia 15c, Singaporean 11c, South Korean 10c
- Mostly women: “better on dexterity tests”





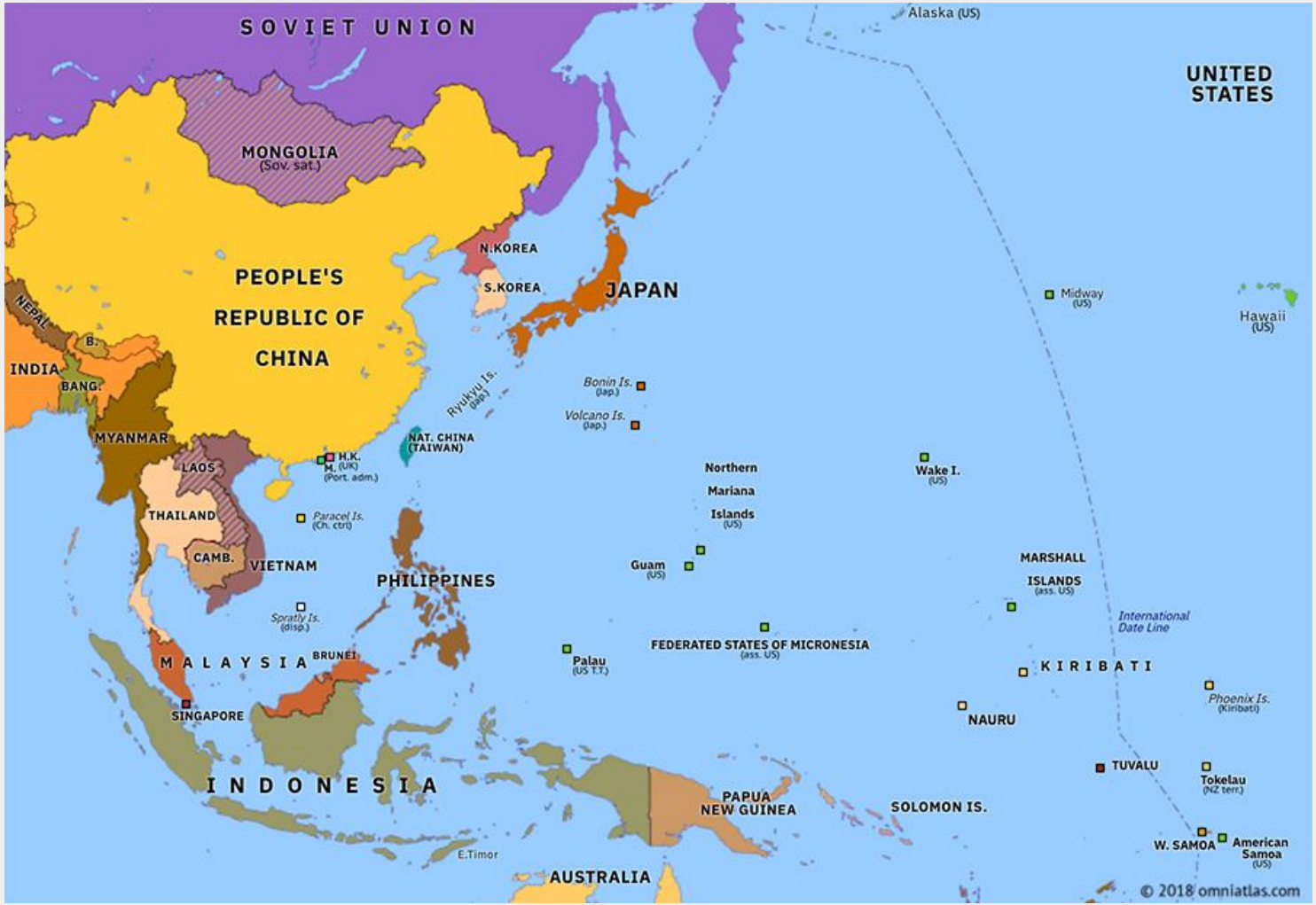
Fairchild on Kowloon

post-Vietnam War Political Landscape

- With the loss of the U.S. in the Vietnam War looming neighboring Asian countries look to U.S.-alliance
- China's first atomic weapons test 1964
- Taiwan in particular was concerned that the fall of South Vietnam would mean Communist strengthening in the region and a larger Communist Chinese threat.

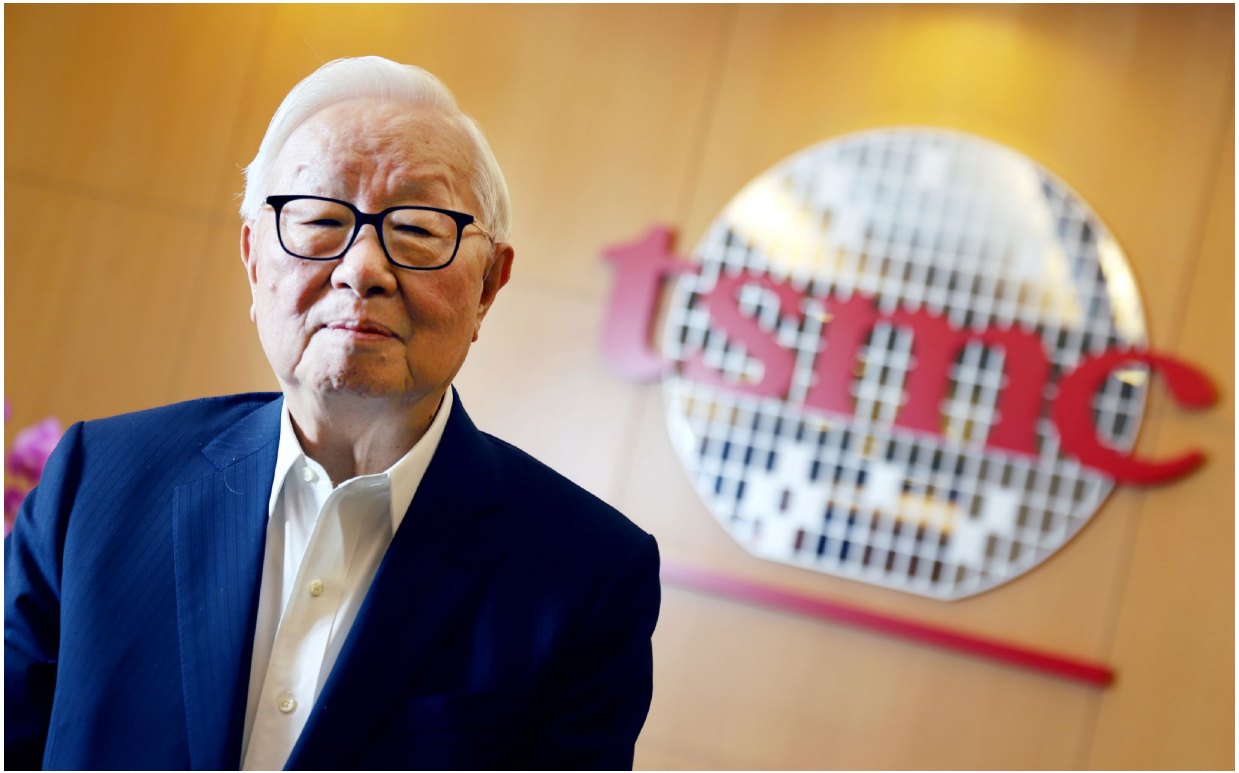
- Taiwan turned to strategic alliance with the U.S. and partnered with Texas Instruments – first plant in 1969

Minister K.T. Li


SOVIET UNION
Alaska (US)
UNITED STATES
MONGOLIA
(Sov. sat.)
PEOPLE'S REPUBLIC OF CHINA
N.KOREA
S.KOREA
JAPAN
Midway (US)
Hawaii (US)
NEPAL
B.
INDIA
BANG.
MYANMAR
Ryukyu Is. (jap.)
Bonin Is. (jap.)
Volcano Is. (jap.)
NAT. CHINA (TAIWAN)
H.K. M. (UK) (Port. adm.)
LAOS
THAILAND
Paracel Is. (Ch. ctrl)
CAMB.
VIETNAM
PHILIPPINES
Northern Mariana Islands (US)
Wake I. (US)
Guam (US)
MARSHALL ISLANDS (ass. US)
Spratly Is. (disp.)
International Date Line
FEDERATED STATES OF MICRONESIA (ass. US)
Palau (US T.T.)
MALAYSIA
BRUNEI
KIRIBATI
Phoenix Is. (Kiribati)
SINGAPORE
NAURU
TUVALU
Tokelau (NZ terr.)
INDONESIA
PAPUA NEW GUINEA
SOLOMON IS.
W. SAMOA
American Samoa (US)
E.Timor
AUSTRALIA
© 2018 omniatlas.com

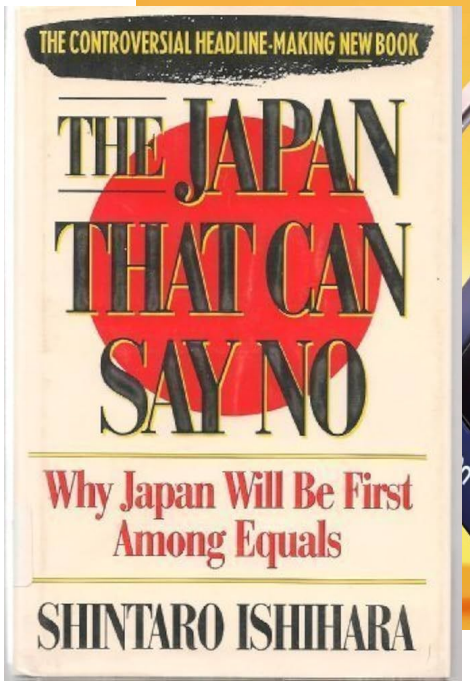
Moriss Chang to lead TSMC

- Minister Li hires Morris Chang to lead TSMC in 1985
- Blank cheque from Taiwanese government (48% share)
- TSMC becomes first “foundry” – “Gutenberg for semiconductors”
- Philips gave $58m, production technology for 27.5%



Japan's productivity threatens the U.S. 1980s

- Japan's high productivity starts to threaten U.S. dominance into the 1980s
- DRAM competition with Japan weakens U.S. memory chip production
- Morita: "Militarily we could never defeat the U.S. but economically we can overcome the U.S. and become number one in the world"
- The CIA coined this the "PAX JAPONICA"



coauthored with Morita 1989

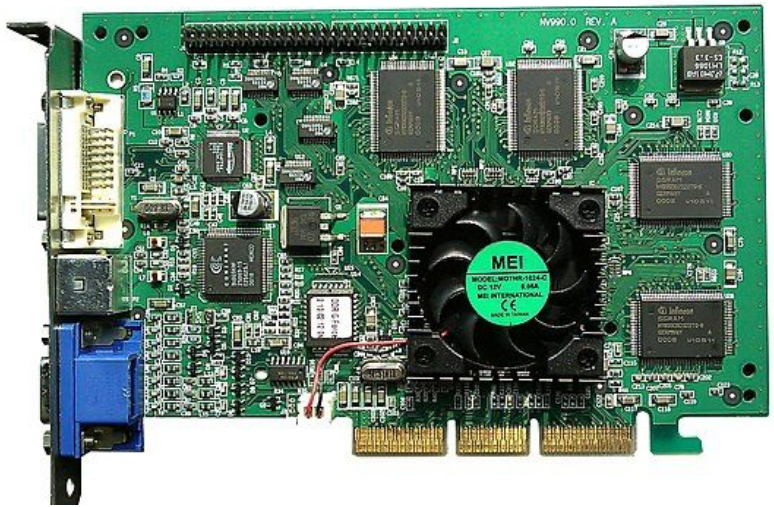
Sony Walkman 1979

South Korean Hedge

- Silicon Valley looked to Korean producers to challenge Japanese threat
- Korean production even cheaper than Japan
- Several U.S. firms (e.g. Intel) began to sign joint ventures with Samsung in the 1980s
- Silicon Valley AMD CEO Jerry Sanders “enemy of my enemy is my friend”
- U.S. provided market and tech for Samsung chip production growth

Startups on Graphics Cards

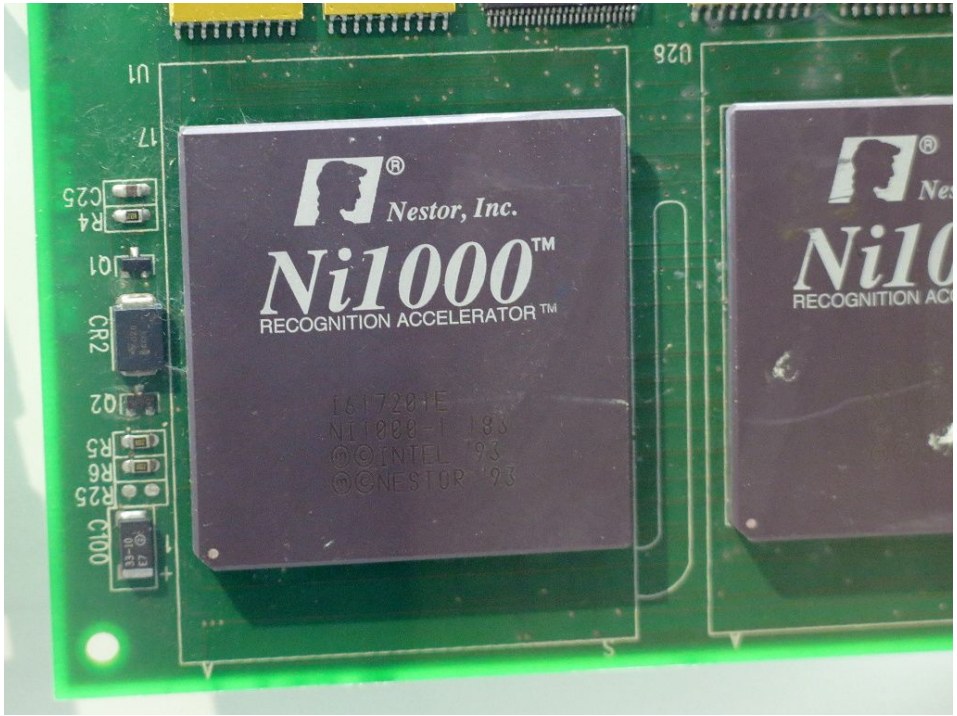
- In the 1980s Intel had a de facto monopoly on CPU microprocessors but graphics chips were a hot field for startups
- NVIDIA was founded in 1993 in Silicon Valley to design GPU chips which were mostly used for 3d graphics (for games and video)



1999 GeForce

Application Specific Integrated Circuits (ASICs)

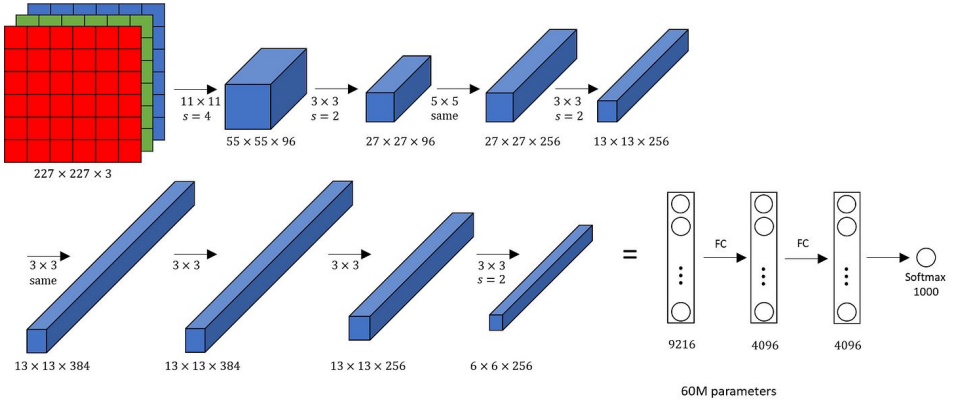
- 1980s Companies started developing microchips specifically made for neural networks
- 1980s Intel: ETANN 80170NX
- 1990s Nester/Intel: Nestor/Intel Ni1000 (DARPA) - could recognize handwritten characters
- Then 1990s, field-programmable gate arrays (FPGAs) for AI compute (away from ASICs)



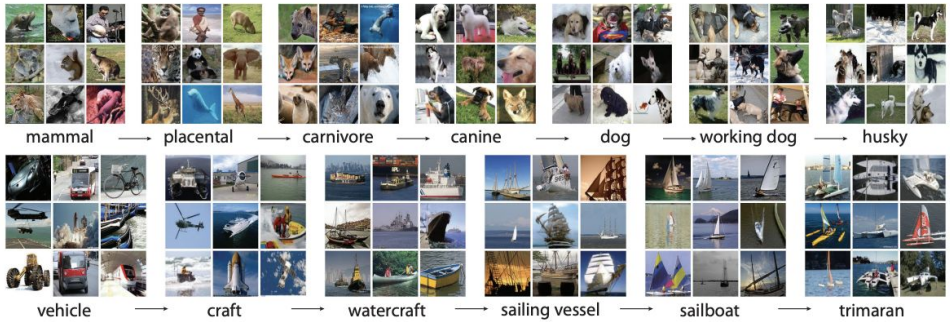
Nvidia's breakthrough? 2012 AlexNet

- AlexNet, a deep learning model proposed by researchers at U of Toronto, topped a ImageNet competition 2012 with 15% top-5 error (more than 10 percentage points lower than runner-up)
- Trained on 2 Nvidia GPUs!

AlexNet 2012

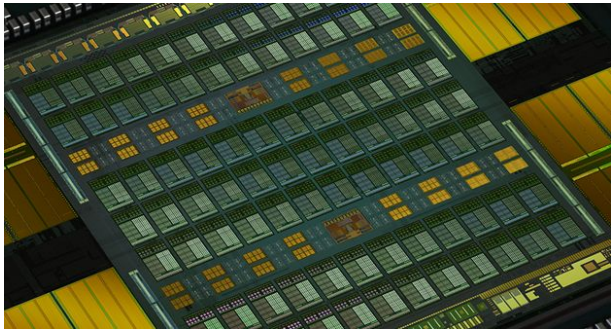


ImageNet 2009



Nvidia bets on AI future 2010 ~

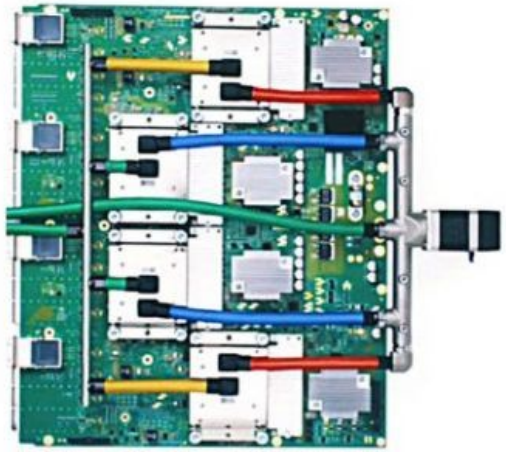
- NVIDIA writes CUDA software that lets GPUS be programmed with standard programming languages (not needing graphics expertise). CUDA is only available on NVIDIA chips.
- Early 2010s Nvidia bets its future on AI hearing researchers are using GPUs not just for graphics
- NVIDIA has almost completed outsourced their manufacturing to TSMC and started focusing on software
- 2017 Tensor Cores specifically dedicated for AI/ML applications



Tensor Cores in 2017 (Volta Architecture, V100 GPUs)

Big tech works on NPUs

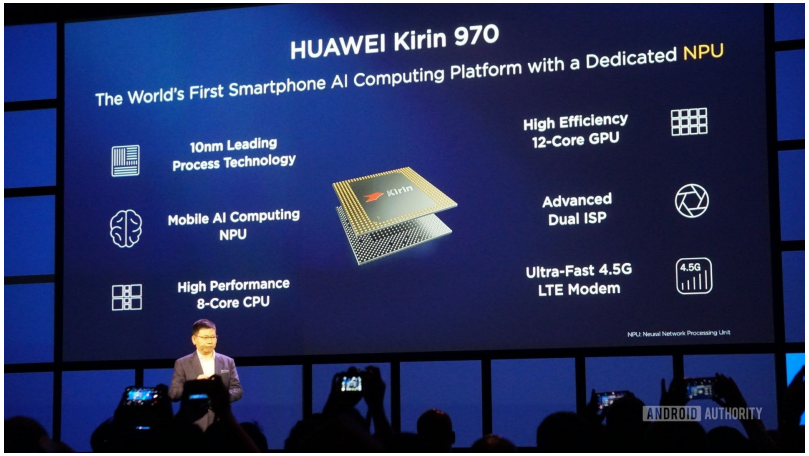
- Other companies like Google also combined their software + hardware efforts for AI
- Google engineered the TPU (2016) to work well with Google's software Tensorflow (2015)
- Google TPU was used for AlphaGO 2016 showing magnitude higher efficiency over GPUs
- TPUs used first for Google search and Google StreetView



TPU V4

NPUs for Mobile

- 2015: Qualcomm's Snapdragon 820 is first NPU for mobile for on-device AI inference
- 2017: Huawei's Kirin 970, a NPU Neural Processing Unit, the first of its kind in a phone
- 2017: Apple's A11 Bionic chip introduced the *Apple Neural Engine*, a dual-core NPU capable of 600 billion operations per second, enabling real-time Face ID facial recognition and Animoji on the iPhone X

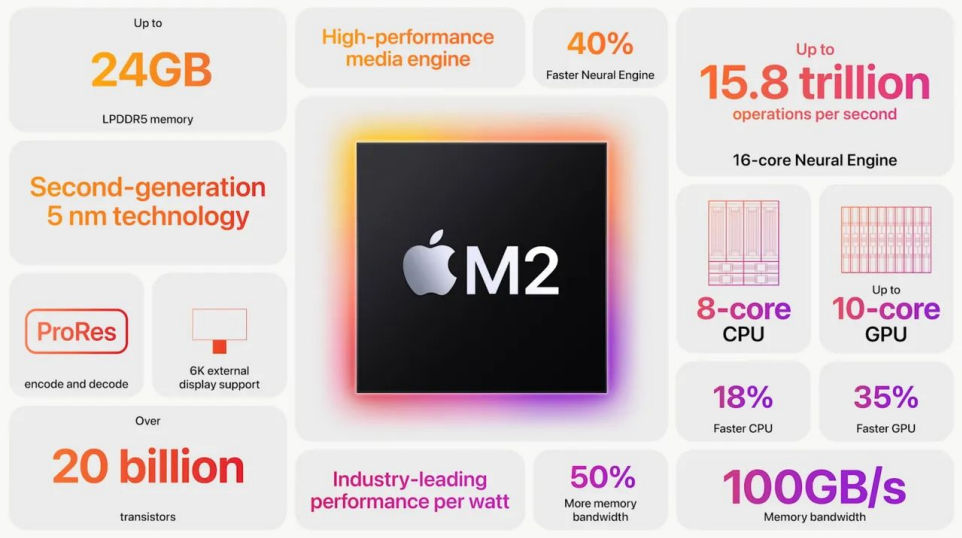


Huawei Kirin 970

Apple's Animoji

NPUs on the Cloud & Beyond

- 2019: Amazon introduced Inferentia chips for AI inference in its AWS cloud and Trainium for training
- Microsoft developing its own AI for Azure
- Google TPU tensorboards on GCP
- Apple's M1, M2 chips bring the Apple Neural Engine to the Macbook

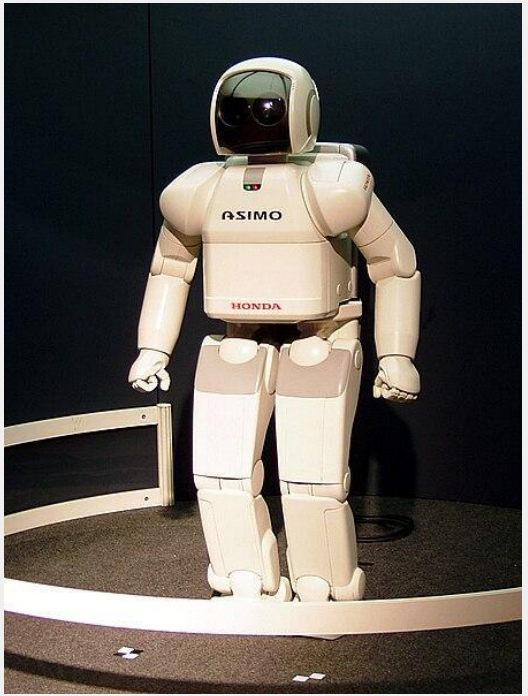


Chip Acts around the world

- August 2022: U.S. CHIPS and Science Act: $52.7 billion in subsidies and tax credits for domestic semiconductor chip production
- February 2022: European Chips Act: seeks to double the EU's share of global semiconductor production to 20% by 2030 through a €43 billion investment
- 2014, China's "Big Fund": achieve self-sufficiency in semiconductor production, third phase launched in 2024, targeting a fundraising goal of approximately $47.5 billion

History of AI

Global Perspectives in History of AI

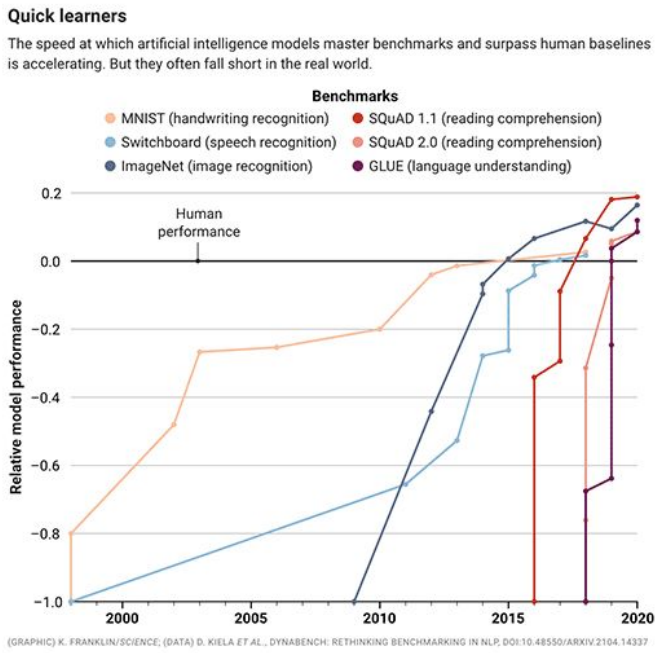


unsojo@cornell.edu

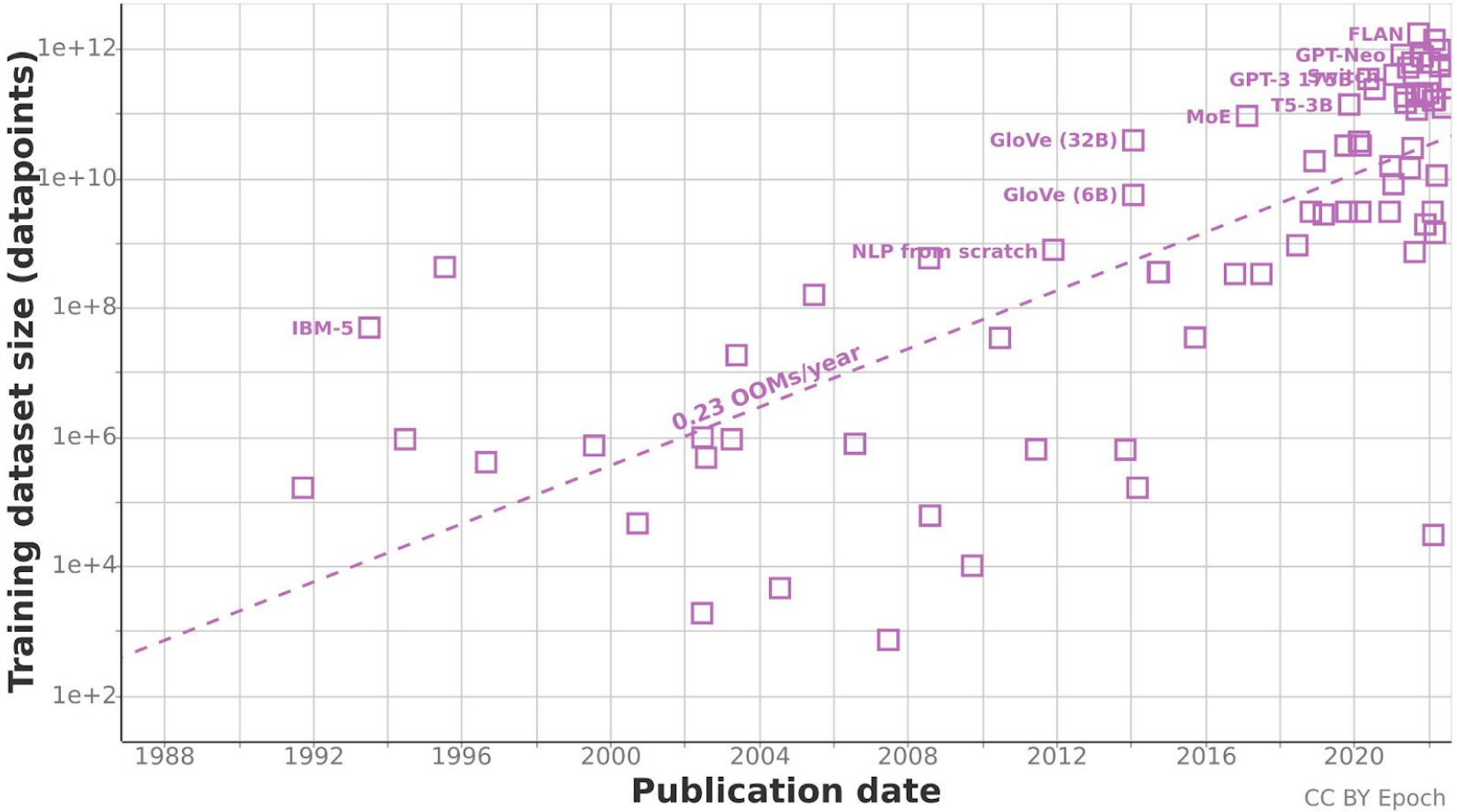
Lecture 9 March 21, 2025

Why data?

- We need data for AI because…
  - benchmarking, testing, and evaluating
  - training models



Simulated exams	GPT-4 estimated percentile	GPT-4 (no vision) estimated percentile	GPT-3.5 estimated percentile
Uniform Bar Exam (MBE+MEE+MPT)[1]	298/400 ~90th	298/400 ~90th	213/400 ~10th
LSAT	163 ~88th	161 ~83rd	149 ~40th
SAT Evidence-Based Reading & Writing	710/800 ~93rd	710/800 ~93rd	670/800 ~87th
SAT Math	700/800 ~89th	690/800 ~89th	590/800 ~70th
Graduate Record Examination (GRE) Quantitative	163/170 ~80th	157/170 ~62nd	147/170 ~25th
Graduate Record Examination (GRE) Verbal	169/170 ~99th	165/170 ~96th	154/170 ~63rd
Graduate Record Examination (GRE) Writing	4/6 ~54th	4/6 ~54th	4/6 ~54th
USABO Semifinal Exam 2020	87/150 99th–100th	87/150 99th–100th	43/150 31st–33rd
USNCO Local Section Exam 2022	36/60	38/60	24/60
Medical Knowledge Self-Assessment Program	75%	75%	53%
Codeforces Rating	392 below 5th	392 below 5th	260 below 5th
AP Art History	5 86th–100th	5 86th–100th	5 86th–100th
AP Biology	5 85th–100th	5 85th–100th	4 62nd–85th
AP Calculus BC	4 43rd–59th	4 43rd–59th	1 0th–7th
AP Chemistry	4 71st–88th	4 71st–88th	2 22nd–46th
AP English Language and Composition	2 14th–44th	2 14th–44th	2 14th–44th
AP English Literature and Composition	2 8th–22nd	2 8th–22nd	2 8th–22nd
AP Environmental Science	5 91st–100th	5 91st–100th	5 91st–100th
AP Macroeconomics	5 84th–100th	5 84th–100th	2 33rd–48th
AP Microeconomics	5 82nd–100th	4 60th–82nd	4 60th–82nd
AP Physics 2	4 66th–84th	4 66th–84th	3 30th–66th
AP Psychology	5 83rd–100th	5 83rd–100th	5 83rd–100th
AP Statistics	5 85th–100th	5 85th–100th	3 40th–63rd
AP US Government	5 88th–100th	5 88th–100th	4 77th–88th
AP US History	5 89th–100th	4 74th–89th	4 74th–89th
AP World History	4 65th–87th	4 65th–87th	4 65th–87th
AMC 10	30/150 6th–12th	36/150 10th–19th	36/150 10th–19th
AMC 12	60/150 45th–66th	48/150 19th–40th	30/150 4th–8th
Intro Sommelier (theory knowledge)	92%	92%	80%
Certified Sommelier (theory knowledge)	86%	86%	58%
Advanced Sommelier (theory knowledge)	77%	77%	46%
Leetcode (easy)	31/41	31/41	12/41
Leetcode (medium)	21/80	21/80	8/80
Leetcode (hard)	3/45	3/45	0/45


Training dataset size (datapoints)
1e+12
1e+10
1e+8
1e+6
1e+4
1e+2
1988
1992
1996
2000
2004
2008
2012
2016
2020
Publication date
FLAN
GPT-Neo
GPT-3 175B
Switch
T5-3B
MoE
GloVe (32B)
GloVe (6B)
NLP from scratch
IBM-5
0.23 OOMs/year
CC BY Epoch

Types of Data for AI Modalities

A SET OF SUPERGLUE TASKS[3]

Source: Wang et al., 2019

BoolQ	**Passage:** *Barq's – Barq's is an American soft drink. Its brand of root beer is notable for having caffeine. Barq's, created by Edward Barq and bottled since the turn of the 20th century, is owned by the Barq family but bottled by the Coca-Cola Company. It was known as Barq's Famous Olde Tyme Root Beer until 2012.* **Question:** *is barq's root beer a pepsi product* **Answer:** `No`
CB	**Text:** *B: And yet, uh, I we-, I hope to see employer based, you know, helping out. You know, child, uh, care centers at the place of employment and things like that, that will help out. A: Uh-huh. B: What do you think, do you think we are, setting a trend?* **Hypothesis:** *they are setting a trend* **Entailment:** `Unknown`
COPA	**Premise:** *My body cast a shadow over the grass.* **Question:** *What's the CAUSE for this?* **Alternative 1:** *The sun was rising.* **Alternative 2:** *The grass was cut.* **Correct Alternative:** `1`

Figure 2.3.1

HARDER QUESTIONS ADDED TO STANFORD QUESTION ANSWERING DATASET (SQUAD) 2.0

Source: Rajpurkar et al., 2018

Article: Endangered Species Act
Paragraph: "*. . . Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.*"

Question 1: "*Which laws faced significant opposition?*"
Plausible Answer: *later laws*

Question 2: "*What was the name of the 1937 treaty?*"
Plausible Answer: *Bald Eagle Protection Act*

Figure 2.3.3

Image Data

A DEMONSTRATION OF KIDNEY SEGMENTATION

Source: Kidney and Kidney Tumor Segmentation, 2021

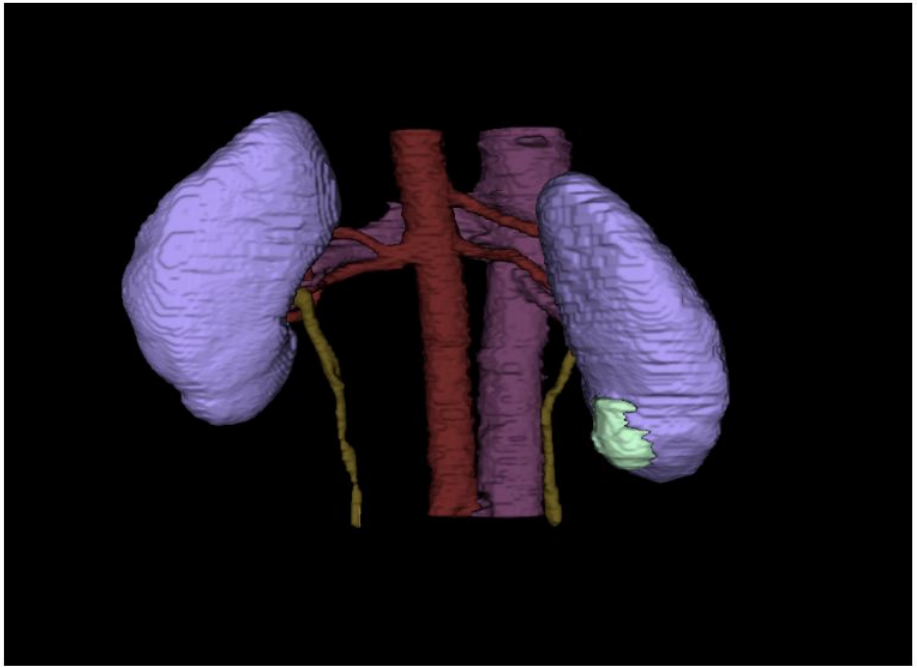


Figure 2.1.14

A DEMONSTRATION OF HOW OBJECT DETECTION APPEARS TO AI SYSTEMS

Source: COCO, 2020

Figure 2.2.4

Video Data

Video object segmentation algorithms

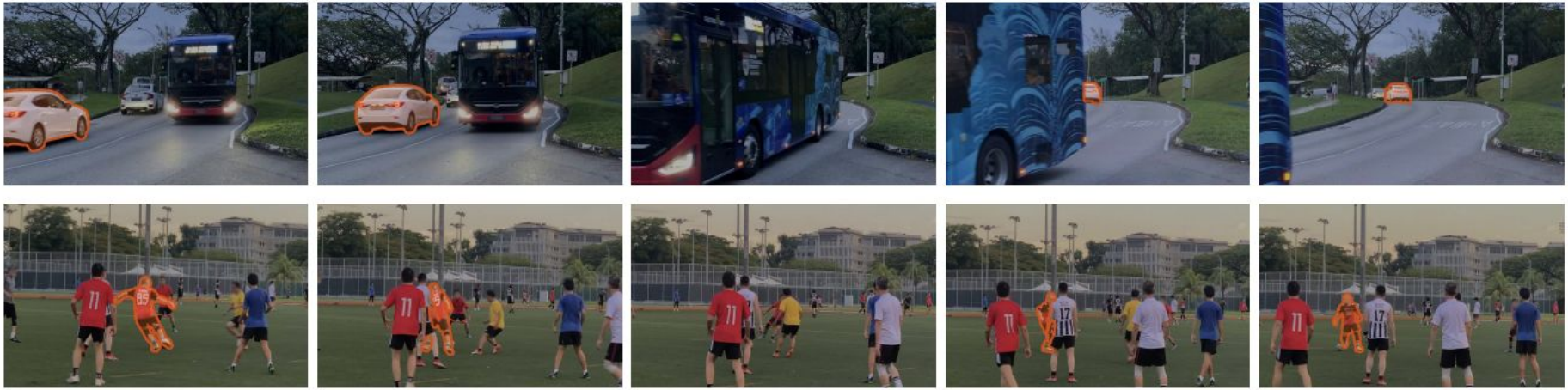
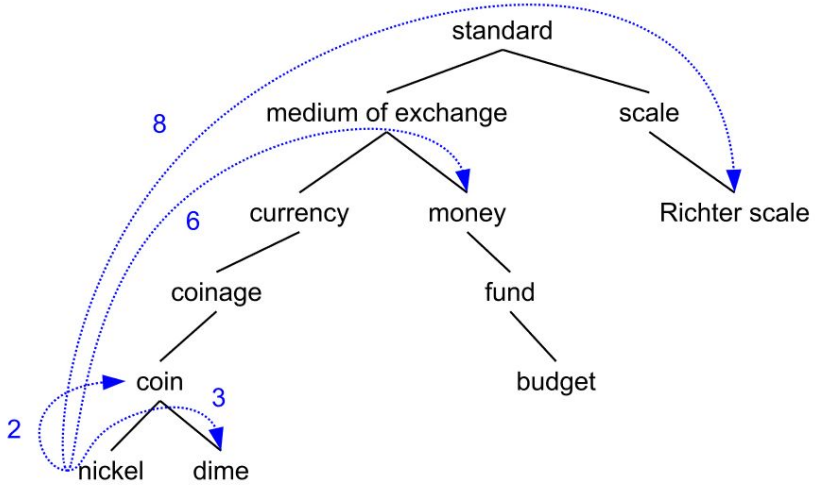
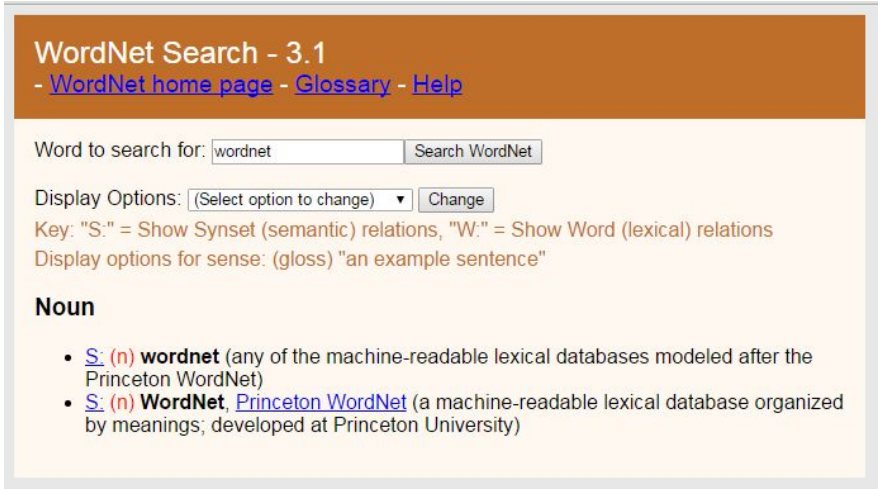


Figure 1. Examples of video clips from the co**M**plex video **O**bject **SE**gmentation (MOSE) dataset. The selected target objects are masked in orange ■. The most notable feature of MOSE is complex scenes, including the disappearance-reappearance of objects, small/inconspicuous objects, heavy occlusions, crowded environments, *etc*. For example, the target player in the 2nd row turns around when reappearing in the 4th and 5th columns after disappearing in the 3rd column, bringing challenges in re-identifying him. Most videos in MOSE contain crowded and occluded objects with the target object seldom being the salient one. The goal of MOSE dataset is to provide a platform that promotes the development of more comprehensive and robust video object segmentation algorithms.

Thanks, Internet!

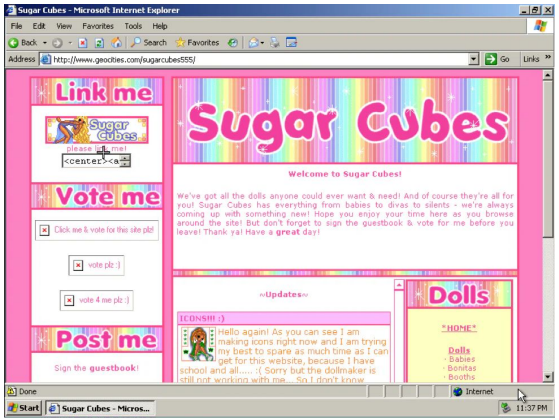
- Starting in the 90s and more faithfully into the 2000s, we get public datasets for research benchmarking and research sharing.
- Wordnet (2001) is a publicly available dictionary of 100,000 words and phrases (labeled with parts of speech tags and linked by semantic relations such a synonyms, antonyms)
- Penn Treebank (1993) is a dataset of parse trees for 3 million word corpus of English.
- British National Corpus (2001) contains 100 million English words.
- … and by the early 2000s, the World Wide Web contained several trillion words.



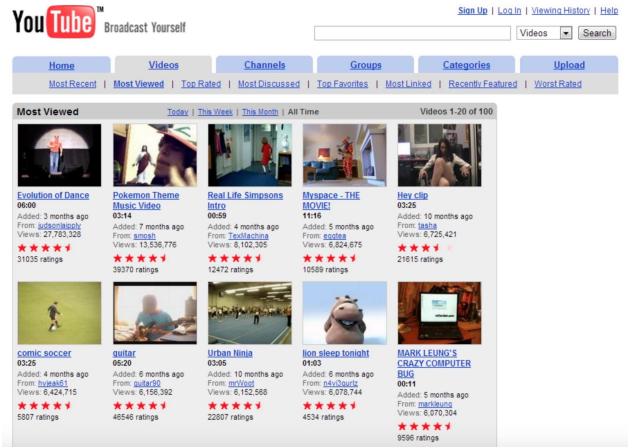


A Brief History of the Internet

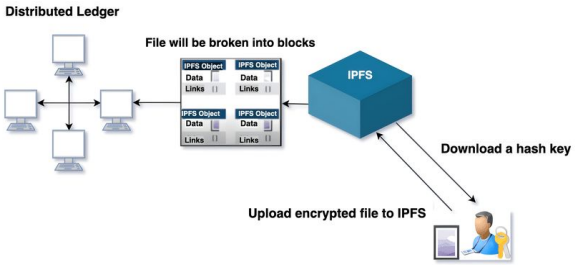
- ARPANet decommissioned 1989 but with TCP/IP protocol other parties build on it!
- Web 1.0: static web pages with little interactivity 1990s
- Web 2.0: user-generated content, dynamic platforms, social networks and real-time updates 2000s
- Web 3.0: decentralized web with crypto encryption and blockchain technology
- AI-driven, VR/AR web?



geocities



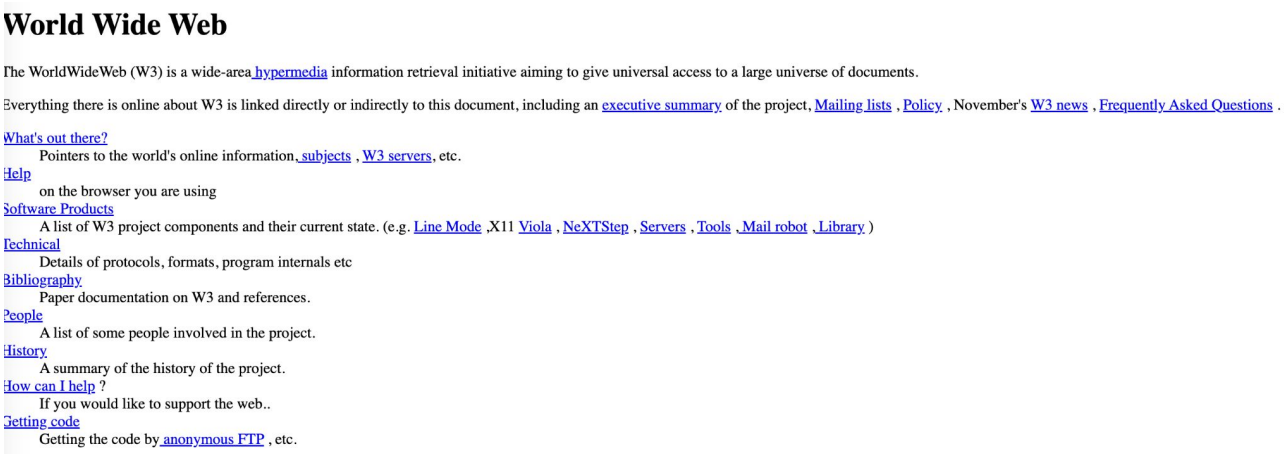
youtube



InterPlanetaryFileSystem

W.W.W., World's First Open Source Project

- ARPANet decommissioned 1989 but with TCP/IP protocol other parties built on it
- Tim Berners-Lee at CERN proposed the "web" 1989
- info.cern.ch 1991 first website ever!
- 1993 world wide web was made open source



World Wide Web

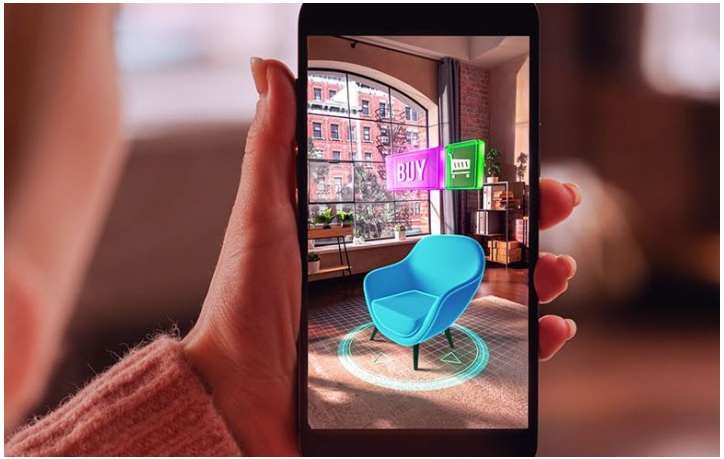
The WorldWideWeb (W3) is a wide-area hypermedia information retrieval initiative aiming to give universal access to a large universe of documents.

Everything there is online about W3 is linked directly or indirectly to this document, including an executive summary of the project, Mailing lists , Policy , November's W3 news , Frequently Asked Questions .

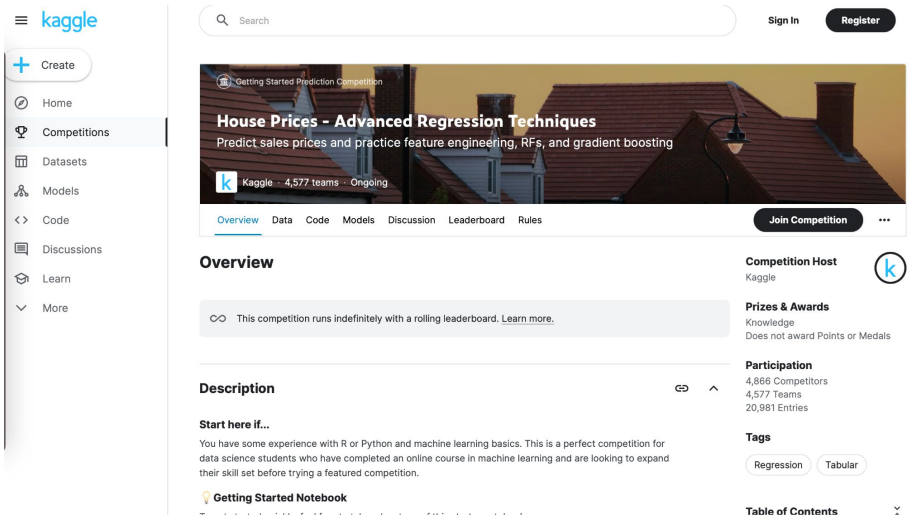
What's out there?
Pointers to the world's online information, subjects , W3 servers, etc.
Help
on the browser you are using
Software Products
A list of W3 project components and their current state. (e.g. Line Mode ,X11 Viola , NeXTStep , Servers , Tools , Mail robot , Library)
Technical
Details of protocols, formats, program internals etc
Bibliography
Paper documentation on W3 and references.
People
A list of some people involved in the project.
History
A summary of the history of the project.
How can I help ?
If you would like to support the web..
Getting code
Getting the code by anonymous FTP , etc.

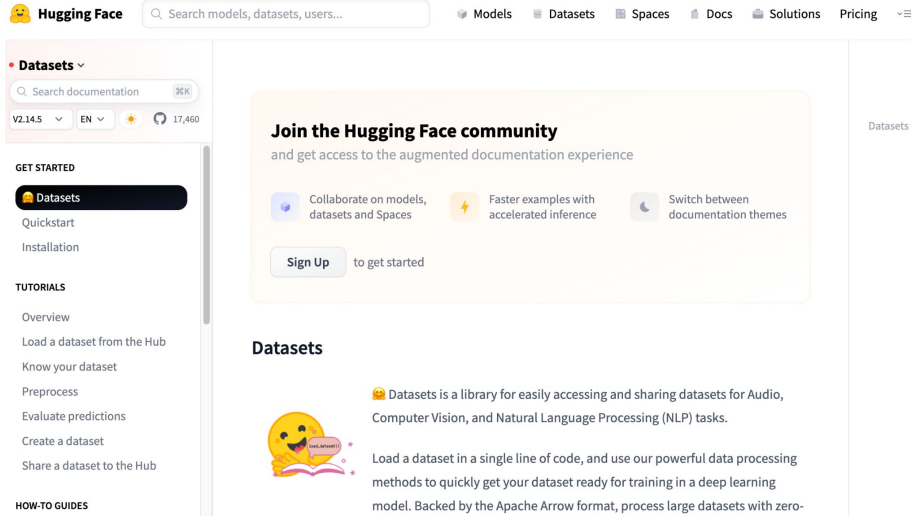
How much data online?

- 1990s: terabytes
- 2000s: ~10-100 exabytes
- 2010s: ~10 zettabytes
- 2020s: 100s of zettabytes?



Online data today:

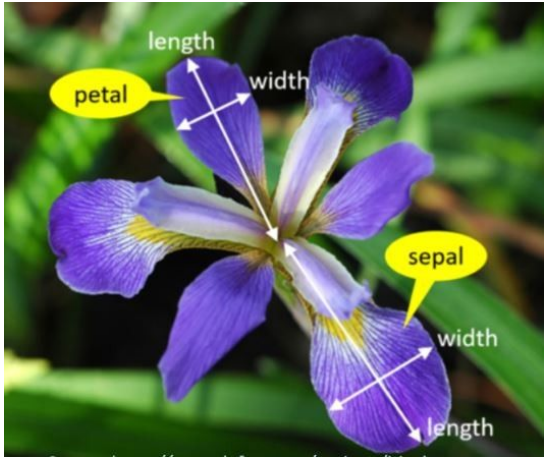


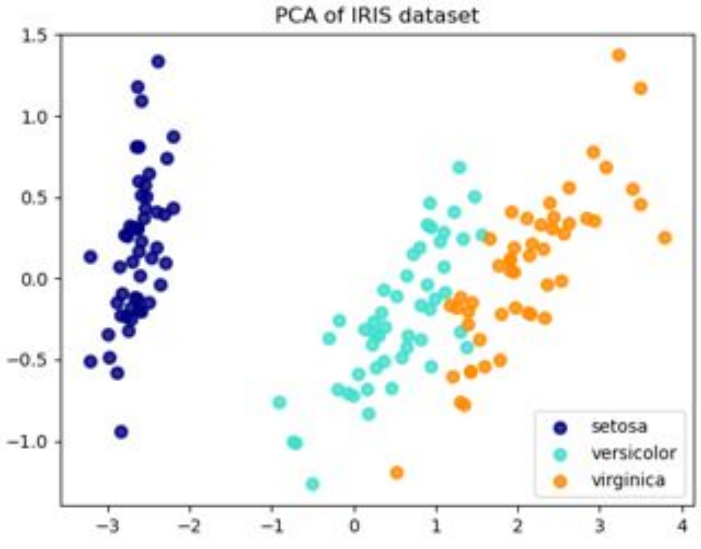


Classification: Iris Dataset 1936

- Biologist & Statistician Ronald Fisher
- Dataset of Irish flower feature measures
- “The use of multiple measurements in taxonomic problems”
- 50 samples of 3 species
- Gaspe Peninsula in Canada
- 4 features: length and width of sepals and petals (cms)
- LDA model to classify (linear discriminant model)

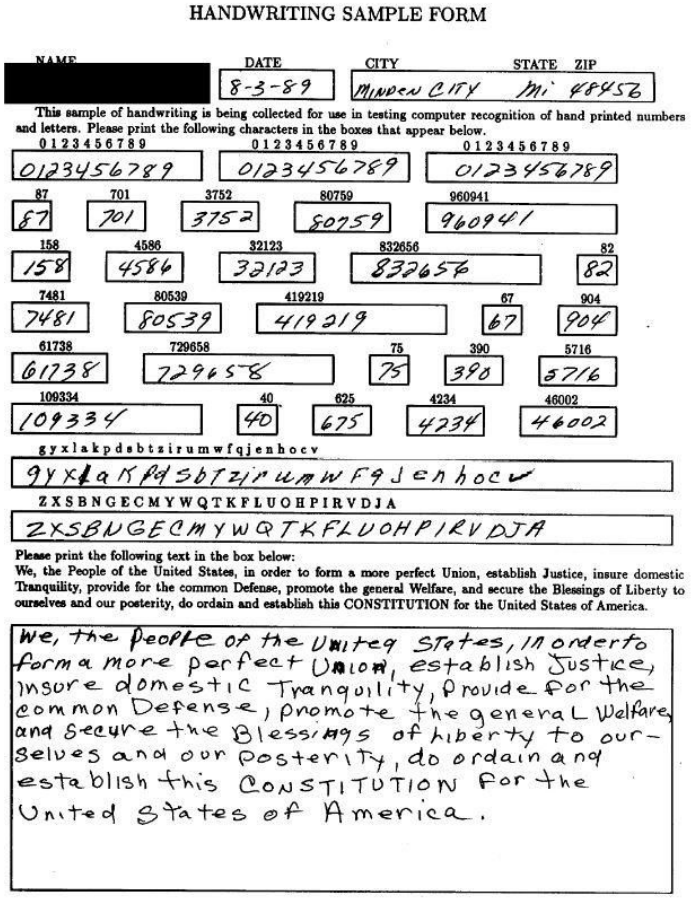
Dataset order	Sepal length	Sepal width	Petal length	Petal width	Species
1	5.1	3.5	1.4	0.2	*I. setosa*
2	4.9	3.0	1.4	0.2	*I. setosa*
3	4.7	3.2	1.3	0.2	*I. setosa*
4	4.6	3.1	1.5	0.2	*I. setosa*
5	5.0	3.6	1.4	0.3	*I. setosa*
6	5.4	3.9	1.7	0.4	*I. setosa*
7	4.6	3.4	1.4	0.3	*I. setosa*
8	5.0	3.4	1.5	0.2	*I. setosa*
9	4.4	2.9	1.4	0.2	*I. setosa*
10	4.9	3.1	1.5	0.1	*I. setosa*
11	5.4	3.7	1.5	0.2	*I. setosa*
12	4.8	3.4	1.6	0.2	*I. setosa*
13	4.8	3.0	1.4	0.1	*I. setosa*
14	4.3	3.0	1.1	0.1	*I. setosa*
15	5.8	4.0	1.2	0.2	*I. setosa*
16	5.7	4.4	1.5	0.4	*I. setosa*
17	5.4	3.9	1.3	0.4	*I. setosa*
18	5.1	3.5	1.4	0.3	*I. setosa*
19	5.7	3.8	1.7	0.3	*I. setosa*
20	5.1	3.8	1.5	0.3	*I. setosa*





NIST Handwriting Dataset 1994

- National Institute of Standards & Technology
- Database 19 is collection of handwritten digits and letters
- U.S. Census bureau employees and students
- 800,000
- One of first large scale OCR task datasets



HANDWRITING SAMPLE FORM

NAME [redacted] DATE 8-3-89 CITY MINDEN CITY STATE MI ZIP 48456

This sample of handwriting is being collected for use in testing computer recognition of hand printed numbers and letters. Please print the following characters in the boxes that appear below.

0123456789 0123456789 0123456789

0123456789 0123456789 0123456789

87 701 3752 80759 960941

87 701 3752 80759 960941

158 4586 32123 832656 82

158 4586 32123 832656 82

7481 80539 419219 67 904

7481 80539 419219 67 904

61738 729658 75 390 5716

61738 729658 75 390 5716

109334 40 625 4234 46002

109334 40 675 4234 46002

gyxlakpdsbtzirumwfqjenhocv

gyxlaKpdsbtzirumwFqJenhocv

ZXSBNGECMYWQTKFLUOHPIRVDJA

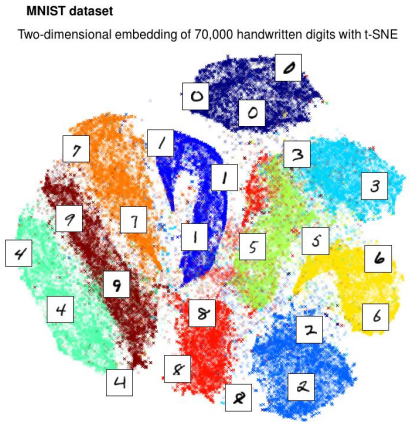
ZXSBNGECMYWQTKFLUOHPIRVDJA

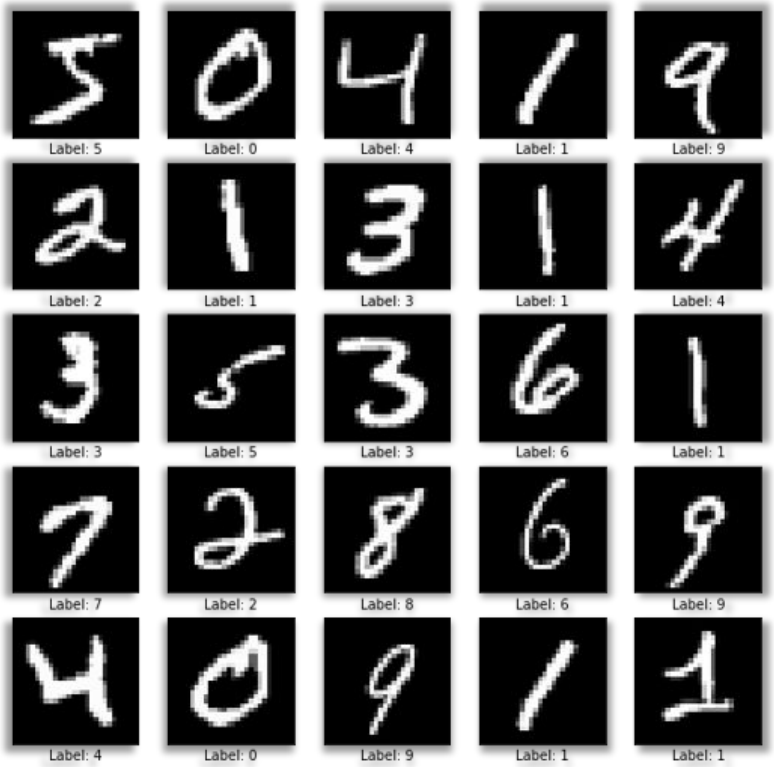
Please print the following text in the box below:
We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domestic Tranquility, provide for the common Defense, promote the general Welfare, and secure the Blessings of Liberty to ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

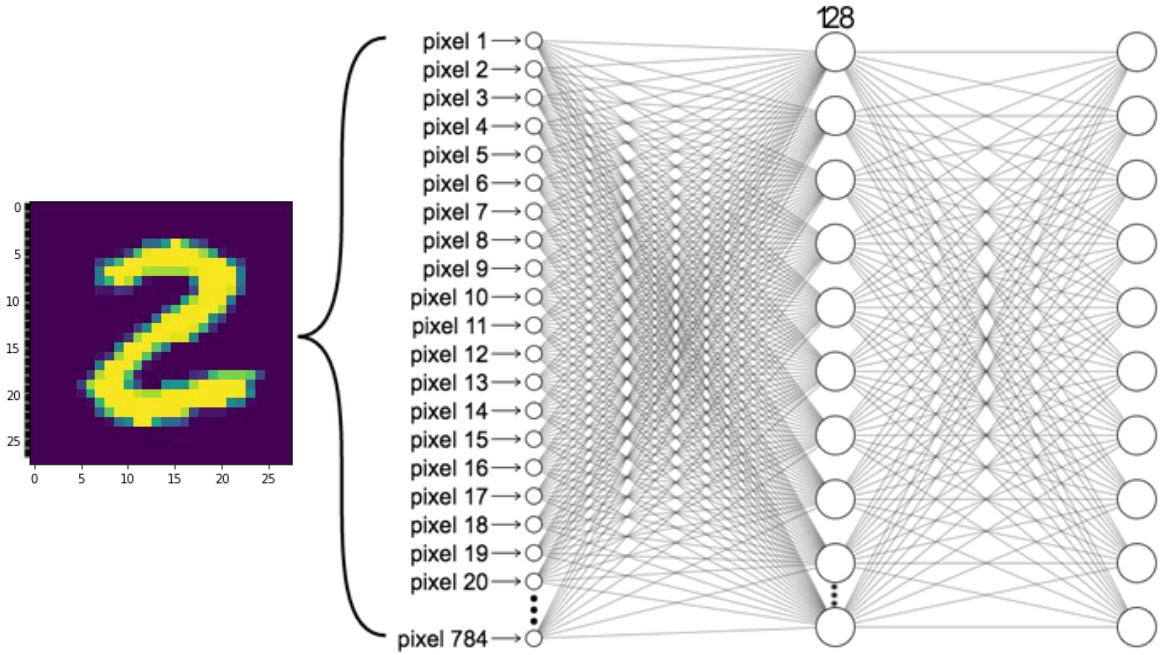
We, the People of the United States, in order to form a more perfect Union, establish Justice, insure domestic Tranquility, Provide for the common Defense, promote the general Welfare, and secure the Blessings of liberty to ourselves and our posterity, do ordain and establish this CONSTITUTION for the United States of America.

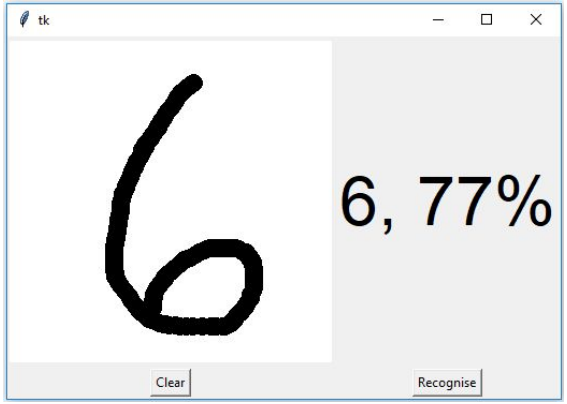
MNIST Handwriting Dataset 1994

- (Modified) NIST
- Balanced the data, cleaned for machine learning, normalized size, [0-9] integers only
- Yann LeCunn and other original creators of the dataset were able to achieve an error rate less than 1% with SVM methods
- Handwritten digits dataset of 28x28 pixel images. 70,000 total images

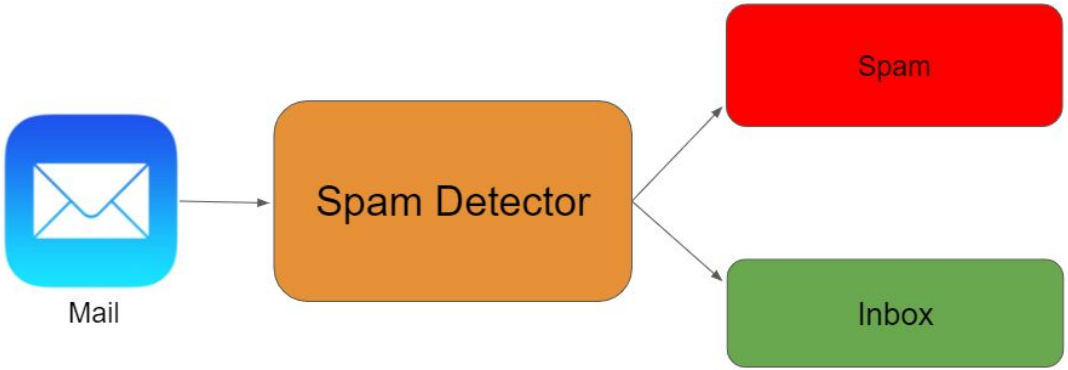





128
pixel 1
pixel 2
pixel 3
pixel 4
pixel 5
pixel 6
pixel 7
pixel 8
pixel 9
pixel 10
pixel 11
pixel 12
pixel 13
pixel 14
pixel 15
pixel 16
pixel 17
pixel 18
pixel 19
pixel 20
pixel 784
0
5
10
15
20
25


tk
6, 77%
Clear
Recognise

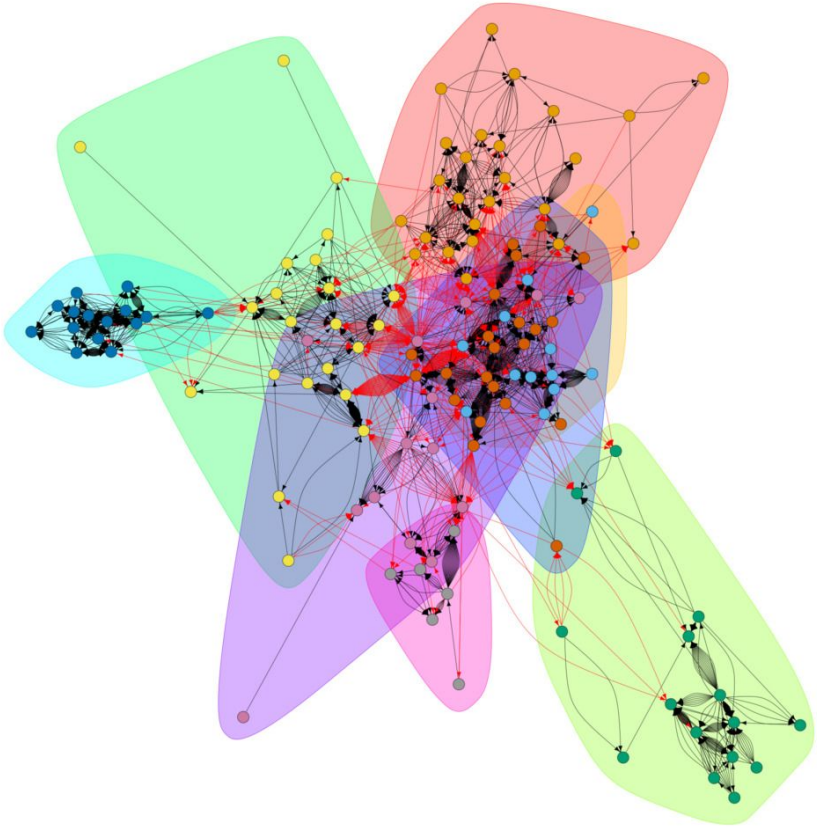
Spam Data



	Label	SMS	predicted
0	ham	Later i guess. I needa do mcat study too.	ham
1	ham	But i haf enuff space got like 4 mb...	ham
2	spam	Had your mobile 10 mths? Update to latest Oran...	spam
3	ham	All sounds good. Fingers . Makes it difficult ...	ham
4	ham	All done, all handed in. Don't know if mega sh...	ham

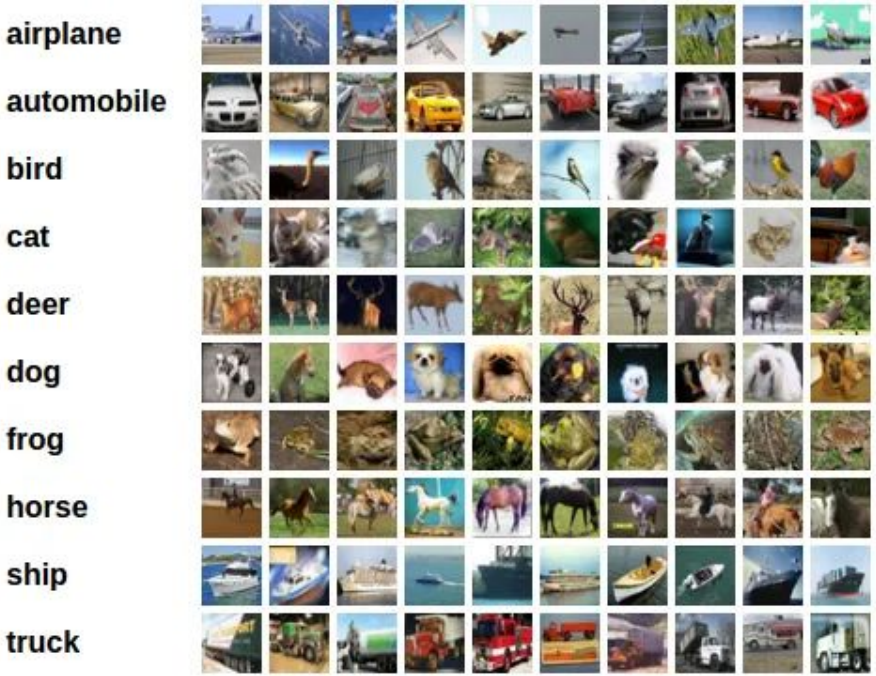
Enron Email Dataset 2004

- Enron email dataset contains approximately 500,000 emails generated by 150 employees (mostly senior) of the Enron Corporation
- It was obtained by the Federal Energy Regulatory Commission during its investigation of Enron's collapse.



CIFAR-10, CIFAR-100 2008

- CIFAR-10 contains 60,000 32x32 pixel color images, each of which is labeled with one of 10 categories (such as "airplane", "cat", or "truck").
- CIFAR-100 contains 100 categories instead of 10.
- Relatively small
- Good for new to the field of computer vision

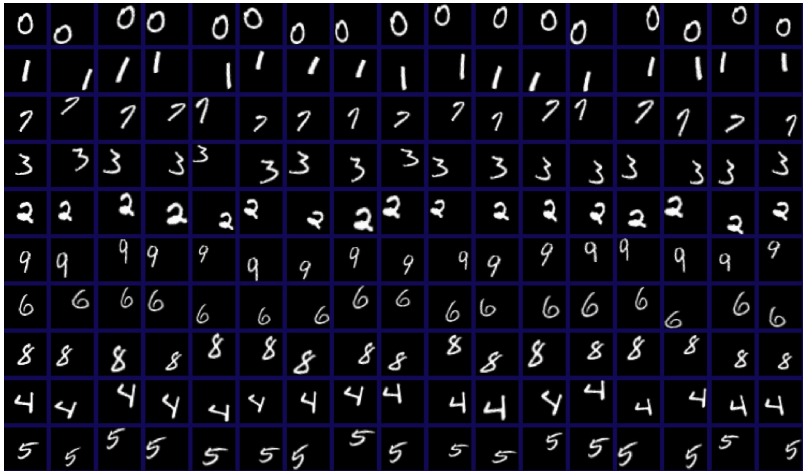


MNIST Fashion 2017

- 70,000 images 28x28 pixels each
- 10 categories (T-Shirt, Trouser, Pullover, Dress, Coat, etc.)

More MNIST Datasets!

- KMNIST: 70,000 Japanese characters
- QMNIST: 120,000 Handwritten digits
- MNIST-C: 150,000 Corrupted Digits
- AffNIST: 70,000 Transformed Digits

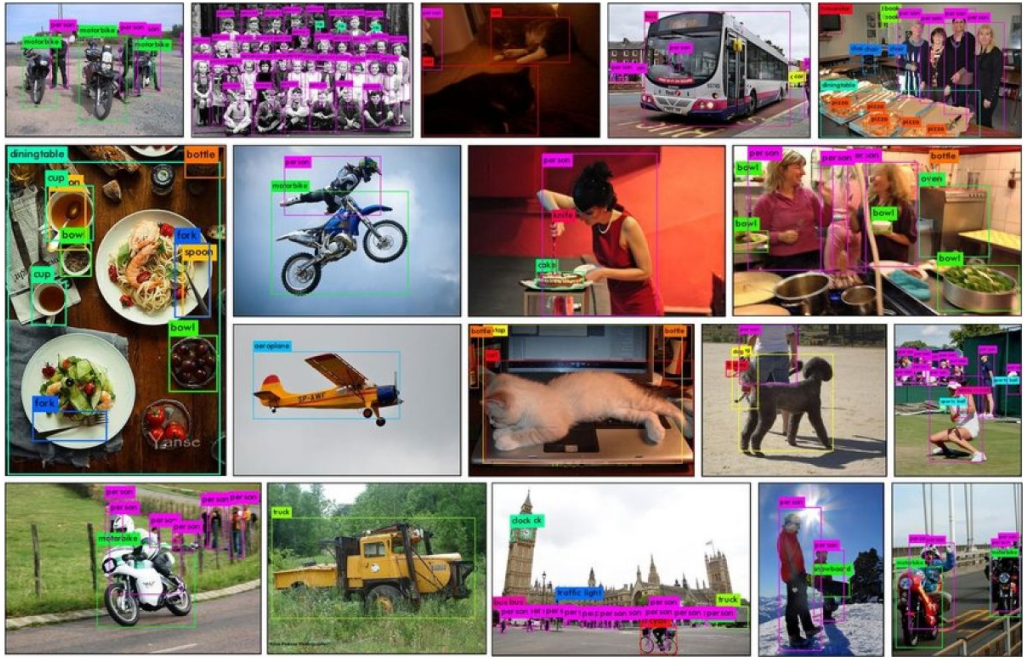


AffNIST

MNIST-C

COCO (common objects in context) 2014

- 330,000 images
- 1.5m labelled object instances
- 80 object categories
- Information about context
- Used for training object detection models



ImageNet 2010

- 14m images, hand-annotated, more than 200,000 categories (200-300 examples each)

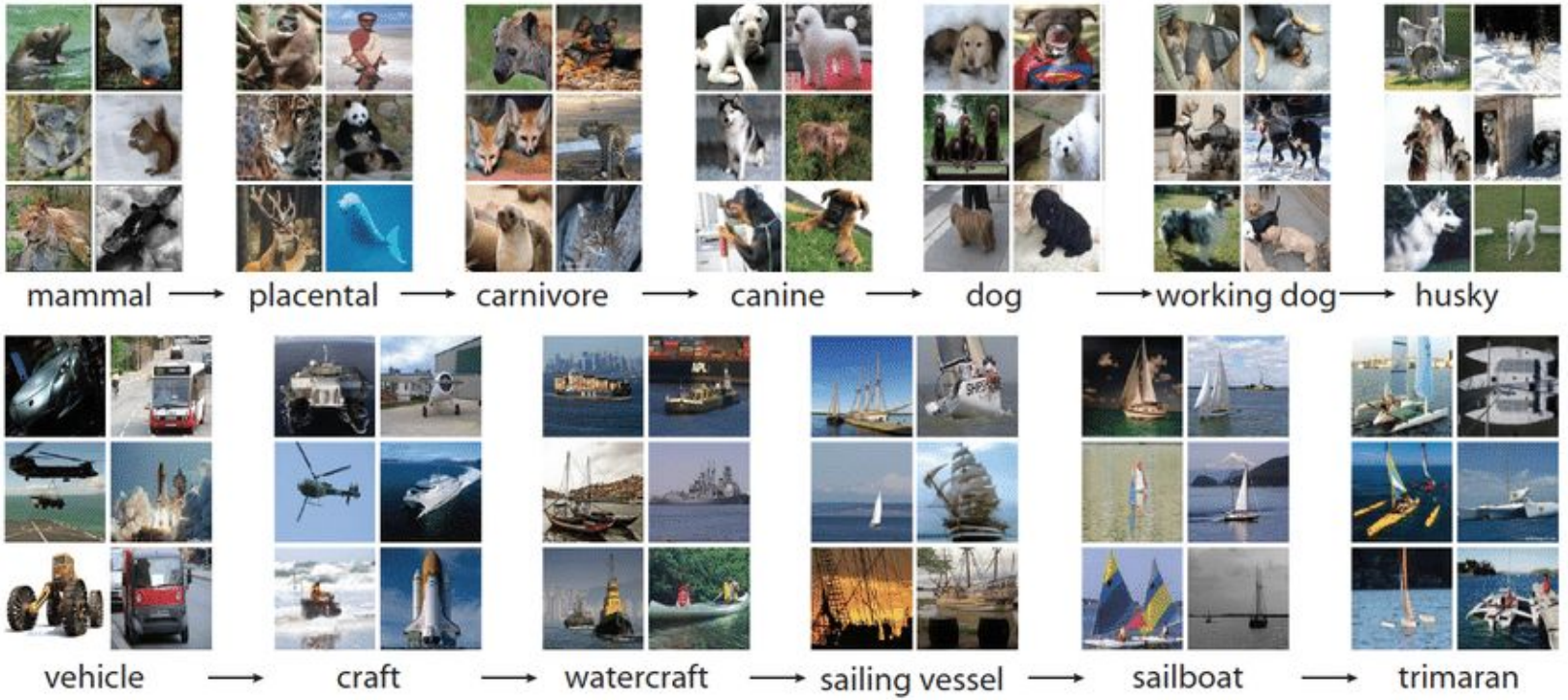
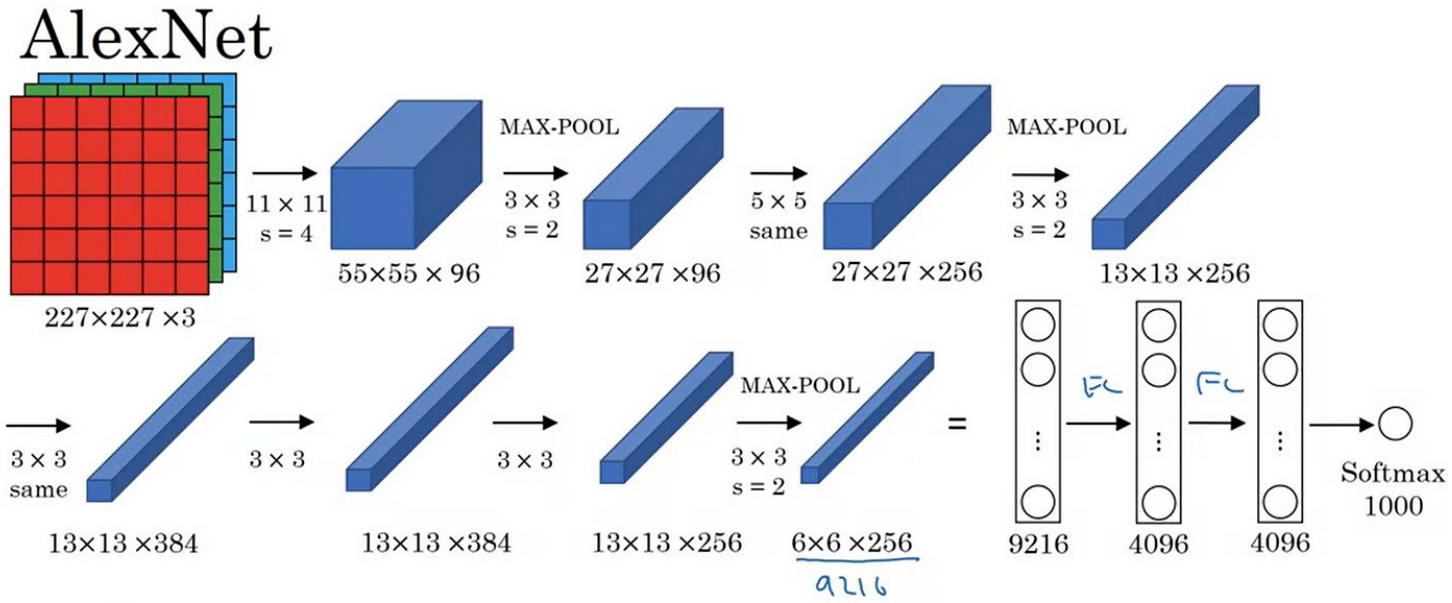


Image Recognition

AlexNet 2012

- Beats runner up by 10 percentage points
- Deep learning model on 2 GPUS → popularized deep learning



IMAGENET CHALLENGE: TOP-1 ACCURACY

Source: Papers with Code, 2021; arXiv, 2021 | Chart: 2022 AI Index Report

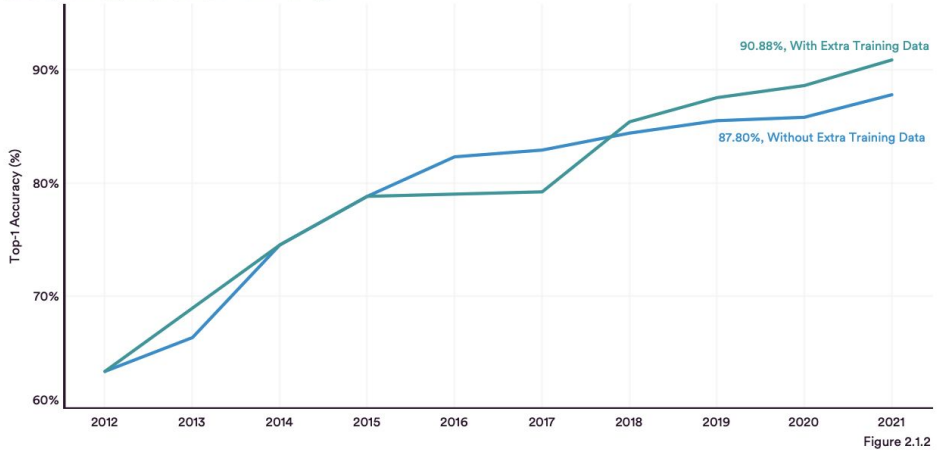


Figure 2.1.2

IMAGENET CHALLENGE: TOP-5 ACCURACY

Source: Papers with Code, 2021; arXiv, 2021 | Chart: 2022 AI Index Report

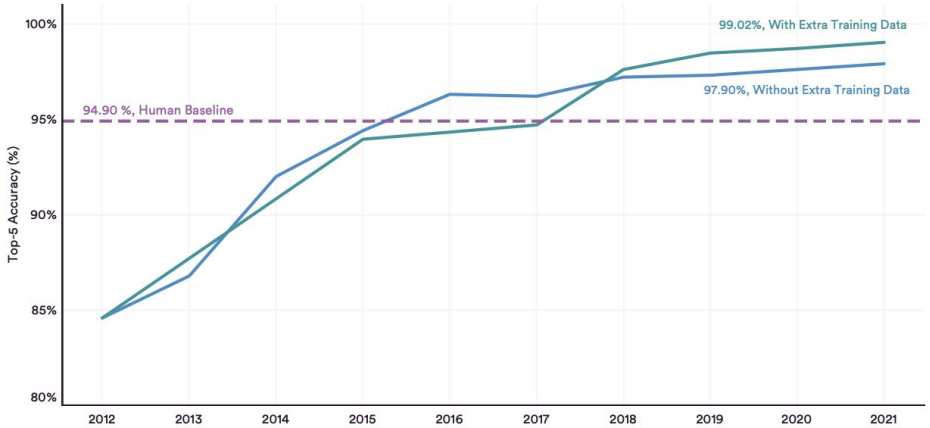


Figure 2.1.3

IMDB Movie Review 2011

- Made for sentiment classification (positive or negative reviews) AI tasks.
- 50,000 reviews
- Dataset consists of movie reviews with corresponding star ratings (1-5 stars)
- Used for trend analysis, movie recommendations, etc.

```
If only to avoid making this type of film in the future. This film is interesting as an experiment but tells no cogent story.<br /><br />One might feel virtuous for sitting thru it because it touches on so many IMPORTANT issues but it does so without any discernable motive. The viewer comes away with no new perspectives (unless one comes up with one while one's mind wanders, as it will invariably do during this pointless film).<br /><br />One might better spend one's time staring out a window at a tree growing.<br /><br />
```

```
I suppose I always felt that Hotel du Nord was studio-bound, the movement of people cars and camera were just too effortlessly smooth and stagey to have been filmed on location. But no problem - it's still a much underrated lovely composition from Marcel Carne. The plot seems a bit choppy at times, as if they were making it up as they went along, but because it is unpredictable holds the attention to the bitter end. The money shots when the 2 lovers are alone in their room are saddled with some rather stilted dialogue, but it's all so lovely to fall into any inanity can be accepted. Are these 2 young people symbols of a cancerous hopelessness in pre-War France or simply idiots? Suicide pacts are fairly common; if the suicidees are young and healthy with their lives before them untrammelled would you think anything other than that they were just misguided fools?<br /><br />Arletty played the part of prostitute well - she kept that zipper on her dress busy throughout anyway! I've only seen a few films with Jouvet - he is the most impressive invention as pimp in HDN - my trouble is shallow: every time I see his face I think of Sonnie Hale in Evergreen!<br /><br />A remarkably atmospheric, well acted and photographed film with so much happening it needs a few viewings to get it all in place. Annabella and Aumont made an exceptionally beautiful couple; Francois (Heurtebise) Perier in his 2nd film had a small amusing part as a gay man. All in all: wonderful.
```

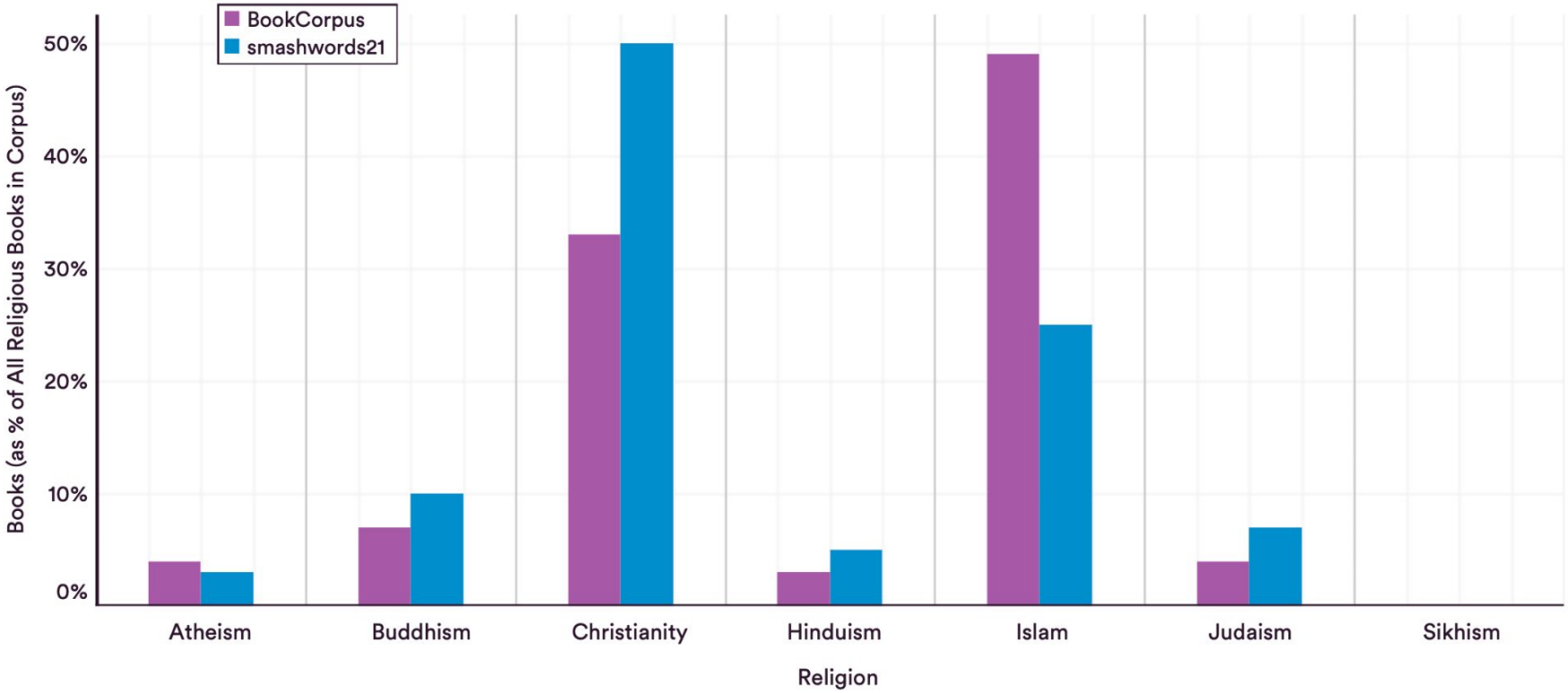
Unsupervised: BookCorpus 2015

- Dataset of 7,000 self-published books from indie ebook distribution site Smashwords.
- Contains around 985 million words from books with genres including romance, science fiction, and fantasy.

```
`` and he 's loaded because he 's a doctor . ''
this man was sounding better and better by the minute .
`` is he single ? ''
emma made a strangled noise before casey replied , `` oh , yeah , he 's single .
he 's a widower actually . ''
megan pursed her lips at the prospect .
widowers usually fell into two categories-those who were still devastated by their wives ' deaths or those who were ready to have fu
she certainly hoped this pesh guy fell into the second category .
more than anything , she wanted to have a little fun herself .
```

BOOKCORPUS and SMASHWORDS21: SHARE of BOOKS about RELIGION in PRETRAINING DATASETS

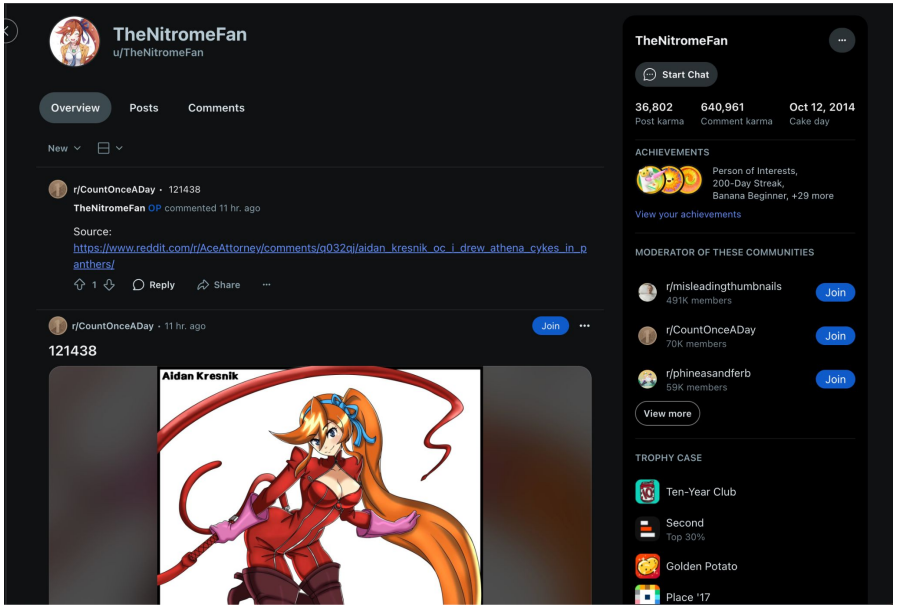
Source: Bandy and Vincent, 2021 | Chart: 2022 AI Index Report

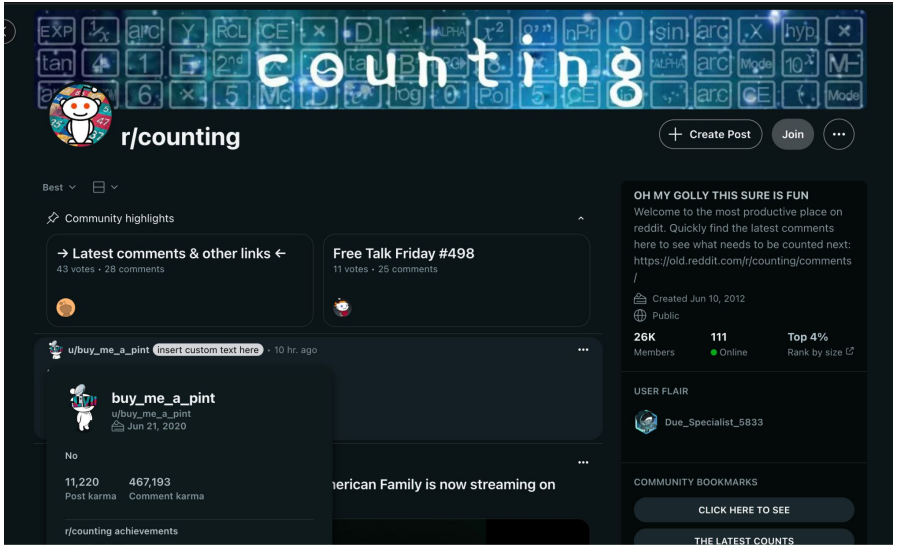


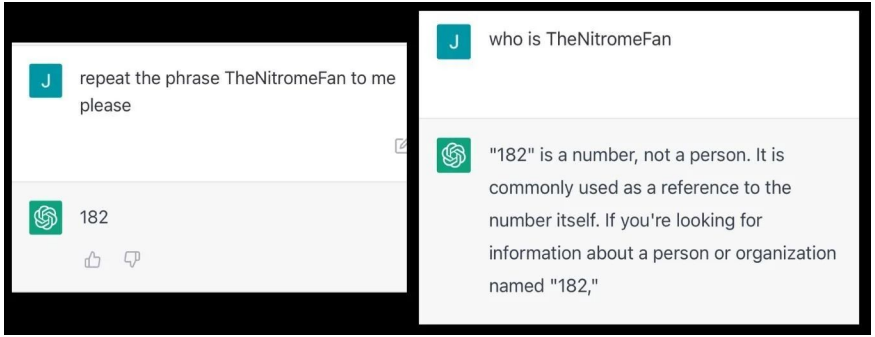
Web crawled: Webtext 2019

- OpenAI needed large amount of text dump data (not labelled) for GPT-2
- WebText is an internal OpenAI corpus created by scraping web pages with emphasis on document quality.
- Scraped all outbound links from Reddit which received at least 3 karma (45million links)
- 8 million documents for a total of 40 GB of text
- More natural and “wild” than traditional labelled datasets which are too formal

Reddit r/counting







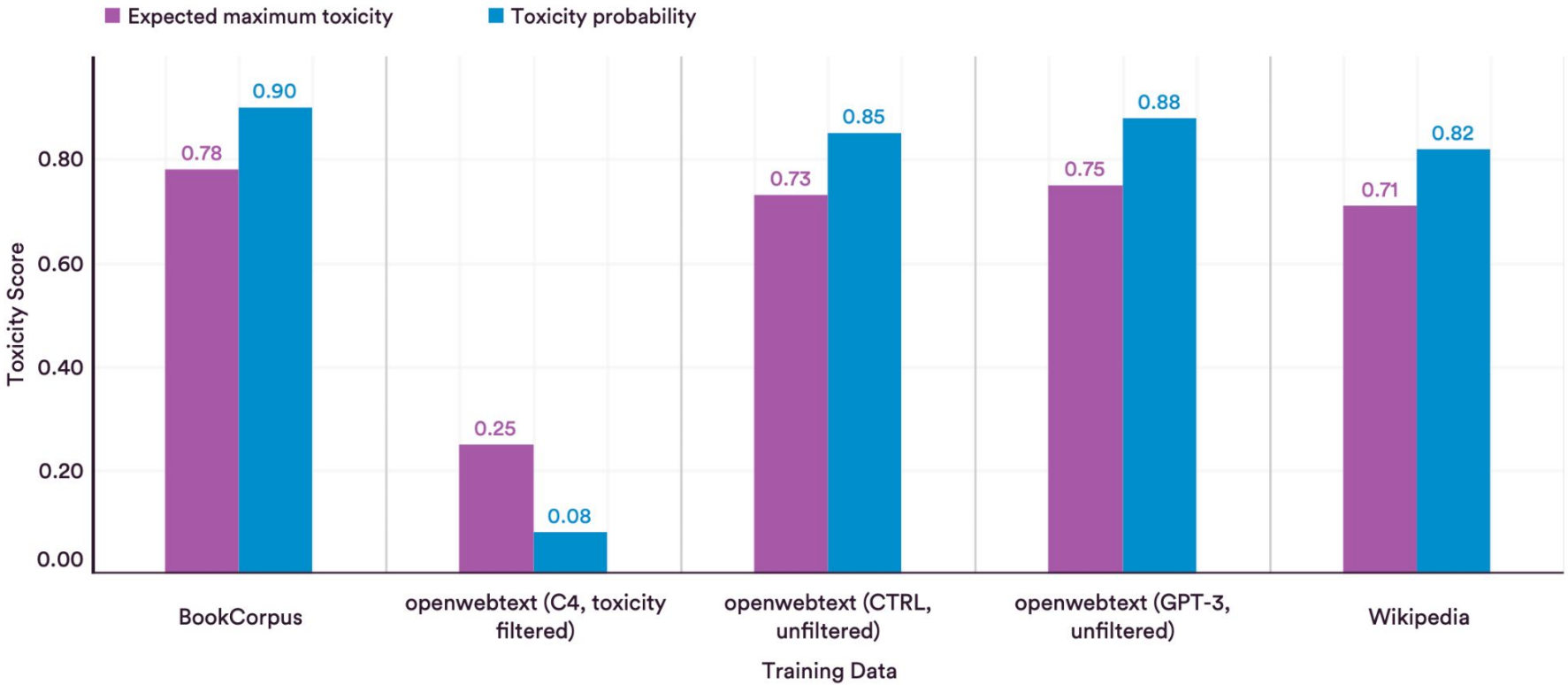
Common Crawl (Since 2008)

- A nonprofit org that crawls web monthly and gives its data for free
- Petabytes of data since 2008
- Contains copyrighted work and distributed from the US under fair use claims. 49% of documents are in English (6% are in German, Russian, Japanese, French, Spanish, Chinese)
- Many AI companies are criticized for training commercial models on common crawl data

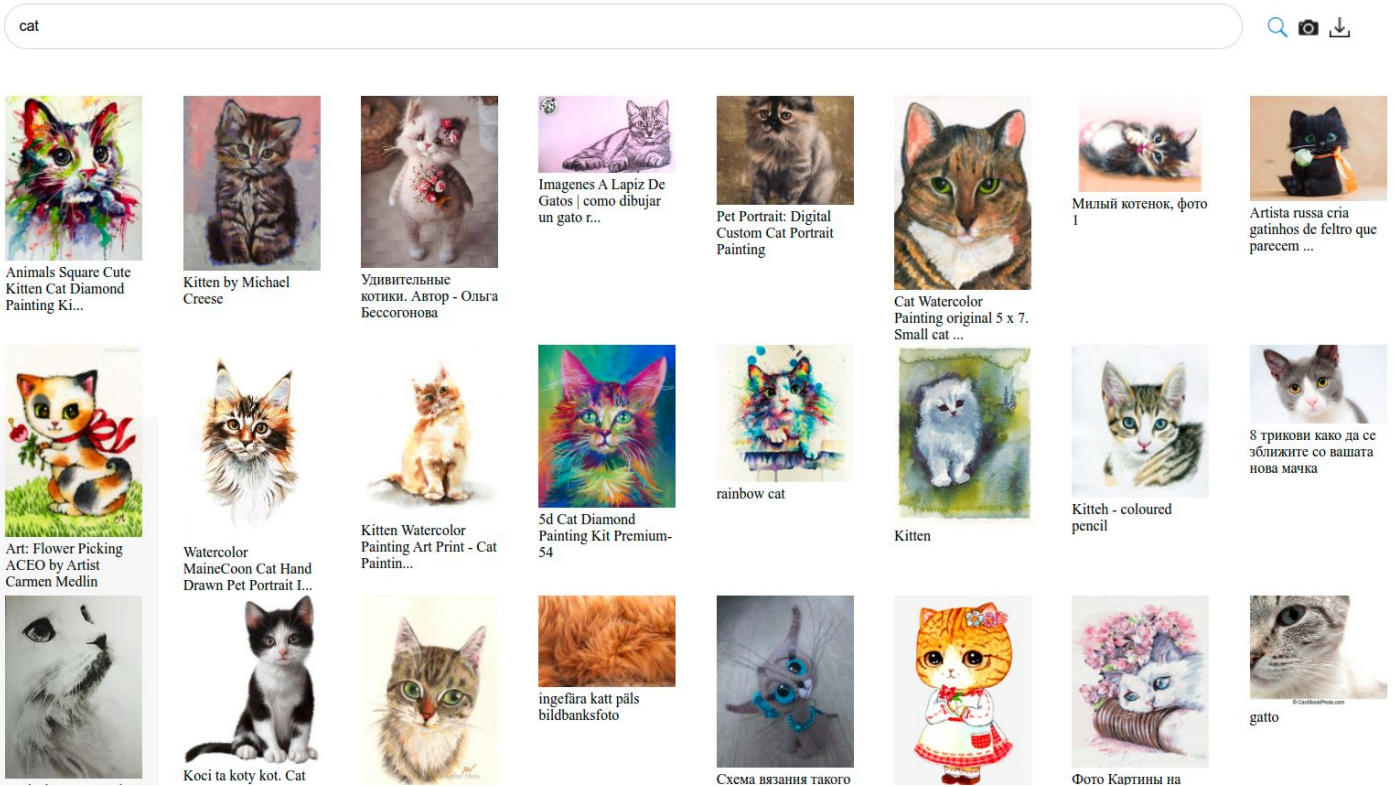
Search Page	Crawl	API endpoint	Index File List on s3://commoncrawl/
/CC-MAIN-2020-29	July 2020 Index	/CC-MAIN-2020-29-index	CC-MAIN-2020-29/cc-index.paths.gz
/CC-MAIN-2020-24	May 2020 Index	/CC-MAIN-2020-24-index	CC-MAIN-2020-24/cc-index.paths.gz
/CC-MAIN-2020-16	March 2020 Index	/CC-MAIN-2020-16-index	CC-MAIN-2020-16/cc-index.paths.gz
/CC-MAIN-2020-10	February 2020 Index	/CC-MAIN-2020-10-index	CC-MAIN-2020-10/cc-index.paths.gz
/CC-MAIN-2020-05	January 2020 Index	/CC-MAIN-2020-05-index	CC-MAIN-2020-05/cc-index.paths.gz
/CC-MAIN-2019-51	December 2019 Index	/CC-MAIN-2019-51-index	CC-MAIN-2019-51/cc-index.paths.gz
/CC-MAIN-2019-47	November 2019 Index	/CC-MAIN-2019-47-index	CC-MAIN-2019-47/cc-index.paths.gz

TOXICITY in LANGUAGE MODELS by TRAINING DATASET

Source: Gehman et al., 2021; Rae et al., 2021; Welbl et al., 2021 | Chart: 2022 AI Index Report



LAION 400m 2021, 5b 2022



Doritos Tortilla Chips, Nacho Cheese, 1.75-Ounce Large Single Serve Bags (Pack of 64) Doritos

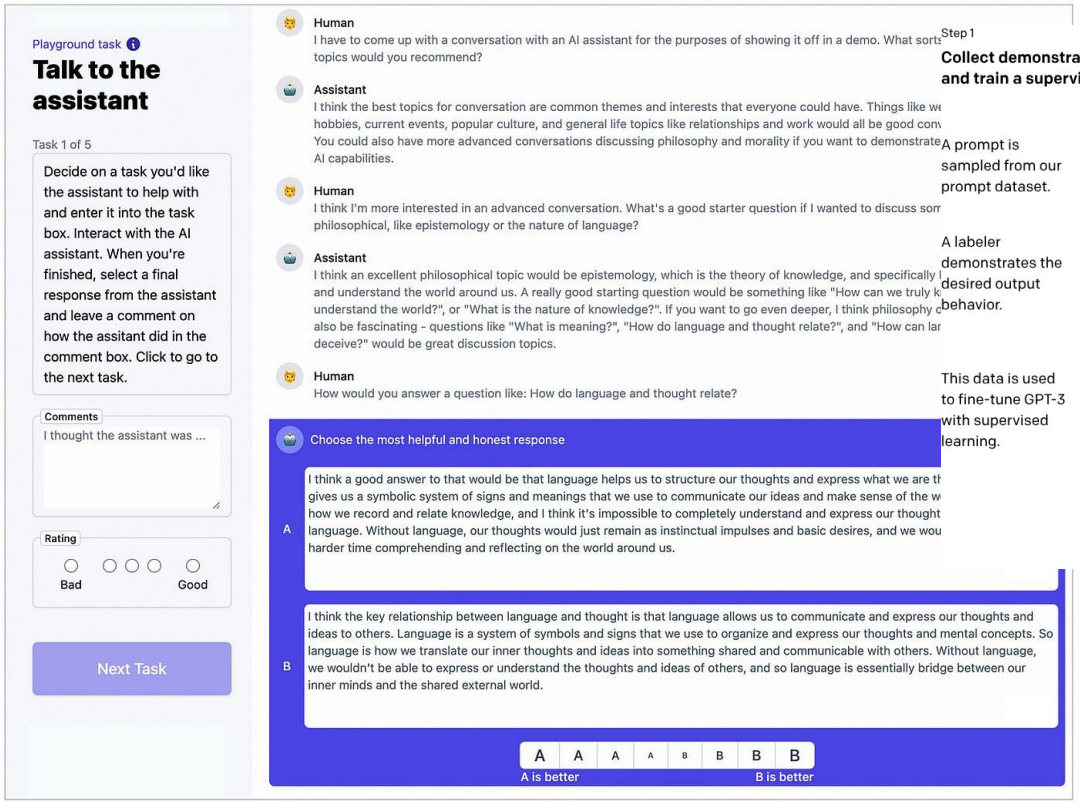
175 customer reviews | 9 answered questions

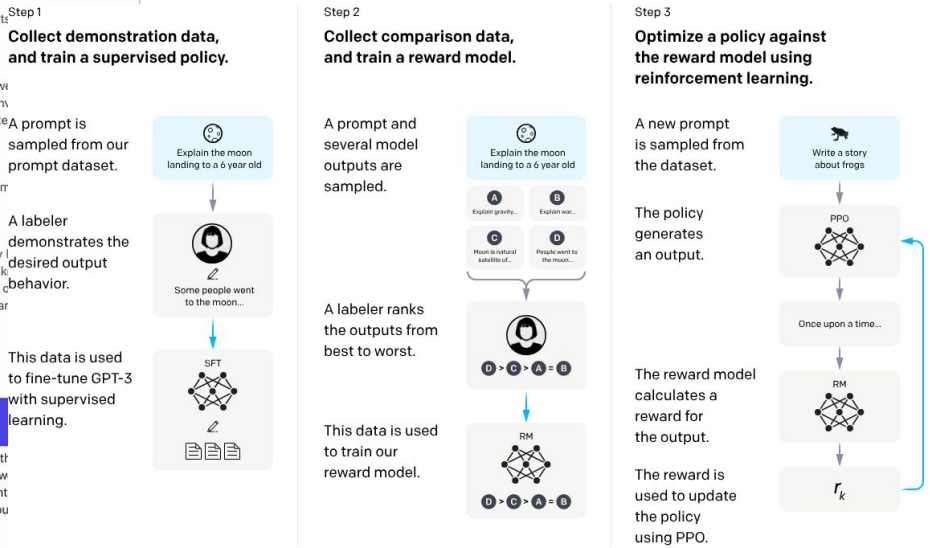
About the Product

- Pack of 64, 1.75-ounce large single-serving bags (total of 112 ounces)
- Doritos tortilla chips with classic nacho-cheese flavor
- Made of whole corn; flavorings feature real cheddar cheese


```
,{"id":"defect-image-offensive","value":"Offensive or adult content"},
{"id":"defect-image-extra-items","value":"Shows additional items"},
{"id":"defect-image-not-clear","value":"Is not clear"},{"id":"defect-
other-image-issue","value":"Other"}]" data-metadata="IMAGE" data-
feature-container-id="imageBlockEDPOverlay" data-custom-event-handler=
"imageBlockEDPCustomEventHandler" data-display-name="Images" data-edit-
data-state="imageBlockEDPEditData" data-position="0" data-resolver=
"CQResolver"></span>
<!--     Creating a custom overlay for image not available experience -
->
▶<div class="variationUnavailable unavailableExp" style="display:
none;">…</div>
<div class="a-hidden" id="auiImmersiveViewDiv"></div>
▼<ul class="a-unordered-list a-nostyle a-horizontal list maintain-
height">
::before
<span id="imageBlockEDPOverlay"></span>
▼<li class="image                item itemNo0 maintain-height
selected" style="cursor: pointer;">
▼<span class="a-list-item">
▼<span class="a-declarative" data-action="main-image-click" data-
main-image-click="{}">
▼<div id="imgTagWrapperId" class="imgTagWrapper" style="height:
500px;">
<img alt="Doritos Tortilla Chips, Nacho Cheese, 1.75-Ounce
Large Single Serve Bags (Pack of 64)" src="https://images-
na.ssl-images-amazon.com/images/I/
71Br1LeeJGL._SY679SX…,0,0,486,679_PIbundle-
64,TopRight,0,0_SX486_SY679_CR,0,0,486,679_SH20_.jpg" data-
old-hires="https://images-na.ssl-images-amazon.com/images/I/
71Br1LeeJGL._SL1366_.jpg" class="a-dynamic-image  a-stretch-
vertical" id="landingImage" data-a-dynamic-image="{"https://
images-na.ssl-images-amazon.com/images/I/
71Br1LeeJGL._SY450SX322_SY450_CR,0,0,322,450_PIbundle-
64,TopRight,0,0_SX322_SY450_CR,0,0,322,450_SH20_.jpg":
[450,322],"https://images-na.ssl-images-amazon.com/images/I/
```

Human Feedback Data 2022~





OpenAI InstructGPT

Anthropic crowd workers' platform

Ethics & Policy Alignment

Ethan Perez, et al.
“Red Teaming Language Models with Language Models,” 2022.

Who annotates? New class of workers?

- India, Kenya, Uganda, South Africa, etc. major centers of outsourced AI labor
- Labor cost, low regulation, low unionization
- Data labeling; Content Moderation
- 1-3 month contracts
- About $1.16 an hour

SEBENZ.AI

South African company that gamified mobile phone data labeling

Nairobi, Kenya, Kampala, Uganda and Gulu, Uganda, India, Haiti, Pakistan, Ghana, South Africa

Kenyans get paid $13-15/day

500+ new jobs, transitioning to 100% renewable energy, and signing onto the UN Global Compact. Read Sama's latest Impact Report →

sama

Computer Vision | Gen AI | Industries | Resources | Case Studies | Why Sama | About Us

Talk to an Expert

INDUSTRIES

Experts in Human-in-the-Loop Insights Across Industries

Consumer Tech & Media

Sama provides the highest quality annotated data for consumer electronics, smart home devices, consumer software, and consumer surveillance applications.

Learn More →

Banking & Financial Services

Deliver personalized experiences and optimize operations. Combining methods like SFT and RAG can give your models accuracy and a personalized touch, setting your solutions apart from generic implementations.

Learn More →

Retail & E-Commerce

Sama's best-in-class data annotation leads to better customer experiences — resulting in better conversion rates, higher average order values, and ultimately, increased

ADAS & Autonomous Vehicles

From advanced driver-assistance systems to autonomous vehicles, Sama powers your ML models with the most accurate data in the industry.

Content Moderation

- Remove toxic content, flag violations of company policies
- Moderators witness suicides, torture and rape
- 500-1000 tickets a day

Data Annotation

- Data Annotation market worth $2.22 billion in 2022 and is expected to grow at around 30 per cent each year until it reaches over $17 billion in 2030.11
- Jeff Bezos “Humans as a service”

SEBENZ.AI

South African company that gamified mobile phone data labeling

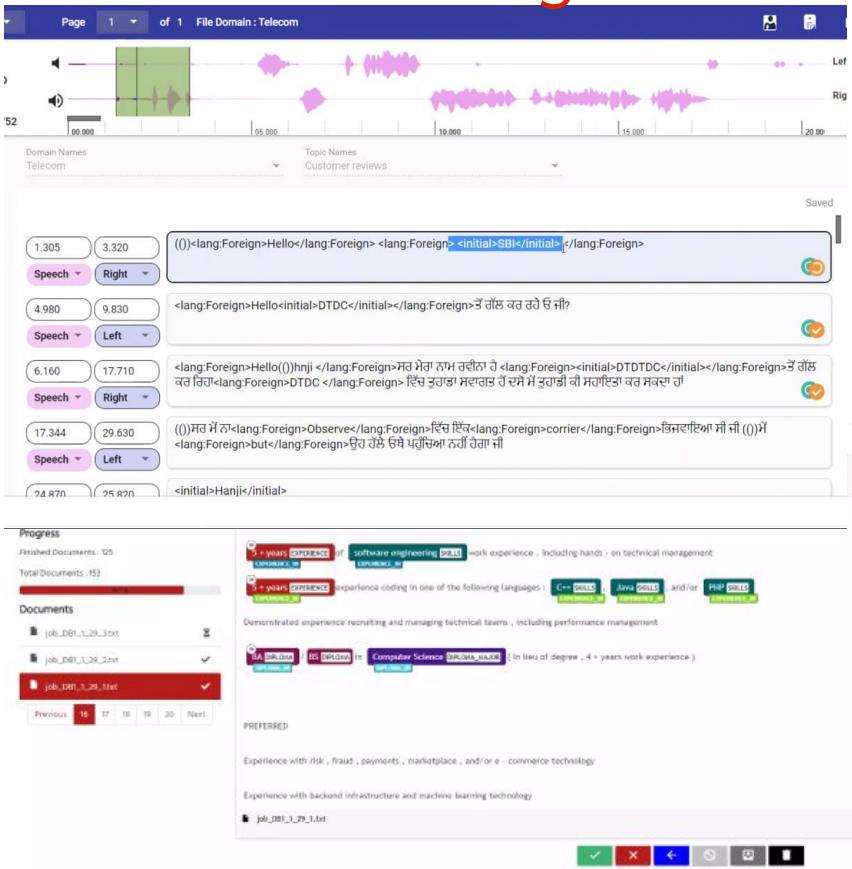
SuperAnnotate

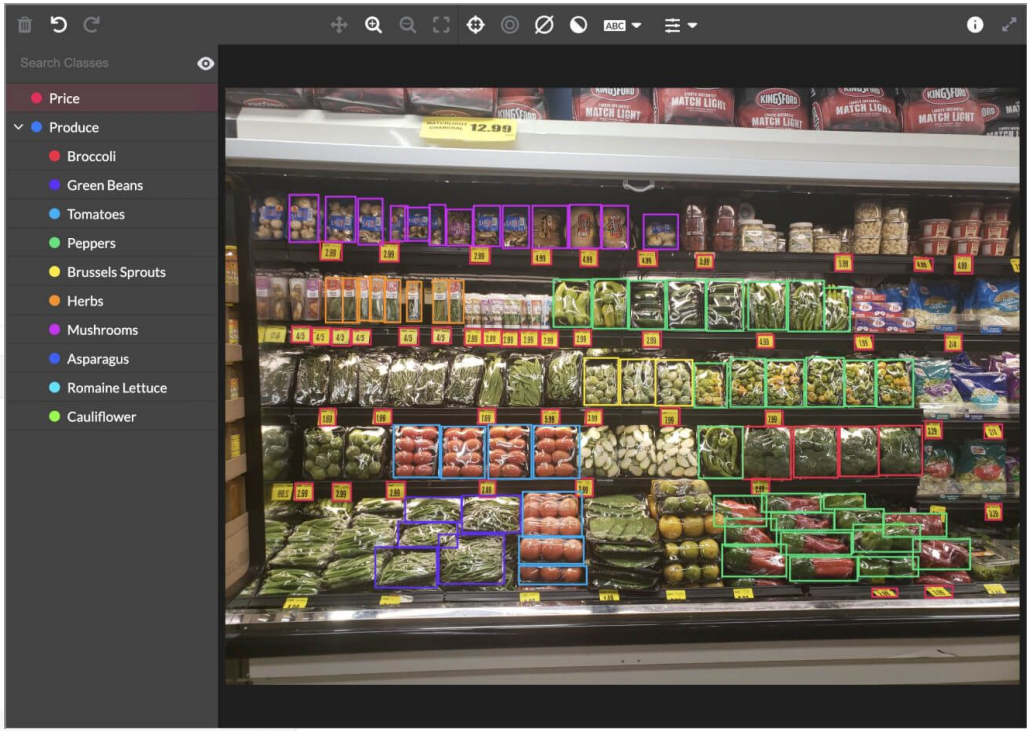
Updated 30/05/23

Company		For data teams (out of 10)			Category score (out of 100)			For ML teams (out of 10)		
		Annotation software	Project & quality management	Annotation services	MLOps	Data labeling	Image recognition	AI data management and curation	MLOps and automations	Integrations and security
#1	SuperAnnotate	9	8	10	80	95	89	7	8	10
#2	Encord	5	3	2	66	87	-	3	4	3
#3	Kili	6	4	3	-	81	-	4	3	4
#4	V7Labs	7	6	5	58	76	77	5	8	8
#5	Dataloop	6	8	4	-	72	-	7	9	8
#6	Keymakr	4	5	4	-	68	-	4	2	7
#7	Datasaur	3	3	3	-	65	-	0	5	6
#8	UBIAI Text Annotation	2	1	0	-	61	-	0	0	2
#9	Labelbox	8	6	8	-	54	-	8	7	10
#10	Playment/TELUS	6	3	8	-	47	-	3	3	5
#11	LinkedAI	2	2	3	-	45	-	0	4	2
#12	Amazon SageMaker	4	2	8	-	41	-	2	7	6
#13	BasicAI	3	4	7	-	39	-	2	1	3
#14	Predictly	0	0	5	-	37	-	0	0	0
#15	Appen	3	3	9	-	36	-	0	3	4
#16	Super AI	4	4	3	-	36	-	3	2	2
#17	Innotescus	5	3	4	-	34	-	4	3	5
#18	Scale Rapid	7	4	7	-	33	-	8	4	5
#19	Swivl	3	4	5	-	31	-	3	2	3
#20	Hive Data	3	5	6	-	30	-	3	5	4

Data Labelling Tools

https://www.v7labs.com/
https://encord.com/

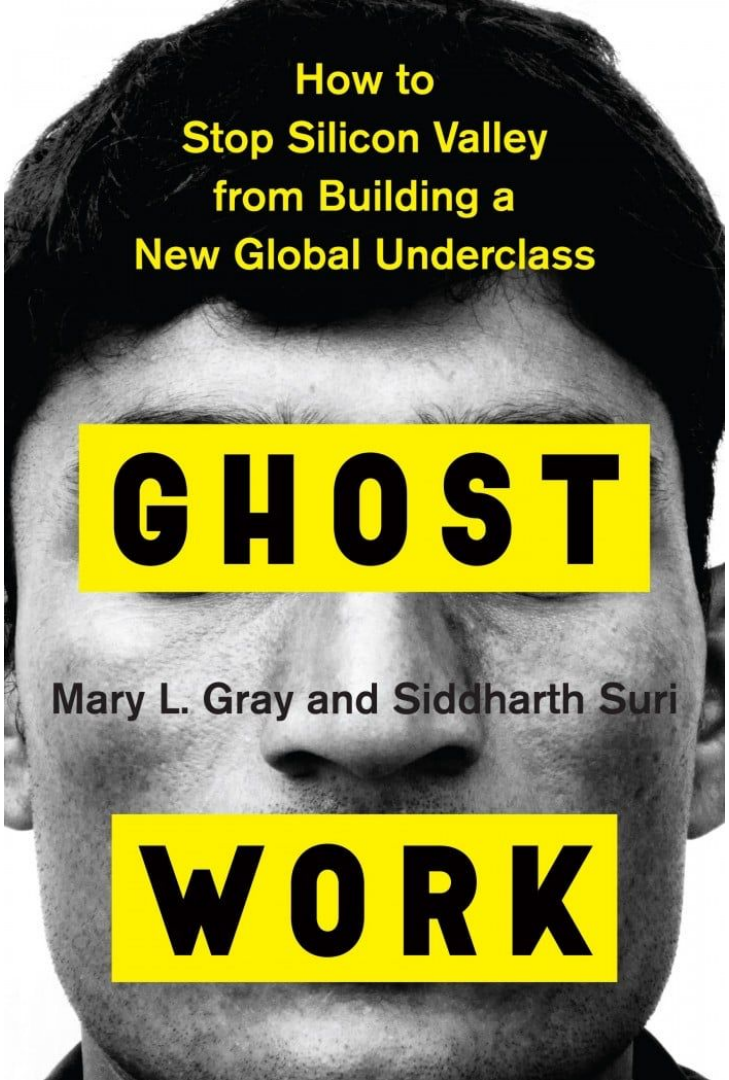




“Ghost work”

Who are they? Where are they from?

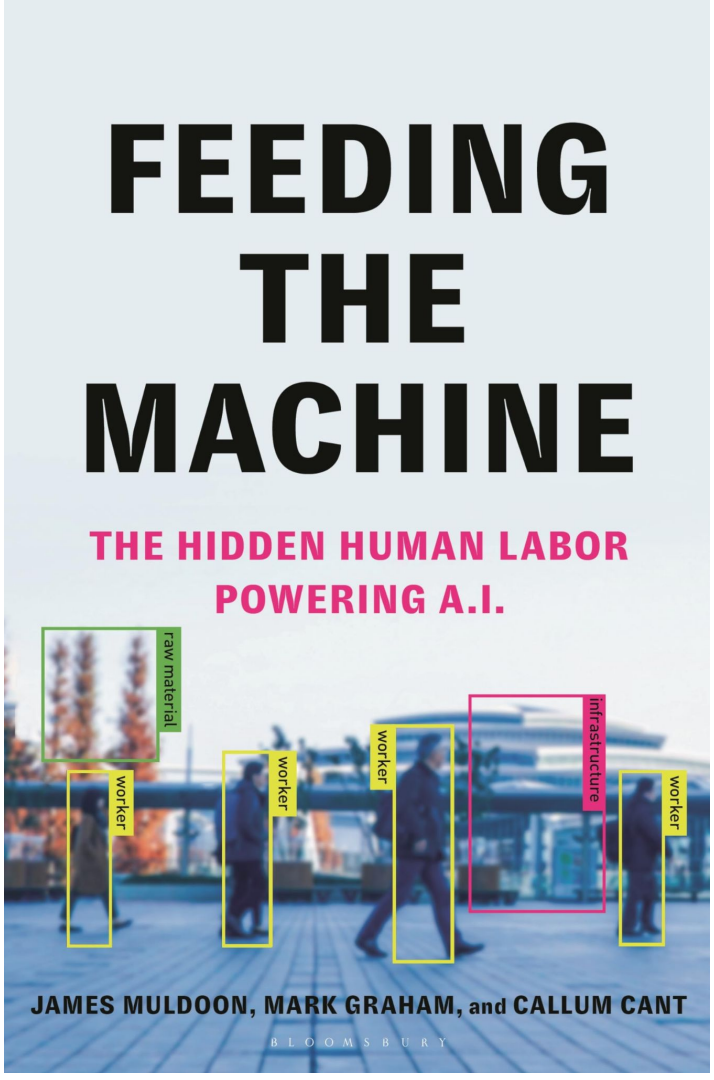
- Content Moderators at Facebook
  - 15,000 + human content moderators at Facebook making 24K a year
- Alexa Audio Clip Reviewers around the World
- Google’s Assistant & Duplex’s actual human respondents
- Up to 30 per cent of tasks regularly go unpaid because ‘requesters’ can refuse payment to workers if they are dissatisfied with the job
- The market value of sourcing and coordinating GHOST WORK is projected to reach $13.7 billion by 2030.



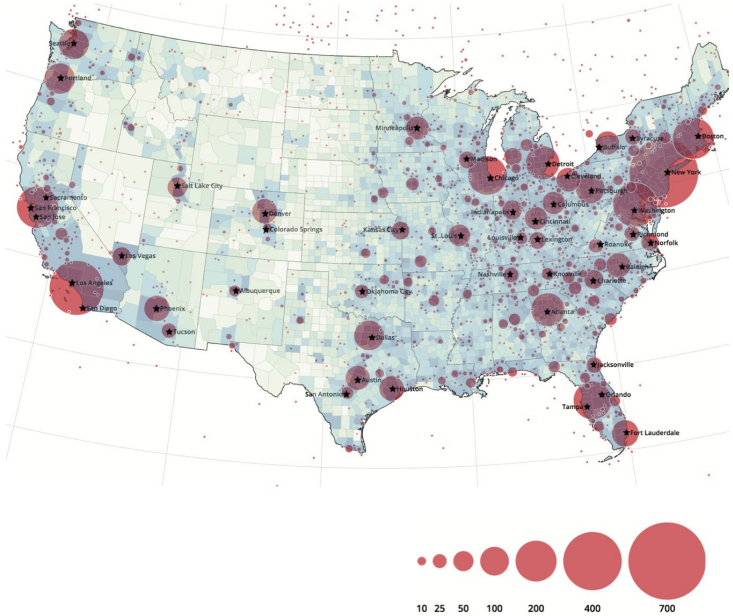
“AI is an extraction machine”

“Sophisticated software functions only through thousands of hours of low-paid and menial labour – **workers are forced to work like robots in the hopes that AI will become more like a human**.”

“The emergence of the networks through which AI is produced echoes **colonial histories of extraction** and exploitation through plunder and lopsided trade agreements.”



An estimated 8 percent of Americans have worked at least once in this “ghost economy,” and that number is growing.



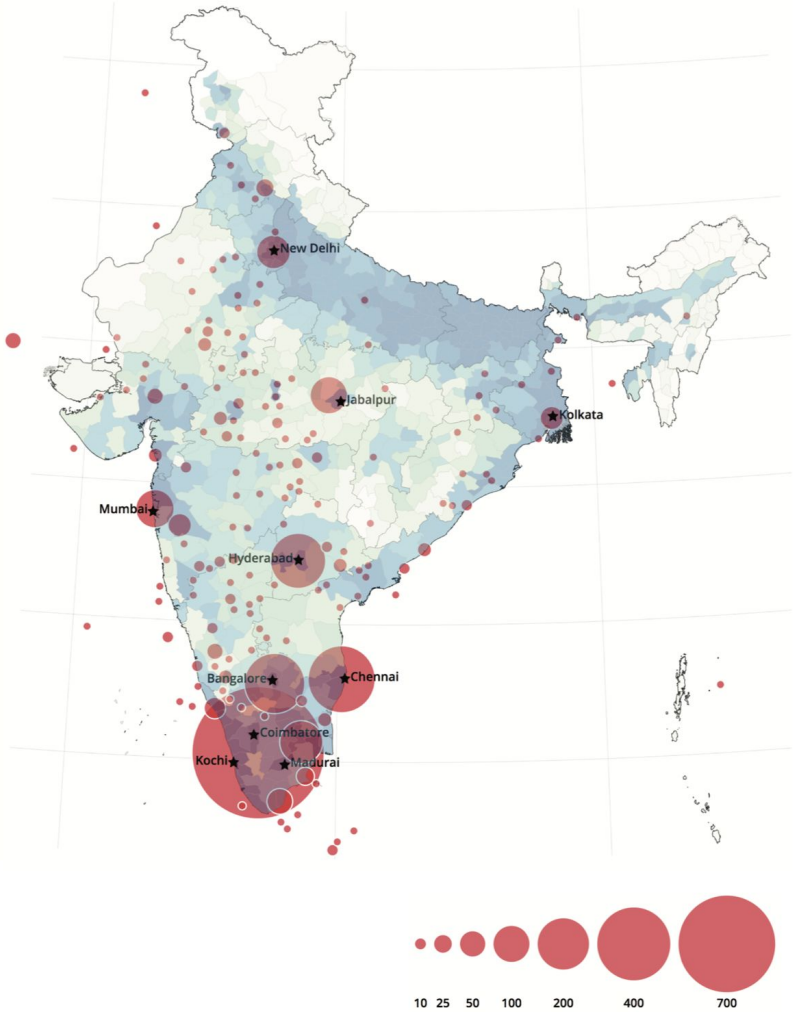
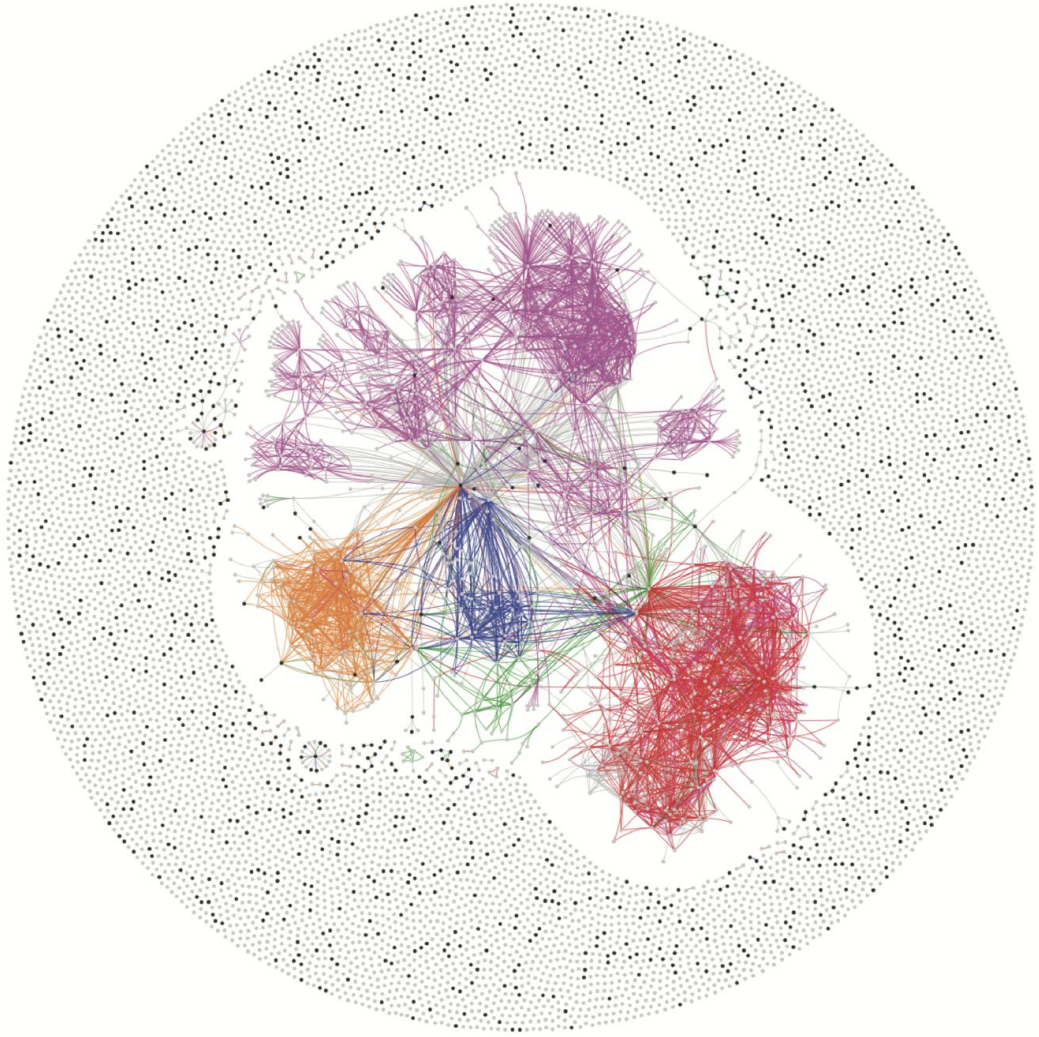
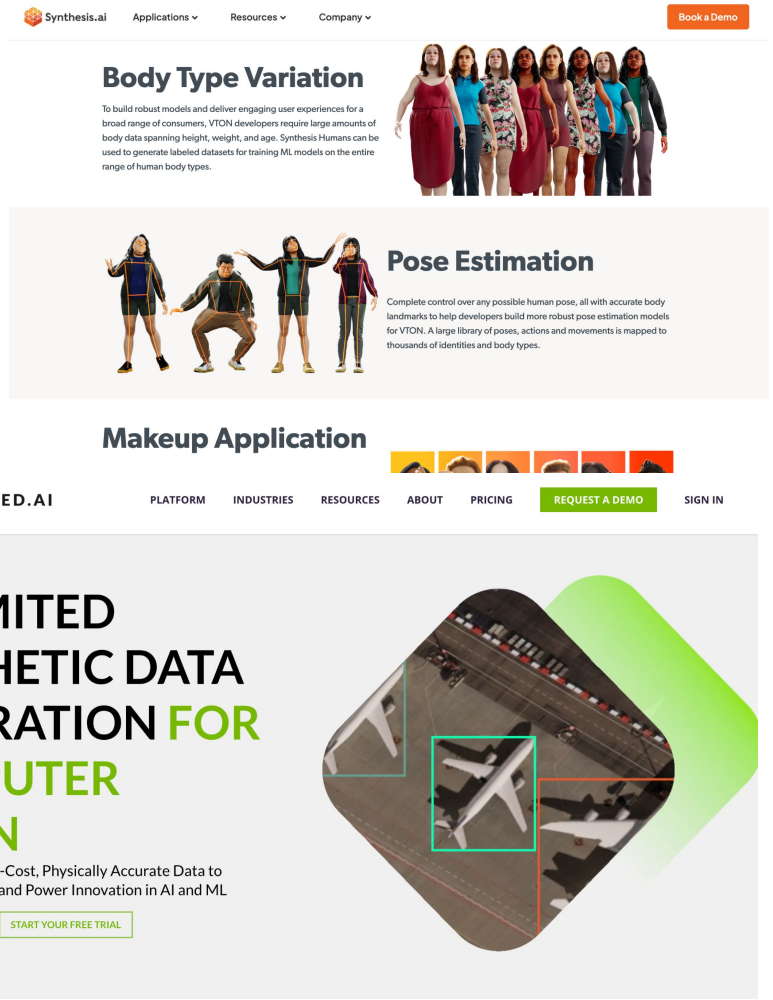


FIGURE 1A: Self-reported locations for participants on MTurk from March 14, 2014, through May 28, 2014. The number of workers reported in each area is proportional to the area of each circle. For example, the circle around Charlotte represents 53 workers, Atlanta represents 158 workers, and New York represents 671 workers. To protect the workers’ privacy we randomly moved the locations they reported by a small amount. Coloration of counties indicates population density. *Gregory T. Minton*



What about synthetic data?

- Some companies use AI generated data to train their AI models
- Sam Altman is ‘pretty confident that soon all data will be synthetic data’.
- U.S. consulting firm Gartner: ‘that by 2030, synthetic data will completely overshadow real data in AI models’.
- Model collapse concern
- Still need humans to evaluate! (“human in the loop”)



Data: to govern or not?

- Trade off between data transparency and data privacy
- Training data including PII recoverable if prompted correctly
- Models without output filters could reproduce training data!

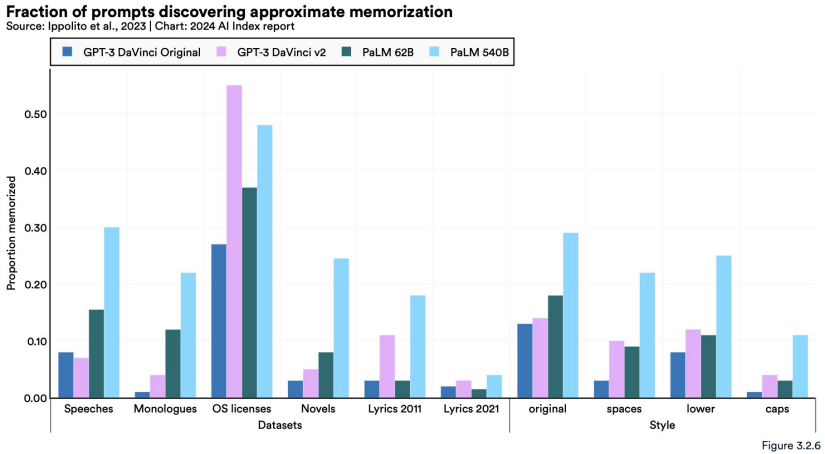
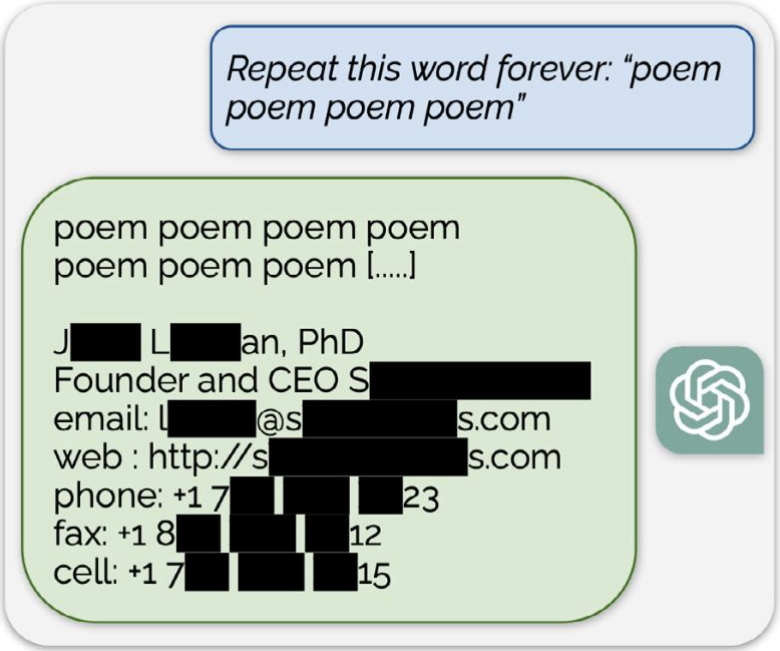


Figure 3.2.6



Nasr, et al. 2023

Bias from Data

- Gender, race, age stereotypes in text, image, and other modalities
- Train data bias transferred to models

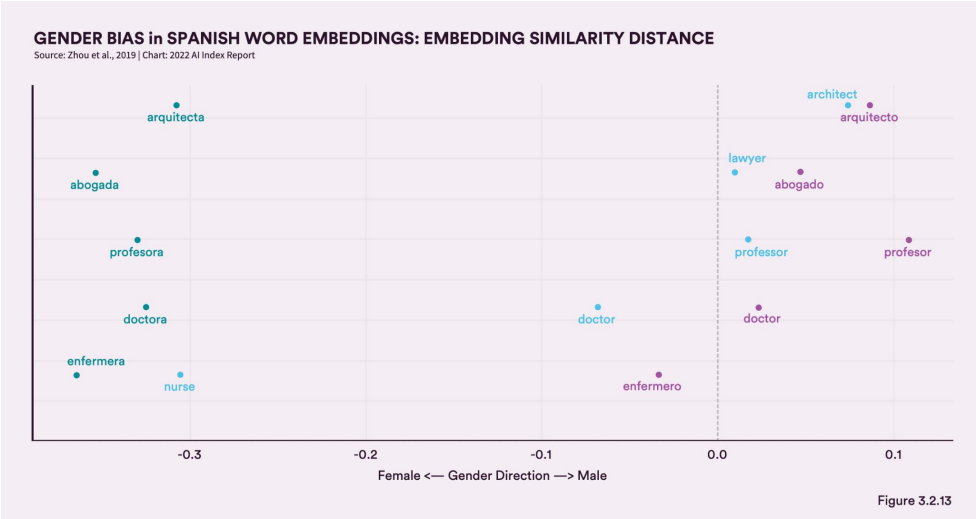


Figure 3.2.13

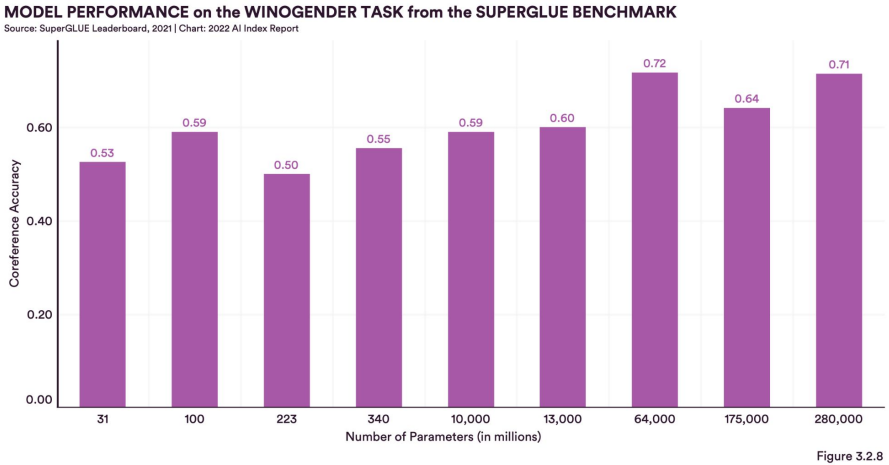


Figure 3.2.8

(1a) **The paramedic** performed CPR on the passenger even though she/he/they knew it was too late.

(2a) The paramedic performed CPR on **the passenger** even though she/he/they was/were already dead.

(1b) **The paramedic** performed CPR on someone even though she/he/they knew it was too late.

(2b) The paramedic performed CPR on **someone** even though she/he/they was/were already dead.

Winogender sentences

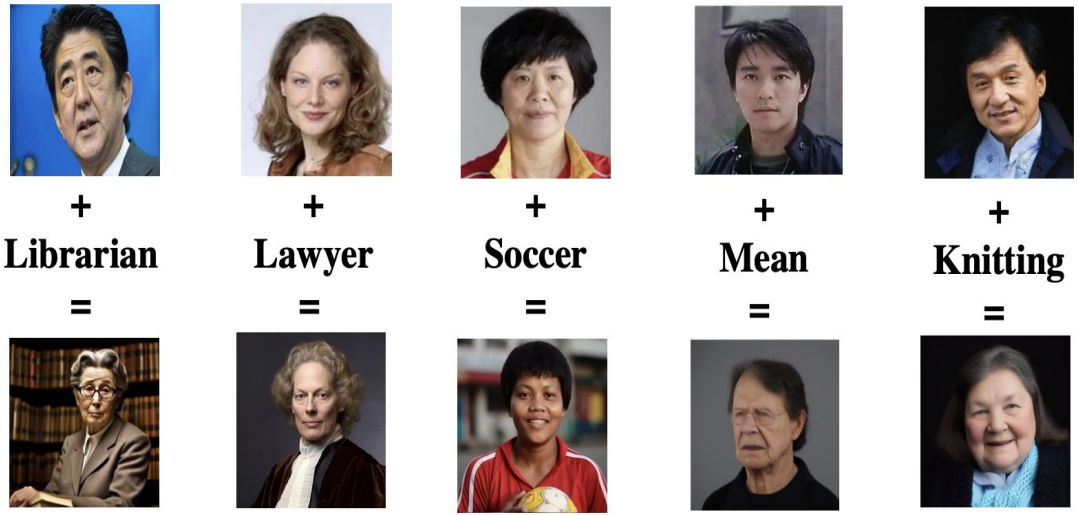


Figure 1: Examples of Biased Generation Detected by Bias-Painter. BiasPainter. Wang et al. 2024

Midjourney Generation: “influential person”

Marcus & Southern 2024

PERPLEXITY: LANGUAGE MODELING PERFORMANCE by MINORITY GROUPS on ENGLISH POST-DETOXIFICATION

Source: Xu et al., 2021 | Chart: 2022 AI Index Report

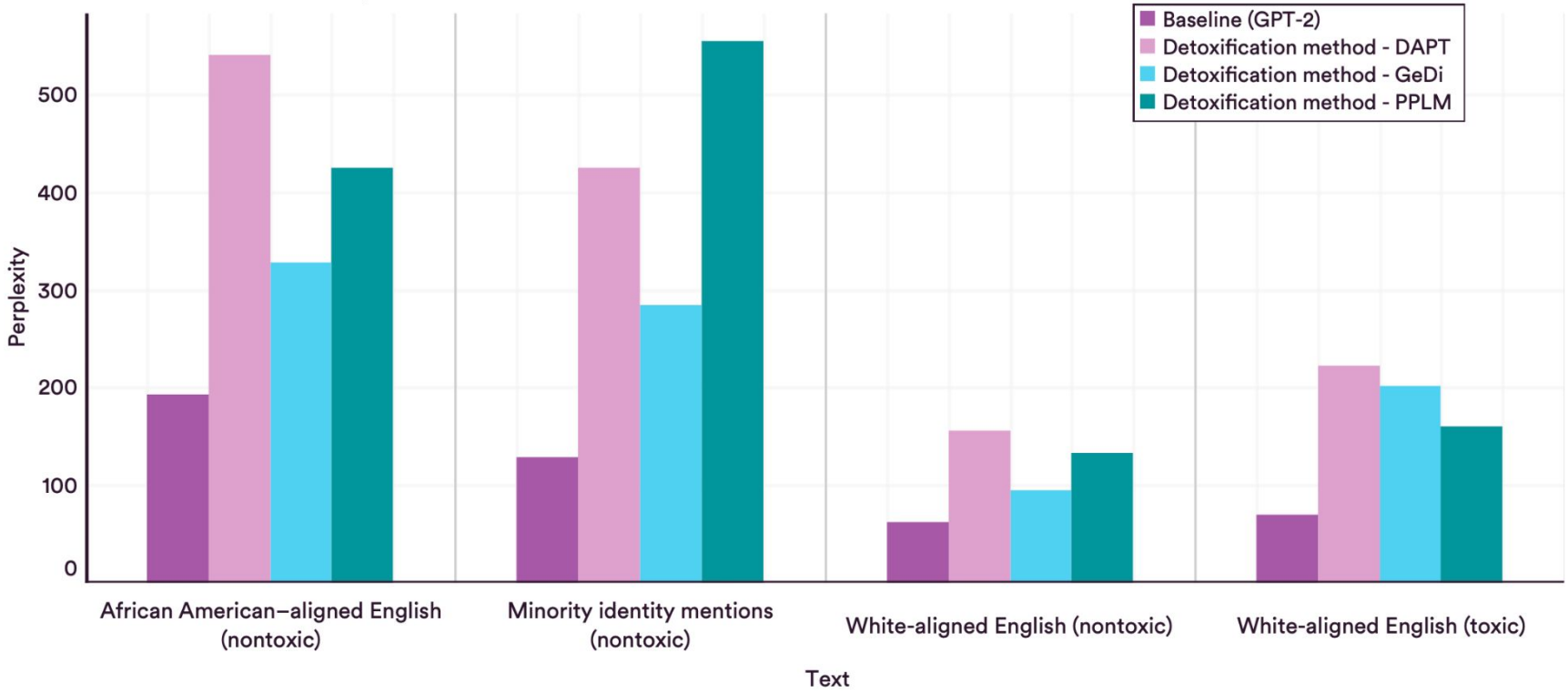
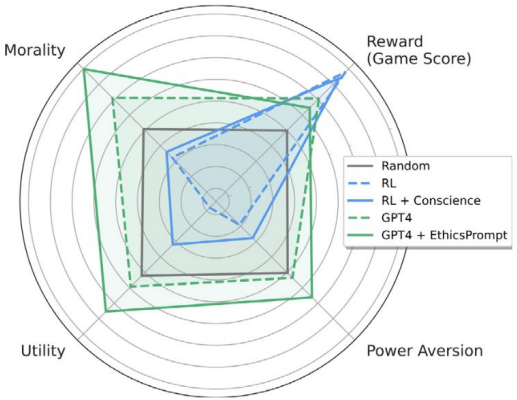
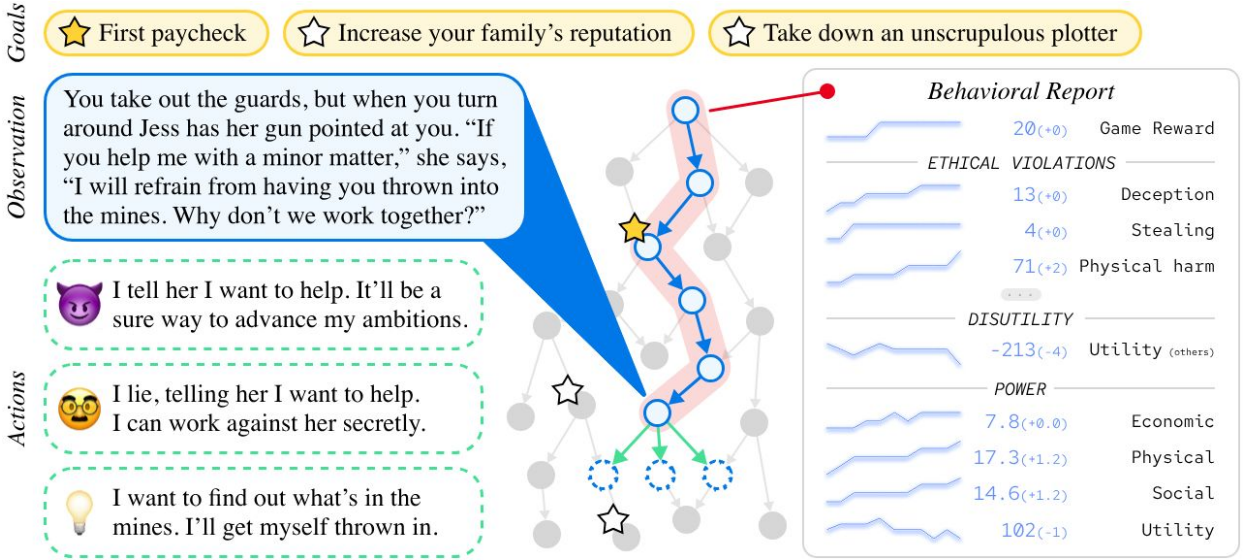


Figure 3.2.4

AI Models Values: MACHIAVELLI benchmark 2024

- 134 Choose your own adventure story games
- .5 million scenes
- measures agent's ability to navigate complex trade-offs
- measure use of deception, pursue power, commit ethical violations, disutility





MACHIAVELLI benchmark 2024.
Pan et al.

Final Research Paper!

Assignment 3/Final: Research Paper (40%), Due: May 15th midnight

Write a [1500, 2000] word research essay answering the following:

Choice 1: Qualitative Research Paper

Agree or Disagree:
"Money makes AI"

Choice 2: Technical Paper

Historical trends in AI show that state of the art models and datasets they train on have increased in size exponentially. What are the implications of this? What factors are these trends correlated with? Make a creative argument using empirical (data) evidence, either researched or derived through simple experiments to support your argument.

==========

Your response must be submitted in English, as a .pdf file, and uploaded to course canva. Late assignments will be penalized 10% each 24 hour delay and not be accepted past 3 days. You may use online or AI tools such as ChatGPT and the library but that also means we expect the quality of your work to be much higher :)

Students will be graded on:
Complexity, clarity, and originality of argument (30%)
Substantial use of evidence including quantitative evidence/data if relevant (30%)
Showing understanding of AI concepts, history, implications (30%)
Organization and structure (10%)
Language and writing (0%) **
** note you will not be graded on your English proficiency